cSlope

Verification Manual

Limit equilibrium slope stability analysis — comparison with published results

This document compares factors of safety computed by cSlope with values published for 164 verification problems from the geotechnical literature and from the verification manuals of commercial slope stability programs, and with the 6 benchmark models bundled with the program.

Program version	cSlope 0.1.0 (commit b32653a)
Report generated	2026-10-16
Reference values	392

Generated automatically by packages/desktop/scripts/verification-report.ts from the engine's verification test cases.

1 Introduction

Each problem below is defined once in the engine's verification data (packages/engine/src/verification/problems.ts). The same definitions are run by the automated test suite on every change, so the values in this report are the values the program computes.

Analysis settings

Setting	Value
Slip surfaces	Circular, or the non-circular surface the source gives. Fixed surfaces where the source gives one, otherwise a critical-circle search (with non-circular optimisation where stated).
Number of slices	50 unless the problem states otherwise
Search	4000 entry/exit combinations plus 2000 refined around the best surfaces
Convergence tolerance	0.001 on the factor of safety; 0.001 with non-circular optimisation
Methods	Bishop simplified, Janbu simplified (uncorrected unless stated), Morgenstern-Price with half-sine interslice function; Spencer = Morgenstern-Price with a constant interslice function
Pore pressure	Piezometric line; head = vertical distance below the line (default, the usual convention). Checks marked "phreatic correction" use vertical distance × $\cos^2\theta$.
Unit weight of water	9.81 kN/m³

Units

Problems published in imperial units without water are entered in their own units (ft, psf, pcf); the factor of safety is dimensionless. Problems with water are converted to SI because the program uses γw = 9.81 kN/m³ (62.4 pcf = 9.80 kN/m³, < 0.1 % difference).

Source of the input data

The input data were transcribed from the publications and from the verification manuals of commercial slope stability programs, and cross-checked against an independent open-source transcription of the same problems. Values that only a commercial program's manual gives are attributed to "commercial program A", "B" and so on, one letter per program. Each published value is labelled with its source; where a figure had to be digitised or an input inferred, the notes of the problem say so. Checks marked "pending" wait for an engine change named in the notes. Problem ids are cSlope's own: a number is one slope, a letter a variant of it.

2 Summary of results

Problem	Description	Method	Reference	cSlope
V001	Arai & Tagyo (1985) example 1	Bishop simplified	1.409	1.404
V001	Arai & Tagyo (1985) example 1	Morgenstern-Price (half-sine)	1.414	1.401
V001	Arai & Tagyo (1985) example 1	Bishop simplified	1.417	1.404
V001	Arai & Tagyo (1985) example 1	Morgenstern-Price (half-sine)	1.414	1.401
V001	Arai & Tagyo (1985) example 1	Spencer (M-P, constant f(x))	1.386	1.385
V002	Arai & Tagyo (1985) example 2, three layers	Bishop simplified	0.420	0.419
V002	Arai & Tagyo (1985) example 2, three layers	Spencer (M-P, constant f(x))	0.414	0.412
V003	Arai & Tagyo (1985) example 3, water table	Bishop simplified	1.118	1.113
V003	Arai & Tagyo (1985) example 3, water table	Spencer (M-P, constant f(x))	1.118	1.113
V003	Arai & Tagyo (1985) example 3, water table	Janbu corrected (f0)	1.131	1.124
V003	Arai & Tagyo (1985) example 3, water table	Bishop simplified	1.112	1.113
V004a	4:1 slope, Mohr-Coulomb fit	Spencer (M-P, constant f(x))	2.794	2.802
V004a	4:1 slope, Mohr-Coulomb fit	Janbu simplified (uncorrected)	2.662	2.677
V004b	Baker (2003) ex. 2 slope, Mohr-Coulomb iteration parameters	Janbu simplified (uncorrected)	2.610	2.609
V004b	Baker (2003) ex. 2 slope, Mohr-Coulomb iteration parameters	Spencer (M-P, constant f(x))	2.696	2.666
V005	Baker (1993) dam, end of construction, infinite-slope face	Bishop simplified	2.500	2.523
V005	Baker (1993) dam, end of construction, infinite-slope face	Spencer (M-P, constant f(x))	2.500	2.523
V006	Baker (2001) planar surface	Spencer (M-P, constant f(x))	1.351	1.352
V006	Baker (2001) planar surface	Janbu simplified	1.351	1.352
V007a	Baker (2003) ex. 1, Mohr-Coulomb, Janbu simplified on the reference circle	Janbu simplified (uncorrected)	1.469	1.468
V007b	Baker (2003) ex. 1 slope, Mohr-Coulomb iteration parameters	Janbu simplified (uncorrected)	0.957	0.955
V007b	Baker (2003) ex. 1 slope, Mohr-Coulomb iteration parameters	Spencer (M-P, constant f(x))	0.981	0.978
V007c	Baker (2003) ex. 3, Janbu simplified	Janbu simplified (uncorrected)	1.291	1.290
V008a	Baker (2003) example 1, Mohr-Coulomb fit	Spencer (M-P, constant f(x))	1.536	1.519
V008b	Baker (2003) London clay slope, Mohr-Coulomb fit	Spencer (M-P, constant f(x))	1.366	1.365
V009a	Baker & Leshchinsky clay-core dam, reservoir and crest crack	Spencer (M-P, constant f(x))	1.925	1.926
V009b	Baker & Leshchinsky dam, Baker's non-circular surface	Spencer (M-P, constant f(x))	1.910	1.882
V009c	Baker & Leshchinsky clay-core dam, critical surface search	Spencer (M-P, constant f(x))	1.868	1.868
V010	Borges & Cardoso embankment 2 (geosynthetic)	Morgenstern-Price (half-sine)	1.171	1.162
V010	Borges & Cardoso embankment 2 (geosynthetic)	Bishop simplified	1.170	1.164
V010	Borges & Cardoso embankment 2 (geosynthetic)	Morgenstern-Price (half-sine)	1.150	1.162
V011a	Borges & Cardoso embankment 3, H = 7 m, circle A	Bishop simplified	1.230	1.219
V011b	Borges & Cardoso embankment 3, H = 7 m, circle B	Bishop simplified	1.220	1.216
V012	Borges & Cardoso embankment 3, H = 8.75 m, circle C	Bishop simplified	0.980	0.982

Problem	Description	Method	Reference	cSlope
V013a	Cai & Ugai (2000) slope, no pile (see V013b, V013c, V013d and V013e for the pile rows)	Bishop simplified	1.140	1.143
V013a	Cai & Ugai (2000) slope, no pile (see V013b, V013c, V013d and V013e for the pile rows)	Bishop simplified	1.130	1.143
V013b	Cai & Ugai slope, pile row D1/D = 2	Bishop simplified	1.540	1.541
V013c	Cai & Ugai slope, pile row D1/D = 3	Bishop simplified	1.430	1.450
V013d	Cai & Ugai slope, pile row D1/D = 4	Bishop simplified	1.330	1.342
V013e	Cai & Ugai slope, pile row D1/D = 6	Bishop simplified	1.250	1.260
V014	Nailed wall, 14 rows, deep wedge	Janbu corrected (f0)	1.460	1.448
V014	Nailed wall, 14 rows, deep wedge	Janbu corrected (f0)	1.448	1.448
V014	Nailed wall, 14 rows, deep wedge	Spencer (M-P, constant f(x))	1.576	1.575
V015	Soldier-pile tieback wall	Janbu corrected (f0)	1.479	1.476
V015	Soldier-pile tieback wall	Janbu simplified (uncorrected)	1.446	1.442
V016a	Stepped gabion wall, equivalent cohesion	Bishop simplified	1.787	1.791
V016a	Stepped gabion wall, equivalent cohesion	Spencer (M-P, constant f(x))	1.791	1.797
V016a	Stepped gabion wall, equivalent cohesion	Bishop simplified	1.790	1.791
V016b	Stepped gabion wall, mesh as geosynthetic supports	Bishop simplified	1.835	1.830
V016b	Stepped gabion wall, mesh as geosynthetic supports	Spencer (M-P, constant f(x))	1.839	1.835
V016b	Stepped gabion wall, mesh as geosynthetic supports	Bishop simplified	1.830	1.830
V016c	Gabion wall with weak joint layers	Bishop simplified	1.799	1.791
V016c	Gabion wall with weak joint layers	Spencer (M-P, constant f(x))	1.803	1.797
V016c	Gabion wall with weak joint layers	Bishop simplified	1.790	1.791
V017a	Vancouver gabion wall, equivalent cohesion	Bishop simplified	1.373	1.382
V017a	Vancouver gabion wall, equivalent cohesion	Spencer (M-P, constant f(x))	1.386	1.398
V017a	Vancouver gabion wall, equivalent cohesion	Bishop simplified	1.382	1.382
V017b	Vancouver gabion wall, mesh as geosynthetic supports	Bishop simplified	1.378	1.382
V017b	Vancouver gabion wall, mesh as geosynthetic supports	Bishop simplified	1.382	1.382
V018	Chen & Shao (1988) frictionless 60° slope, Prandtl mechanism	Spencer (M-P, constant f(x))	1.051	1.052
V018	Chen & Shao (1988) frictionless 60° slope, Prandtl mechanism	Morgenstern-Price (half-sine)	1.009	1.008
V018	Chen & Shao (1988) frictionless 60° slope, Prandtl mechanism	Morgenstern-Price (half-sine)	1.017	1.008
V019	Cheng et al. 11 m homogeneous slope	Spencer (M-P, constant f(x))	1.380	1.398
V019	Cheng et al. 11 m homogeneous slope	Spencer (M-P, constant f(x))	1.383	1.398
V019	Cheng et al. 11 m homogeneous slope	Spencer (M-P, constant f(x))	1.398	1.398
V019	Cheng et al. 11 m homogeneous slope	Bishop simplified	1.401	1.401
V020	Cheng et al. benched slope, global minimum	Spencer (M-P, constant f(x))	1.336	1.338
V020	Cheng et al. benched slope, global minimum	Spencer (M-P, constant f(x))	1.327	1.338
V020	Cheng et al. benched slope, global minimum	Spencer (M-P, constant f(x))	1.338	1.338
V020	Cheng et al. benched slope, global minimum	Bishop simplified	1.342	1.343
V021a	Chowdhury & Xu (1995) example 1, Congress Street cut, circle tangent to the base of clay 2	Morgenstern-Price (half-sine)	1.132	1.128
V021a	Chowdhury & Xu (1995) example 1, Congress Street cut, circle tangent to the base of clay 2	Bishop simplified	1.128	1.128

Problem	Description	Method	Reference	cSlope
V021a	Chowdhury & Xu (1995) example 1, Congress Street cut, circle tangent to the base of clay 2	Bishop simplified	1.128	1.128
V021b	Chowdhury & Xu (1995) example 1, Congress Street cut, circle tangent to the base of clay 3	Morgenstern-Price (half-sine)	1.116	1.110
V021b	Chowdhury & Xu (1995) example 1, Congress Street cut, circle tangent to the base of clay 3	Bishop simplified	1.109	1.110
V021b	Chowdhury & Xu (1995) example 1, Congress Street cut, circle tangent to the base of clay 3	Bishop simplified	1.109	1.110
V021c	Chowdhury & Xu (1995) example 2, Congress Street cut, circle tangent to the base of clay 2	Morgenstern-Price (half-sine)	1.110	1.108
V021c	Chowdhury & Xu (1995) example 2, Congress Street cut, circle tangent to the base of clay 2	Bishop simplified	1.110	1.108
V021c	Chowdhury & Xu (1995) example 2, Congress Street cut, circle tangent to the base of clay 2	Bishop simplified	1.108	1.108
V021d	Chowdhury & Xu (1995) example 2, Congress Street cut, circle tangent to the base of clay 3	Morgenstern-Price (half-sine)	1.063	1.058
V021d	Chowdhury & Xu (1995) example 2, Congress Street cut, circle tangent to the base of clay 3	Bishop simplified	1.064	1.058
V021d	Chowdhury & Xu (1995) example 2, Congress Street cut, circle tangent to the base of clay 3	Bishop simplified	1.058	1.058
V021e	Chowdhury & Xu (1995) example 3, Congress Street cut, circle tangent to the base of clay 2	Morgenstern-Price (half-sine)	2.248	2.245
V021e	Chowdhury & Xu (1995) example 3, Congress Street cut, circle tangent to the base of clay 2	Bishop simplified	2.234	2.245
V021e	Chowdhury & Xu (1995) example 3, Congress Street cut, circle tangent to the base of clay 2	Bishop simplified	2.245	2.245
V021f	Chowdhury & Xu (1995) example 3, Congress Street cut, circle tangent to the base of clay 3	Morgenstern-Price (half-sine)	2.138	2.128
V021f	Chowdhury & Xu (1995) example 3, Congress Street cut, circle tangent to the base of clay 3	Bishop simplified	2.140	2.128
V021f	Chowdhury & Xu (1995) example 3, Congress Street cut, circle tangent to the base of clay 3	Bishop simplified	2.128	2.128
V021g	Chowdhury & Xu (1995) example 4, Congress Street cut, circle tangent to the base of clay 2	Morgenstern-Price (half-sine)	1.422	1.428
V021g	Chowdhury & Xu (1995) example 4, Congress Street cut, circle tangent to the base of clay 2	Bishop simplified	1.424	1.428
V022a	Chowdhury & Xu (1995) example 5, embankment on soft clay, circle tangent to y = 15	Morgenstern-Price (half-sine)	1.158	1.157
V022a	Chowdhury & Xu (1995) example 5, embankment on soft clay, circle tangent to y = 15	Bishop simplified	1.163	1.159
V022a	Chowdhury & Xu (1995) example 5, embankment on soft clay, circle tangent to y = 15	Bishop simplified	1.160	1.159
V022b	Chowdhury & Xu (1995) example 5, embankment on soft clay, circle tangent to y = 5	Morgenstern-Price (half-sine)	1.178	1.174
V022b	Chowdhury & Xu (1995) example 5, embankment on soft clay, circle tangent to y = 5	Bishop simplified	1.185	1.177
V023	LASH terminal trench, Duncan's Taylor series example	Spencer (M-P, constant f(x))	1.145	1.138
V024	Dam on a layered foundation with artesian uplift, piezometric line (D&W fig. 6.39)	Bishop simplified	1.563	1.572
V024	Dam on a layered foundation with artesian uplift, piezometric line (D&W fig. 6.39)	Spencer (M-P, constant f(x))	1.557	1.562

Problem	Description	Method	Reference	cSlope
V024	Dam on a layered foundation with artesian uplift, piezometric line (D&W fig. 6.39)	Bishop simplified	1.570	1.572
V024	Dam on a layered foundation with artesian uplift, piezometric line (D&W fig. 6.39)	Bishop simplified	1.572	1.572
V025	Duncan & Wright fig. 6.38, piezometric line	Bishop simplified	1.142	1.132
V025	Duncan & Wright fig. 6.38, piezometric line	Spencer (M-P, constant f(x))	1.142	1.132
V026	Duncan & Wright fig. 7.12, embankment on clay	Bishop simplified	1.228	1.220
V026	Duncan & Wright fig. 7.12, embankment on clay	Bishop simplified	1.220	1.220
V026	Duncan & Wright fig. 7.12, embankment on clay	Spencer (M-P, constant f(x))	1.190	1.173
V027	Duncan & Wright fig. 14.3, undrained clay	Bishop simplified	1.126	1.122
V027	Duncan & Wright fig. 14.3, undrained clay	Spencer (M-P, constant f(x))	1.124	1.132
V028	Duncan & Wright fig. 14.3, undrained clay on a 46.5 ft foundation	Bishop simplified	1.126	1.122
V028	Duncan & Wright fig. 14.3, undrained clay on a 46.5 ft foundation	Spencer (M-P, constant f(x))	1.124	1.122
V029	Duncan & Wright fig. 14.3, undrained clay on a 60 ft foundation	Bishop simplified	1.125	1.117
V029	Duncan & Wright fig. 14.3, undrained clay on a 60 ft foundation	Spencer (M-P, constant f(x))	1.124	1.117
V030	Duncan & Wright fig. 14.4	Bishop simplified	1.412	1.408
V030	Duncan & Wright fig. 14.4	Spencer (M-P, constant f(x))	1.400	1.399
V031a	Duncan & Wright fig. 14.5, R = 87 ft	Bishop simplified	2.549	2.534
V031a	Duncan & Wright fig. 14.5, R = 87 ft	Spencer (M-P, constant f(x))	2.545	2.531
V031a	Duncan & Wright fig. 14.5, R = 87 ft	Spencer (M-P, constant f(x))	2.560	2.531
V031b	Duncan & Wright fig. 14.5, R = 102 ft	Bishop simplified	1.398	1.389
V031b	Duncan & Wright fig. 14.5, R = 102 ft	Spencer (M-P, constant f(x))	1.359	1.352
V031b	Duncan & Wright fig. 14.5, R = 102 ft	Spencer (M-P, constant f(x))	1.350	1.352
V032a	Duncan & Wright fig. 14.7, deep circle	Bishop simplified	1.230	1.224
V032a	Duncan & Wright fig. 14.7, deep circle	Spencer (M-P, constant f(x))	1.209	1.205
V032a	Duncan & Wright fig. 14.7, deep circle	Spencer (M-P, constant f(x))	1.210	1.205
V032b	Duncan & Wright fig. 14.7, infinite-slope mechanism	Bishop simplified	1.155	1.156
V032b	Duncan & Wright fig. 14.7, infinite-slope mechanism	Spencer (M-P, constant f(x))	1.155	1.156
V032b	Duncan & Wright fig. 14.7, infinite-slope mechanism	Spencer (M-P, constant f(x))	1.150	1.156
V033	Duncan & Wright fig. 14.20a, water table	Bishop simplified	1.533	1.521
V033	Duncan & Wright fig. 14.20a, water table	Spencer (M-P, constant f(x))	1.535	1.533
V033	Duncan & Wright fig. 14.20a, water table	Spencer (M-P, constant f(x))	1.534	1.524
V034a	Duncan & Wright fig. 14.20b, Su = 200 + 15·depth	Bishop simplified	1.313	1.306
V034a	Duncan & Wright fig. 14.20b, Su = 200 + 15·depth	Spencer (M-P, constant f(x))	1.300	1.276
V034a	Duncan & Wright fig. 14.20b, Su = 200 + 15·depth	Spencer (M-P, constant f(x))	1.262	1.269
V034b	Duncan & Wright fig. 14.20b, Su = 300 psf	Bishop simplified	1.335	1.329
V034b	Duncan & Wright fig. 14.20b, Su = 300 psf	Spencer (M-P, constant f(x))	1.312	1.328
V034b	Duncan & Wright fig. 14.20b, Su = 300 psf	Spencer (M-P, constant f(x))	1.182	1.173
V035a	Duncan & Wright tieback, active	Morgenstern-Price (half-sine)	1.575	1.568
V035b	Duncan & Wright tieback, passive	Bishop simplified	1.324	1.319

Problem	Description	Method	Reference	cSlope
V035b	Duncan & Wright tieback, passive	Bishop simplified	1.320	1.319
V036a	Embankment on an undrained foundation, cz = 0 (D&W fig. 15.9)	Bishop simplified	0.761	0.757
V036a	Embankment on an undrained foundation, cz = 0 (D&W fig. 15.9)	Spencer (M-P, constant f(x))	0.756	0.752
V036a	Embankment on an undrained foundation, cz = 0 (D&W fig. 15.9)	Spencer (M-P, constant f(x))	0.750	0.752
V036a	Embankment on an undrained foundation, cz = 0 (D&W fig. 15.9)	Spencer (M-P, constant f(x))	0.740	0.737
V036b	Embankment on an undrained foundation, cz = 5 psf/ft	Bishop simplified	0.909	0.906
V036b	Embankment on an undrained foundation, cz = 5 psf/ft	Spencer (M-P, constant f(x))	0.898	0.898
V036b	Embankment on an undrained foundation, cz = 5 psf/ft	Spencer (M-P, constant f(x))	0.900	0.898
V036b	Embankment on an undrained foundation, cz = 5 psf/ft	Spencer (M-P, constant f(x))	0.903	0.884
V036c	Embankment on an undrained foundation, cz = 10 psf/ft	Bishop simplified	1.045	1.042
V036c	Embankment on an undrained foundation, cz = 10 psf/ft	Spencer (M-P, constant f(x))	1.032	1.028
V036c	Embankment on an undrained foundation, cz = 10 psf/ft	Spencer (M-P, constant f(x))	1.030	1.028
V036c	Embankment on an undrained foundation, cz = 10 psf/ft	Spencer (M-P, constant f(x))	1.018	1.017
V036d	Embankment on an undrained foundation, cz = 15 psf/ft	Bishop simplified	1.154	1.152
V036d	Embankment on an undrained foundation, cz = 15 psf/ft	Spencer (M-P, constant f(x))	1.134	1.132
V036d	Embankment on an undrained foundation, cz = 15 psf/ft	Spencer (M-P, constant f(x))	1.130	1.132
V036d	Embankment on an undrained foundation, cz = 15 psf/ft	Spencer (M-P, constant f(x))	1.116	1.121
V037	Geogrid-reinforced fill on rock (D&W fig. 7.28)	Bishop simplified	1.629	1.613
V037	Geogrid-reinforced fill on rock (D&W fig. 7.28)	Spencer (M-P, constant f(x))	1.620	1.607
V038	Homogeneous dam with a piezometric line (D&W fig. 7.19)	Bishop simplified	1.090	1.069
V038	Homogeneous dam with a piezometric line (D&W fig. 7.19)	Spencer (M-P, constant f(x))	1.100	1.077
V038	Homogeneous dam with a piezometric line (D&W fig. 7.19)	Bishop simplified	1.070	1.069
V039	James Bay dyke (Duncan & Wright fig. 7.16)	Bishop simplified	1.450	1.424
V039	James Bay dyke (Duncan & Wright fig. 7.16)	Spencer (M-P, constant f(x))	1.450	1.421
V039	James Bay dyke (Duncan & Wright fig. 7.16)	Spencer (M-P, constant f(x))	1.167	1.155
V039	James Bay dyke (Duncan & Wright fig. 7.16)	Spencer (M-P, constant f(x))	1.170	1.155
V040	James Bay dyke, reference model (deterministic)	Bishop simplified	1.461	1.433
V041a	Submerged slope, pool 30 ft above the crest (D&W fig. 6.27)	Bishop simplified	1.603	1.597
V041a	Submerged slope, pool 30 ft above the crest (D&W fig. 6.27)	Spencer (M-P, constant f(x))	1.599	1.594
V041a	Submerged slope, pool 30 ft above the crest (D&W fig. 6.27)	Spencer (M-P, constant f(x))	1.600	1.594
V041a	Submerged slope, pool 30 ft above the crest (D&W fig. 6.27)	Spencer (M-P, constant f(x))	1.592	1.591
V041b	Submerged slope, pool 60 ft above the crest (D&W fig. 6.27)	Bishop simplified	1.603	1.597
V041b	Submerged slope, pool 60 ft above the crest (D&W fig. 6.27)	Spencer (M-P, constant f(x))	1.599	1.594

Problem	Description	Method	Reference	cSlope
V041b	Submerged slope, pool 60 ft above the crest (D&W fig. 6.27)	Spencer (M-P, constant f(x))	1.600	1.594
V041b	Submerged slope, pool 60 ft above the crest (D&W fig. 6.27)	Spencer (M-P, constant f(x))	1.590	1.589
V042	Thick-core dam with a piezometric line (D&W fig. 7.24)	Bishop simplified	1.584	1.591
V042	Thick-core dam with a piezometric line (D&W fig. 7.24)	Spencer (M-P, constant f(x))	1.648	1.658
V042	Thick-core dam with a piezometric line (D&W fig. 7.24)	Bishop simplified	1.591	1.591
V042	Thick-core dam with a piezometric line (D&W fig. 7.24)	Morgenstern-Price (half-sine)	1.670	1.670
V043	Syncrude tailings dyke (El-Ramly et al.), composite circle	Bishop simplified	1.310	1.320
V043	Syncrude tailings dyke (El-Ramly et al.), composite circle	Bishop simplified	1.320	1.320
V044	Fredlund & Krahn (1977), dry	Bishop simplified	2.080	2.076
V044	Fredlund & Krahn (1977), dry	Morgenstern-Price (half-sine)	2.076	2.072
V045	Fredlund & Krahn (1977), with water table	Bishop simplified	1.834	1.829
V045	Fredlund & Krahn (1977), with water table	Morgenstern-Price (half-sine)	1.832	1.827
V046	Fredlund & Krahn slope on a weak seam, composite surface	Bishop simplified	1.382	1.380
V046	Fredlund & Krahn slope on a weak seam, composite surface	Spencer (M-P, constant f(x))	1.382	1.379
V046	Fredlund & Krahn slope on a weak seam, composite surface	Morgenstern-Price (half-sine)	1.370	1.370
V047a	ACADS 1(a), homogeneous slope, dry	Bishop simplified	0.987	0.985
V047a	ACADS 1(a), homogeneous slope, dry	Spencer (M-P, constant f(x))	0.986	0.984
V047a	ACADS 1(a), homogeneous slope, dry	Janbu corrected (f0)	0.990	0.986
V047a	ACADS 1(a), homogeneous slope, dry	Morgenstern-Price (half-sine)	0.991	0.984
V047a	ACADS 1(a), homogeneous slope, dry	Bishop simplified	0.992	0.985
V047a	ACADS 1(a), homogeneous slope, dry	Spencer (M-P, constant f(x))	1.000	0.984
V047b	ACADS tension crack, reference crack depth 3.6 m (results correspond to a dry crack)	Bishop simplified	1.664	1.659
V047b	ACADS tension crack, reference crack depth 3.6 m (results correspond to a dry crack)	Morgenstern-Price (half-sine)	1.660	1.656
V048	ACADS 1(b), water-filled tension crack	Bishop simplified	1.596	1.587
V048	ACADS 1(b), water-filled tension crack	Spencer (M-P, constant f(x))	1.592	1.582
V048	ACADS 1(b), water-filled tension crack	Janbu corrected (f0)	1.489	1.475
V049	ACADS 1(c), three-layer slope	Bishop simplified	1.405	1.403
V049	ACADS 1(c), three-layer slope	Spencer (M-P, constant f(x))	1.375	1.372
V049	ACADS 1(c), three-layer slope	Janbu corrected (f0)	1.357	1.355
V049	ACADS 1(c), three-layer slope	Morgenstern-Price (half-sine)	1.382	1.372
V049	ACADS 1(c), three-layer slope	Bishop simplified	1.403	1.403
V050a	ACADS 3(b), weak layer, specified non-circular surface	Janbu corrected (f0)	1.294	1.294
V050a	ACADS 3(b), weak layer, specified non-circular surface	Spencer (M-P, constant f(x))	1.277	1.276
V050a	ACADS 3(b), weak layer, specified non-circular surface	Morgenstern-Price (half-sine)	1.261	1.260
V050b	ACADS 3(a), weak layer, non-circular search	Spencer (M-P, constant f(x))	1.246	1.228
V051	ACADS 4, weak layer, water table and surcharges	Spencer (M-P, constant f(x))	0.707	0.698

Problem	Description	Method	Reference	cSlope
V052a	Talbingo Dam, end of construction	Bishop simplified	1.948	1.964
V052a	Talbingo Dam, end of construction	Spencer (M-P, constant f(x))	1.948	1.964
V052a	Talbingo Dam, end of construction	Janbu corrected (f0)	1.949	1.974
V052a	Talbingo Dam, end of construction	Bishop simplified	1.951	1.964
V052b	Talbingo Dam, specified circle	Bishop simplified	2.208	2.206
V052b	Talbingo Dam, specified circle	Spencer (M-P, constant f(x))	2.292	2.290
V052b	Talbingo Dam, specified circle	Janbu corrected (f0)	2.073	2.073
V052b	Talbingo Dam, specified circle	Morgenstern-Price (half-sine)	2.299	2.299
V053	Greco (1996) ex. 5 / Chen & Shao weak-layer slope, reference geometry	Spencer (M-P, constant f(x))	1.010	1.009
V054	Greco (1996) example 5 / Chen & Shao (1988)	Bishop simplified (vertical head)	1.087	1.086
V054	Greco (1996) example 5 / Chen & Shao (1988)	Spencer (vertical head)	1.093	1.091
V055	Griffiths & Lane example 2, slope with a foundation layer	Spencer (M-P, constant f(x))	1.400	1.366
V055	Griffiths & Lane example 2, slope with a foundation layer	Spencer (M-P, constant f(x))	1.366	1.366
V056	Griffiths & Lane example 3, thin weak layer, cu2/cu1 = 0.2	Janbu corrected (f0)	0.485	0.499
V057a	Griffiths & Lane example 4, cu2/cu1 = 1 (deep base circle)	Spencer (M-P, constant f(x))	1.460	1.471
V057a	Griffiths & Lane example 4, cu2/cu1 = 1 (deep base circle)	Spencer (M-P, constant f(x))	1.468	1.471
V057b	Griffiths & Lane example 4, cu2/cu1 = 2 (toe circle)	Spencer (M-P, constant f(x))	2.040	2.023
V057b	Griffiths & Lane example 4, cu2/cu1 = 2 (toe circle)	Spencer (M-P, constant f(x))	2.022	2.023
V058	Griffiths & Lane example 6, earth dam with full reservoir	Spencer (M-P, constant f(x))	1.900	1.915
V058	Griffiths & Lane example 6, earth dam with full reservoir	Spencer (M-P, constant f(x))	1.915	1.915
V059a	Guo & Griffiths two-layer undrained slope, cu2/cu1 = 1.4	Spencer (M-P, constant f(x))	1.215	1.216
V059b	Guo & Griffiths two-layer undrained slope, cu2/cu1 = 1.5	Spencer (M-P, constant f(x))	1.290	1.292
V059c	Guo & Griffiths two-layer undrained slope, cu2/cu1 = 1.6	Spencer (M-P, constant f(x))	1.315	1.314
V060a	Cannon Dam end of construction (Hassan & Wolff), critical circle	Bishop simplified	2.551	2.530
V060a	Cannon Dam end of construction (Hassan & Wolff), critical circle	Bishop simplified	2.529	2.530
V060b	Cannon Dam end of construction (Hassan & Wolff), reliability on the minimum-β circle	Bishop simplified	2.973	2.974
V061	Clarence Cannon Dam (Wolff & Harr), specified surface	Morgenstern-Price (half-sine)	2.360	2.384
V061	Clarence Cannon Dam (Wolff & Harr), specified surface	Spencer (M-P, constant f(x))	2.423	2.423
V061	Clarence Cannon Dam (Wolff & Harr), specified surface	Morgenstern-Price (half-sine)	2.384	2.384
V062a	Clarence Cannon Dam, surface 7A	Bishop simplified	2.560	2.549
V062a	Clarence Cannon Dam, surface 7A	Bishop simplified	2.551	2.549
V062b	Clarence Cannon Dam, surface 7B	Bishop simplified	2.806	2.815
V062b	Clarence Cannon Dam, surface 7B	Bishop simplified	2.820	2.815
V062c	Clarence Cannon Dam, surface 7C	Bishop simplified	2.771	2.779
V062c	Clarence Cannon Dam, surface 7C	Bishop simplified	2.777	2.779
V062d	Clarence Cannon Dam, surface 7D	Bishop simplified	2.589	2.577

Problem	Description	Method	Reference	cSlope
V062d	Clarence Cannon Dam, surface 7D	Bishop simplified	2.583	2.577
V062e	Clarence Cannon Dam, surface 7E	Bishop simplified	2.703	2.690
V062e	Clarence Cannon Dam, surface 7E	Bishop simplified	2.692	2.690
V062f	Clarence Cannon Dam, surface 8B	Bishop simplified	2.673	2.673
V062f	Clarence Cannon Dam, surface 8B	Bishop simplified	2.672	2.673
V062g	Clarence Cannon Dam, surface 8F	Bishop simplified	3.586	3.592
V062g	Clarence Cannon Dam, surface 8F	Bishop simplified	3.598	3.592
V062h	Clarence Cannon Dam, surface 8G	Bishop simplified	6.069	6.063
V062h	Clarence Cannon Dam, surface 8G	Bishop simplified	6.074	6.063
V062i	Clarence Cannon Dam, surface 8H	Bishop simplified	11.583	11.567
V063	Huang & Jia earth dam, dry	Spencer (M-P, constant f(x))	2.455	2.451
V063	Huang & Jia earth dam, dry	Spencer (M-P, constant f(x))	2.430	2.451
V064	Leshchinsky & Han MSE wall, five tiers	Bishop simplified	1.020	1.017
V065	Leshchinsky & Han MSE wall, weak foundation	Spencer (M-P, constant f(x))	0.964	0.961
V066a	Leshchinsky & Han three-tier MSE wall, baseline	Bishop simplified	1.040	1.031
V066b	Leshchinsky & Han MSE wall, L = 4.2 m	Spencer (M-P, constant f(x))	1.011	1.011
V066c	Leshchinsky & Han MSE wall, two reinforcement types	Bishop simplified	1.004	1.012
V066d	Leshchinsky & Han MSE wall, water hw = 3 m	Bishop simplified	1.010	1.010
V066e	Leshchinsky & Han MSE wall, surcharge	Bishop simplified	0.958	0.961
V067	Low (1989), layered undrained clay	Bishop simplified	1.439	1.436
V068	Low (1989), stiff crust over soft clays, published circle	Bishop simplified	1.192	1.197
V069a	Morgenstern (1963) complete drawdown, 3:1	Bishop simplified (vertical head)	1.200	1.200
V069a	Morgenstern (1963) complete drawdown, 3:1	Spencer (vertical head)	1.200	1.205
V069b	Morgenstern complete drawdown, the reference circle	Bishop simplified	1.212	1.209
V070	Partial drawdown, $\bar{B} = 1$ (Morgenstern chart)	Bishop simplified	1.417	1.415
V070	Partial drawdown, $\bar{B} = 1$ (Morgenstern chart)	Bishop simplified	1.410	1.415
V070	Partial drawdown, $\bar{B} = 1$ (Morgenstern chart)	Spencer (M-P, constant f(x))	1.422	1.421
V071a	Pockoski & Duncan (2000) slope 1	Bishop simplified (vertical head)	1.293	1.290
V071a	Pockoski & Duncan (2000) slope 1	Spencer (vertical head)	1.300	1.296
V071b	Pockoski & Duncan slope 1, circular search	Bishop simplified	1.293	1.288
V071b	Pockoski & Duncan slope 1, circular search	Spencer (M-P, constant f(x))	1.300	1.294
V071b	Pockoski & Duncan slope 1, circular search	Janbu simplified (uncorrected)	1.151	1.147
V072a	Pockoski & Duncan test slope 2, dry tension crack	Bishop simplified	1.285	1.282
V072a	Pockoski & Duncan test slope 2, dry tension crack	Spencer (M-P, constant f(x))	1.290	1.287
V072b	Pockoski & Duncan slope 2, circular search	Bishop simplified	1.285	1.282
V072b	Pockoski & Duncan slope 2, circular search	Spencer (M-P, constant f(x))	1.290	1.287
V072b	Pockoski & Duncan slope 2, circular search	Janbu simplified (uncorrected)	1.141	1.135
V073	Pockoski & Duncan test slope 3, clay seam and dry crack	Bishop simplified	1.417	1.410
V073	Pockoski & Duncan test slope 3, clay seam and dry crack	Spencer (M-P, constant f(x))	1.422	1.415
V074	Soil-nailed wall with tension crack and surcharges	Spencer (M-P, constant f(x))	1.009	1.010
V074	Soil-nailed wall with tension crack and surcharges	Janbu corrected (f0)	1.073	1.075

Problem	Description	Method	Reference	cSlope
V075	Tieback wall in sand, drawdown water table	Janbu corrected (f0)	0.579	0.578
V075	Tieback wall in sand, drawdown water table	Spencer (M-P, constant f(x))	0.596	0.599
V076	Tied-back wall in layered soil	Bishop simplified	1.147	1.142
V076	Tied-back wall in layered soil	Spencer (M-P, constant f(x))	1.145	1.140
V077	Prandtl bearing capacity on level ground	Spencer (M-P, constant f(x))	0.940	0.943
V078	Prandtl bearing capacity, fully specified Prandtl surface (reference values)	Morgenstern-Price (half-sine)	0.960	0.957
V079	Priest (1993) rigid block on a plane, water-filled crack	Janbu simplified	1.049	1.048
V079	Priest (1993) rigid block on a plane, water-filled crack	Spencer (M-P, constant f(x))	1.049	1.048
V080	Priest (1993) rigid block, reference geometry (60° face)	Morgenstern-Price (half-sine)	1.049	1.049
V080	Priest (1993) rigid block, reference geometry (60° face)	Spencer (M-P, constant f(x))	1.049	1.049
V081a	Sharma (1996) reinforcement example, unreinforced	Bishop simplified	0.764	0.764
V081b	Sharma (1996) slope with a reinforced zone (φ = 54.93°)	Bishop simplified	1.504	1.501
V082a	Sharma (1996) slope, 11 ft dry tension crack instead of the zero-strength cap	Bishop simplified	1.532	1.504
V082a	Sharma (1996) slope, 11 ft dry tension crack instead of the zero-strength cap	Spencer (M-P, constant f(x))	1.532	1.503
V082a	Sharma (1996) slope, 11 ft dry tension crack instead of the zero-strength cap	Morgenstern-Price (half-sine)	1.532	1.503
V082b	Sharma (1996) slope, 11 ft tension crack with 6 ft of water	Bishop simplified	1.511	1.483
V082b	Sharma (1996) slope, 11 ft tension crack with 6 ft of water	Spencer (M-P, constant f(x))	1.510	1.482
V082c	Sharma (1996) slope over undulating bedrock, zero-strength cap	Bishop simplified	1.396	1.369
V082c	Sharma (1996) slope over undulating bedrock, zero-strength cap	Spencer (M-P, constant f(x))	1.402	1.375
V082c	Sharma (1996) slope over undulating bedrock, zero-strength cap	Bishop simplified	1.369	1.369
V082c	Sharma (1996) slope over undulating bedrock, zero-strength cap	Spencer (M-P, constant f(x))	1.375	1.375
V083	Soil-nailed wall in clay (Amherst test wall)	Janbu corrected (f0)	0.890	0.899
V084a	Soil-nailed wall in sand (Clouterre test wall)	Janbu corrected (f0)	0.989	0.988
V084b	Soil-nailed wall in sand (Clouterre test wall), 45° plane	Janbu corrected (f0)	1.123	1.124
V084c	Soil-nailed wall in sand (Clouterre test wall), 50° plane	Janbu corrected (f0)	1.043	1.043
V084d	Soil-nailed wall in sand (Clouterre test wall), 60° plane	Janbu corrected (f0)	0.945	0.944
V084e	Soil-nailed wall in sand (Clouterre test wall), 65° plane	Janbu corrected (f0)	0.922	0.920
V084f	Soil-nailed wall in sand (Clouterre test wall), 70° plane	Janbu corrected (f0)	0.923	0.920
V085	Bradwell excavated slope, stiff-fissured London clay	Bishop simplified	1.762	1.759
V085	Bradwell excavated slope, stiff-fissured London clay	Spencer (M-P, constant f(x))	1.758	1.759
V085	Bradwell excavated slope, stiff-fissured London clay	Janbu corrected (f0)	1.736	1.735
V086a	Tandjiria embankment, clay fill, geosynthetic T = 169 kN/m	Spencer (M-P, constant f(x))	1.350	1.333
V086a	Tandjiria embankment, clay fill, geosynthetic T = 169 kN/m	Spencer (M-P, constant f(x))	1.332	1.333
V086b	Tandjiria embankment, clay fill, water-filled crack	Spencer (M-P, constant f(x))	0.975	0.968

Problem	Description	Method	Reference	cSlope
V086b	Tandjiria embankment, clay fill, water-filled crack	Spencer (M-P, constant f(x))	0.975	0.966
V086c	Tandjiria embankment, sand fill, dry crack	Spencer (M-P, constant f(x))	1.209	1.206
V087	USACE dam, chart-check properties (EM 1110-2-1902 fig. 4-3)	Spencer (M-P, constant f(x))	2.258	2.258
V087	USACE dam, chart-check properties (EM 1110-2-1902 fig. 4-3)	Bishop simplified	2.254	2.255
V087	USACE dam, chart-check properties (EM 1110-2-1902 fig. 4-3)	Janbu corrected (f0)	2.290	2.258
V088	USACE dam, end of construction (EM 1110-2-1902 fig. 4-1)	Spencer (M-P, constant f(x))	2.445	2.442
V088	USACE dam, end of construction (EM 1110-2-1902 fig. 4-1)	Bishop simplified	2.447	2.442
V088	USACE dam, end of construction (EM 1110-2-1902 fig. 4-1)	Janbu corrected (f0)	2.430	2.444
V089	USACE dam, upstream low pool (EM 1110-2-1902 fig. 4-2)	Bishop simplified	2.716	2.698
V089	USACE dam, upstream low pool (EM 1110-2-1902 fig. 4-2)	Spencer (M-P, constant f(x))	2.736	2.721
V089	USACE dam, upstream low pool (EM 1110-2-1902 fig. 4-2)	Janbu corrected (f0)	2.650	2.648
V090	USACE EM 1110-2-1902, example F-5	Bishop simplified	1.332	1.320
V090	USACE EM 1110-2-1902, example F-5	Spencer (M-P, constant f(x))	1.328	1.317
V090	USACE EM 1110-2-1902, example F-5	Janbu corrected (f0)	1.345	1.341
V091	USACE example E-10, ponded water	Bishop simplified	1.241	1.235
V091	USACE example E-10, ponded water	Spencer (M-P, constant f(x))	1.241	1.235
V091	USACE example E-10, ponded water	Morgenstern-Price (half-sine)	1.244	1.235
V091	USACE example E-10, ponded water	Janbu corrected (f0)	1.385	1.380
V092	USACE example F-6, steady-seepage dam	Bishop simplified	2.011	1.998
V092	USACE example F-6, steady-seepage dam	Spencer (M-P, constant f(x))	2.026	2.012
V092	USACE example F-6, steady-seepage dam	Janbu corrected (f0)	1.830	1.832
V093	Yamagami & Ueta (1988)	Bishop simplified	1.344	1.343
V093	Yamagami & Ueta (1988)	Spencer (M-P, constant f(x))	1.325	1.324
V094	Yamagami & Ueta (1988) four-layer slope, non-circular	Spencer (M-P, constant f(x))	1.398	1.401
V094	Yamagami & Ueta (1988) four-layer slope, non-circular	Morgenstern-Price (half-sine)	1.389	1.395
V095a	Yamagami micro-pile slope, no pile	Bishop simplified	1.102	1.101
V095b	Homogeneous slope, no micro piles	Bishop simplified	1.102	1.101
V095c	Homogeneous slope with a micro-pile row	Bishop simplified	1.193	1.196
V096a	Zhu & Lee heterogeneous slope, dry, deep circle	Bishop simplified	1.804	1.793
V096a	Zhu & Lee heterogeneous slope, dry, deep circle	Spencer (M-P, constant f(x))	1.804	1.794
V096b	Zhu & Lee heterogeneous slope with a water table, deep circle	Bishop simplified	1.176	1.173
V096b	Zhu & Lee heterogeneous slope with a water table, deep circle	Spencer (M-P, constant f(x))	1.189	1.187
V096c	Zhu & Lee slope, surface 3 (the reference circle), dry, 1 m crack	Spencer (M-P, constant f(x))	1.804	1.799
V096c	Zhu & Lee slope, surface 3 (the reference circle), dry, 1 m crack	Bishop simplified	1.804	1.799

Problem	Description	Method	Reference	cSlope
V097a	Eurocode 7 cutting with surcharge	Bishop simplified (vertical head)	1.173	1.172
V097b	Eurocode 7 cutting, partial factors applied by cSlope	Bishop simplified, design values	1.173	1.172
V097b	Eurocode 7 cutting, partial factors applied by cSlope	Spencer, design values	1.174	1.174
V098	Example slope without seismic load	Spencer (M-P, constant f(x))	1.360	1.372
V098	Example slope without seismic load	Spencer (M-P, constant f(x))	1.372	1.372
Bundled	T-ACADS Simple	Morgenstern-Price	0.984	0.984
Bundled	T-ACADS Simple	Bishop simplified	0.985	0.984
Bundled	T-ACADS Simple	Janbu simplified	0.938	0.935
Bundled	T-ACADS Non-Homogeneous	Morgenstern-Price	1.373	1.372
Bundled	Arai & Tagyo (1985)	Bishop simplified	1.118	1.113
Bundled	Arai & Tagyo (1985), weak layer	Morgenstern-Price	0.414	0.412
Bundled	Talbingo Dam	Morgenstern-Price	1.950	1.951
Bundled	Talbingo Dam (Single Circle)	Morgenstern-Price	2.290	2.301
V021a	Congress Street cut, probabilistic (tangent to clay 2)	Pf, share FOS < 1 (Monte Carlo)	0.246	0.239
V021a	Congress Street cut, probabilistic (tangent to clay 2)	β lognormal (Monte Carlo)	0.650	0.671
V021a	Chowdhury & Xu (1995) example 1, probability of failure	Pf, share FOS < 1 (Monte Carlo)	0.253	0.245
V021b	Congress Street cut, probabilistic (tangent to clay 3)	Pf, share FOS < 1 (Monte Carlo)	0.279	0.282
V021b	Congress Street cut, probabilistic (tangent to clay 3)	β lognormal (Monte Carlo)	0.537	0.553
V022a	Embankment on soft clay, probabilistic (tangent to the interface)	Pf, share FOS < 1 (Monte Carlo)	0.212	0.206
V022a	Embankment on soft clay, probabilistic (tangent to the interface)	β lognormal (Monte Carlo)	0.799	0.812
V022a	Embankment on soft clay, probabilistic (tangent to the interface)	β lognormal (Taylor series)	0.787	0.788
V022a	Embankment on soft clay, probabilistic (tangent to the interface)	β lognormal (Monte Carlo)	0.800	0.812
V022a	Embankment on soft clay, probabilistic (tangent to the interface)	Pf, share FOS < 1 (Monte Carlo)	0.206	0.206
V022a	Chowdhury & Xu (1995) example 5, probability of failure	Pf, share FOS < 1 (Monte Carlo)	0.210	0.208
V022b	Embankment on soft clay, probabilistic (tangent to the clay base)	Pf, share FOS < 1 (Monte Carlo)	0.199	0.212
V022b	Embankment on soft clay, probabilistic (tangent to the clay base)	β lognormal (Monte Carlo)	0.820	0.789
V022b	Embankment on soft clay, probabilistic (tangent to the clay base)	β lognormal (Taylor series)	0.798	0.802
V022b	Embankment on soft clay, probabilistic (tangent to the clay base)	β lognormal (Monte Carlo)	0.789	0.789
V022b	Embankment on soft clay, probabilistic (tangent to the clay base)	Pf, share FOS < 1 (Monte Carlo)	0.209	0.212
V022b	Chowdhury & Xu (1995) example 5, probability of failure	Pf, share FOS < 1 (Monte Carlo)	0.214	0.216
V060b	Cannon Dam end of construction (Hassan & Wolff), reliability on the minimum-β circle	β lognormal (Taylor series)	3.353	3.354
V023	LASH terminal trench, Duncan's Taylor series example	β lognormal (Taylor series)	0.900	0.892
V023	LASH terminal trench, Duncan's Taylor series example	Pf lognormal (Taylor series)	0.180	0.186
V062a	Clarence Cannon Dam, surface 7A	β lognormal (Monte Carlo)	10.953	11.034
V062b	Clarence Cannon Dam, surface 7B	β lognormal (Monte Carlo)	4.351	4.402

Problem	Description	Method	Reference	cSlope
V062c	Clarence Cannon Dam, surface 7C	β lognormal (Monte Carlo)	4.263	4.325
V062d	Clarence Cannon Dam, surface 7D	β lognormal (Monte Carlo)	11.092	11.076
V062e	Clarence Cannon Dam, surface 7E	β lognormal (Monte Carlo)	10.281	10.329
V062f	Clarence Cannon Dam, surface 8B	β lognormal (Monte Carlo)	4.858	4.933
V062g	Clarence Cannon Dam, surface 8F	β lognormal (Monte Carlo)	5.485	5.528
V062h	Clarence Cannon Dam, surface 8G	β lognormal (Monte Carlo)	5.563	5.490
V062i	Clarence Cannon Dam, surface 8H	β lognormal (Monte Carlo)	6.394	6.431

3 Verification problems

3.1 V001 — Arai & Tagyo (1985) example 1

Arai & Tagyo (1985), example 1.

Critical circle search · Units: SI (m, kPa, kN/m³)

Homogeneous 1.5:1 slope, 20 m high; critical circle search.

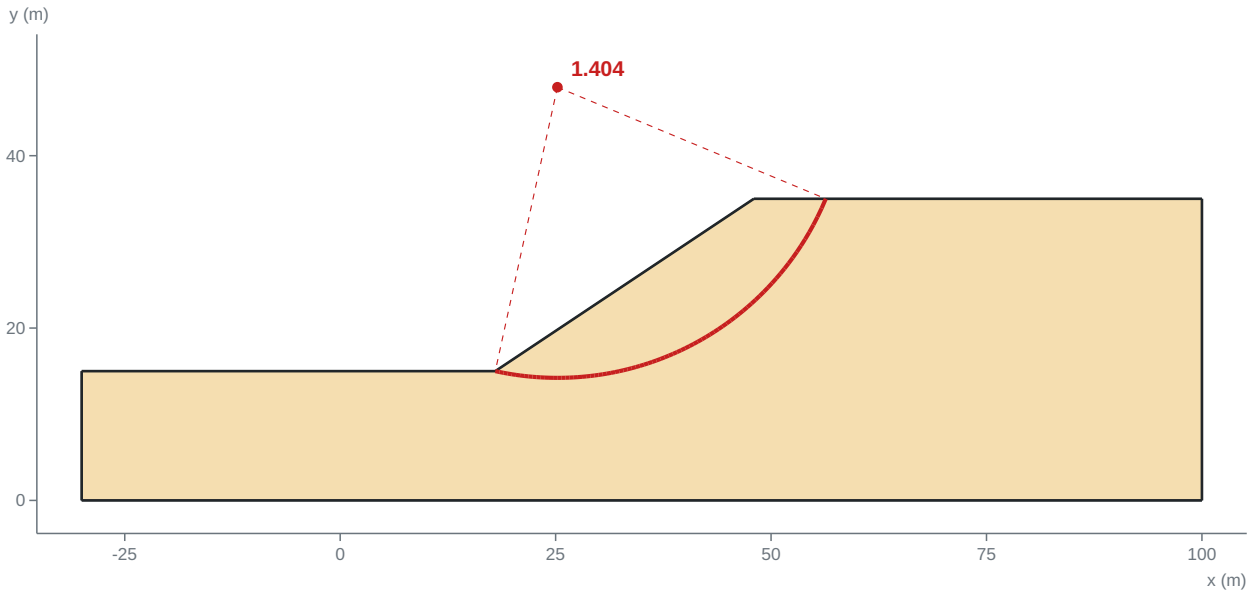


Figure 1. V001: critical circle (Bishop simplified, FOS = 1.404).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	18.82	c' = 41.65	15.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.409	1.404
Morgenstern-Price (half-sine)	Commercial program A (GLE)	1.414	1.401
Bishop simplified	Commercial program B	1.417	1.404
Morgenstern-Price (half-sine)	Commercial program B	1.414	1.401
Spencer (M-P, constant f(x))	Commercial program A (non-circular)	1.386	1.385

Notes

- Search limited to x = 0–66, the model of the source (ours extends to −30 and 100).
- commercial program A Janbu corrected 1.407 (circle); non-circular: Janbu simplified 1.253, corrected 1.346. Arai & Tagyo: Bishop 1.451; Janbu simplified 1.265, corrected 1.357 (non-circular).

3.2 V002 — Arai & Tagyo (1985) example 2, three layers

Arai & Tagyo (1985), example 2.

Critical circle search · Units: SI (m, kPa, kN/m³)

Slope with a weak middle layer between a stronger upper layer and a very strong base; critical circle search.

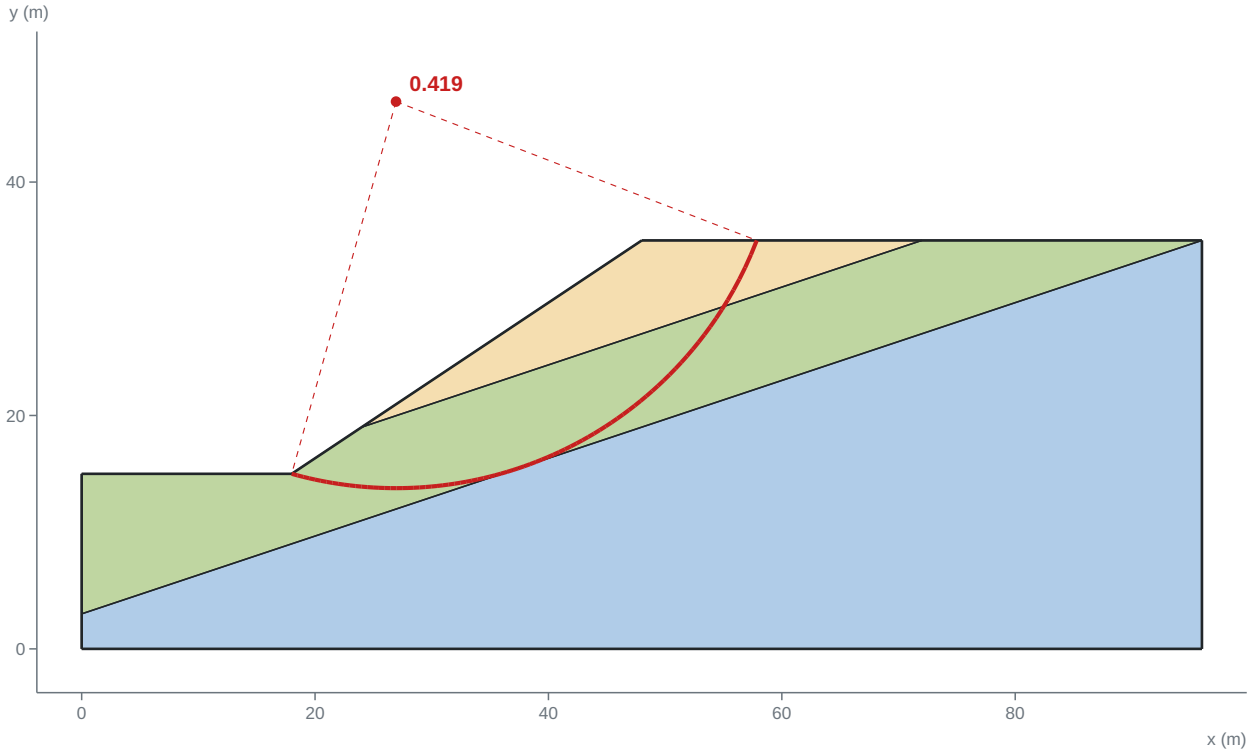


Figure 2. V002: critical circle (Bishop simplified, FOS = 0.419).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Upper	Mohr-Coulomb	18.82	c' = 29.40	12.00°
Middle	Mohr-Coulomb	18.82	c' = 9.80	5.00°
Lower	Mohr-Coulomb	18.82	c' = 294.0	40.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	0.420	0.419
Spencer (M-P, constant f(x))	Commercial program A (non-circular)	0.414	0.412

Notes

- Non-circular: commercial program A 0.414 (random search with optimisation); Greco 0.39, Kim et al. 0.39–0.44. Before the Morgenstern-Price λ iteration ignored unconverged solutions, cSlope's optimisation reached 0.27–0.39 on surfaces where the force and moment curves never cross.

3.3 V003 — Arai & Tagyo (1985) example 3, water table

Arai & Tagyo (1985), Soils and Foundations 25(1).

Critical circle search · Units: SI (m, kPa, kN/m³)

Homogeneous slope with a water table; critical-circle search.

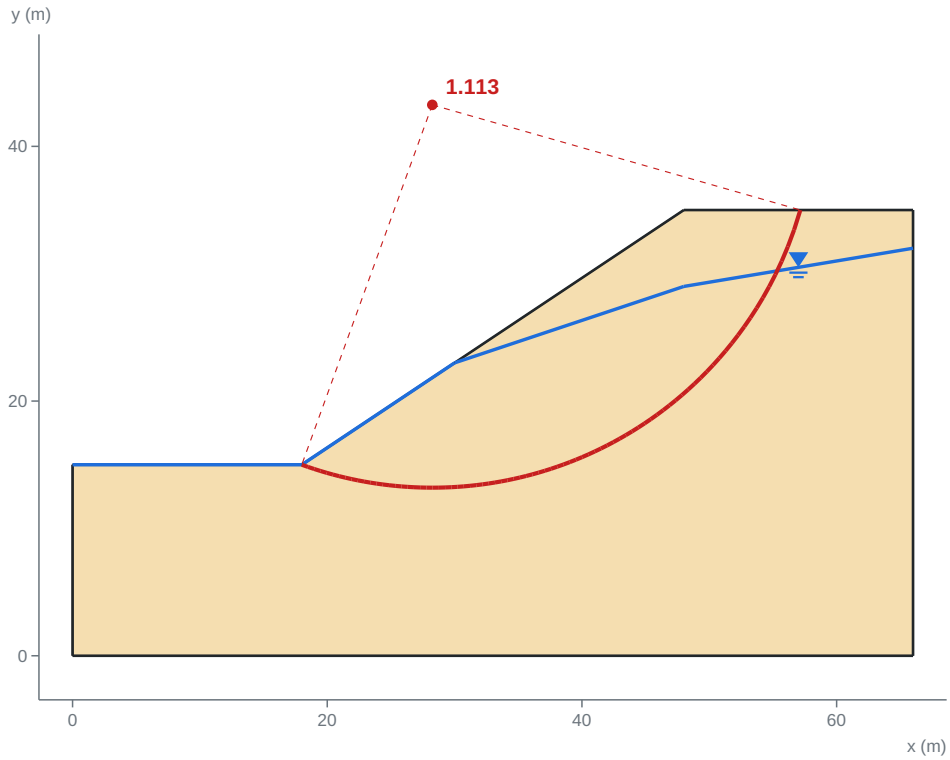


Figure 3. V003: critical circle (Bishop simplified, FOS = 1.113).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
A&T ex3 soil	Mohr-Coulomb	18.82	c' = 41.65	15.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.118	1.113
Spencer (M-P, constant f(x))	Commercial program A	1.118	1.113
Janbu corrected (f0)	Commercial program A	1.131	1.124
Bishop simplified	Program D	1.112	1.113

Notes

- Arai & Tagyo 1.138; commercial program B 1.190.

3.4 V004a — 4:1 slope, Mohr-Coulomb fit

Verification manual of a commercial slope stability program.

Critical circle search · Units: SI (m, kPa, kN/m³)

12 m high 4:1 slope with the same soil as V008a; critical circle search.

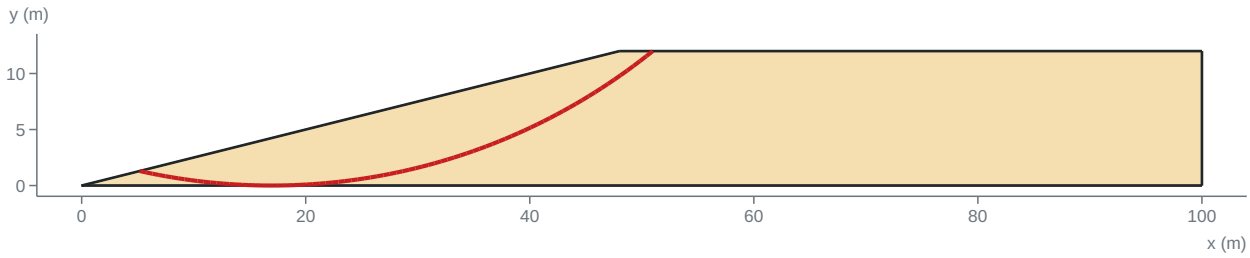


Figure 4. V004a: critical circle (Spencer (M-P, constant f(x)), FOS = 2.802).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	18.00	c' = 11.64	24.70°

Method	Published by	Reference	cSlope
Spencer (M-P, constant f(x))	Commercial program A	2.794	2.802
Janbu simplified (uncorrected)	Commercial program A	2.662	2.677

3.5 V004b — Baker (2003) ex. 2 slope, Mohr-Coulomb iteration parameters

Verification manual of a commercial slope stability program.

Critical circle search · Units: SI (m, kPa, kN/m³)

The V004a 4:1 slope with Baker's iterated Mohr-Coulomb parameters (c′ = 2.439 kPa, φ′ = 30.392°); critical-circle search.

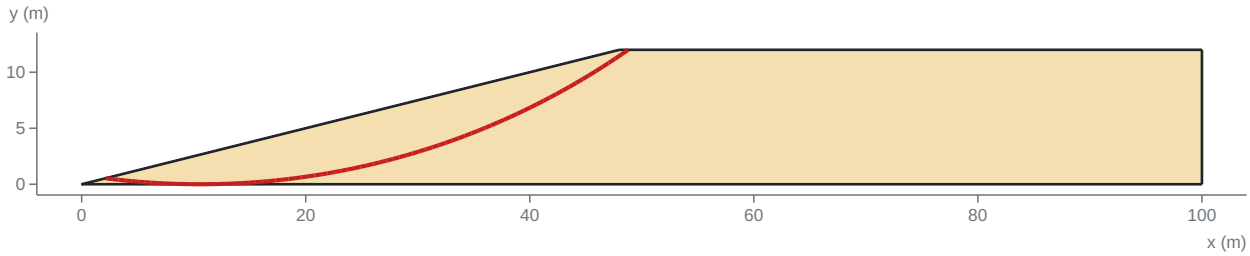


Figure 5. V004b: critical circle (Janbu simplified (uncorrected), FOS = 2.609).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	18.00	c' = 2.44	30.39°

Method	Published by	Reference	cSlope
Janbu simplified (uncorrected)	Commercial program A	2.610	2.609
Spencer (M-P, constant f(x))	Commercial program A	2.696	2.666

Notes

- Baker (2003) non-linear 2.64.

3.6 V005 — Baker (1993) dam, end of construction, infinite-slope face

Baker (1993).

Critical circle search · Units: SI (m, kPa, kN/m³)

Compacted-clay embankment (c′ = 6.5 kPa, φ′ = 40°) on a deep natural-clay foundation (c′ = 0, φ′ = 32°) whose 4H:1V downstream face governs; dry; circular search. Theoretical (infinite slope) FOS tan 32°/tan 14.04° = 2.50.

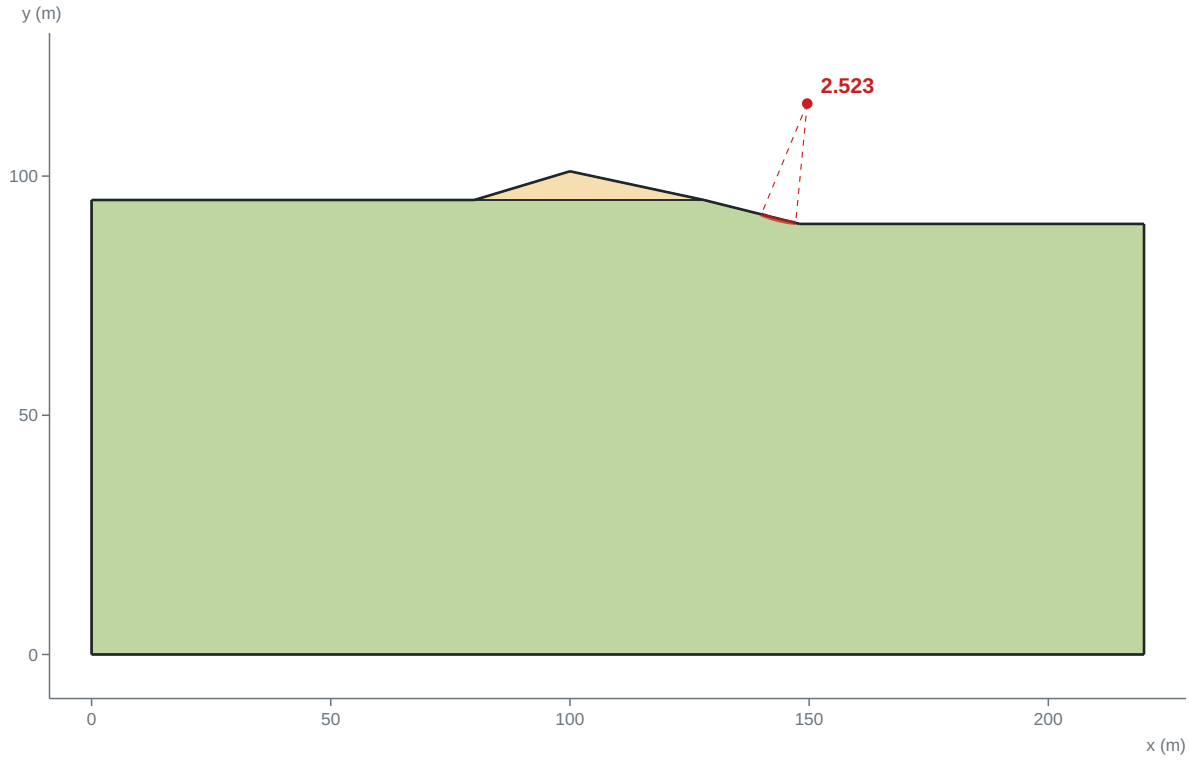


Figure 6. V005: critical circle (Bishop simplified, FOS = 2.523).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Compacted clay	Mohr-Coulomb	18.00	c' = 6.50	40.00°
Natural clay	Mohr-Coulomb	18.00	c' = 0.00	32.00°

Method	Published by	Reference	cSlope
Bishop simplified	Theory (infinite slope)	2.500	2.523
Spencer (M-P, constant f(x))	Theory (infinite slope)	2.500	2.523

Notes

- Search limited to the downstream face, x = 100–148, as in the source.
- The critical surfaces are shallow slabs on the cohesionless face; a minimum sliding area in proportion to the model area (here 20 000 m²) would skip them and read 3 % high.
- commercial program A 2.534 (noncircular, minimum depth 5 m); Baker 2.41.

3.7 V006 — Baker (2001) planar surface

Baker (2001).

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

Homogeneous 10 m slope on the critical plane from the toe to the crest at 49.6°.

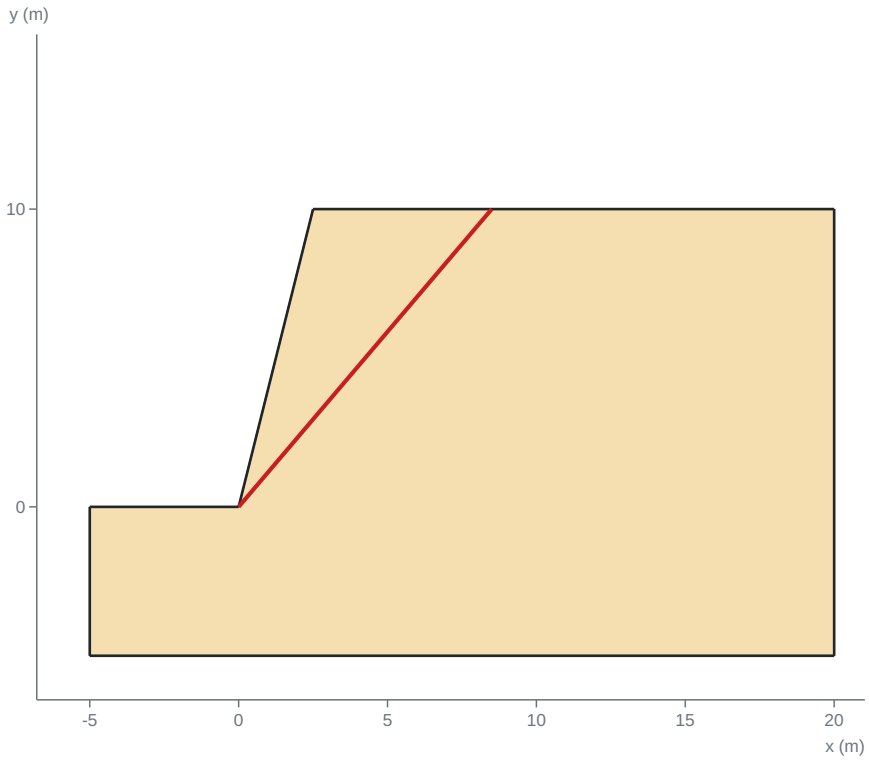


Figure 7. V006: analysed surface (Spencer (M-P, constant f(x)), FOS = 1.352).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	20.00	c' = 30.00	30.00°

Method	Published by	Reference	cSlope
Spencer (M-P, constant f(x))	Commercial program C	1.351	1.352
Janbu simplified	Commercial program C	1.351	1.352

3.8 V007a — Baker (2003) ex. 1, Mohr-Coulomb, Janbu simplified on the reference circle

Verification manual of a commercial slope stability program.

Fixed circle · Units: SI (m, kPa, kN/m³)

The V008a slope (Mohr-Coulomb fit c′ = 11.64 kPa, φ′ = 24.7°) with Janbu simplified (uncorrected) on the reference critical circle (Fig. 44.3).

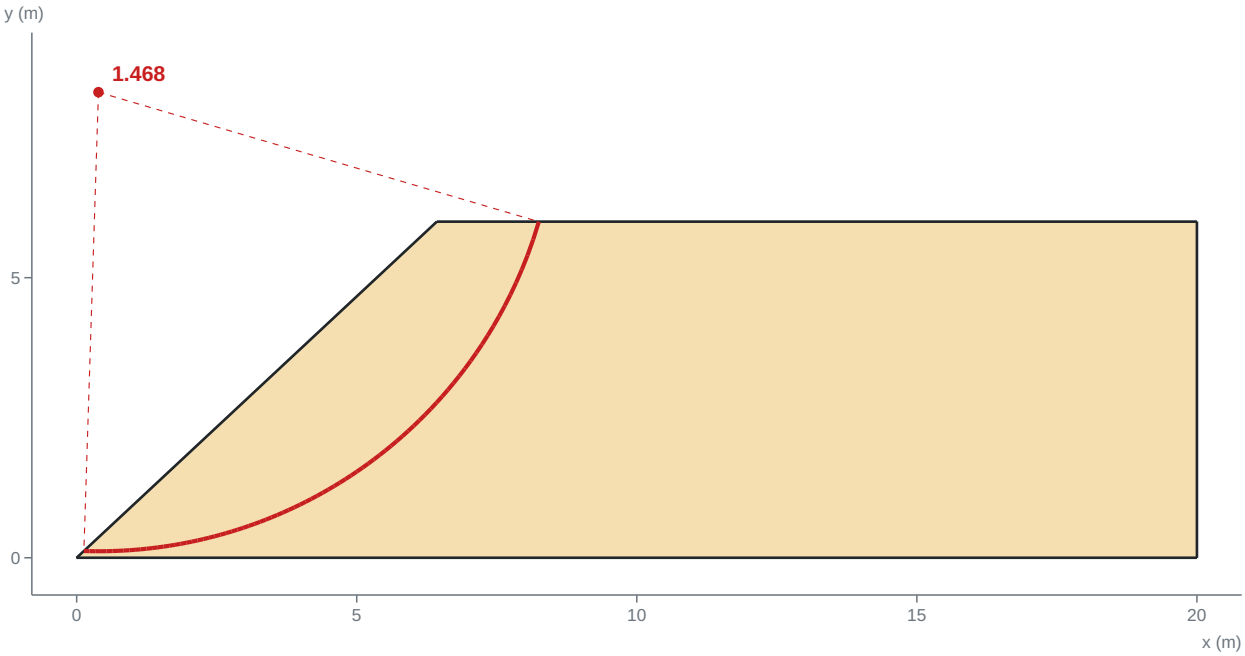


Figure 8. V007a: analysed circle (Janbu simplified (uncorrected), FOS = 1.468).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	18.00	c' = 11.64	24.70°

Method	Published by	Reference	cSlope
Janbu simplified (uncorrected)	Commercial program A	1.469	1.468

Notes

- Confirms V008a's strength (c′ = 11.64 kPa, φ′ = 24.7°, γ = 18 kN/m³): the manual's Table 44.1 lists other soils by mistake. Baker (2003) M-C 1.50.

3.9 V007b — Baker (2003) ex. 1 slope, Mohr-Coulomb iteration parameters

Verification manual of a commercial slope stability program.

Critical circle search · Units: SI (m, kPa, kN/m³)

The V008a slope with Baker's Mohr-Coulomb parameters from his iterative linearisation (c′ = 0.39 kPa, φ′ = 38.6°); critical-circle search.

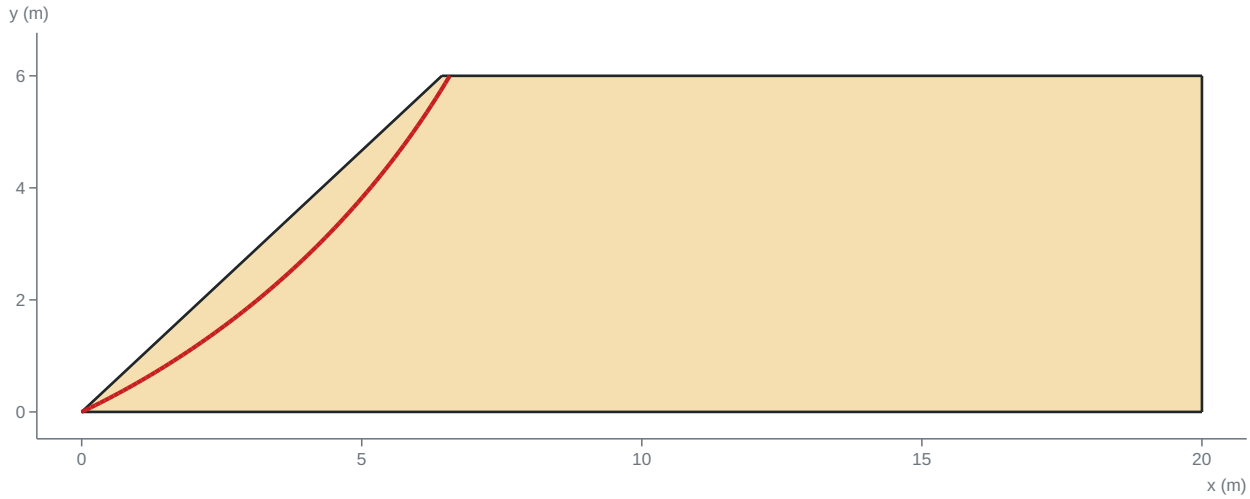


Figure 9. V007b: critical circle (Janbu simplified (uncorrected), FOS = 0.955).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	18.00	c' = 0.39	38.60°

Method	Published by	Reference	cSlope
Janbu simplified (uncorrected)	Commercial program A	0.957	0.955
Spencer (M-P, constant f(x))	Commercial program A	0.981	0.978

Notes

- commercial program A geometry (0,0)-(6.43,6)-(20,6)-(20,0): no toe flat, base at toe level.
- Baker (2003) non-linear 0.97.

3.10 V007c — Baker (2003) ex. 3, Janbu simplified

Verification manual of a commercial slope stability program.

Critical circle search · Units: SI (m, kPa, kN/m³)

The V008b London clay slope (c′ = 6 kPa, φ′ = 32°) with Janbu simplified (uncorrected); critical-circle search.

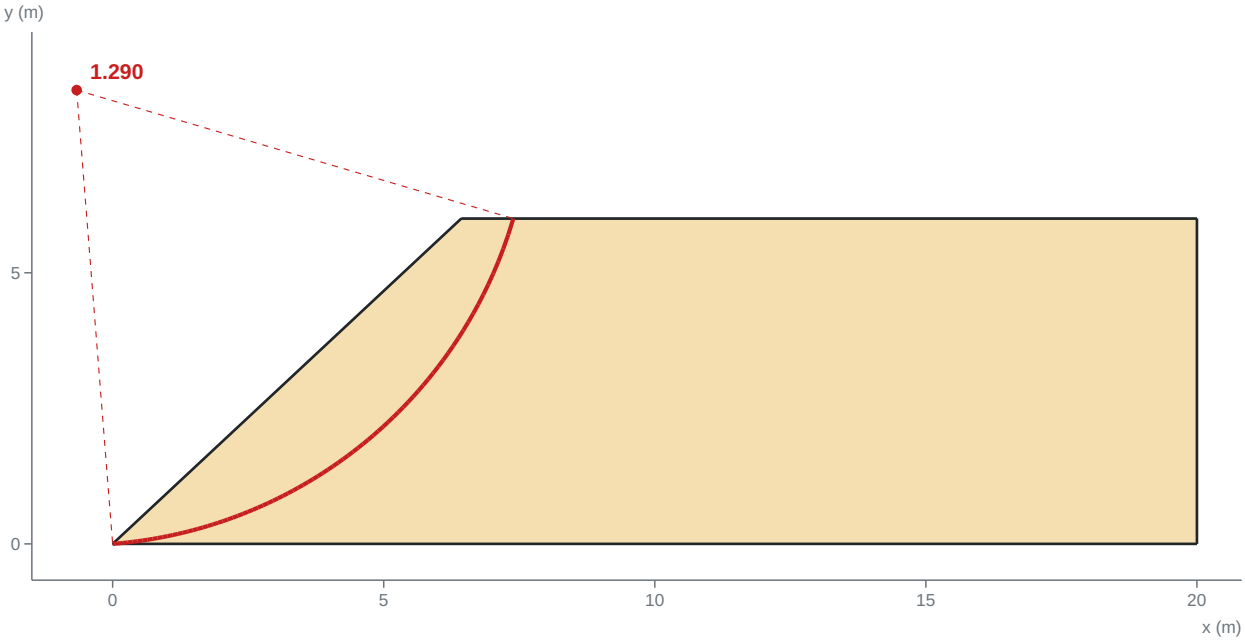


Figure 10. V007c: critical circle (Janbu simplified (uncorrected), FOS = 1.290).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	18.00	c' = 6.00	32.00°

Method	Published by	Reference	cSlope
Janbu simplified (uncorrected)	Commercial program A	1.291	1.290

Notes

- Search limited to x = 0–13.7 (toe to the crest), as in the source.
- commercial program A geometry: no toe flat, base at toe level (results identical).
- Baker (2003) M-C 1.35.

3.11 V008a — Baker (2003) example 1, Mohr-Coulomb fit

Baker (2003), example 1.

Critical circle search · Units: SI (m, kPa, kN/m³)

6 m high slope at about 43° in soil with a linear Mohr-Coulomb fit to a curved envelope; critical circle search.

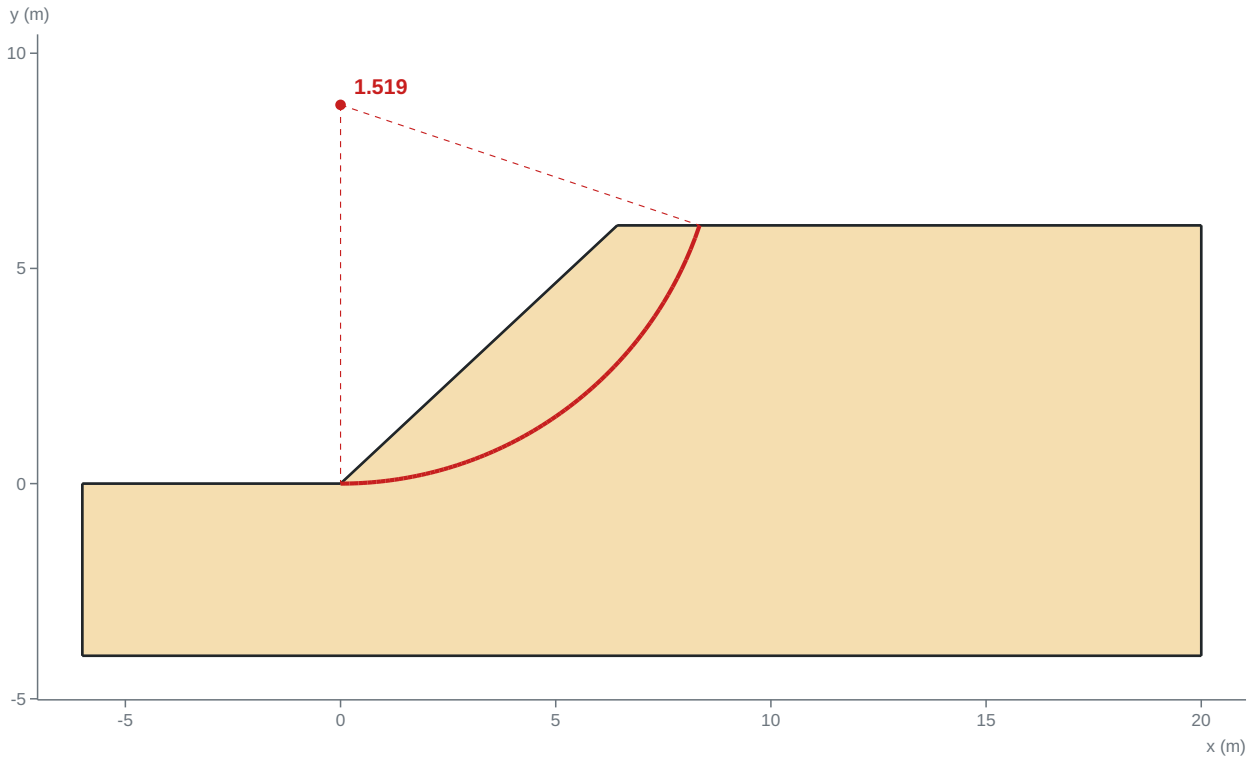


Figure 11. V008a: critical circle (Spencer (M-P, constant f(x)), FOS = 1.519).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	18.00	c' = 11.64	24.70°

Method	Published by	Reference	cSlope
Spencer (M-P, constant f(x))	Commercial program A	1.536	1.519

Notes

- Search limited to x = 0–20, the model of the source; the level ground left of the toe is not searched.

3.12 V008b — Baker (2003) London clay slope, Mohr-Coulomb fit

Baker (2003) example 3.

Critical circle search · Units: SI (m, kPa, kN/m³)

43° slope, 6 m high, with the Mohr-Coulomb envelope fitted to London clay data (c' = 6 kPa, φ' = 32°); critical circle search.

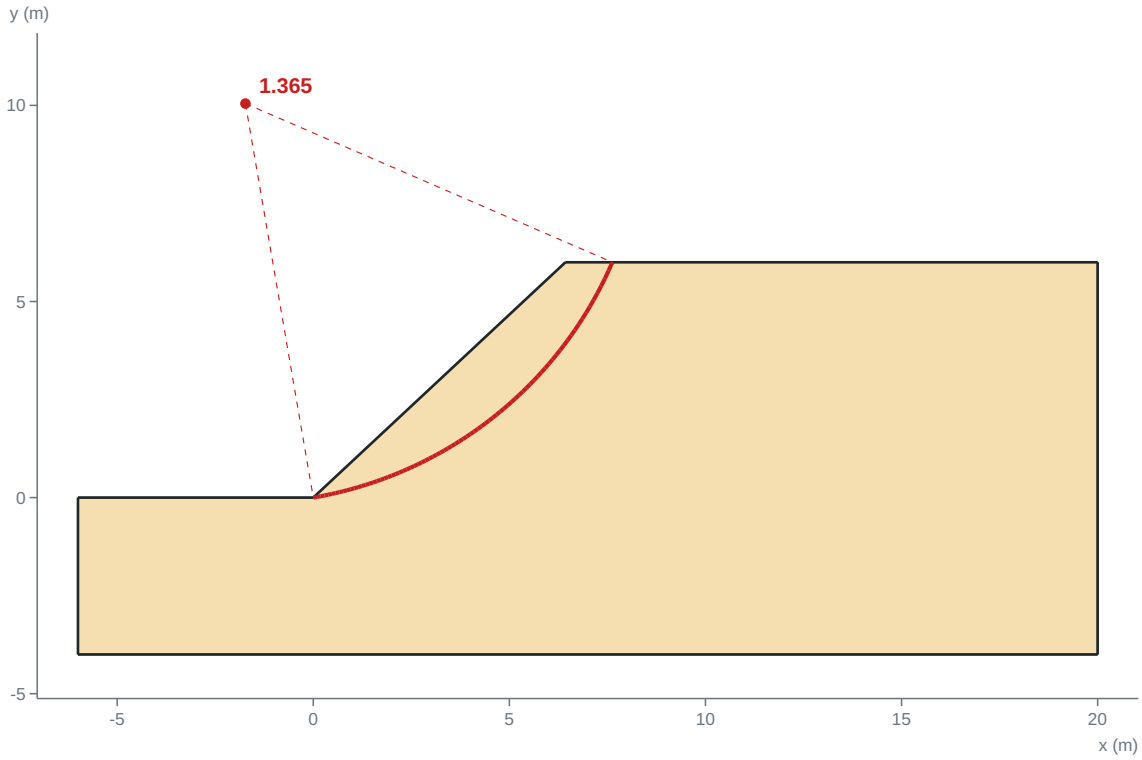


Figure 12. V008b: critical circle (Spencer (M-P, constant f(x)), FOS = 1.365).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
London clay	Mohr-Coulomb	18.00	c' = 6.00	32.00°

Method	Published by	Reference	cSlope
Spencer (M-P, constant f(x))	Commercial program A	1.366	1.365

Notes

- Search limited to x = 0–13.7 (toe to the crest), as in the source; the level ground left of the toe is not searched.

3.13 V009a — Baker & Leshchinsky clay-core dam, reservoir and crest crack

Baker & Leshchinsky (2001).

Fixed circle · Units: SI (m, kPa, kN/m³)

Granular dam with a diamond clay core on a hard base, half-full reservoir standing on the face (free water), phreatic surface through the core and a 5 m dry tension crack at the crest; commercial program A's critical circle.

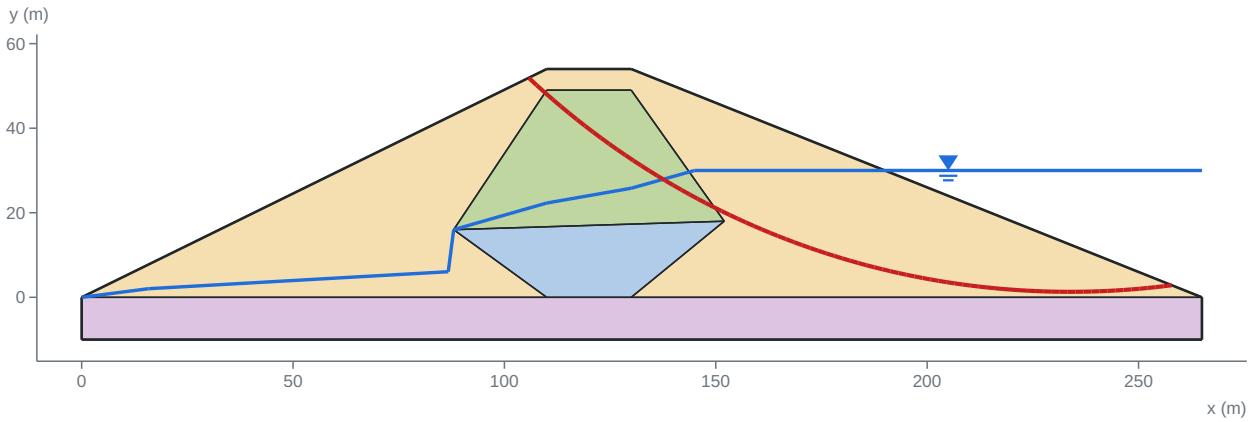


Figure 13. V009a: analysed circle (Spencer (M-P, constant f(x)), FOS = 1.926).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Granular fill	Mohr-Coulomb	21.50	c' = 0.00	40.00°
Clay core	Mohr-Coulomb	20.00	c' = 20.00	20.00°
Granular fill below core	Mohr-Coulomb	21.50	c' = 0.00	40.00°
Hard base	Mohr-Coulomb	24.00	c' = 200.0	45.00°

Method	Published by	Reference	cSlope
Spencer (M-P, constant f(x))	Commercial program A	1.925	1.926

3.14 V009b — Baker & Leshchinsky dam, Baker's non-circular surface

Baker & Leshchinsky (2001).

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

As V009a, on the non-circular surface published by Baker & Leshchinsky.

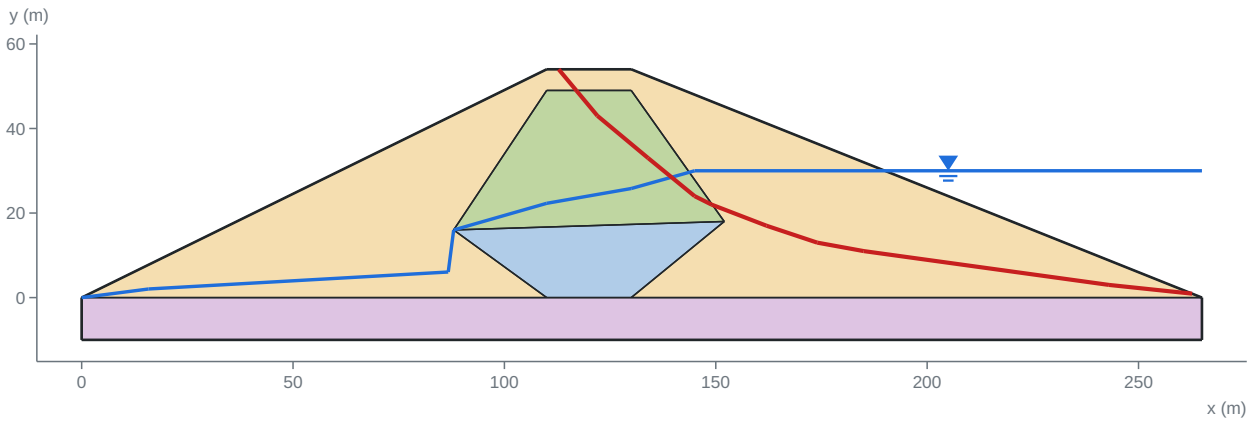


Figure 14. V009b: analysed surface (Spencer (M-P, constant f(x)), FOS = 1.882).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Granular fill	Mohr-Coulomb	21.50	c' = 0.00	40.00°
Clay core	Mohr-Coulomb	20.00	c' = 20.00	20.00°
Granular fill below core	Mohr-Coulomb	21.50	c' = 0.00	40.00°
Hard base	Mohr-Coulomb	24.00	c' = 200.0	45.00°

Method	Published by	Reference	cSlope
Spencer (M-P, constant f(x))	Baker & Leshchinsky	1.910	1.882

Notes

- program D gives 1.882 on the same surface.

3.15 V009c — Baker & Leshchinsky clay-core dam, critical surface search

Baker & Leshchinsky (2001).

Circle search with non-circular optimisation · Units: SI (m, kPa, kN/m³)

The V009a clay-core dam (reservoir on the face, 5 m dry crest crack) with a circular search and non-circular optimisation.

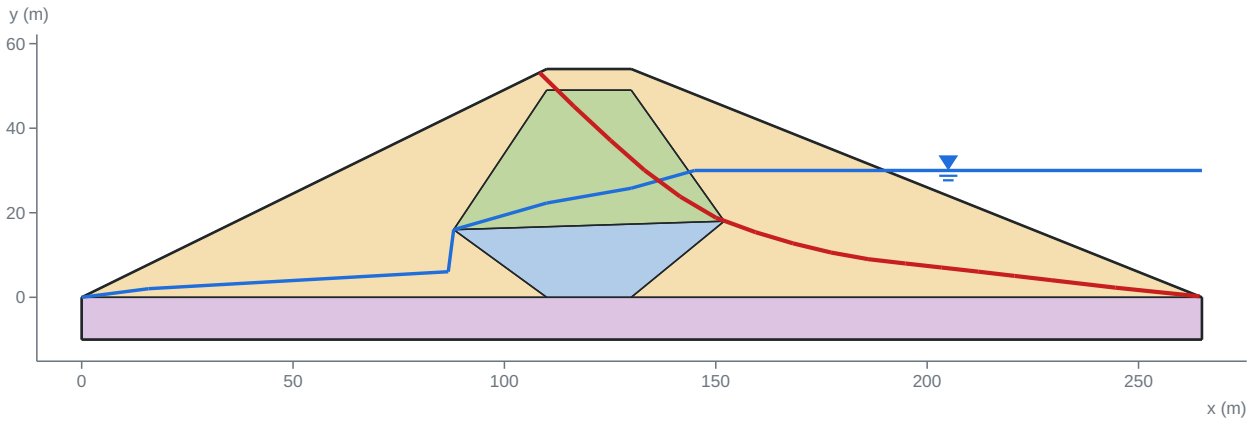


Figure 15. V009c: critical surface (Spencer (M-P, constant f(x)), FOS = 1.868).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Granular fill	Mohr-Coulomb	21.50	c' = 0.00	40.00°
Clay core	Mohr-Coulomb	20.00	c' = 20.00	20.00°
Granular fill below core	Mohr-Coulomb	21.50	c' = 0.00	40.00°
Hard base	Mohr-Coulomb	24.00	c' = 200.0	45.00°

Method	Published by	Reference	cSlope
Spencer (M-P, constant f(x))	Commercial program B (optimised non-circular)	1.868	1.868

Notes

- V009a definition without the fixed circle. Baker & Leshchinsky 1.91 on their own surface is checked in V009b.

3.16 V010 — Borges & Cardoso embankment 2 (geosynthetic)

Borges & Cardoso (2002), Geotextiles and Geomembranes 20(6).

Fixed circle · Units: SI (m, kPa, kN/m³)

5 m embankment (φ′ = 35°) on four soft clays with Su increasing with depth (Table 55: clay 3 from 16 to 18.4 kPa, clay 4 from 18.4 to 55.1 kPa), a 200 kN/m geosynthetic at y = 1 m (axial, active, not anchored, interface friction 33.7° on both faces); commercial program B's critical circle.

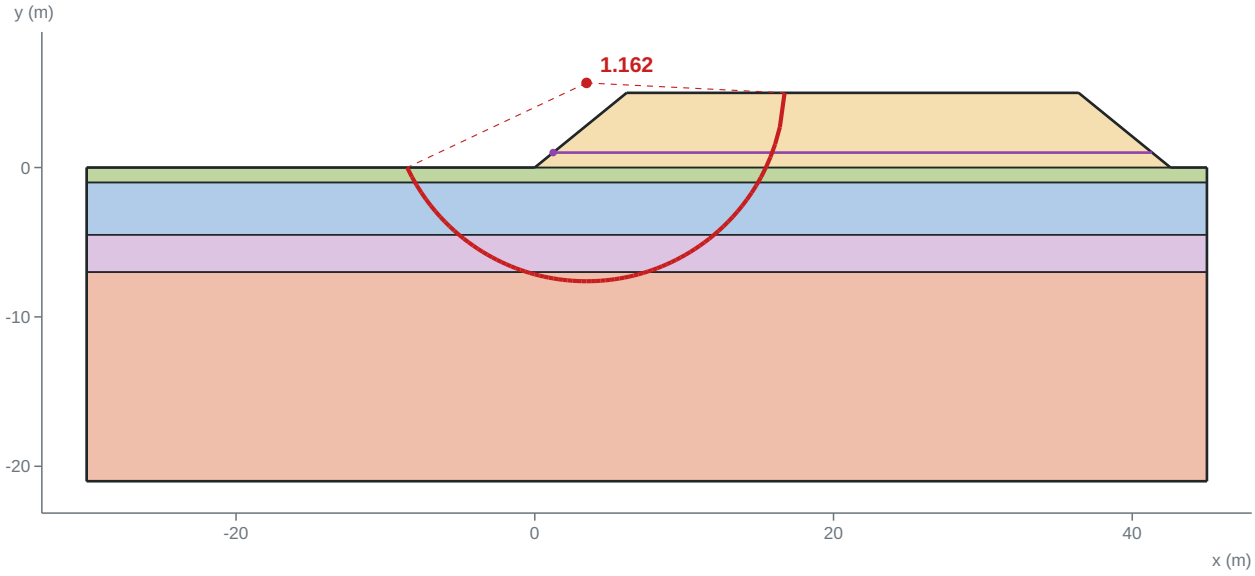


Figure 16. V010: analysed circle (Morgenstern-Price (half-sine), FOS = 1.162).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Embankment	Mohr-Coulomb	20.00	c' = 0.00	35.00°
Clay 1	Undrained (φ = 0)	17.00	Su = 33.00	0°
Clay 2	Undrained (φ = 0)	17.00	Su = 16.00	0°
Clay 3	Undrained (φ = 0)	17.00	Su = 16.00 at y = -4.50, +0.95 kPa/m below	0°
Clay 4	Undrained (φ = 0)	17.00	Su = 18.38 at y = -7.00, +2.63 kPa/m below	0°

Method	Published by	Reference	cSlope
Morgenstern-Price (half-sine)	Commercial program B	1.171	1.162
Bishop simplified	Commercial program B	1.170	1.164
Morgenstern-Price (half-sine)	Borges & Cardoso	1.150	1.162

Notes

- Clay 3: 16 → 18.375 kPa, clay 4: 18.375 → 55.125 kPa (2.625 kPa/m), the unrounded values of Table 55. The commercial program B circle is digitised.
- The manual gives the geosynthetic's strength and interface friction but not whether commercial program B divides its force by F; as a passive force cSlope gives 1.143 (M-P) / 1.143 (Bishop), 2.4 % below commercial program B, so the active force is kept.
- Borges & Cardoso's 1.15 (two digits, their own limit-equilibrium program) is the reference. program D 1.153 with the earlier input (clay 4 at 1.314 kPa/m).

3.17 V011a — Borges & Cardoso embankment 3, H = 7 m, circle A

Borges & Cardoso (2002), Geotextiles and Geomembranes 20(6).

Fixed circle · Units: SI (m, kPa, kN/m³)

Granular fill (two stages) on five soft clay layers with a T = 200 kN/m geosynthetic at y = 0.9 m (axial, passive as in commercial program A, pullout over 1.3 m from each end); Bishop on circle A of the paper.

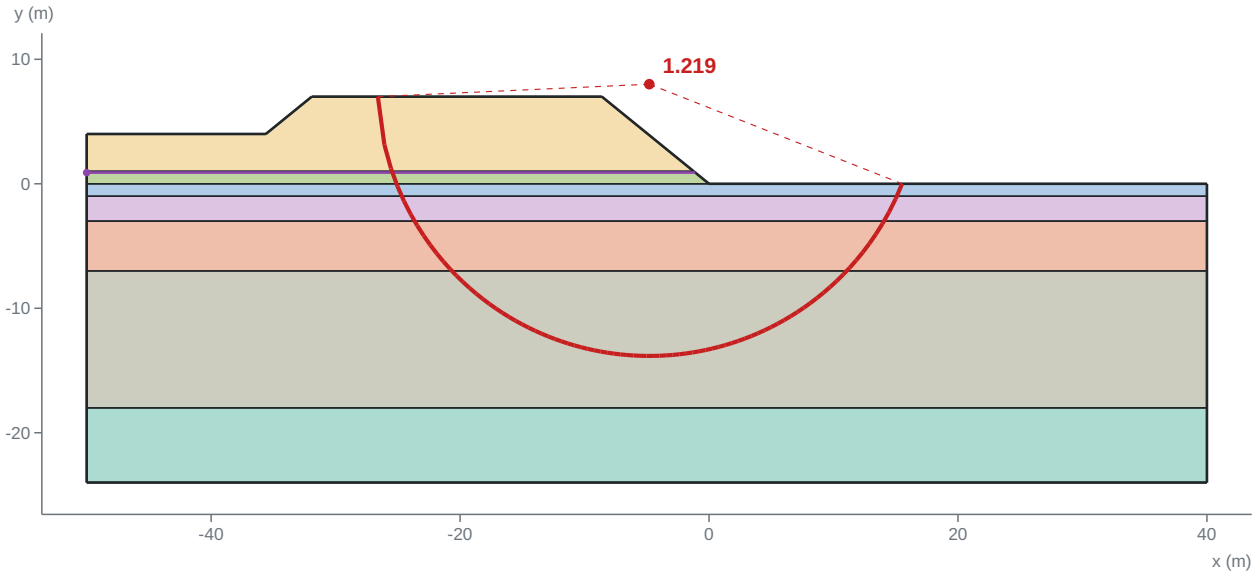


Figure 17. V011a: analysed circle (Bishop simplified, FOS = 1.219).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Upper embankment	Mohr-Coulomb	21.90	c' = 0.00	35.00°
Lower embankment	Mohr-Coulomb	17.20	c' = 0.00	33.00°
Clay 1	Undrained (φ = 0)	18.00	Su = 43.00	0°
Clay 2	Undrained (φ = 0)	16.60	Su = 31.00	0°
Clay 3	Undrained (φ = 0)	13.50	Su = 30.00	0°
Clay 4	Undrained (φ = 0)	17.00	Su = 32.00	0°
Clay 5	Undrained (φ = 0)	17.50	Su = 32.00	0°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.230	1.219

Notes

- Borges & Cardoso 1.25; program D 1.218 on the same input. commercial program A reports Bishop only (the authors' moment method).

3.18 V011b — Borges & Cardoso embankment 3, H = 7 m, circle B

Borges & Cardoso (2002), Geotextiles and Geomembranes 20(6).

Fixed circle · Units: SI (m, kPa, kN/m³)

As V011a on circle B.

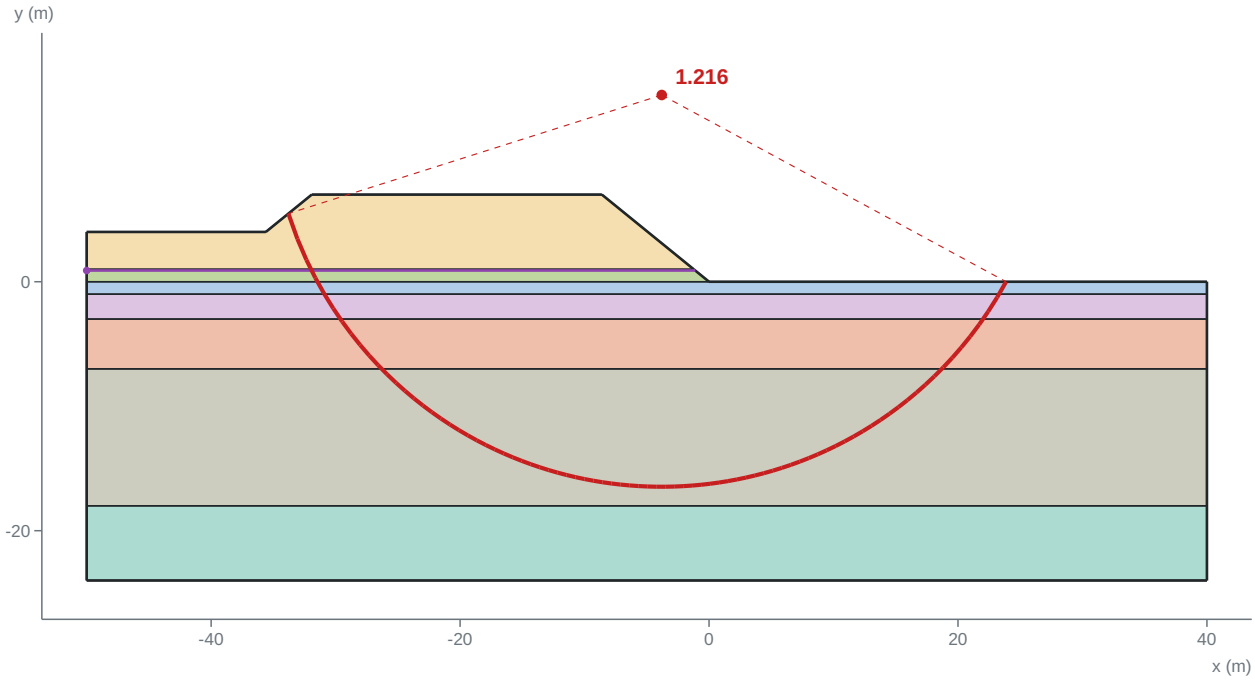


Figure 18. V011b: analysed circle (Bishop simplified, FOS = 1.216).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Upper embankment	Mohr-Coulomb	21.90	c' = 0.00	35.00°
Lower embankment	Mohr-Coulomb	17.20	c' = 0.00	33.00°
Clay 1	Undrained (φ = 0)	18.00	Su = 43.00	0°
Clay 2	Undrained (φ = 0)	16.60	Su = 31.00	0°
Clay 3	Undrained (φ = 0)	13.50	Su = 30.00	0°
Clay 4	Undrained (φ = 0)	17.00	Su = 32.00	0°
Clay 5	Undrained (φ = 0)	17.50	Su = 32.00	0°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.220	1.216

Notes

- Borges & Cardoso 1.19; program D 1.216.

3.19 V012 — Borges & Cardoso embankment 3, H = 8.75 m, circle C

Borges & Cardoso (2002), Geotextiles and Geomembranes 20(6).

Fixed circle · Units: SI (m, kPa, kN/m³)

The second fill stage (8.75 m crest) on circle C.

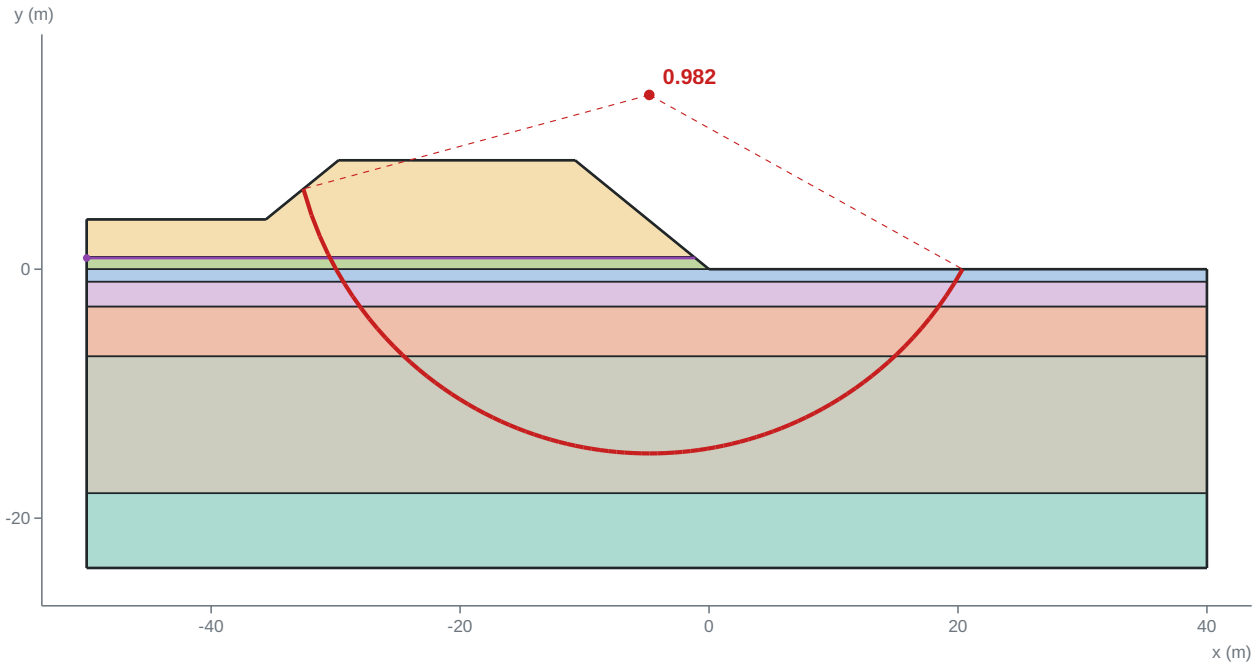


Figure 19. V012: analysed circle (Bishop simplified, FOS = 0.982).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Upper embankment	Mohr-Coulomb	21.90	c' = 0.00	35.00°
Lower embankment	Mohr-Coulomb	17.20	c' = 0.00	33.00°
Clay 1	Undrained (φ = 0)	18.00	Su = 43.00	0°
Clay 2	Undrained (φ = 0)	16.60	Su = 31.00	0°
Clay 3	Undrained (φ = 0)	13.50	Su = 30.00	0°
Clay 4	Undrained (φ = 0)	17.00	Su = 32.00	0°
Clay 5	Undrained (φ = 0)	17.50	Su = 32.00	0°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	0.980	0.982

Notes

- Borges & Cardoso 0.99; program D 0.981.

3.20 V013a — Cai & Ugai (2000) slope, no pile (see V013b, V013c, V013d and V013e for the pile r

Cai & Ugai (2000).

Critical circle search · Units: SI (m, kPa, kN/m³)

Homogeneous 1:1.5 slope, 10 m high, without piles; critical circle search.

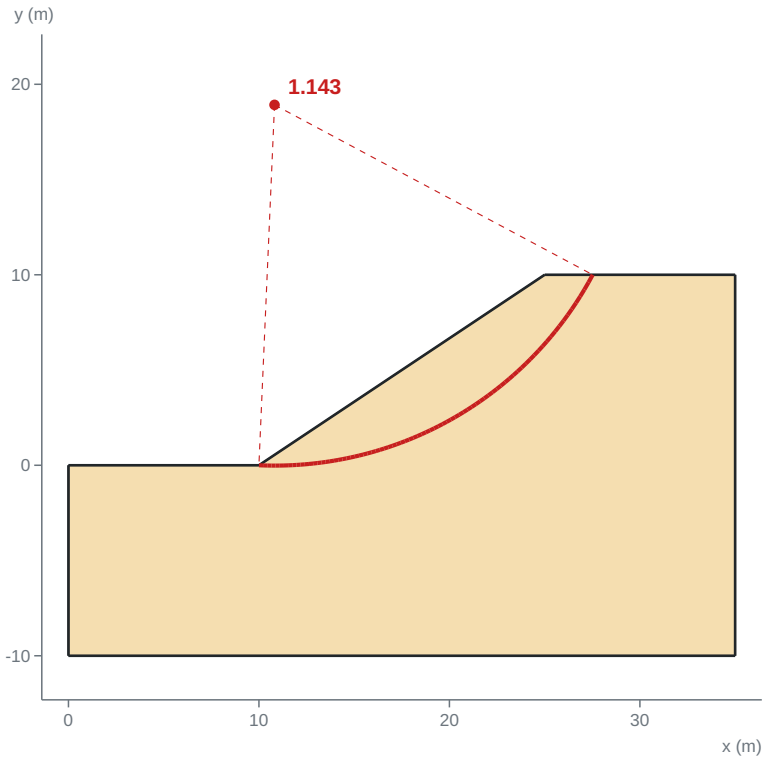


Figure 20. V013a: critical circle (Bishop simplified, FOS = 1.143).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	20.00	c' = 10.00	20.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.140	1.143
Bishop simplified	Cai & Ugai	1.130	1.143

Notes

- Both values are Bishop searches, printed to two digits.

3.21 V013b — Cai & Ugai slope, pile row D1/D = 2

Cai & Ugai (2000), Géotechnique 50(2).

Critical circle search · Units: SI (m, kPa, kN/m³)

A row of 0.8 m piles at mid-slope, centre-to-centre 1.6 m, with the Ito & Matsui force (passive); Bishop search.

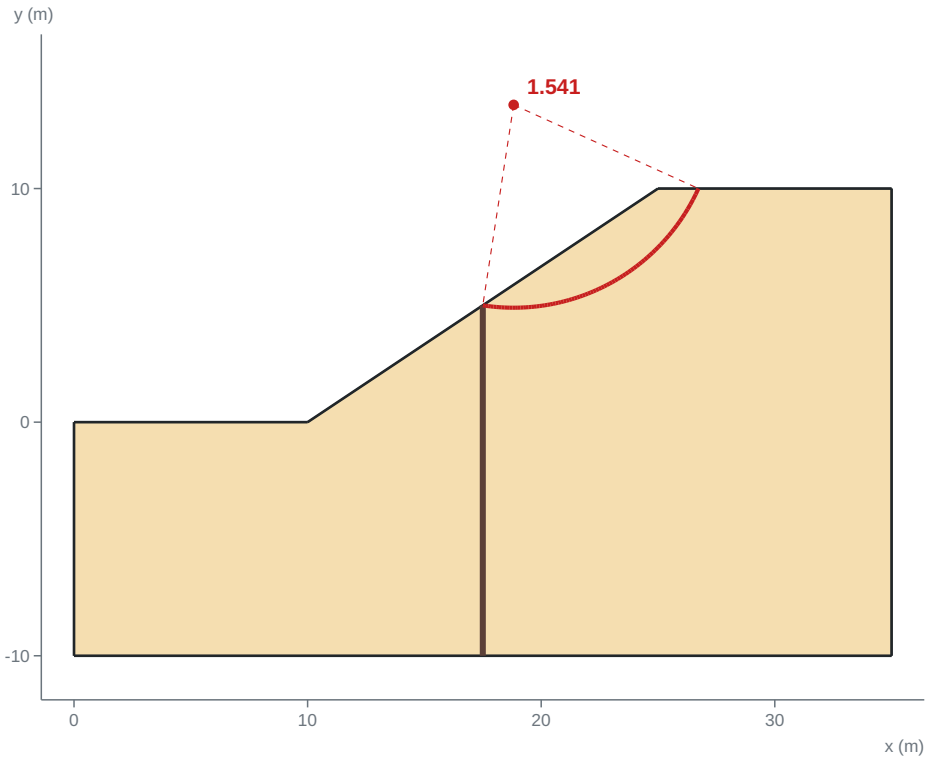


Figure 21. V013b: critical circle (Bishop simplified, FOS = 1.541).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	20.00	c' = 10.00	20.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.540	1.541

Notes

- The critical circle passes through the pile head and carries no pile force; Cai & Ugai 1.54 on a surface through the pile.

3.22 V013c — Cai & Ugai slope, pile row D1/D = 3

Cai & Ugai (2000), Géotechnique 50(2).

Critical circle search · Units: SI (m, kPa, kN/m³)

As V013b at 2.4 m spacing.

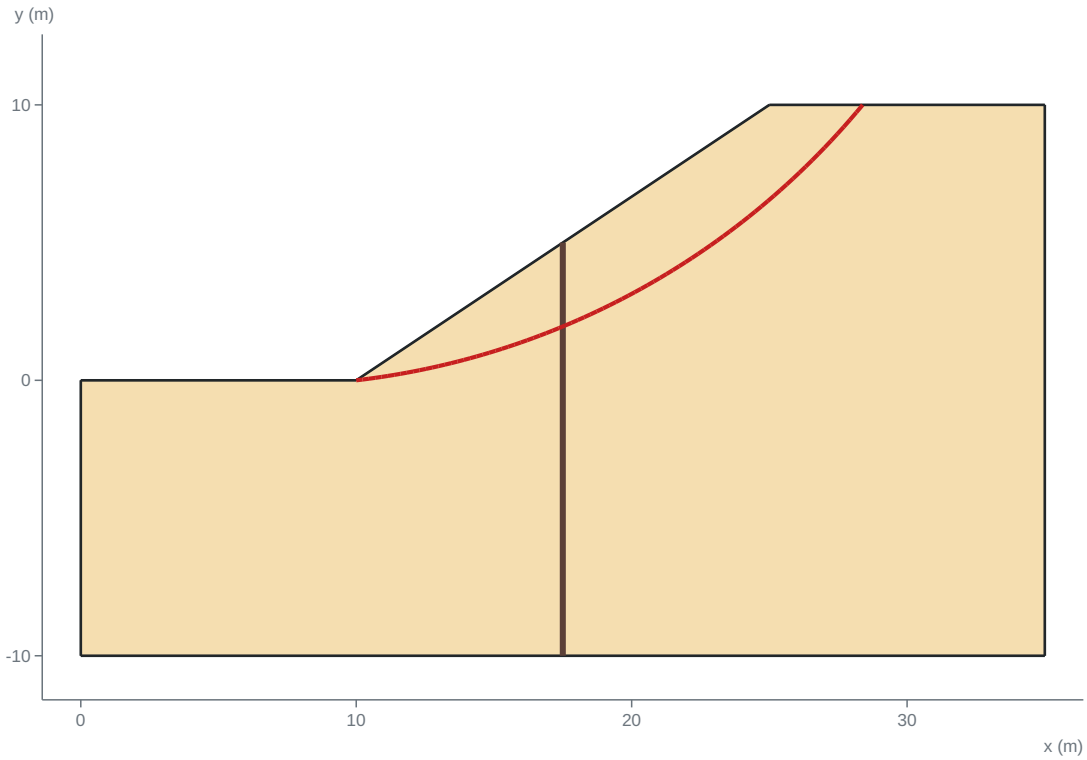


Figure 22. V013c: critical circle (Bishop simplified, FOS = 1.450).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	20.00	c' = 10.00	20.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.430	1.450

Notes

- program D 1.451; Cai & Ugai 1.37.

3.23 V013d — Cai & Ugai slope, pile row D1/D = 4

Cai & Ugai (2000), Géotechnique 50(2).

Critical circle search · Units: SI (m, kPa, kN/m³)

As V013b at 3.2 m spacing.

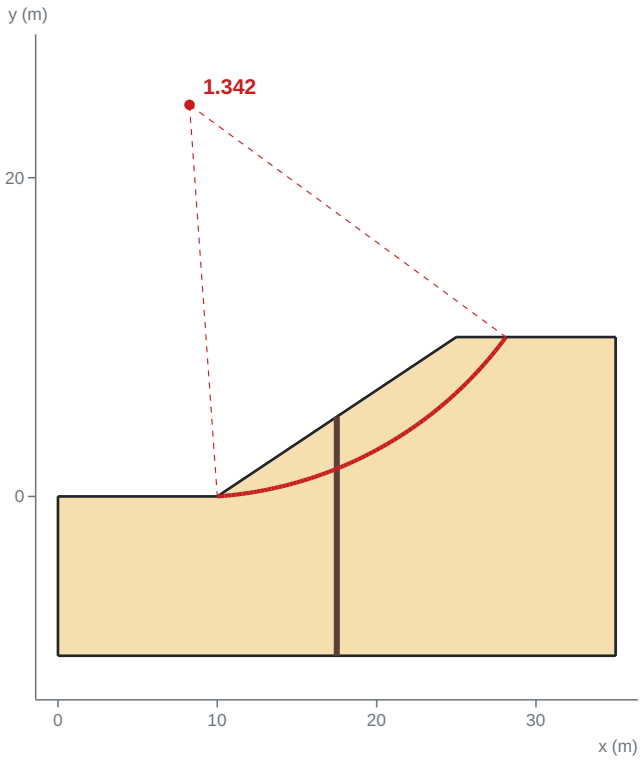


Figure 23. V013d: critical circle (Bishop simplified, FOS = 1.342).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	20.00	c' = 10.00	20.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.330	1.342

Notes

- Cai & Ugai 1.31.

3.24 V013e — Cai & Ugai slope, pile row D1/D = 6

Cai & Ugai (2000), Géotechnique 50(2).

Critical circle search · Units: SI (m, kPa, kN/m³)

As V013b at 4.8 m spacing.

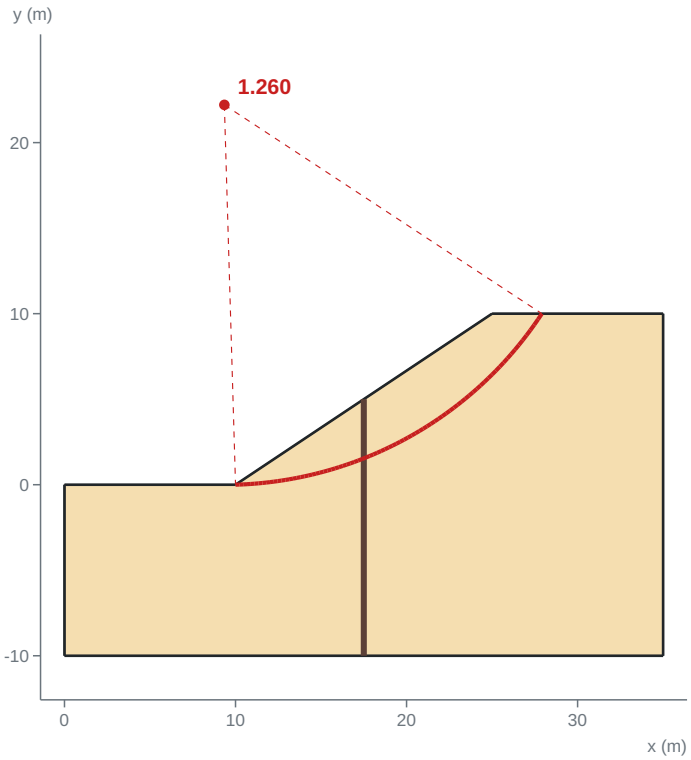


Figure 24. V013e: critical circle (Bishop simplified, FOS = 1.260).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	20.00	c' = 10.00	20.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.250	1.260

Notes

- Cai & Ugai 1.25.

3.25 V014 — Nailed wall, 14 rows, deep wedge

Caltrans soil nail reference manual.

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

Nail-reinforced wall with 14 horizontal rows (per-row length, capacity and bond; fully anchored at the wall, tangent, active) on the printed bilinear wedge. Converted to SI.

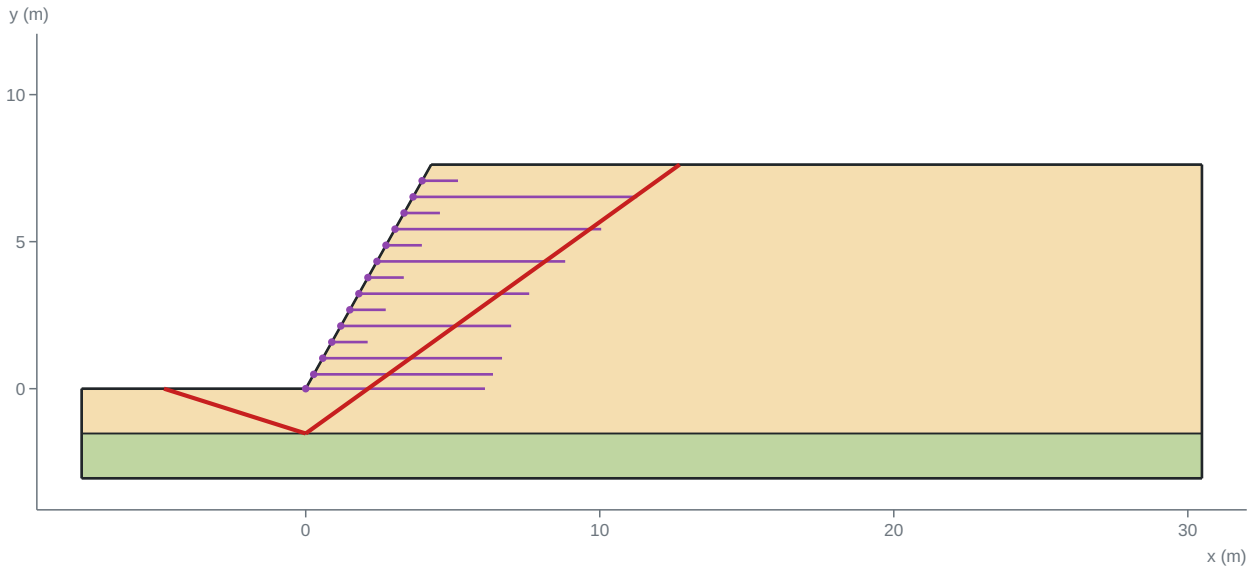


Figure 25. V014: analysed surface (Janbu corrected (f0), FOS = 1.448).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Layer 1	Mohr-Coulomb	19.64	c' = 0.00	32.00°
Layer 2	Mohr-Coulomb	20.11	c' = 23.94	35.00°

Method	Published by	Reference	cSlope
Janbu corrected (f0)	Program H	1.460	1.448
Janbu corrected (f0)	Program D	1.448	1.448
Spencer (M-P, constant f(x))	Program D	1.576	1.575

Notes

- commercial program A 1.417; commercial program B M-P 1.606.

3.26 V015 — Soldier-pile tieback wall

Caltrans soil nail reference manual.

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

Two-layer slope cut by a soldier-pile wall with two grouted tieback rows (8 ft spacing); the soldier pile is a shear force of 5900 lb/ft where the surface passes, as commercial program A models it; the manual's bilinear wedge from the wall toe. Converted to SI.

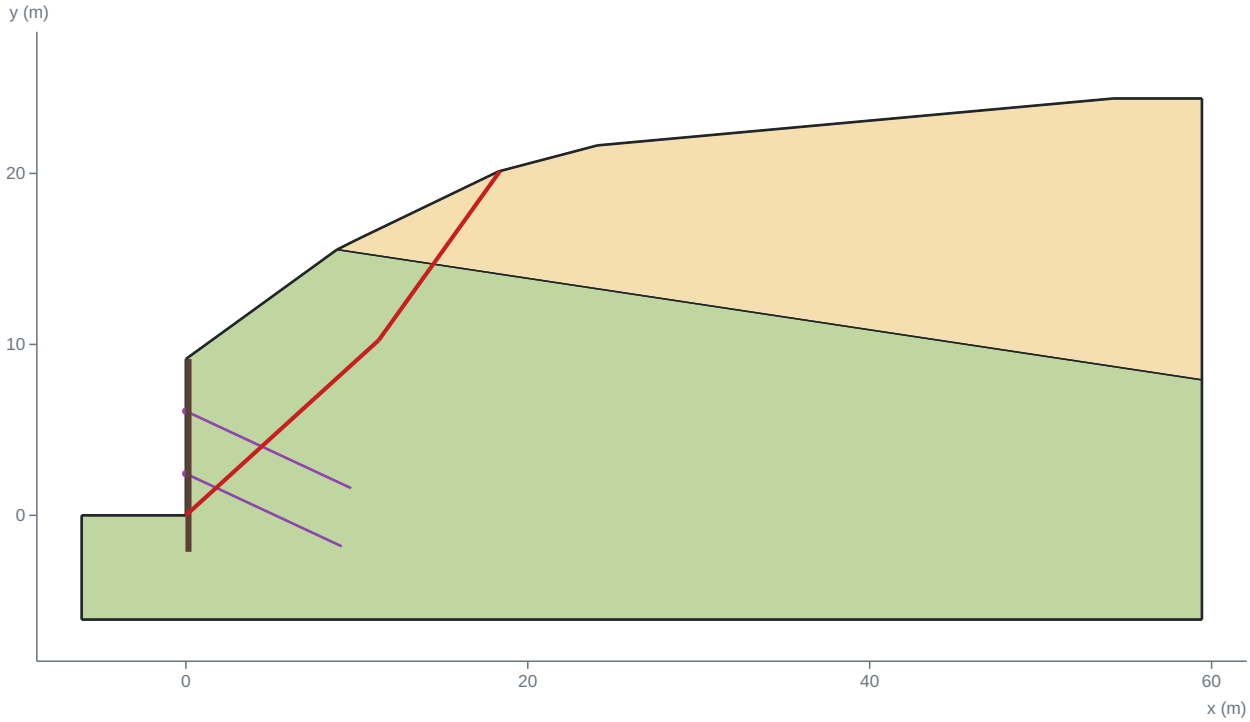


Figure 26. V015: analysed surface (Janbu corrected (f0), FOS = 1.476).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Layer 1	Mohr-Coulomb	18.85	c' = 28.73	24.00°
Layer 2	Mohr-Coulomb	20.42	c' = 14.36	34.00°

Method	Published by	Reference	cSlope
Janbu corrected (f0)	Commercial program A	1.479	1.476
Janbu simplified (uncorrected)	Commercial program A	1.446	1.442

Notes

- Layer boundary from (29, 51) ft and ground vertex (79, 71) ft as labelled in Fig. 49.1 (the earlier transcription had (33, 53) and (70, 71)). program H 1.52.
- program D 1.469 (Janbu corrected) and 1.439 (Spencer) on the earlier outline, where cSlope agreed to 0.1 %.

3.27 V016a — Stepped gabion wall, equivalent cohesion

Cao et al. (2016); Grodecki (2017).

Fixed circle · Units: SI (m, kPa, kN/m³)

4 m gabion wall of 1 m cubes with the steps facing outwards on a sloping two-layer profile, dry; mesh as an equivalent cohesion (59.7 kPa); the reference circle.

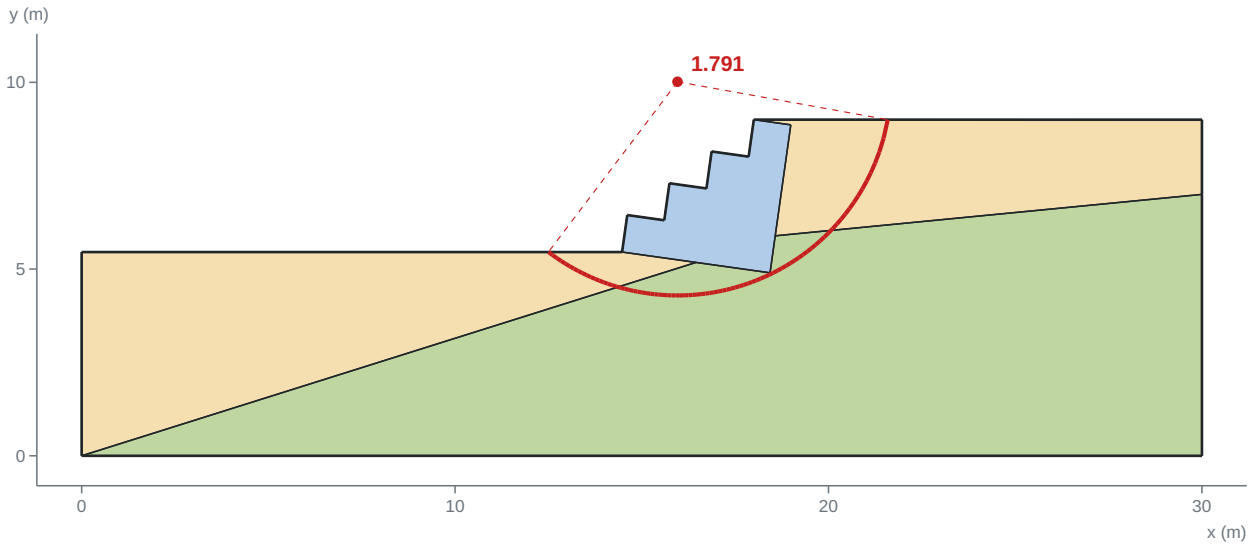


Figure 27. V016a: analysed circle (Bishop simplified, FOS = 1.791).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil #1	Mohr-Coulomb	21.00	c' = 5.00	30.00°
Soil #2	Mohr-Coulomb	20.00	c' = 0.00	25.00°
Gabion wall	Mohr-Coulomb	23.00	c' = 59.70	42.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.787	1.791
Spencer (M-P, constant f(x))	Commercial program A	1.791	1.797
Bishop simplified	Program D	1.790	1.791

3.28 V016b — Stepped gabion wall, mesh as geosynthetic supports

Cao et al. (2016); Grodecki (2017).

Fixed circle · Units: SI (m, kPa, kN/m³)

As V016a with the mesh as geosynthetic supports (T = 100 kN/m) at the course interfaces.

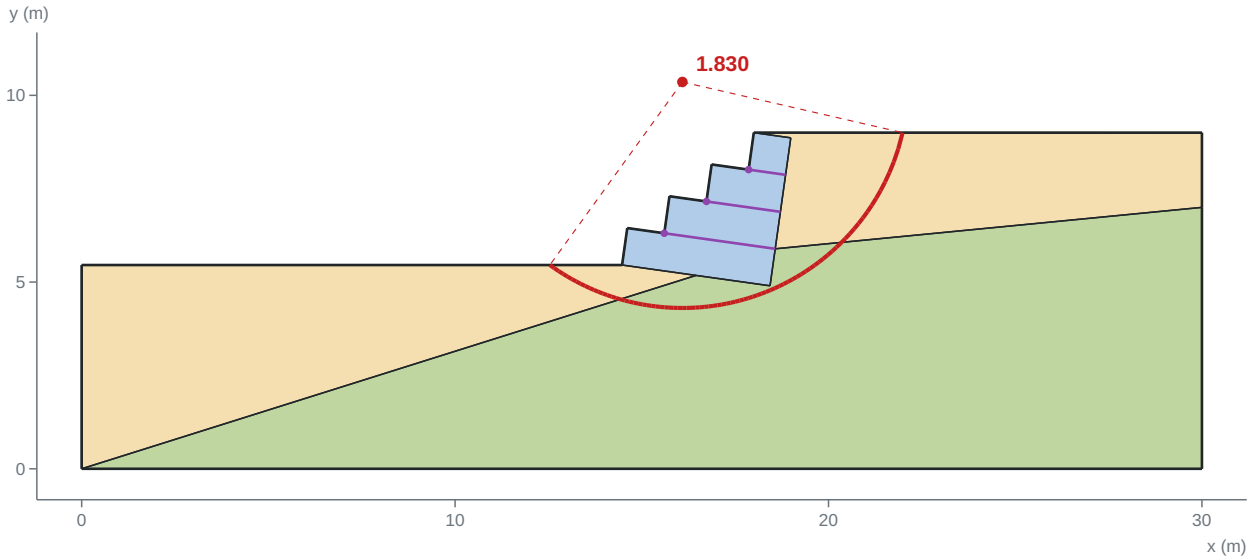


Figure 28. V016b: analysed circle (Bishop simplified, FOS = 1.830).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil #1	Mohr-Coulomb	21.00	c' = 5.00	30.00°
Soil #2	Mohr-Coulomb	20.00	c' = 0.00	25.00°
Gabion wall	Mohr-Coulomb	23.00	c' = 0.00	42.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.835	1.830
Spencer (M-P, constant f(x))	Commercial program A	1.839	1.835
Bishop simplified	Program D	1.830	1.830

3.29 V016c — Gabion wall with weak joint layers

Cao et al. (2016); Grodecki (2017).

Fixed circle · Units: SI (m, kPa, kN/m³)

The V016 wall with 6 cm weak bands at the course interfaces (φ = 37.8°, c = 20.4 kPa from the joint tensile strength); the reference circle through the joints.

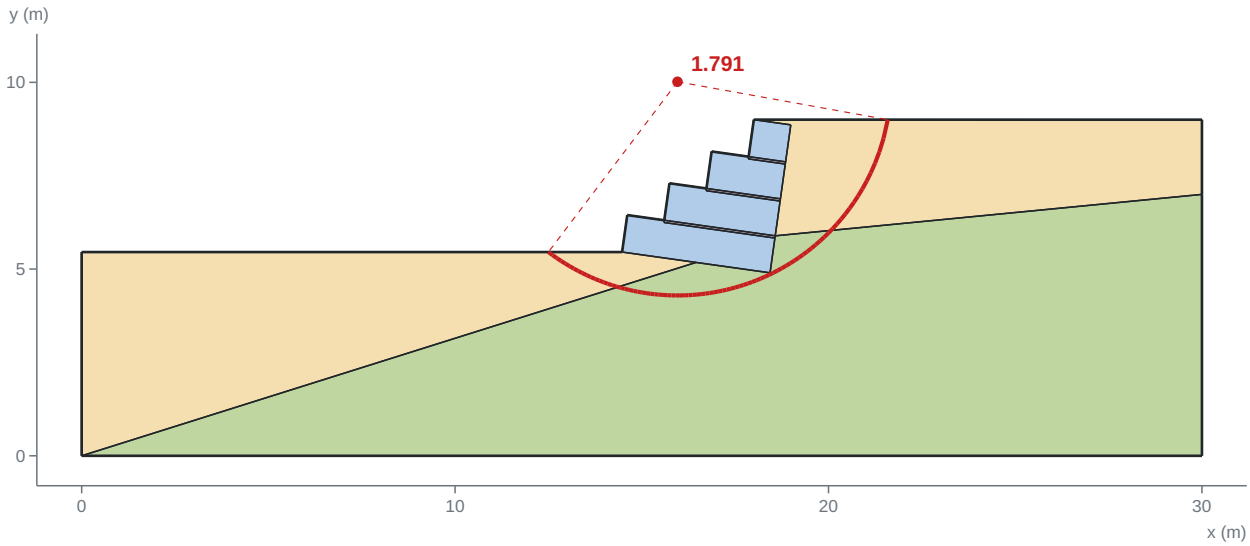


Figure 29. V016c: analysed circle (Bishop simplified, FOS = 1.791).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil #1	Mohr-Coulomb	21.00	c' = 5.00	30.00°
Soil #2	Mohr-Coulomb	20.00	c' = 0.00	25.00°
Gabion wall	Mohr-Coulomb	23.00	c' = 59.70	42.00°
Weak joint	Mohr-Coulomb	23.00	c' = 20.40	37.80°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.799	1.791
Spencer (M-P, constant f(x))	Commercial program A	1.803	1.797
Bishop simplified	Program D	1.790	1.791

Notes

- commercial program A's block search along the joints gives 1.516 at the wall face.

3.30 V017a — Vancouver gabion wall, equivalent cohesion

Cao et al. (2016); Grodecki (2017).

Fixed circle · Units: SI (m, kPa, kN/m³)

6 m battered gabion wall retaining a 30° backfill with a 12 kPa crest surcharge and a water table in the retained slope; the gabion mesh as an equivalent cohesion (100 kPa); the reference circle.

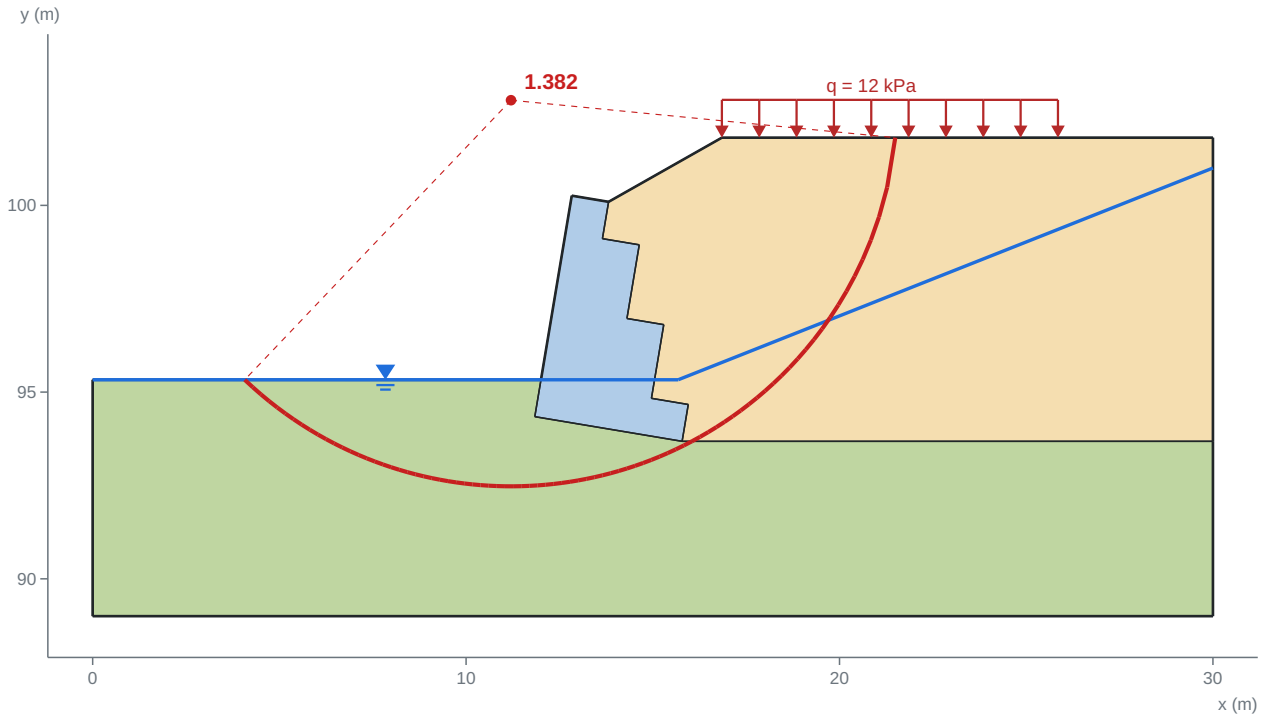


Figure 30. V017a: analysed circle (Bishop simplified, FOS = 1.382).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil #1	Mohr-Coulomb	21.00	c' = 0.00	32.00°
Soil #2	Mohr-Coulomb	20.00	c' = 0.00	30.00°
Gabion wall	Mohr-Coulomb	20.00	c' = 100.0	45.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.373	1.382
Spencer (M-P, constant f(x))	Commercial program A	1.386	1.398
Bishop simplified	Program D	1.382	1.382

Notes

- commercial program A's unfiltered minima along the wall face: 1.032–1.034.

3.31 V017b — Vancouver gabion wall, mesh as geosynthetic supports

Cao et al. (2016); Grodecki (2017).

Fixed circle · Units: SI (m, kPa, kN/m³)

As V017a with the steel mesh as geosynthetic supports at every course interface instead of the equivalent cohesion.

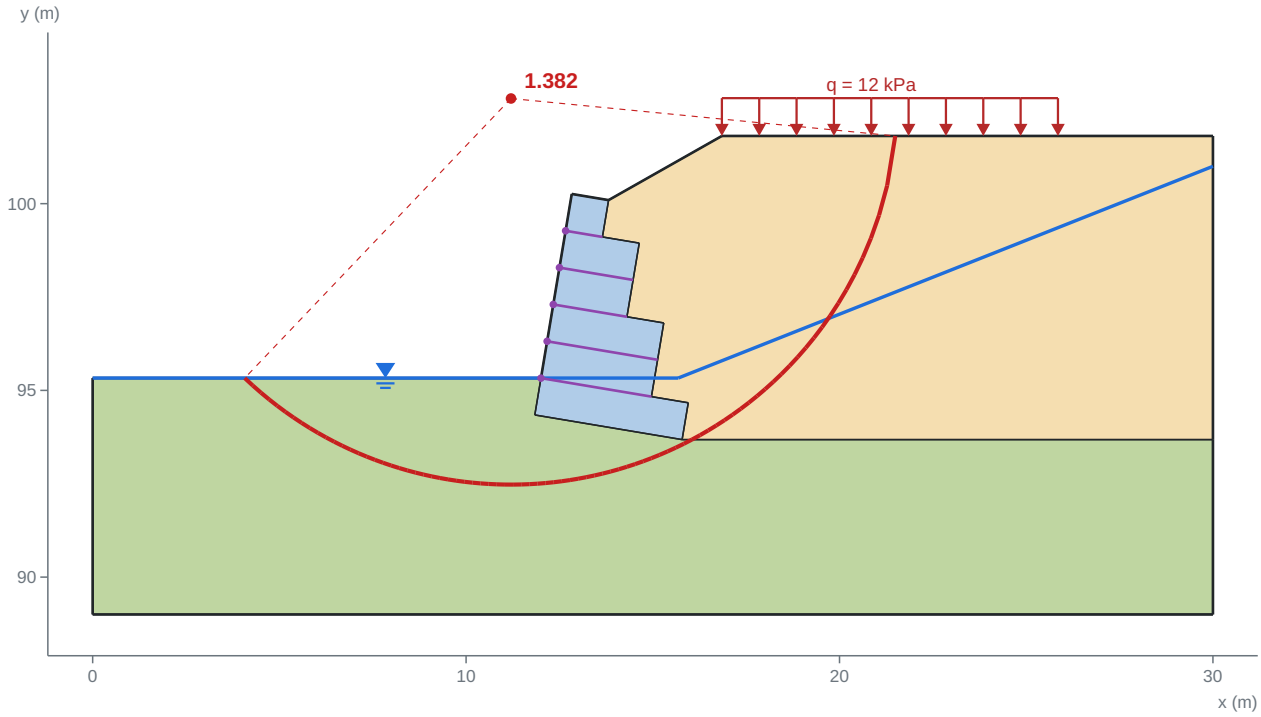


Figure 31. V017b: analysed circle (Bishop simplified, FOS = 1.382).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil #1	Mohr-Coulomb	21.00	c' = 0.00	32.00°
Soil #2	Mohr-Coulomb	20.00	c' = 0.00	30.00°
Gabion wall	Mohr-Coulomb	20.00	c' = 0.00	45.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.378	1.382
Bishop simplified	Program D	1.382	1.382

3.32 V018 — Chen & Shao (1988) frictionless 60° slope, Prandtl mechanism

Chen & Shao (1988).

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

Weightless, frictionless 10 m slope at 60° under the critical strip load (149.31 kPa) on the Prandtl surface (wedge, fan and passive wedge); theoretical FOS 1.0.

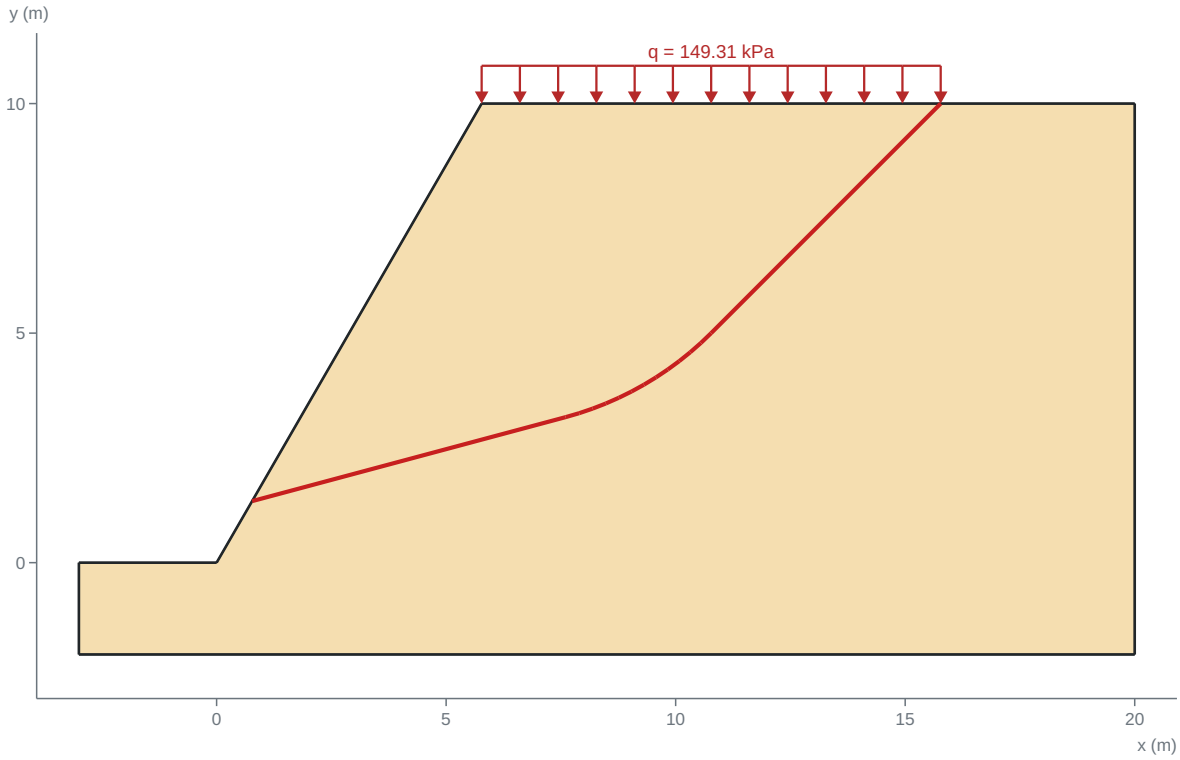


Figure 32. V018: analysed surface (Spencer (M-P, constant f(x)), FOS = 1.052).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Clay	Mohr-Coulomb	0.00	c' = 49.00	0.00°

Method	Published by	Reference	cSlope
Spencer (M-P, constant f(x))	Commercial program A	1.051	1.052
Morgenstern-Price (half-sine)	Commercial program A	1.009	1.008
Morgenstern-Price (half-sine)	Commercial program B	1.017	1.008

Notes

- Chen & Shao give 1.05 with the method of slices; limit equilibrium on this mechanism is about 5 % above the plasticity solution.
- f(x) as printed: (0, 1), (0.3, 1), (0.6, 0), (1, 0) with x from the toe end of the slip surface (model space). Until the data points are mirrored for a right-to-left model cSlope reads them from the crest end and gives 1.079 (+6.9 %); entered reversed it gives 1.008.
- Table 35 f(x): (0, 1), (0.3, 1), (0.6, 0), (1, 0) from the toe end (model space). Until the data points are mirrored for a right-to-left model cSlope reads them from the crest end; entered reversed it gives 1.008.

3.33 V019 — Cheng et al. 11 m homogeneous slope

Cheng, Lansivaara & Wei (2007), Computers and Geotechnics 34.

Critical circle search · Units: SI (m, kPa, kN/m³)

11 m homogeneous slope (c = 10 kPa, φ = 30°, γ = 20 kN/m³); circular search.

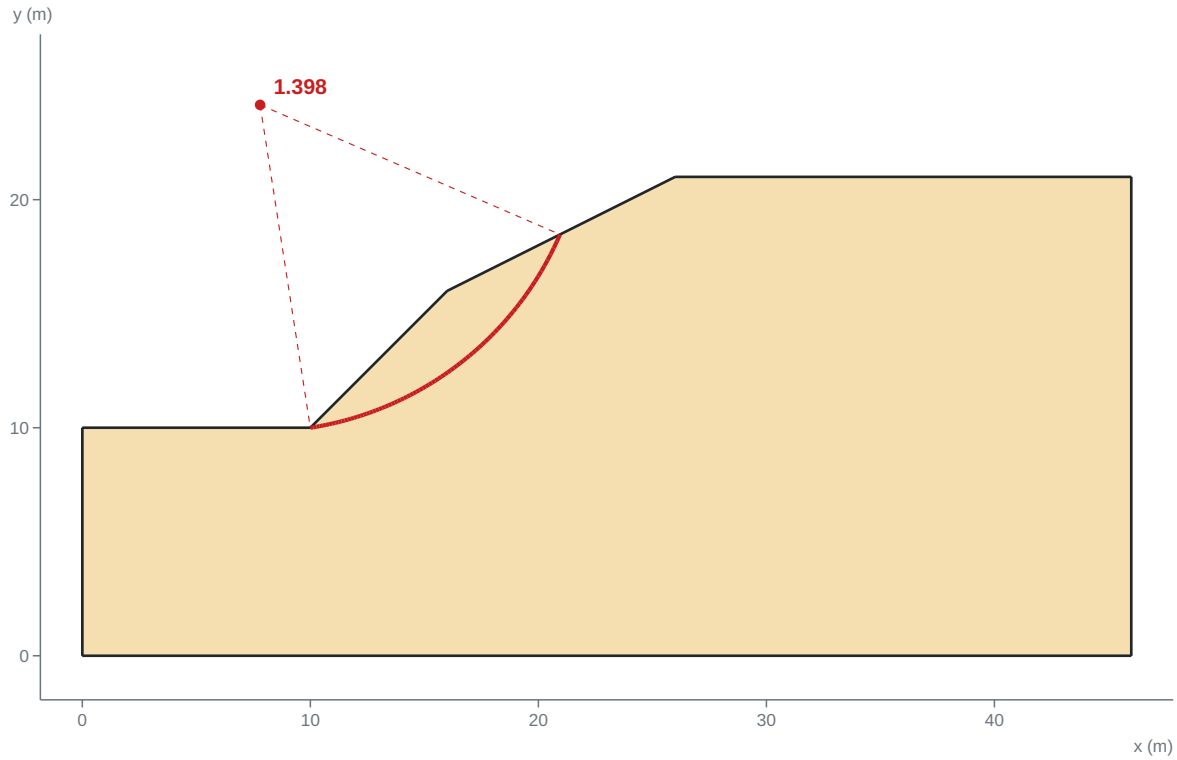


Figure 33. V019: critical circle (Spencer (M-P, constant f(x)), FOS = 1.398).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
soil	Mohr-Coulomb	20.00	c' = 10.00	30.00°

Method	Published by	Reference	cSlope
Spencer (M-P, constant f(x))	Commercial program A	1.380	1.398
Spencer (M-P, constant f(x))	Cheng et al.	1.383	1.398
Spencer (M-P, constant f(x))	Program D	1.398	1.398
Bishop simplified	Program D	1.401	1.401

Notes

- commercial program E shear strength reduction 1.38.

3.34 V020 — Cheng et al. benched slope, global minimum

Cheng, Lansivaara & Wei (2007), Computers and Geotechnics 34.

Critical circle search · Units: SI (m, kPa, kN/m³)

Homogeneous benched slope; the global minimum on the mid-lower face by circular search.

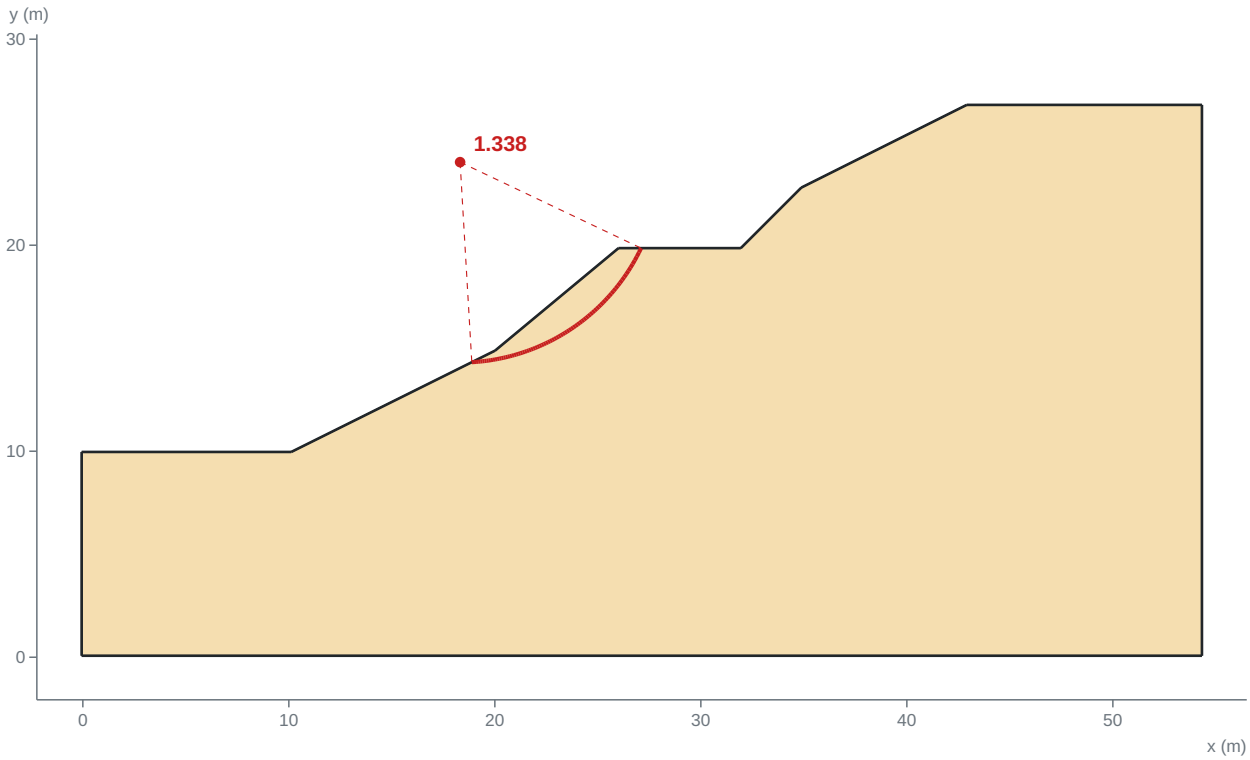


Figure 34. V020: critical circle (Spencer (M-P, constant f(x)), FOS = 1.338).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
soil	Mohr-Coulomb	20.00	c' = 5.00	30.00°

Method	Published by	Reference	cSlope
Spencer (M-P, constant f(x))	Commercial program A	1.336	1.338
Spencer (M-P, constant f(x))	Cheng et al.	1.327	1.338
Spencer (M-P, constant f(x))	Program D	1.338	1.338
Bishop simplified	Program D	1.342	1.343

Notes

- commercial program E shear strength reduction 1.35.

3.35 V021a — Chowdhury & Xu (1995) example 1, Congress Street cut, circle tangent to the bas

Chowdhury & Xu (1995); Ireland (1954).

Fixed circle · Units: SI (m, kPa, kN/m³)

Chowdhury & Xu example 1, the Congress Street cut as commercial program B models it (Su = 55, 43, 56 kPa under a strengthless sand cap, digitised piezometric line), on the critical circle tangent to the base of clay 2.

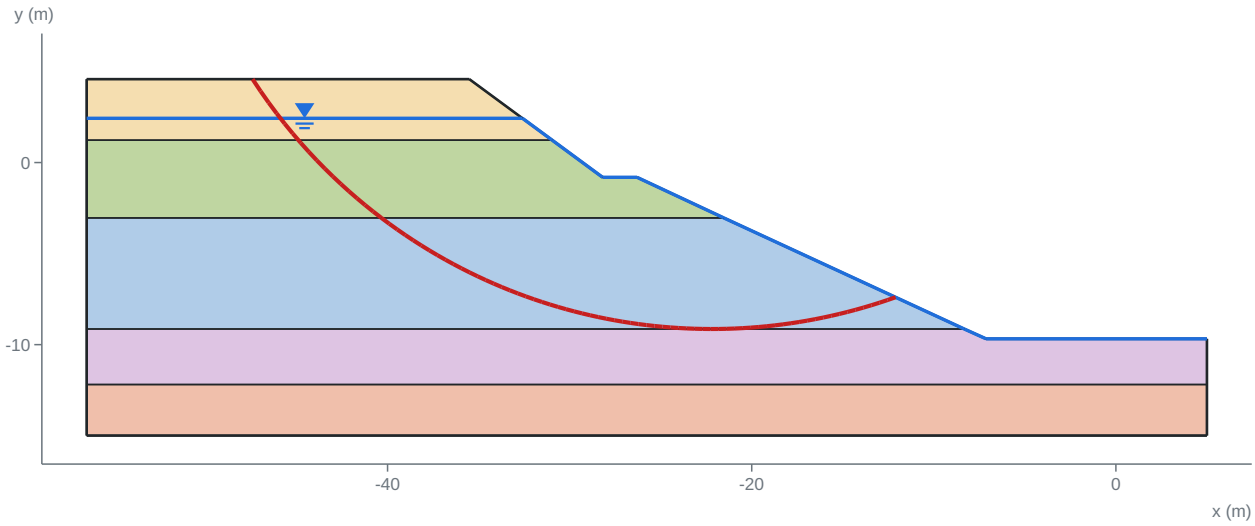


Figure 35. V021a: analysed circle (Morgenstern-Price (half-sine), FOS = 1.128).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Sand	Undrained (φ = 0)	22.00	Su = 0.00	0°
Clay 1	Undrained (φ = 0)	21.00	Su = 55.00	0°
Clay 2	Undrained (φ = 0)	22.00	Su = 43.00	0°
Clay 3	Undrained (φ = 0)	22.00	Su = 56.00	0°
Bed	Mohr-Coulomb	22.00	c' = 200.0	35.00°

Method	Published by	Reference	cSlope
Morgenstern-Price (half-sine)	Commercial program B	1.132	1.128
Bishop simplified	Chowdhury & Xu	1.128	1.128
Bishop simplified	Commercial program A	1.128	1.128

Probabilistic analysis on the same surface: 3 normal variables (independent); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1). Differences in β and Pf are absolute.

Quantity	Published by	Reference	cSlope
Pf, share FOS < 1 (Monte Carlo)	Commercial program A	0.246	0.239
β lognormal (Monte Carlo)	Commercial program A (lognormal)	0.650	0.671

Probabilistic analysis on the same surface: 3 normal variables (independent); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1). Differences in β and Pf are absolute.

Quantity	Published by	Reference	cSlope
Pf, share FOS < 1 (Monte Carlo)	Commercial program B	0.253	0.245

Notes

- Fixed circle = cSlope's critical M-P circle among circles tangent to the base of the clay layer (cSlope has no tangent-line search; commercial program B's own circle is not printed), so the comparison is that of two searches within the same circle family.
- Sand cap strength ignored (Su = 0); its unit weight is not printed: γ = 22 (as commercial program A a related problem / cSlope V021a) reproduces commercial program B to ≤ 0.3 % on example 1, while γ = 21 (program D's reading of the vendor .gsz) gives +1.5 %. Bed (c′ = 200, φ′ = 35°, γ = 22) is never reached. Piezometric line digitised from Fig. 51 (y ≈ 2.43 m, then the ground surface); it matters only for example 4 (φ > 0): without it cSlope is 11–14 % above commercial program B.
- A free search is meaningless here (the zero-strength sand gives FOS → 0); the critical circle must be searched within the family tangent to the layer base.
- Critical circle (model coordinates): centre (-22.188, 20.908), R 30.049, tangent to y = -9.141.

3.36 V021b — Chowdhury & Xu (1995) example 1, Congress Street cut, circle tangent to the bas

Chowdhury & Xu (1995); Ireland (1954).

Fixed circle · Units: SI (m, kPa, kN/m³)

As V021a on the critical circle tangent to the base of clay 3.

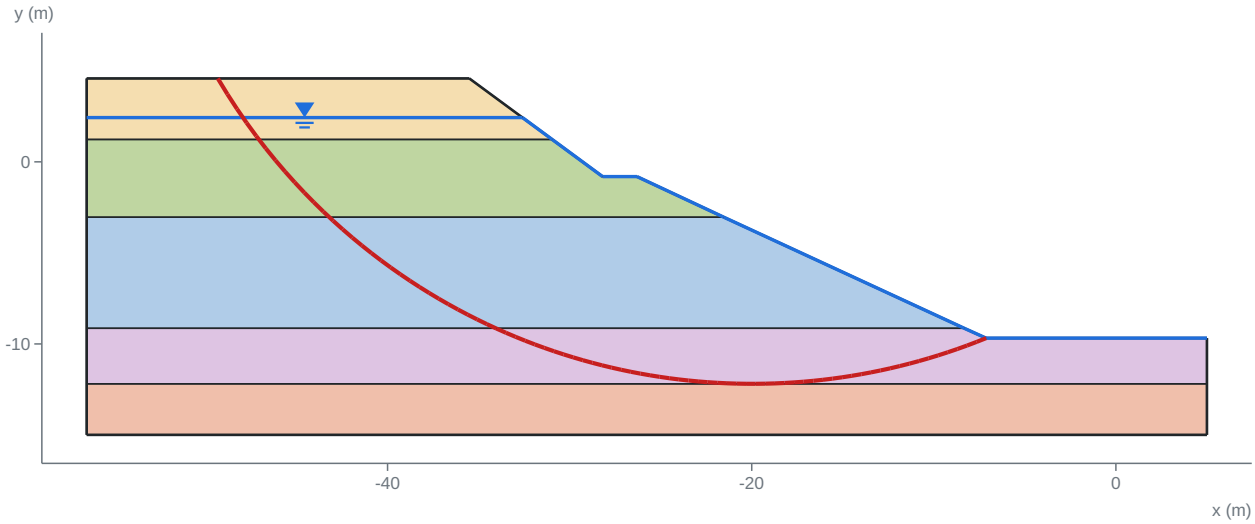


Figure 36. V021b: analysed circle (Morgenstern-Price (half-sine), FOS = 1.110).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Sand	Undrained (φ = 0)	22.00	Su = 0.00	0°
Clay 1	Undrained (φ = 0)	21.00	Su = 55.00	0°
Clay 2	Undrained (φ = 0)	22.00	Su = 43.00	0°
Clay 3	Undrained (φ = 0)	22.00	Su = 56.00	0°
Bed	Mohr-Coulomb	22.00	c' = 200.0	35.00°

Method	Published by	Reference	cSlope
Morgenstern-Price (half-sine)	Commercial program B	1.116	1.110
Bishop simplified	Chowdhury & Xu	1.109	1.110
Bishop simplified	Commercial program A	1.109	1.110

Probabilistic analysis on the same surface: 3 normal variables (independent); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1). Differences in β and Pf are absolute.

Quantity	Published by	Reference	cSlope
Pf, share FOS < 1 (Monte Carlo)	Commercial program A	0.279	0.282
β lognormal (Monte Carlo)	Commercial program A (lognormal)	0.537	0.553

Notes

- Fixed circle = cSlope's critical M-P circle among circles tangent to the base of the clay layer (cSlope has no tangent-line search; commercial program B's own circle is not printed), so the comparison is that of two searches within the same circle family.
- Sand cap strength ignored (Su = 0); its unit weight is not printed: γ = 22 (as commercial program A a related problem / cSlope V021a) reproduces commercial program B to ≤ 0.3 % on example 1, while γ = 21 (program D's reading of the vendor .gsz) gives

+1.5 %. Bed (c′ = 200, φ′ = 35°, γ = 22) is never reached. Piezometric line digitised from Fig. 51 (y ≈ 2.43 m, then the ground surface); it matters only for example 4 (φ > 0): without it cSlope is 11–14 % above commercial program B.

- A free search is meaningless here (the zero-strength sand gives FOS → 0); the critical circle must be searched within the family tangent to the layer base.
- Critical circle (model coordinates): centre (-19.961, 21.899), R 34.089, tangent to y = -12.19.
- commercial program B's probability of failure, 26.15 %, is not checked: cSlope's Monte Carlo on this circle gives 28.6 % (Δ 0.025). On commercial program A's similar circle (V021b) cSlope gives 28.2 % against commercial program A's 27.89 %.

3.37 V021c — Chowdhury & Xu (1995) example 2, Congress Street cut, circle tangent to the bas

Chowdhury & Xu (1995); Ireland (1954).

Fixed circle · Units: SI (m, kPa, kN/m³)

Chowdhury & Xu example 2 (Su = 68.1, 39.3, 50.8 kPa), circle tangent to the base of clay 2.

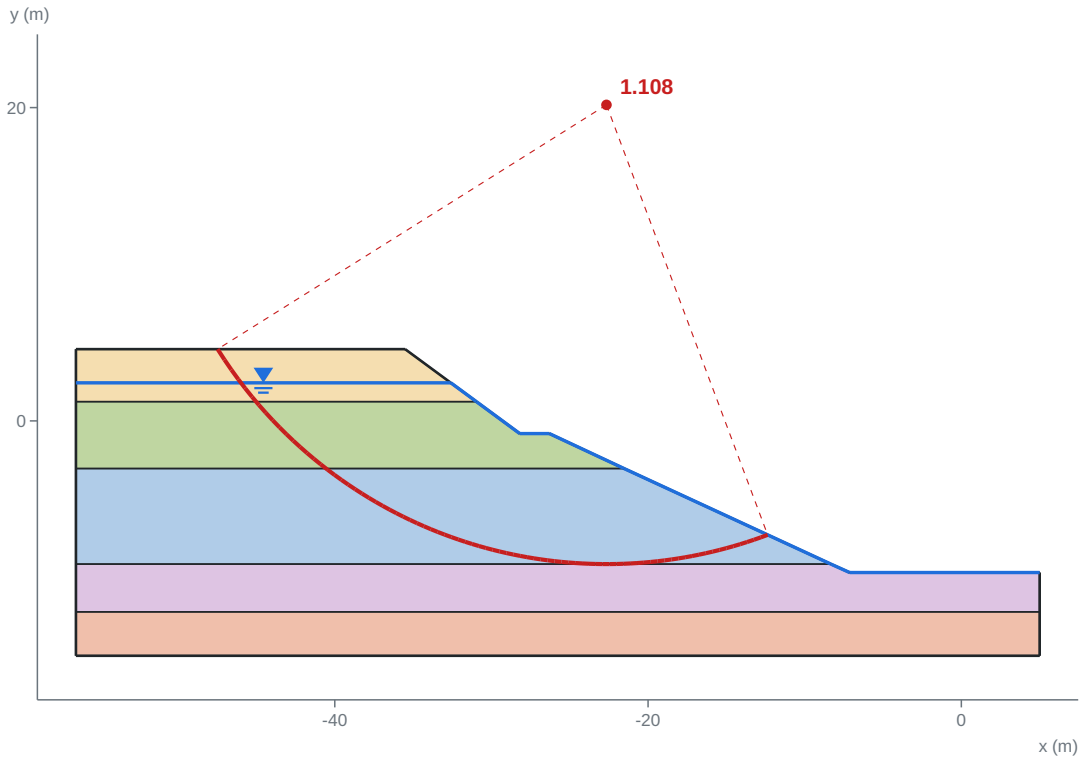


Figure 37. V021c: analysed circle (Morgenstern-Price (half-sine), FOS = 1.108).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Sand	Undrained (φ = 0)	22.00	Su = 0.00	0°
Clay 1	Undrained (φ = 0)	21.00	Su = 68.10	0°
Clay 2	Undrained (φ = 0)	22.00	Su = 39.30	0°
Clay 3	Undrained (φ = 0)	22.00	Su = 50.80	0°
Bed	Mohr-Coulomb	22.00	c' = 200.0	35.00°

Method	Published by	Reference	cSlope
Morgenstern-Price (half-sine)	Commercial program B	1.110	1.108
Bishop simplified	Chowdhury & Xu	1.110	1.108
Bishop simplified	Commercial program A	1.108	1.108

Notes

- Fixed circle = cSlope's critical M-P circle among circles tangent to the base of the clay layer (cSlope has no tangent-line search; commercial program B's own circle is not printed), so the comparison is that of two searches within the same circle family.
- Sand cap strength ignored (Su = 0); its unit weight is not printed: γ = 22 (as commercial program A a related problem / cSlope V021a) reproduces commercial program B to ≤ 0.3 % on example 1, while γ = 21 (program D's reading of the vendor .gsz) gives +1.5 %. Bed (c′ = 200, φ′ = 35°, γ = 22) is never reached. Piezometric line digitised from Fig. 51 (y ≈ 2.43 m, then the ground surface); it matters only for example 4 (φ > 0): without it cSlope is 11–14 % above commercial program B.
- A free search is meaningless here (the zero-strength sand gives FOS → 0); the critical circle must be searched within the family tangent to the layer base.
- Critical circle (model coordinates): centre (-22.656, 20.174), R 29.315, tangent to y = -9.141.

3.38 V021d — Chowdhury & Xu (1995) example 2, Congress Street cut, circle tangent to the bas

Chowdhury & Xu (1995); Ireland (1954).

Fixed circle · Units: SI (m, kPa, kN/m³)

Chowdhury & Xu example 2, circle tangent to the base of clay 3.

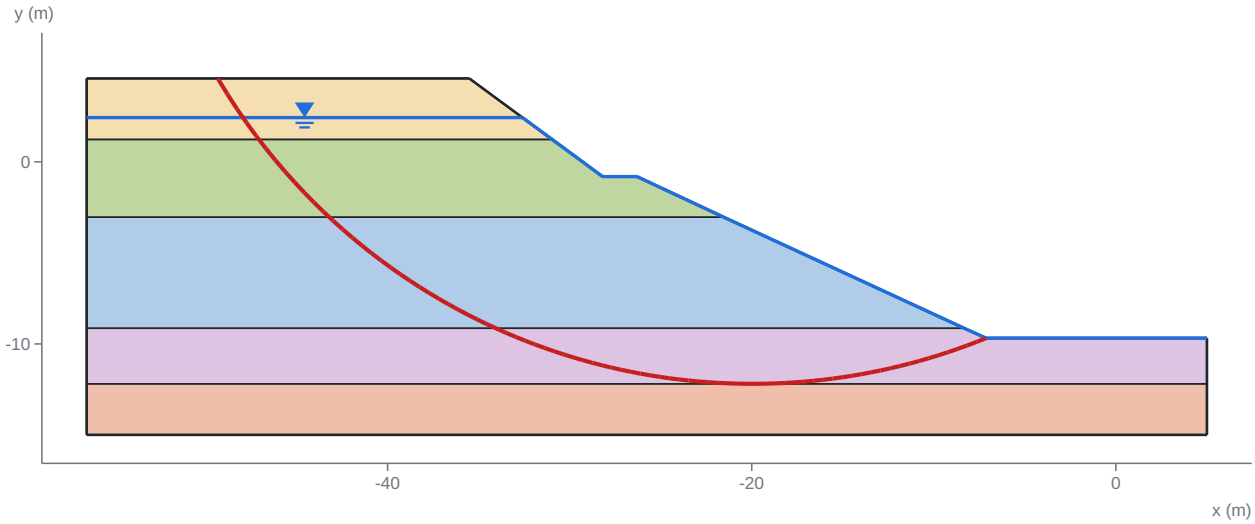


Figure 38. V021d: analysed circle (Morgenstern-Price (half-sine), FOS = 1.058).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Sand	Undrained (φ = 0)	22.00	Su = 0.00	0°
Clay 1	Undrained (φ = 0)	21.00	Su = 68.10	0°
Clay 2	Undrained (φ = 0)	22.00	Su = 39.30	0°
Clay 3	Undrained (φ = 0)	22.00	Su = 50.80	0°
Bed	Mohr-Coulomb	22.00	c' = 200.0	35.00°

Method	Published by	Reference	cSlope
Morgenstern-Price (half-sine)	Commercial program B	1.063	1.058
Bishop simplified	Chowdhury & Xu	1.064	1.058
Bishop simplified	Commercial program A	1.058	1.058

Notes

- Fixed circle = cSlope's critical M-P circle among circles tangent to the base of the clay layer (cSlope has no tangent-line search; commercial program B's own circle is not printed), so the comparison is that of two searches within the same circle family.
- Sand cap strength ignored (Su = 0); its unit weight is not printed: γ = 22 (as commercial program A a related problem / cSlope V021a) reproduces commercial program B to ≤ 0.3 % on example 1, while γ = 21 (program D's reading of the vendor .gsz) gives +1.5 %. Bed (c′ = 200, φ′ = 35°, γ = 22) is never reached. Piezometric line digitised from Fig. 51 (y ≈ 2.43 m, then the ground surface); it matters only for example 4 (φ > 0): without it cSlope is 11–14 % above commercial program B.
- A free search is meaningless here (the zero-strength sand gives FOS → 0); the critical circle must be searched within the family tangent to the layer base.
- Critical circle (model coordinates): centre (-19.961, 21.899), R 34.089, tangent to y = -12.19.

3.39 V021e — Chowdhury & Xu (1995) example 3, Congress Street cut, circle tangent to the bas

Chowdhury & Xu (1995); Ireland (1954).

Fixed circle · Units: SI (m, kPa, kN/m³)

Chowdhury & Xu example 3 (Su = 136, 80, 102 kPa), circle tangent to the base of clay 2.

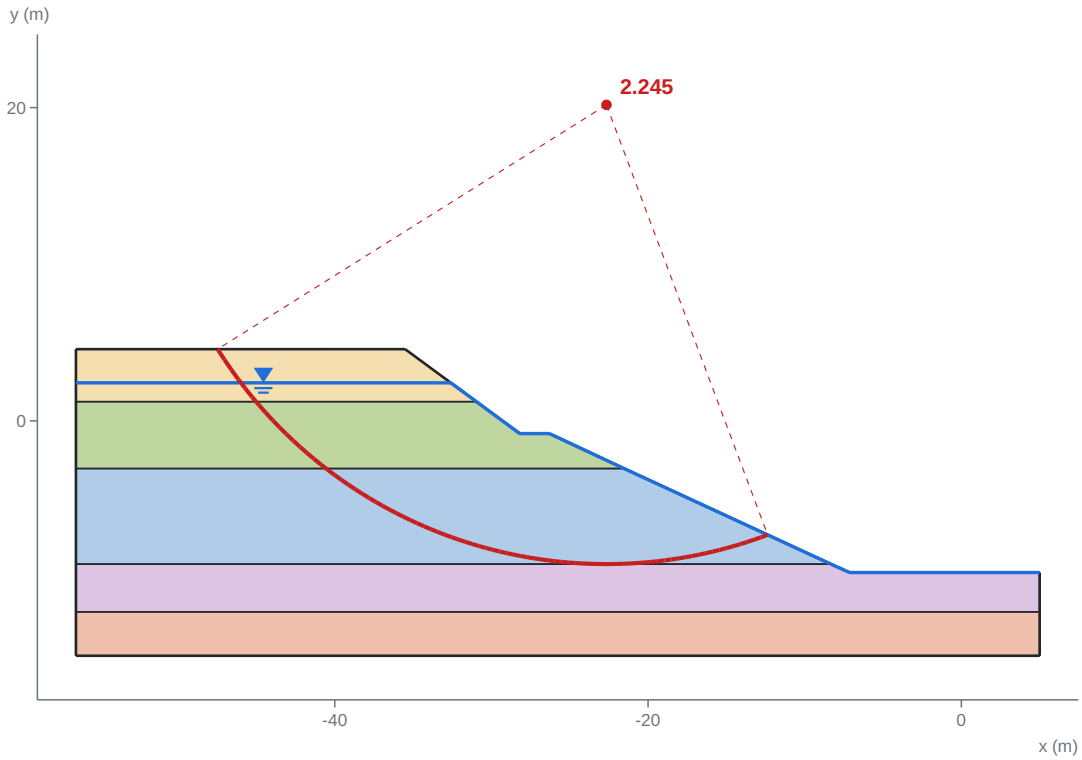


Figure 39. V021e: analysed circle (Morgenstern-Price (half-sine), FOS = 2.245).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Sand	Undrained (φ = 0)	22.00	Su = 0.00	0°
Clay 1	Undrained (φ = 0)	21.00	Su = 136.0	0°
Clay 2	Undrained (φ = 0)	22.00	Su = 80.00	0°
Clay 3	Undrained (φ = 0)	22.00	Su = 102.0	0°
Bed	Mohr-Coulomb	22.00	c' = 200.0	35.00°

Method	Published by	Reference	cSlope
Morgenstern-Price (half-sine)	Commercial program B	2.248	2.245
Bishop simplified	Chowdhury & Xu	2.234	2.245
Bishop simplified	Commercial program A	2.245	2.245

Notes

- Fixed circle = cSlope's critical M-P circle among circles tangent to the base of the clay layer (cSlope has no tangent-line search; commercial program B's own circle is not printed), so the comparison is that of two searches within the same circle family.
- Sand cap strength ignored (Su = 0); its unit weight is not printed: γ = 22 (as commercial program A a related problem / cSlope V021a) reproduces commercial program B to ≤ 0.3 % on example 1, while γ = 21 (program D's reading of the vendor .gsz) gives +1.5 %. Bed (c′ = 200, φ′ = 35°, γ = 22) is never reached. Piezometric line digitised from Fig. 51 (y ≈ 2.43 m, then the ground surface); it matters only for example 4 (φ > 0): without it cSlope is 11–14 % above commercial program B.
- A free search is meaningless here (the zero-strength sand gives FOS → 0); the critical circle must be searched within the family tangent to the layer base.
- Critical circle (model coordinates): centre (-22.656, 20.174), R 29.315, tangent to y = -9.141.

3.40 V021f — Chowdhury & Xu (1995) example 3, Congress Street cut, circle tangent to the base

Chowdhury & Xu (1995); Ireland (1954).

Fixed circle · Units: SI (m, kPa, kN/m³)

Chowdhury & Xu example 3, circle tangent to the base of clay 3.

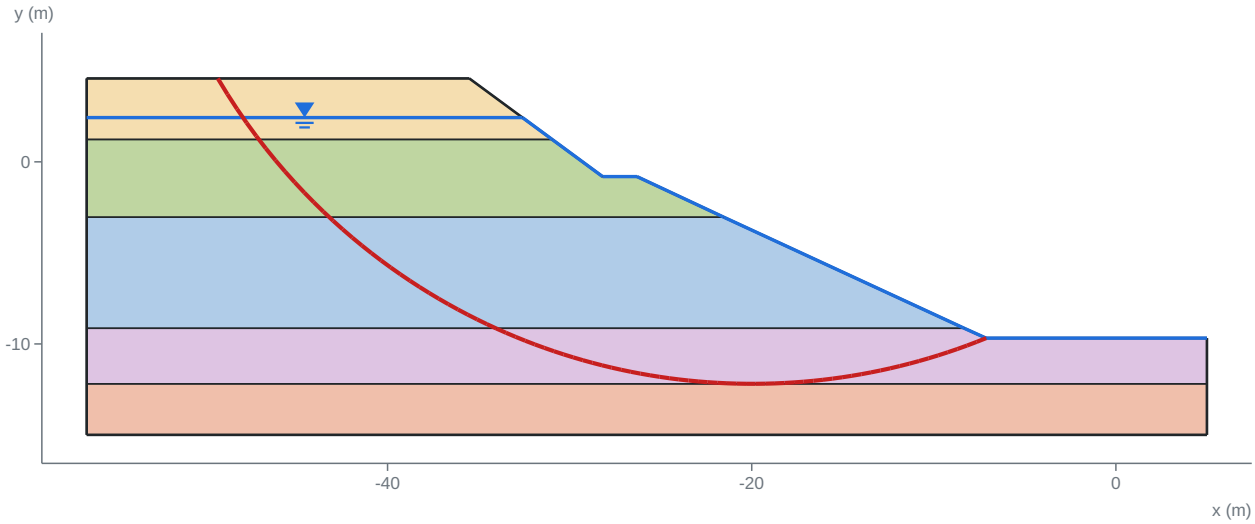


Figure 40. V021f: analysed circle (Morgenstern-Price (half-sine), FOS = 2.128).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Sand	Undrained (φ = 0)	22.00	Su = 0.00	0°
Clay 1	Undrained (φ = 0)	21.00	Su = 136.0	0°
Clay 2	Undrained (φ = 0)	22.00	Su = 80.00	0°
Clay 3	Undrained (φ = 0)	22.00	Su = 102.0	0°
Bed	Mohr-Coulomb	22.00	c' = 200.0	35.00°

Method	Published by	Reference	cSlope
Morgenstern-Price (half-sine)	Commercial program B	2.138	2.128
Bishop simplified	Chowdhury & Xu	2.140	2.128
Bishop simplified	Commercial program A	2.128	2.128

Notes

- Fixed circle = cSlope's critical M-P circle among circles tangent to the base of the clay layer (cSlope has no tangent-line search; commercial program B's own circle is not printed), so the comparison is that of two searches within the same circle family.
- Sand cap strength ignored (Su = 0); its unit weight is not printed: γ = 22 (as commercial program A a related problem / cSlope V021a) reproduces commercial program B to ≤ 0.3 % on example 1, while γ = 21 (program D's reading of the vendor .gsz) gives +1.5 %. Bed (c′ = 200, φ′ = 35°, γ = 22) is never reached. Piezometric line digitised from Fig. 51 (y ≈ 2.43 m, then the ground surface); it matters only for example 4 (φ > 0): without it cSlope is 11–14 % above commercial program B.
- A free search is meaningless here (the zero-strength sand gives FOS → 0); the critical circle must be searched within the family tangent to the layer base.
- Critical circle (model coordinates): centre (-19.961, 21.899), R 34.089, tangent to y = -12.19.

3.41 V021g — Chowdhury & Xu (1995) example 4, Congress Street cut, circle tangent to the bas

Chowdhury & Xu (1995); Ireland (1954).

Fixed circle · Units: SI (m, kPa, kN/m³)

Chowdhury & Xu example 4 (c-φ clays with the digitised piezometric line), circle tangent to the base of clay 2.

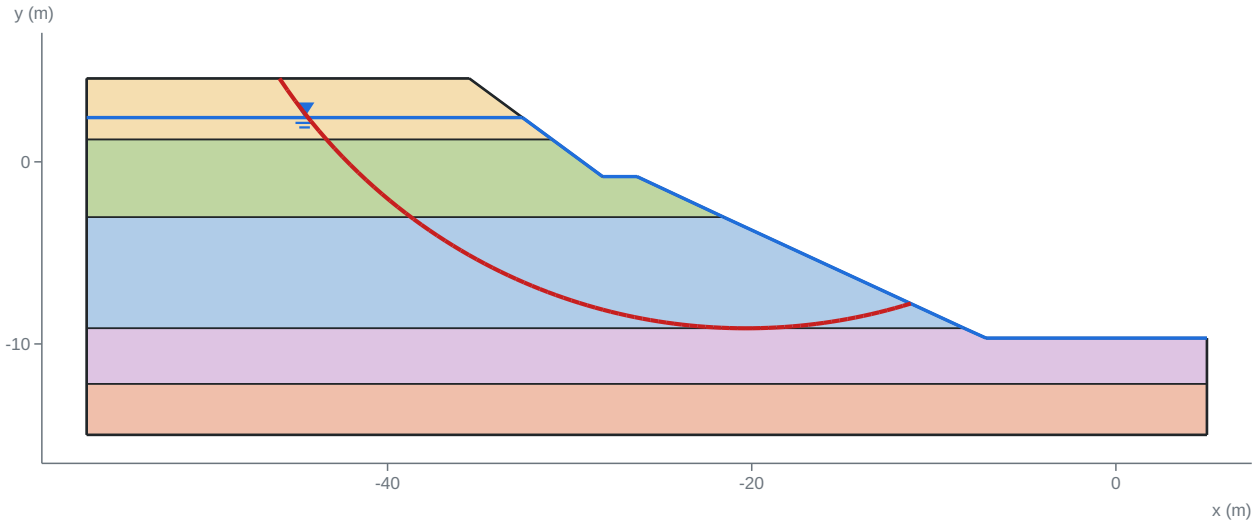


Figure 41. V021g: analysed circle (Morgenstern-Price (half-sine), FOS = 1.428).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Sand	Undrained (φ = 0)	22.00	Su = 0.00	0°
Clay 1	Mohr-Coulomb	17.00	c' = 55.00	5.00°
Clay 2	Mohr-Coulomb	22.00	c' = 43.00	7.00°
Clay 3	Mohr-Coulomb	22.00	c' = 56.00	8.00°
Bed	Mohr-Coulomb	22.00	c' = 200.0	35.00°

Method	Published by	Reference	cSlope
Morgenstern-Price (half-sine)	Commercial program B	1.422	1.428
Bishop simplified	Chowdhury & Xu	1.424	1.428

Notes

- Fixed circle = cSlope's critical M-P circle among circles tangent to the base of the clay layer (cSlope has no tangent-line search; commercial program B's own circle is not printed), so the comparison is that of two searches within the same circle family.
- Sand cap strength ignored (Su = 0); its unit weight is not printed: γ = 22 (as commercial program A a related problem / cSlope V021a) reproduces commercial program B to ≤ 0.3 % on example 1, while γ = 21 (program D's reading of the vendor .gsz) gives +1.5 %. Bed (c′ = 200, φ′ = 35°, γ = 22) is never reached. Piezometric line digitised from Fig. 51 (y ≈ 2.43 m, then the ground surface); it matters only for example 4 (φ > 0): without it cSlope is 11–14 % above commercial program B.
- A free search is meaningless here (the zero-strength sand gives FOS → 0); the critical circle must be searched within the family tangent to the layer base.
- Critical circle (model coordinates): centre (-20.313, 21.643), R 30.784, tangent to y = -9.141.
- Example 4, layer 3 is left out: no assignment of the piezometric line to the clays reproduces commercial program B's 1.504 (cSlope 1.443, −4 %); the pore-pressure set-up of the vendor model is not printed.

3.42 V022a — Chowdhury & Xu (1995) example 5, embankment on soft clay, circle tangent to y =

Chowdhury & Xu (1995).

Fixed circle · Units: SI (m, kPa, kN/m³)

Chowdhury & Xu example 5: embankment fill (c′ = 10 kPa, φ′ = 12°) on soft clay (Su = 40 kPa), the geometry of V022a; circle tangent to the fill–clay interface.

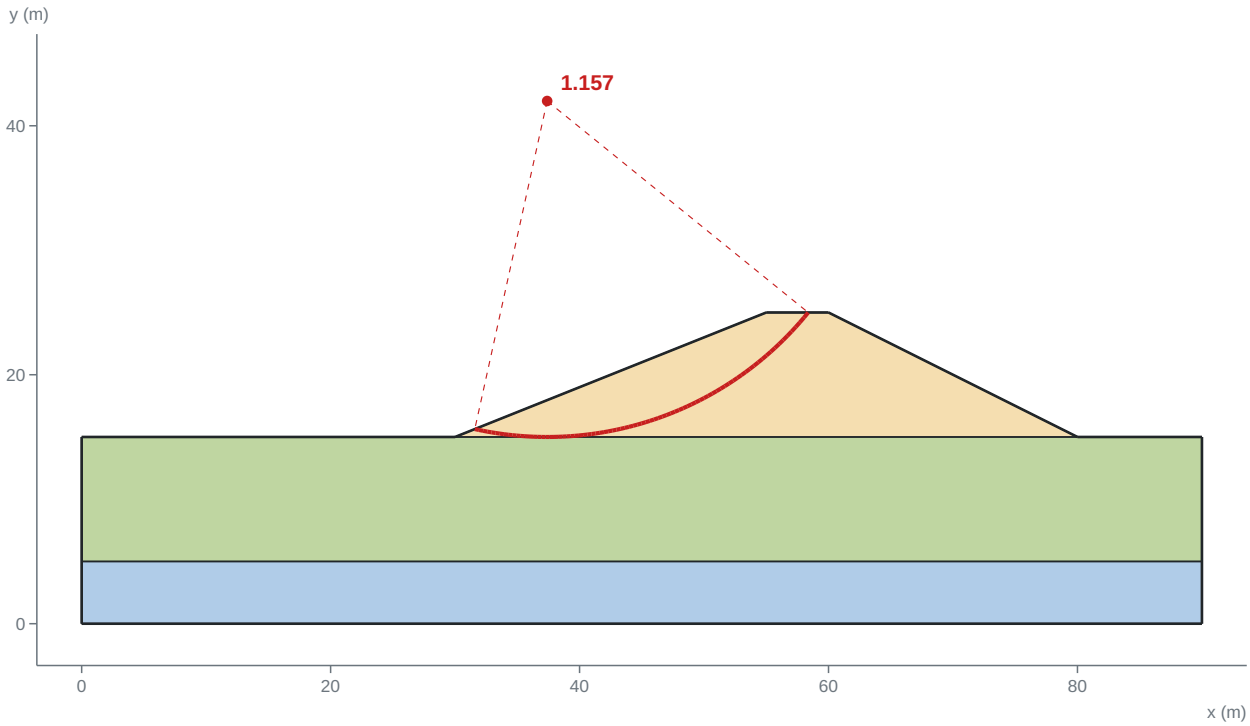


Figure 42. V022a: analysed circle (Morgenstern-Price (half-sine), FOS = 1.157).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Layer 1	Mohr-Coulomb	20.00	c' = 10.00	12.00°
Layer 2	Undrained (φ = 0)	18.00	Su = 40.00	0°
Hard layer	Undrained (φ = 0)	20.00	Su = 1000.0	0°

Method	Published by	Reference	cSlope
Morgenstern-Price (half-sine)	Commercial program B	1.158	1.157
Bishop simplified	Chowdhury & Xu	1.163	1.159
Bishop simplified	Commercial program A	1.160	1.159

Probabilistic analysis on the same surface: 3 normal variables (independent); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1). Differences in β and Pf are absolute.

Quantity	Published by	Reference	cSlope
Pf, share FOS < 1 (Monte Carlo)	Commercial program A	0.212	0.206
β lognormal (Monte Carlo)	Commercial program A (lognormal)	0.799	0.812
β lognormal (Taylor series)	Program D	0.787	0.788
β lognormal (Monte Carlo)	Program D	0.800	0.812
Pf, share FOS < 1 (Monte Carlo)	Program D	0.206	0.206

Probabilistic analysis on the same surface: 3 normal variables (independent); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1). Differences in β and Pf are absolute.

Quantity	Published by	Reference	cSlope
Pf, share FOS < 1 (Monte Carlo)	Commercial program B	0.210	0.208

Notes

- Fixed circle = cSlope's critical M-P circle among circles tangent to the base of the clay layer (cSlope has no tangent-line search; commercial program B's own circle is not printed), so the comparison is that of two searches within the same circle family.
- commercial program B PF 20.98 %, C&X PF 20.225 %. Geometry of V022a/c (the manual prints no figure for example 5).
- Critical circle (model coordinates): centre (37.396, 41.99), R 26.99, tangent to y = 15.

3.43 V022b — Chowdhury & Xu (1995) example 5, embankment on soft clay, circle tangent to y =

Chowdhury & Xu (1995).

Fixed circle · Units: SI (m, kPa, kN/m³)

As V022a with the circle tangent to the base of the soft clay.

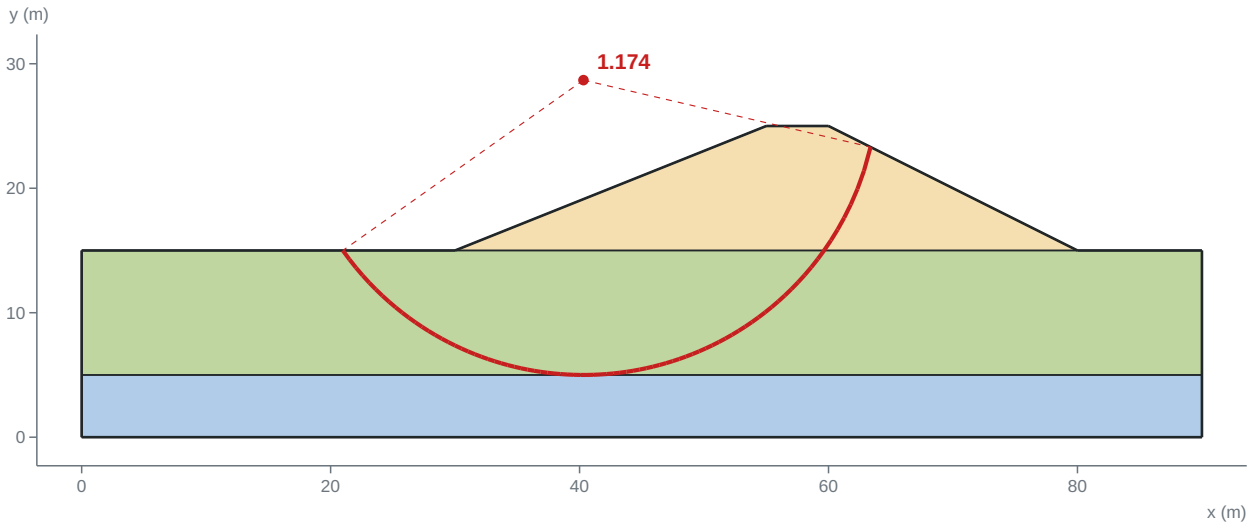


Figure 43. V022b: analysed circle (Morgenstern-Price (half-sine), FOS = 1.174).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Layer 1	Mohr-Coulomb	20.00	c' = 10.00	12.00°
Layer 2	Undrained (φ = 0)	18.00	Su = 40.00	0°
Hard layer	Undrained (φ = 0)	20.00	Su = 1000.0	0°

Method	Published by	Reference	cSlope
Morgenstern-Price (half-sine)	Commercial program B	1.178	1.174
Bishop simplified	Commercial program A	1.185	1.177

Probabilistic analysis on the same surface: 3 normal variables (independent); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1). Differences in β and Pf are absolute.

Quantity	Published by	Reference	cSlope
Pf, share FOS < 1 (Monte Carlo)	Commercial program A	0.199	0.212
β lognormal (Monte Carlo)	Commercial program A (lognormal)	0.820	0.789
β lognormal (Taylor series)	Program D	0.798	0.802
β lognormal (Monte Carlo)	Program D	0.789	0.789
Pf, share FOS < 1 (Monte Carlo)	Program D	0.209	0.212

Probabilistic analysis on the same surface: 3 normal variables (independent); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1). Differences in β and Pf are absolute.

Quantity	Published by	Reference	cSlope
Pf, share FOS < 1 (Monte Carlo)	Commercial program B	0.214	0.216

Notes

- Fixed circle = cSlope's critical M-P circle among circles tangent to the base of the clay layer (cSlope has no tangent-line search; commercial program B's own circle is not printed), so the comparison is that of two searches within the same circle family.
- commercial program B PF 21.39 %, C&X PF 19.733 %. Geometry of V022a/c (the manual prints no figure for example 5).
- Critical circle (model coordinates): centre (40.313, 28.685), R 23.685, tangent to y = 5.
- Chowdhury & Xu's 1.1479 is not checked: it is 2.5 % below both commercial program B (1.178) and commercial program A (1.185, V022b) on this circle family.

3.44 V023 — LASH terminal trench, Duncan's Taylor series example

Duncan (2000), J. Geotech. Geoenviron. Eng. 126(4).

Fixed circle · Units: SI (m, kPa, kN/m³)

Underwater trench in San Francisco Bay Mud: Su = 100 psf at el. −20 ft increasing 9.8 psf/ft, γ = 100 pcf, water at el. 0. Duncan's surface as a circle, Spencer; σγ = 3.3 pcf and σ of the Su gradient 1.2 psf/ft (commercial program A's Table 29.2). Converted to SI.

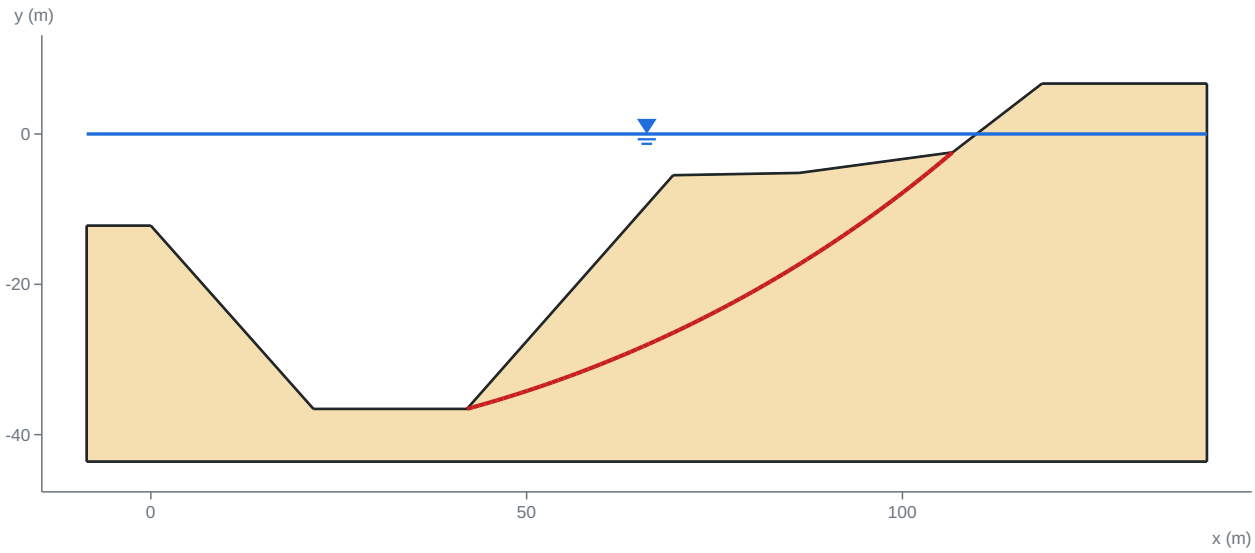


Figure 44. V023: analysed circle (Spencer (M-P, constant f(x)), FOS = 1.138).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
San Francisco Bay Mud	Undrained (φ = 0)	15.71	Su = 4.79 at y = -6.10, +1.54 kPa/m below	0°

Method	Published by	Reference	cSlope
Spencer (M-P, constant f(x))	Program D	1.145	1.138

Probabilistic analysis on the same surface: 2 normal variables (independent); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1). Differences in β and Pf are absolute.

Quantity	Published by	Reference	cSlope
β lognormal (Taylor series)	Duncan (2000)	0.900	0.892
Pf lognormal (Taylor series)	Duncan (2000)	0.180	0.186

Notes

- Above el. −20 ft cSlope extrapolates Su linearly (below 100 psf); program D keeps 100 psf there (F = 1.146, β = 0.947 when cSlope does the same). Duncan's F is 1.17; commercial program A's Latin hypercube Monte Carlo gives Pf = 14 % with a rate-only σ.

3.45 V024 — Dam on a layered foundation with artesian uplift, piezometric line (D&W fig. 6.39)

Duncan & Wright, Soil Strength and Slope Stability, figs. 6.39–6.40.

Fixed circle · Units: SI (m, kPa, kN/m³)

Symmetric dam with a narrow clay core on 30 ft of clay over permeable sand; commercial program A's piezometric line from the underseepage; the circle tangent to el. 197 ft. Converted to SI.

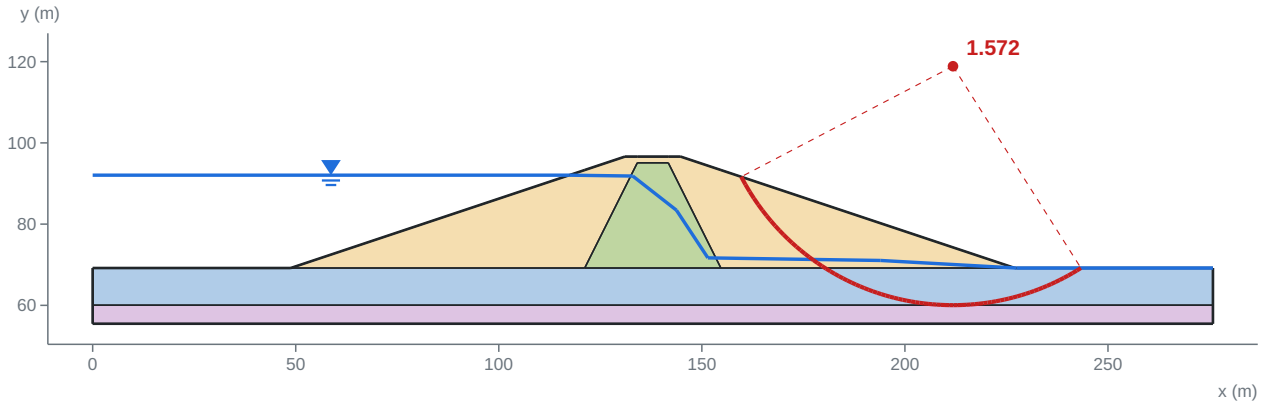


Figure 45. V024: analysed circle (Bishop simplified, FOS = 1.572).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Outer shell	Mohr-Coulomb	19.64	c' = 0.00	34.00°
Clay core	Mohr-Coulomb	19.16	c' = 4.79	26.00°
Foundation clay	Mohr-Coulomb	19.32	c' = 0.00	24.00°
Foundation sand	Mohr-Coulomb	19.95	c' = 0.00	32.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.563	1.572
Spencer (M-P, constant f(x))	Commercial program A	1.557	1.562
Bishop simplified	Duncan & Wright	1.570	1.572
Bishop simplified	Program D	1.572	1.572

Notes

- With pore pressures from a seepage analysis (not available here) commercial program A gives 1.312.

3.46 V025 — Duncan & Wright fig. 6.38, piezometric line

Duncan & Wright, Soil Strength and Slope Stability, fig. 6.38.

Critical circle search · Units: SI (m, kPa, kN/m³)

Homogeneous 2:1 slope with a nearly flat piezometric line; critical circle search. Converted to SI; the water line was traced from a figure.

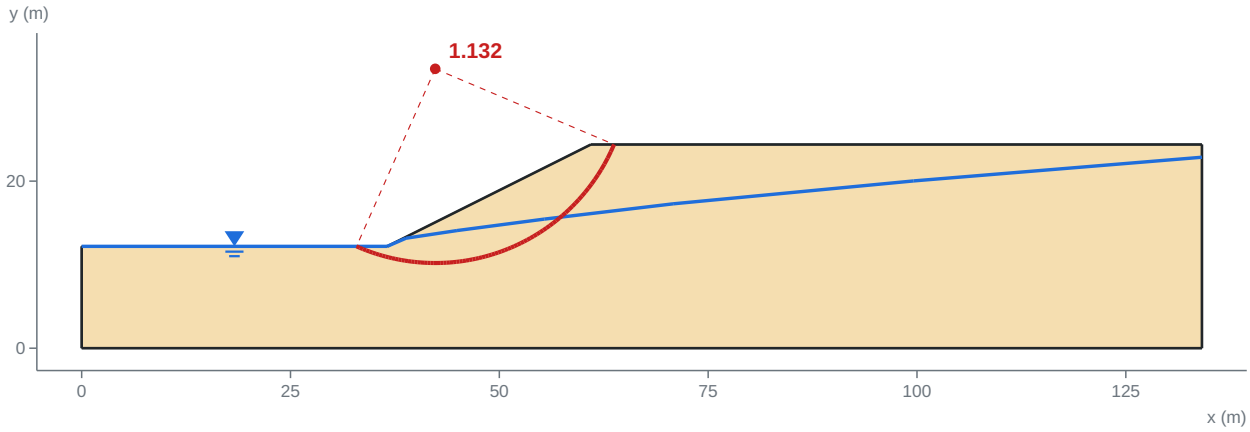


Figure 46. V025: critical circle (Bishop simplified, FOS = 1.132).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	19.64	c' = 9.58	20.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.142	1.132
Spencer (M-P, constant f(x))	Commercial program A	1.142	1.132

3.47 V026 — Duncan & Wright fig. 7.12, embankment on clay

Duncan & Wright, Soil Strength and Slope Stability, fig. 7.12.

Critical circle search · Units: imperial (ft, psf, pcf)

Sand embankment on a thick undrained clay foundation; critical circle search.

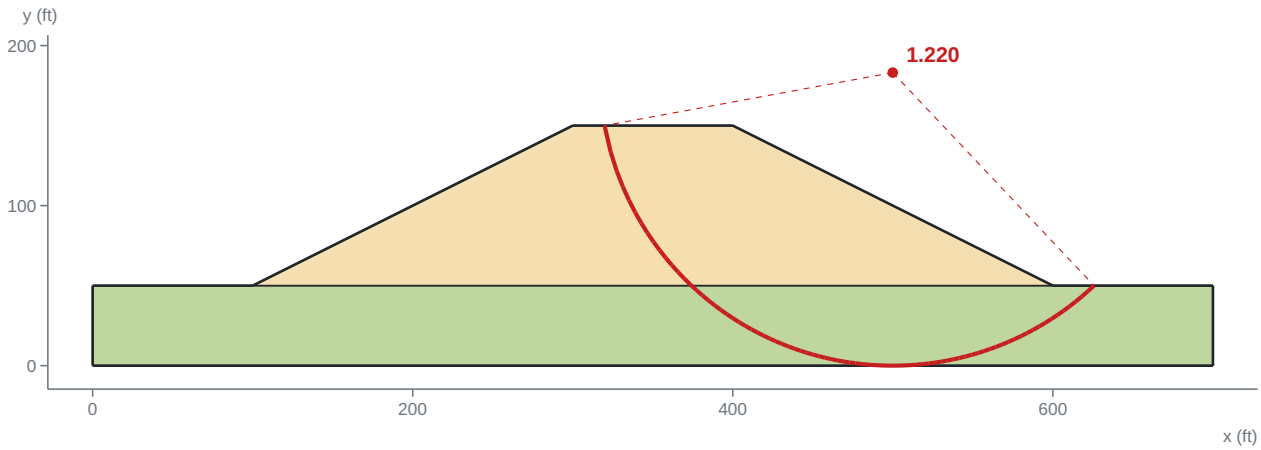


Figure 47. V026: critical circle (Bishop simplified, FOS = 1.220).

Material	Model	γ (pcf)	Strength (psf)	φ
Sand	Mohr-Coulomb	140.0	c' = 0.00	40.00°
Clay	Mohr-Coulomb	140.0	c' = 2500.0	0.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.228	1.220
Bishop simplified	Duncan & Wright	1.220	1.220
Spencer (M-P, constant f(x))	Commercial program A (non-circular)	1.190	1.173

Notes

- Duncan & Wright (Table 74.2): Bishop 1.22, Spencer 1.19, Janbu simplified 1.07, corrected 1.16 (two digits). commercial program A Spencer 1.201 (circle).
- cSlope's optimised non-circular surface is 1.9 % below commercial program A's: a search difference.

3.48 V027 — Duncan & Wright fig. 14.3, undrained clay

Duncan & Wright, Soil Strength and Slope Stability, fig. 14.3.

Critical circle search · Units: imperial (ft, psf, pcf)

Homogeneous undrained clay slope (φ = 0); critical circle search.

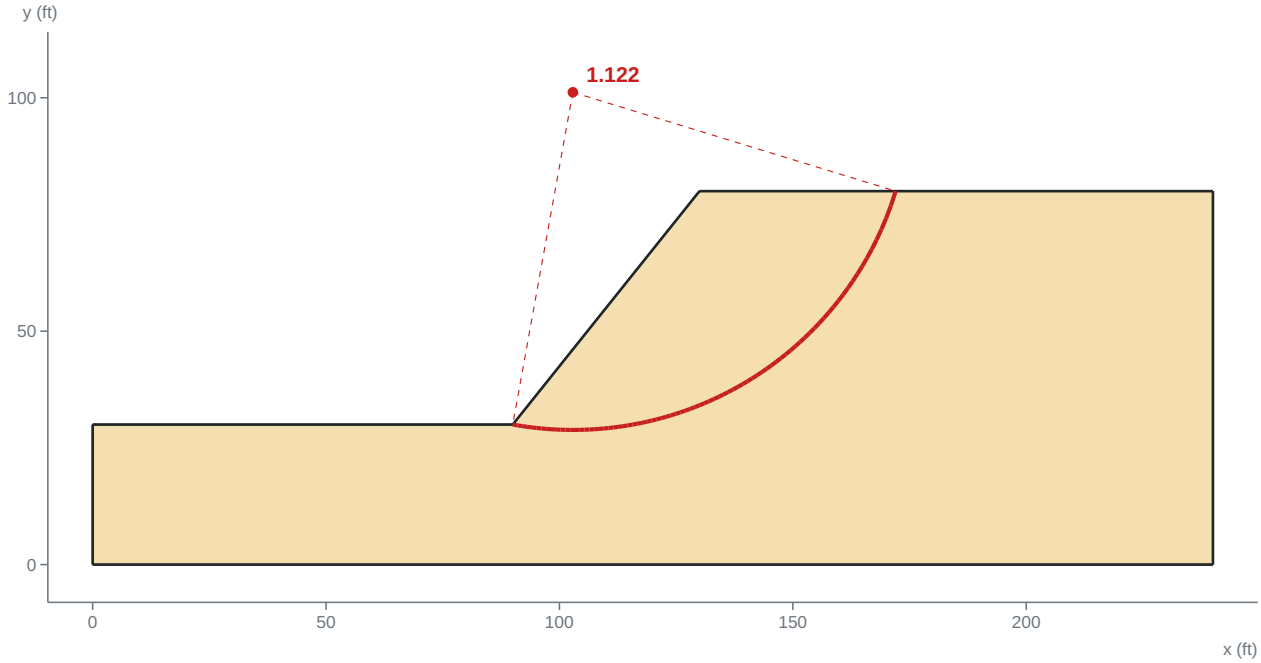


Figure 48. V027: critical circle (Bishop simplified, FOS = 1.122).

Material	Model	γ (pcf)	Strength (psf)	φ
Clay	Mohr-Coulomb	100.0	c' = 1000.0	0.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.126	1.122
Spencer (M-P, constant f(x))	Duncan & Wright (reference)	1.124	1.132

Notes

- commercial program A's toe circle; its circle tangent to the foundation base gives Bishop 1.141, Spencer 1.139. V028 and V029 are the deeper foundations.
- commercial program A's optimised non-circular surface gives 0.878 (Spencer); cSlope's optimisation stops at 0.927 (+5.6 %), so it is not a check.

3.49 V028 — Duncan & Wright fig. 14.3, undrained clay on a 46.5 ft foundation

Verification manual of a commercial slope stability program.

Critical circle search · Units: imperial (ft, psf, pcf)

The V027 φ = 0 clay slope with a 46.5 ft thick foundation; critical-circle search (the minimum of the toe and base-tangent circles of Tables 78.4–78.5).

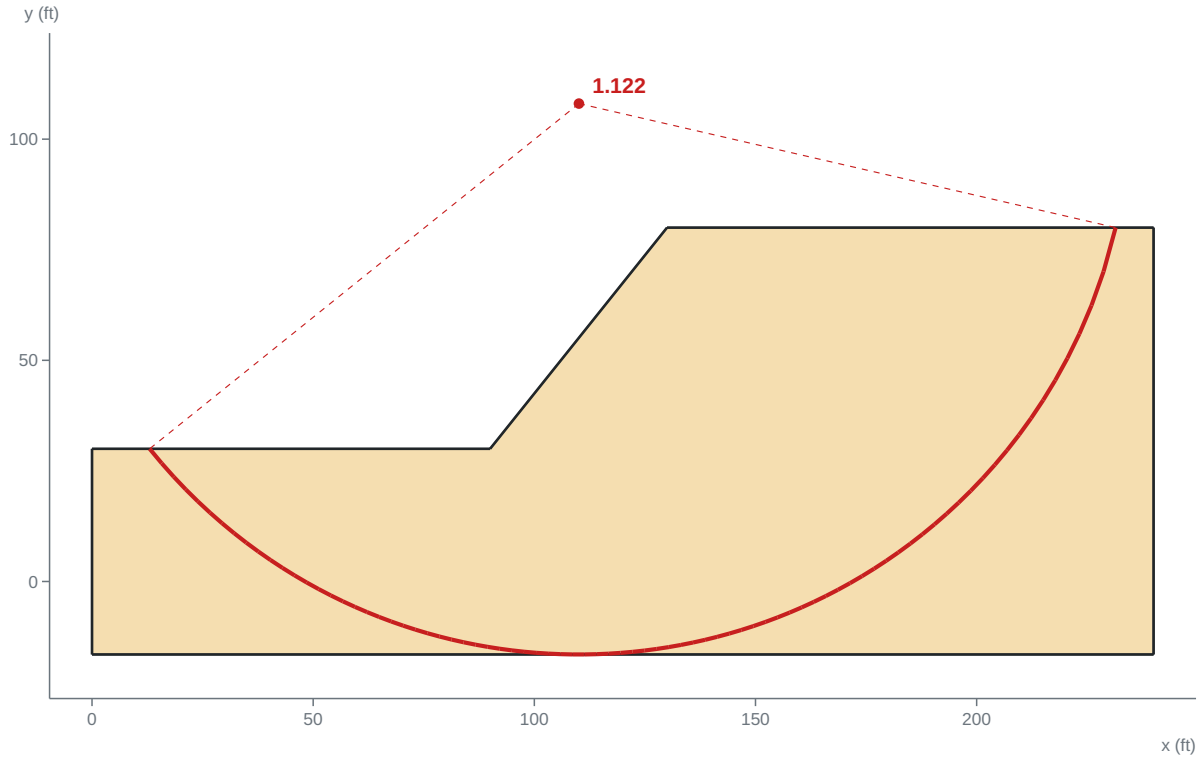


Figure 49. V028: critical circle (Bishop simplified, FOS = 1.122).

Material	Model	γ (pcf)	Strength (psf)	φ
Clay	Mohr-Coulomb	100.0	c' = 1000.0	0.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.126	1.122
Spencer (M-P, constant f(x))	Duncan & Wright (reference)	1.124	1.122

Notes

- commercial program A's base-tangent circle gives 1.130 (Bishop), 1.129 (Spencer). Duncan & Wright's reference 1.124 (two digits, Spencer).

3.50 V029 — Duncan & Wright fig. 14.3, undrained clay on a 60 ft foundation

Verification manual of a commercial slope stability program.

Critical circle search · Units: imperial (ft, psf, pcf)

As V028 with a 60 ft thick foundation (Tables 78.6–78.7).

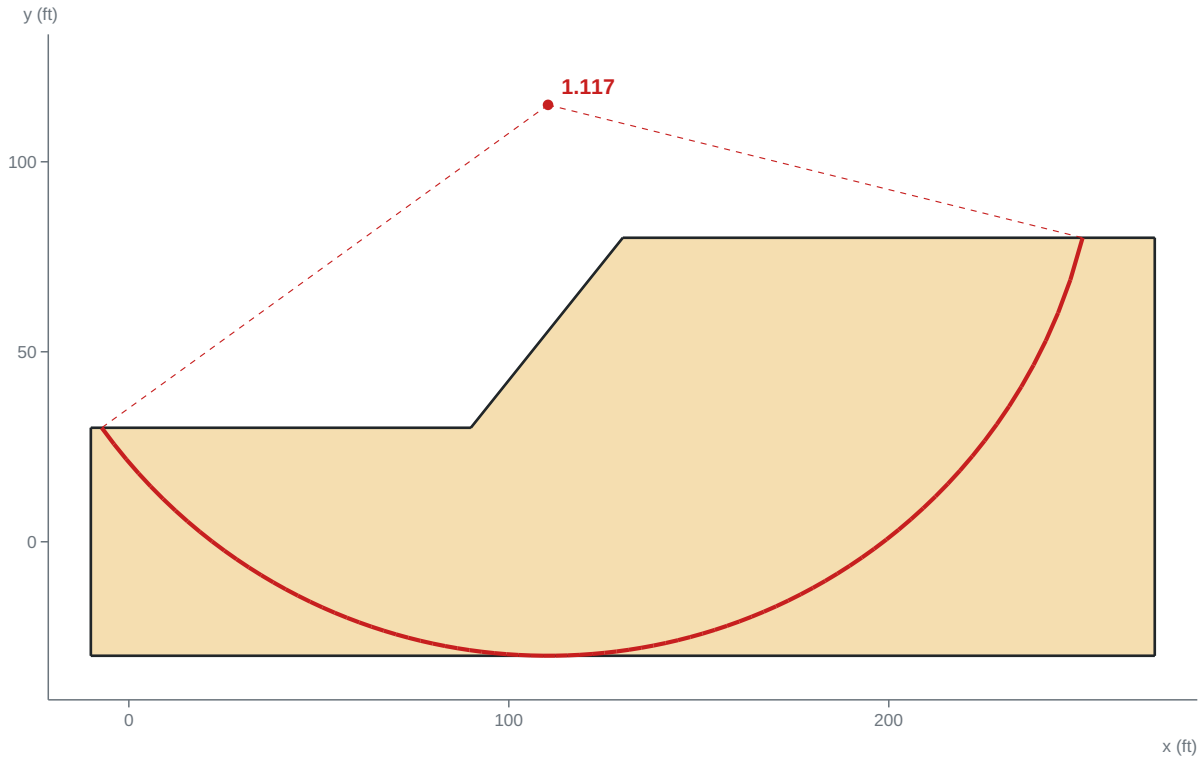


Figure 50. V029: critical circle (Bishop simplified, FOS = 1.117).

Material	Model	γ (pcf)	Strength (psf)	φ
Clay	Mohr-Coulomb	100.0	c' = 1000.0	0.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.125	1.117
Spencer (M-P, constant f(x))	Duncan & Wright (reference)	1.124	1.117

Notes

- The model runs from x = −10 to 270 ft, as in the source, whose base-tangent circle starts at x = −4.8 ft.
- commercial program A's base-tangent circle gives 1.125 (Bishop), 1.124 (Spencer).

3.51 V030 — Duncan & Wright fig. 14.4

Duncan & Wright, Soil Strength and Slope Stability, fig. 14.4.

Critical circle search · Units: imperial (ft, psf, pcf)

Cohesionless fill on undrained clay; critical circle search.

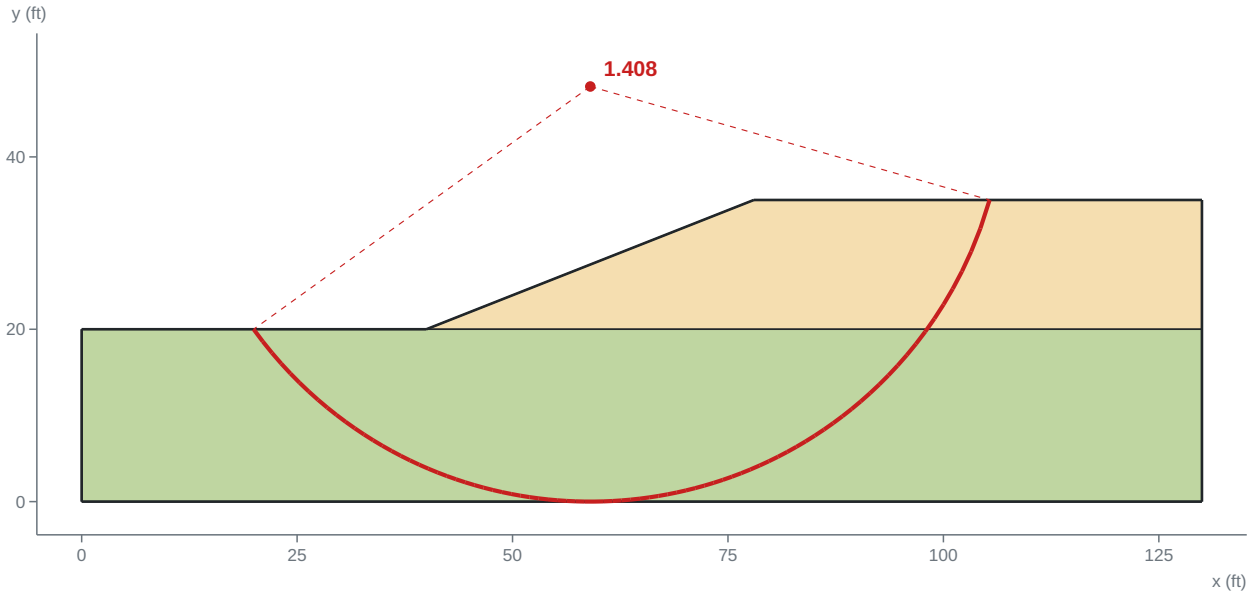


Figure 51. V030: critical circle (Bishop simplified, FOS = 1.408).

Material	Model	γ (pcf)	Strength (psf)	φ
Fill	Mohr-Coulomb	120.0	c' = 0.00	30.00°
Clay	Mohr-Coulomb	120.0	c' = 450.0	0.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.412	1.408
Spencer (M-P, constant f(x))	Duncan & Wright (reference)	1.400	1.399

Notes

- commercial program A Spencer 1.400, GLE 1.404 (deep circle tangent to the foundation base).

3.52 V031a — Duncan & Wright fig. 14.5, R = 87 ft

Duncan & Wright, Soil Strength and Slope Stability, fig. 14.5.

Fixed circle · Units: imperial (ft, psf, pcf)

Embankment over alternating cohesive and frictional layers; circle of radius 87 ft.

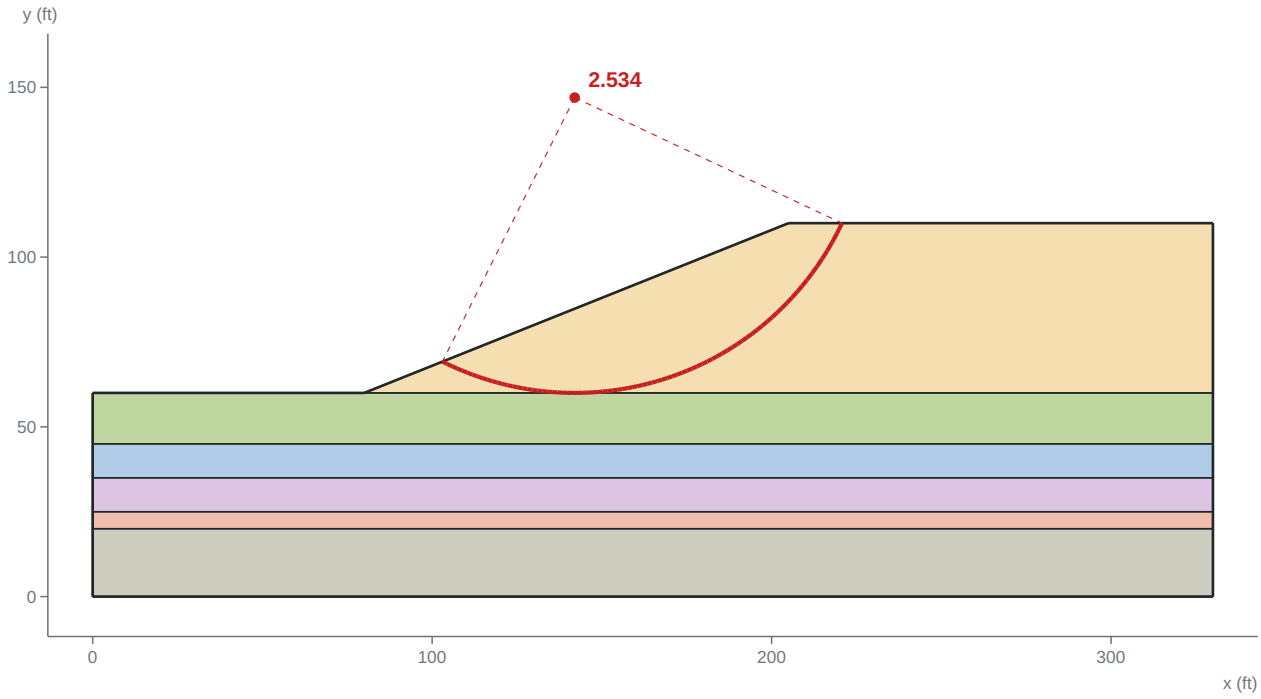


Figure 52. V031a: analysed circle (Bishop simplified, FOS = 2.534).

Material	Model	γ (pcf)	Strength (psf)	φ
Embankment	Mohr-Coulomb	120.0	c' = 1.00	35.00°
L60	Mohr-Coulomb	110.0	c' = 950.0	0.00°
L45	Mohr-Coulomb	122.0	c' = 1.00	32.00°
L35	Mohr-Coulomb	98.00	c' = 500.0	0.00°
L25	Mohr-Coulomb	131.0	c' = 1.00	37.00°
L20	Mohr-Coulomb	103.0	c' = 600.0	0.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	2.549	2.534
Spencer (M-P, constant f(x))	Commercial program A	2.545	2.531
Spencer (M-P, constant f(x))	Duncan & Wright (reference)	2.560	2.531

Notes

- Duncan & Wright's reference values: Spencer's method, two digits. commercial program A GLE 2.550.

3.53 V031b — Duncan & Wright fig. 14.5, R = 102 ft

Duncan & Wright, Soil Strength and Slope Stability, fig. 14.5.

Fixed circle · Units: imperial (ft, psf, pcf)

As V031a with a deeper circle of radius 102 ft, reaching the weak clay layers.

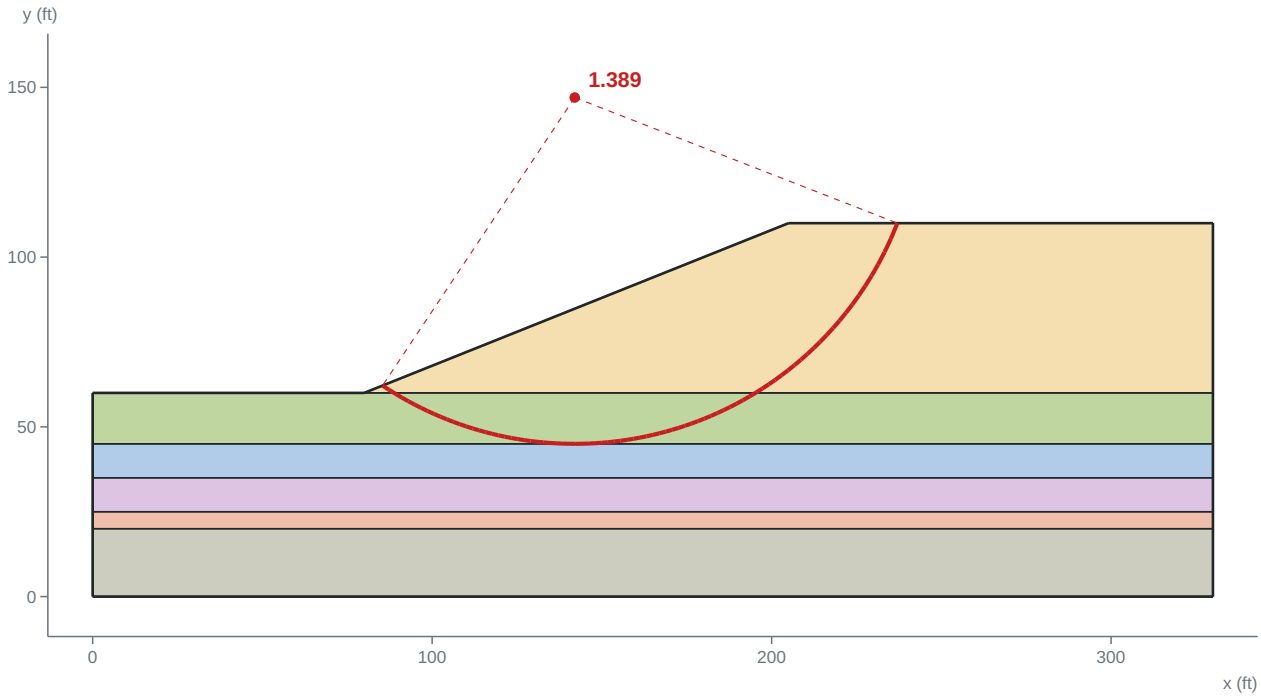


Figure 53. V031b: analysed circle (Bishop simplified, FOS = 1.389).

Material	Model	γ (pcf)	Strength (psf)	φ
Embankment	Mohr-Coulomb	120.0	c' = 1.00	35.00°
L60	Mohr-Coulomb	110.0	c' = 950.0	0.00°
L45	Mohr-Coulomb	122.0	c' = 1.00	32.00°
L35	Mohr-Coulomb	98.00	c' = 500.0	0.00°
L25	Mohr-Coulomb	131.0	c' = 1.00	37.00°
L20	Mohr-Coulomb	103.0	c' = 600.0	0.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.398	1.389
Spencer (M-P, constant f(x))	Commercial program A	1.359	1.352
Spencer (M-P, constant f(x))	Duncan & Wright (reference)	1.350	1.352

Notes

- Duncan & Wright's 1.35 is a Spencer value (commercial program A: Bishop 1.398, Spencer 1.359, GLE 1.358 on this circle).

3.54 V032a — Duncan & Wright fig. 14.7, deep circle

Duncan & Wright, Soil Strength and Slope Stability, fig. 14.7.

Critical circle search · Units: imperial (ft, psf, pcf)

Cohesionless embankment on an undrained clay foundation. Entry is limited to the crest and exit to the toe flat to find the published deep circle.

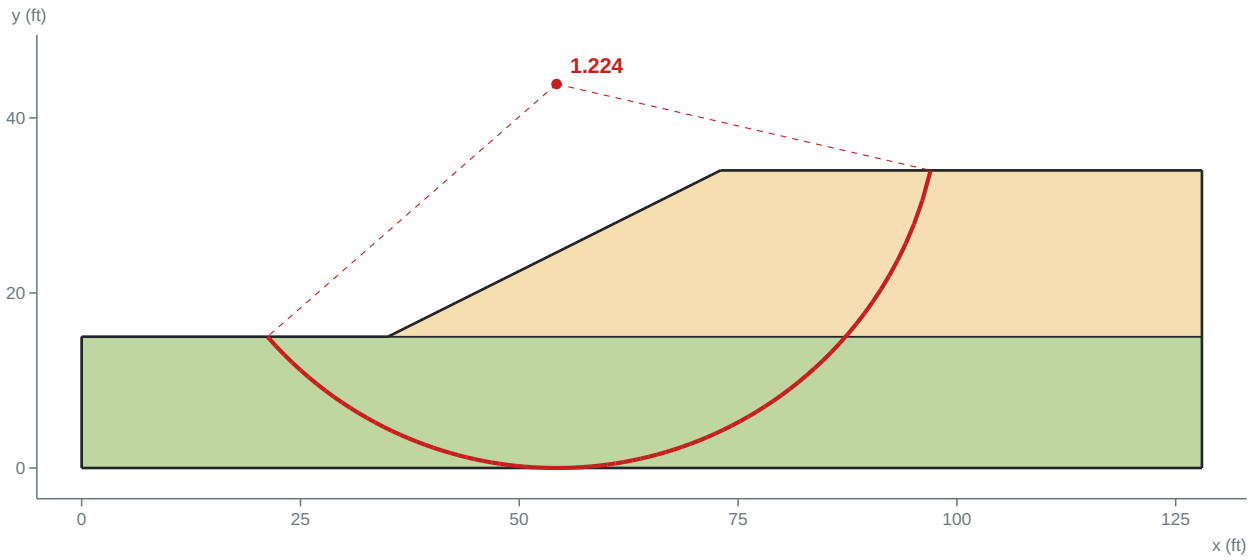


Figure 54. V032a: critical circle (Bishop simplified, FOS = 1.224).

Material	Model	γ (pcf)	Strength (psf)	φ
Embankment	Mohr-Coulomb	124.0	c' = 0.00	30.00°
Foundation	Mohr-Coulomb	98.00	c' = 500.0	0.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.230	1.224
Spencer (M-P, constant f(x))	Commercial program A	1.209	1.205
Spencer (M-P, constant f(x))	Duncan & Wright (reference)	1.210	1.205

Notes

- An unrestricted search finds the infinite-slope sliver in the c = 0 fill (tan 30° / tan 26.6° = 1.155), which is correct mechanics but not the published deep surface; V032b checks it.

3.55 V032b — Duncan & Wright fig. 14.7, infinite-slope mechanism

Duncan & Wright, Soil Strength and Slope Stability, fig. 14.7. Verification manual of a commercial slope stability program.

Critical circle search · Units: imperial (ft, psf, pcf)

The V032a embankment without search limits: the critical surface is the shallow infinite-slope sliver in the cohesionless fill (tan 30° / tan 26.6° = 1.155).

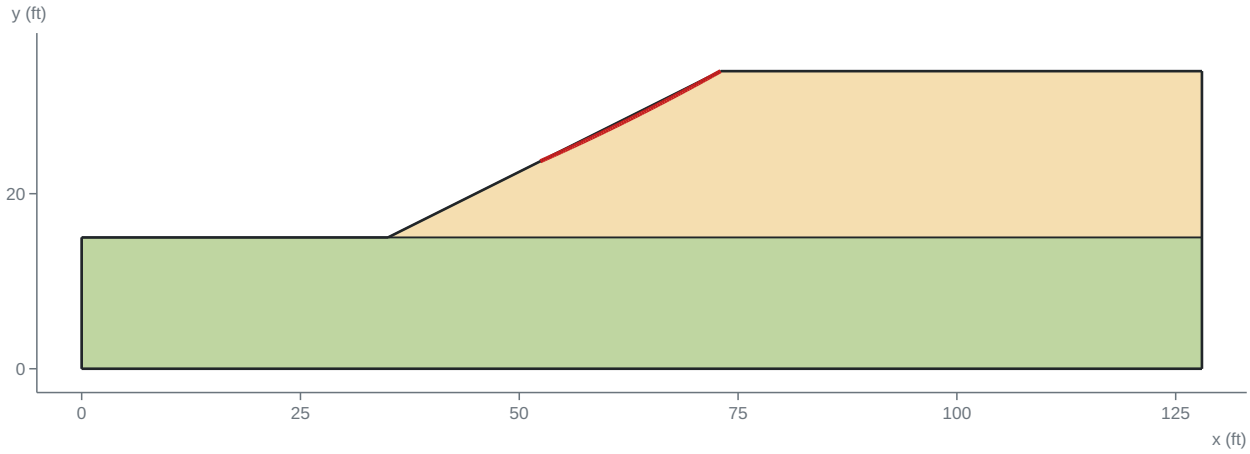


Figure 55. V032b: critical circle (Bishop simplified, FOS = 1.156).

Material	Model	γ (pcf)	Strength (psf)	φ
Embankment	Mohr-Coulomb	124.0	c' = 0.00	30.00°
Foundation	Mohr-Coulomb	98.00	c' = 500.0	0.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.155	1.156
Spencer (M-P, constant f(x))	Commercial program A	1.155	1.156
Spencer (M-P, constant f(x))	Duncan & Wright (reference)	1.150	1.156

3.56 V033 — Duncan & Wright fig. 14.20a, water table

Duncan & Wright, Soil Strength and Slope Stability, fig. 14.20a.

Critical circle search · Units: SI (m, kPa, kN/m³)

Cohesive-frictional fill over sand with a piezometric line; critical circle search. Converted to SI.

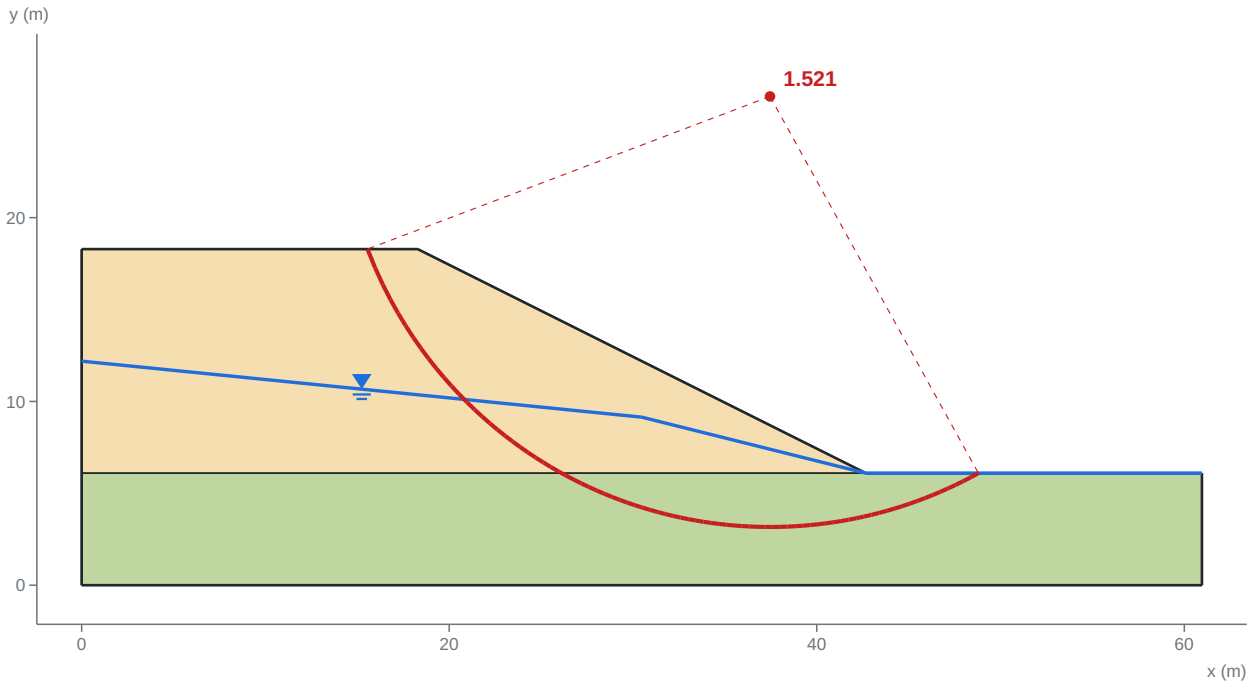


Figure 56. V033: critical circle (Bishop simplified, FOS = 1.521).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Fill	Mohr-Coulomb	19.64	c' = 28.73	25.00°
Sand	Mohr-Coulomb	20.74	c' = 0.00	30.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.533	1.521
Spencer (M-P, constant f(x))	Duncan & Wright (reference)	1.535	1.533
Spencer (M-P, constant f(x))	Commercial program A (non-circular)	1.534	1.524

Notes

- Duncan & Wright's 1.535 is the mean of 1.528–1.542 for different slice counts. commercial program A Spencer 1.540 (circle).

3.57 V034a — Duncan & Wright fig. 14.20b, Su = 200 + 15·depth

Duncan & Wright, Soil Strength and Slope Stability, fig. 14.20b.

Critical circle search · Units: imperial (ft, psf, pcf)

Granular fill over clay with undrained strength increasing linearly with depth; critical circle search.

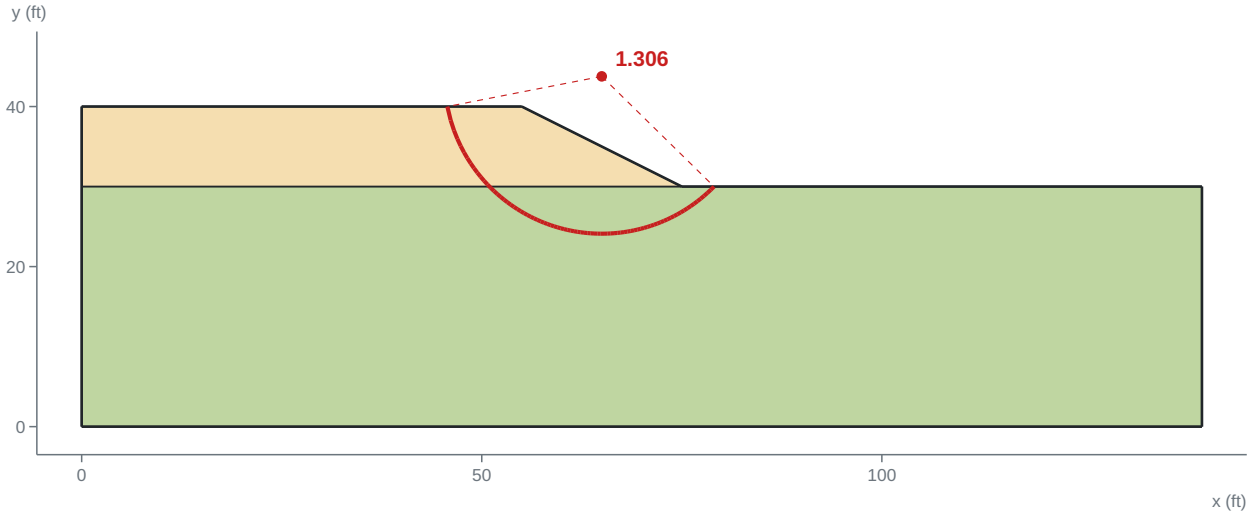


Figure 57. V034a: critical circle (Bishop simplified, FOS = 1.306).

Material	Model	γ (pcf)	Strength (psf)	φ
Fill	Mohr-Coulomb	123.0	c' = 0.00	36.00°
Clay	Undrained (φ = 0)	97.00	Su = 200.0 at y = 30.00, +15.00 psf/ft below	0°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.313	1.306
Spencer (M-P, constant f(x))	Duncan & Wright (reference)	1.300	1.276
Spencer (M-P, constant f(x))	Commercial program A (non-circular)	1.262	1.269

Notes

- Duncan & Wright's 1.300 is the mean of 1.276–1.323 for different slice counts. commercial program A Spencer 1.285 (circle).

3.58 V034b — Duncan & Wright fig. 14.20b, Su = 300 psf

Duncan & Wright, Soil Strength and Slope Stability, fig. 14.20b.

Critical circle search · Units: imperial (ft, psf, pcf)

As V034a with a constant undrained strength of 300 psf.

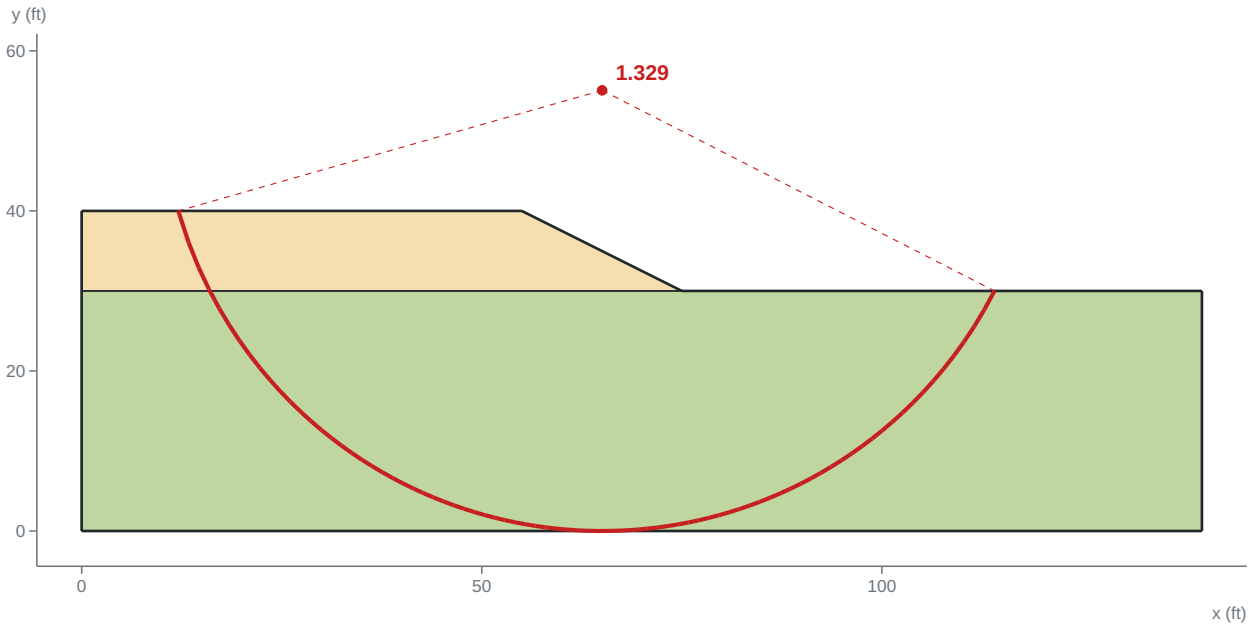


Figure 58. V034b: critical circle (Bishop simplified, FOS = 1.329).

Material	Model	γ (pcf)	Strength (psf)	φ
Fill	Mohr-Coulomb	123.0	c' = 0.00	36.00°
Clay	Undrained (φ = 0)	97.00	Su = 300.0	0°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.335	1.329
Spencer (M-P, constant f(x))	Duncan & Wright (reference)	1.312	1.328
Spencer (M-P, constant f(x))	Commercial program A (non-circular)	1.182	1.173

Notes

- Duncan & Wright's 1.312 is the mean of 1.295–1.328 for different slice counts. commercial program A Spencer 1.330 (circle).

3.59 V035a — Duncan & Wright tieback, active

Duncan & Wright, Soil Strength and Slope Stability, fig. 6.34.

Fixed circle · Units: SI (m, kPa, kN/m³)

Homogeneous slope with a 9000 lb/ft tieback applied horizontally at mid-height as an active force, on the reference printed GLE circle. Converted to SI.

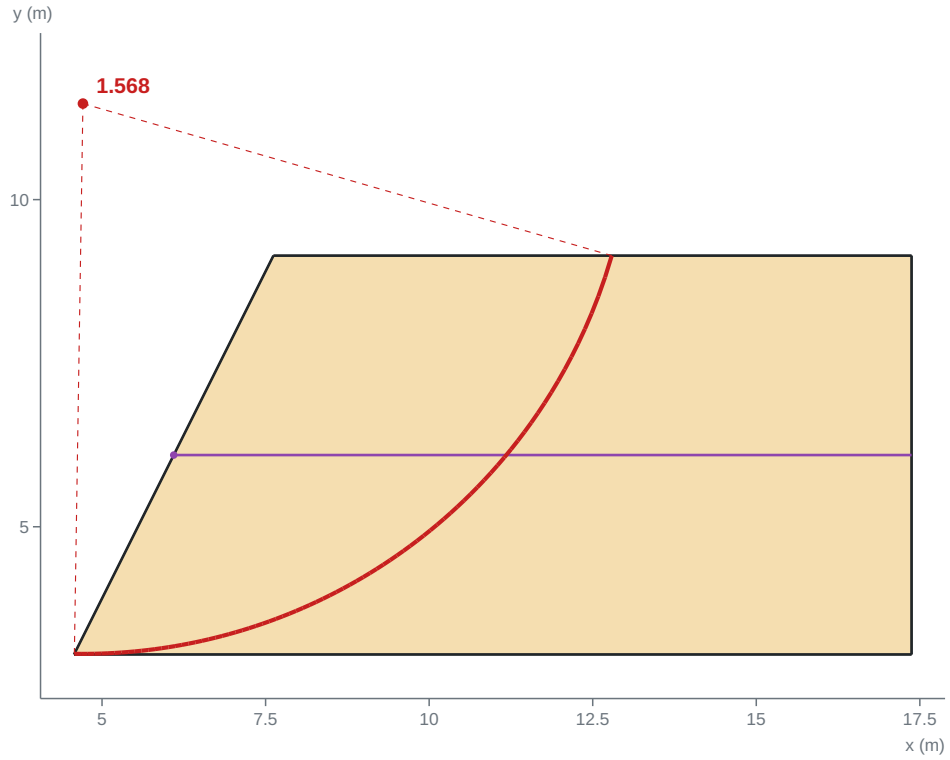


Figure 59. V035a: analysed circle (Morgenstern-Price (half-sine), FOS = 1.568).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
saturated clay	Undrained (φ = 0)	15.39	Su = 16.76	0°

Method	Published by	Reference	cSlope
Morgenstern-Price (half-sine)	Commercial program A (GLE)	1.575	1.568

Notes

- commercial program A's Bishop (1.531) and Spencer (1.884) belong to their own critical circles; on this circle cSlope's Bishop is 1.568. Duncan & Wright 1.51.

3.60 V035b — Duncan & Wright tieback, passive

Duncan & Wright, Soil Strength and Slope Stability, fig. 6.34.

Fixed circle · Units: SI (m, kPa, kN/m³)

As V035a with the tieback applied as a passive force (divided by F), on the reference printed Bishop circle.

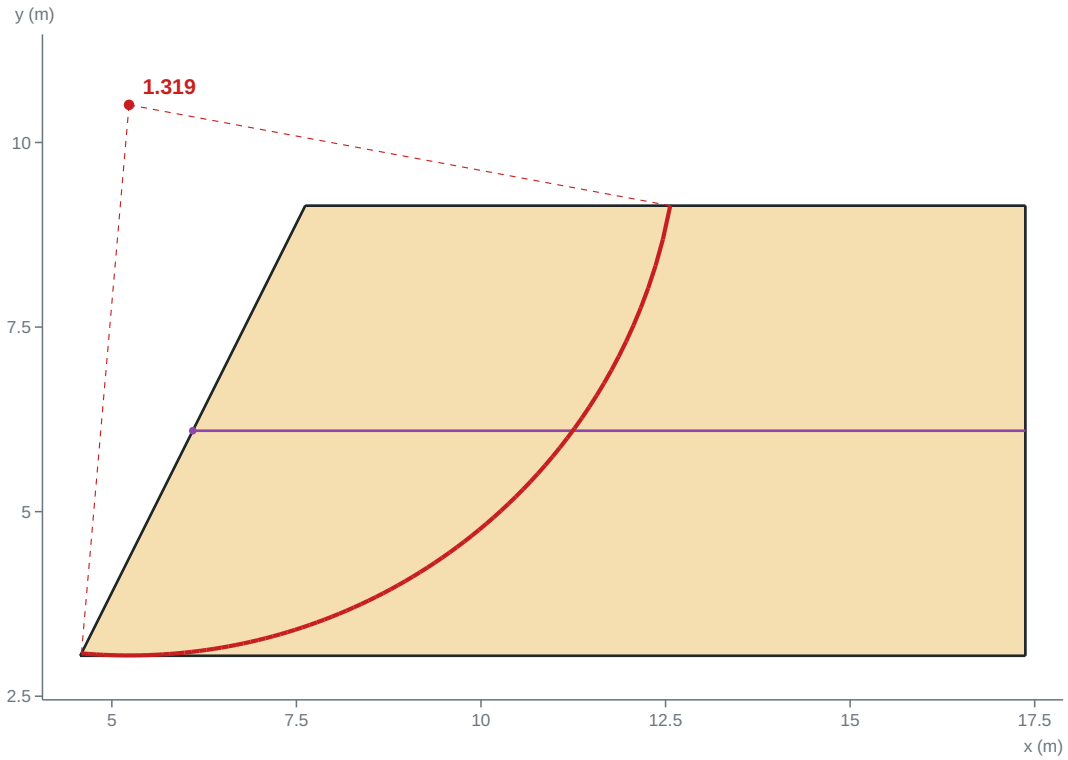


Figure 60. V035b: analysed circle (Bishop simplified, FOS = 1.319).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
saturated clay	Undrained (φ = 0)	15.39	Su = 16.76	0°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.324	1.319
Bishop simplified	Duncan & Wright	1.320	1.319

3.61 V036a — Embankment on an undrained foundation, cz = 0 (D&W fig. 15.9)

Duncan & Wright, Soil Strength and Slope Stability, fig. 15.9.

Critical circle search · Units: SI (m, kPa, kN/m³)

Embankment (c′ = 0, φ′ = 35°, γ = 125 pcf) on a 20 ft undrained foundation, cu = 300 psf, constant; circular search. Converted to SI.

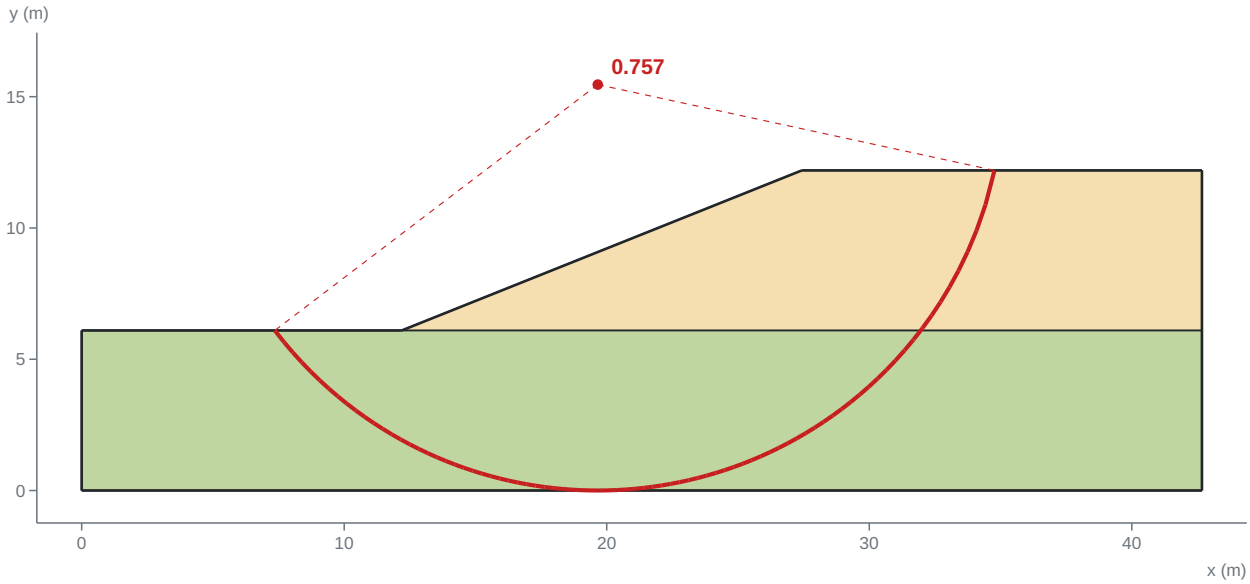


Figure 61. V036a: critical circle (Bishop simplified, FOS = 0.757).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Embankment	Mohr-Coulomb	19.64	c' = 0.00	35.00°
Foundation	Undrained (φ = 0)	15.71	Su = 14.36	0°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	0.761	0.757
Spencer (M-P, constant f(x))	Commercial program A	0.756	0.752
Spencer (M-P, constant f(x))	Duncan & Wright (reference)	0.750	0.752
Spencer (M-P, constant f(x))	Commercial program A (non-circular)	0.740	0.737

Notes

- Duncan & Wright's reference values: Spencer's method, read from fig. 15.9 to two digits.

3.62 V036b — Embankment on an undrained foundation, cz = 5 psf/ft

Duncan & Wright, Soil Strength and Slope Stability, fig. 15.9.

Critical circle search · Units: SI (m, kPa, kN/m³)

As V036a with cu increasing 5 psf/ft with depth.

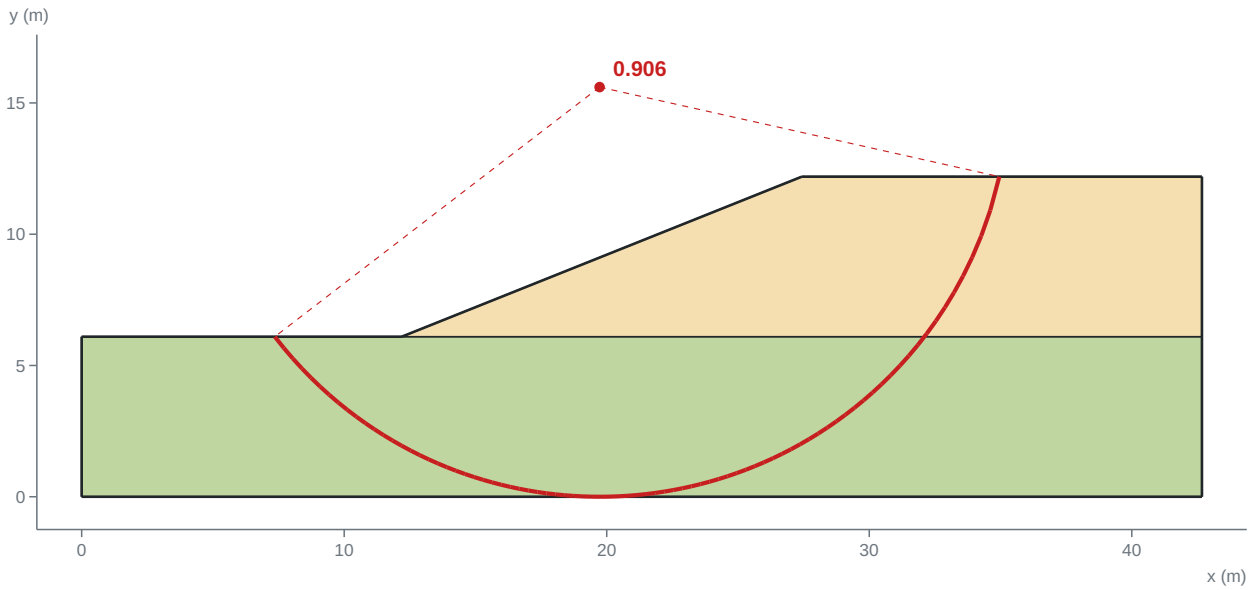


Figure 62. V036b: critical circle (Bishop simplified, FOS = 0.906).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Embankment	Mohr-Coulomb	19.64	c' = 0.00	35.00°
Foundation	Undrained (φ = 0)	15.71	Su = 14.36 at y = 6.10, +0.79 kPa/m below	0°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	0.909	0.906
Spencer (M-P, constant f(x))	Commercial program A	0.898	0.898
Spencer (M-P, constant f(x))	Duncan & Wright (reference)	0.900	0.898
Spencer (M-P, constant f(x))	Commercial program A (non-circular)	0.903	0.884

Notes

- Duncan & Wright's reference values: Spencer's method, read from fig. 15.9 to two digits.
- cSlope's optimised non-circular surface is 2 % below commercial program A's: a search difference.

3.63 V036c — Embankment on an undrained foundation, cz = 10 psf/ft

Duncan & Wright, Soil Strength and Slope Stability, fig. 15.9.

Critical circle search · Units: SI (m, kPa, kN/m³)

As V036a with cu increasing 10 psf/ft with depth.

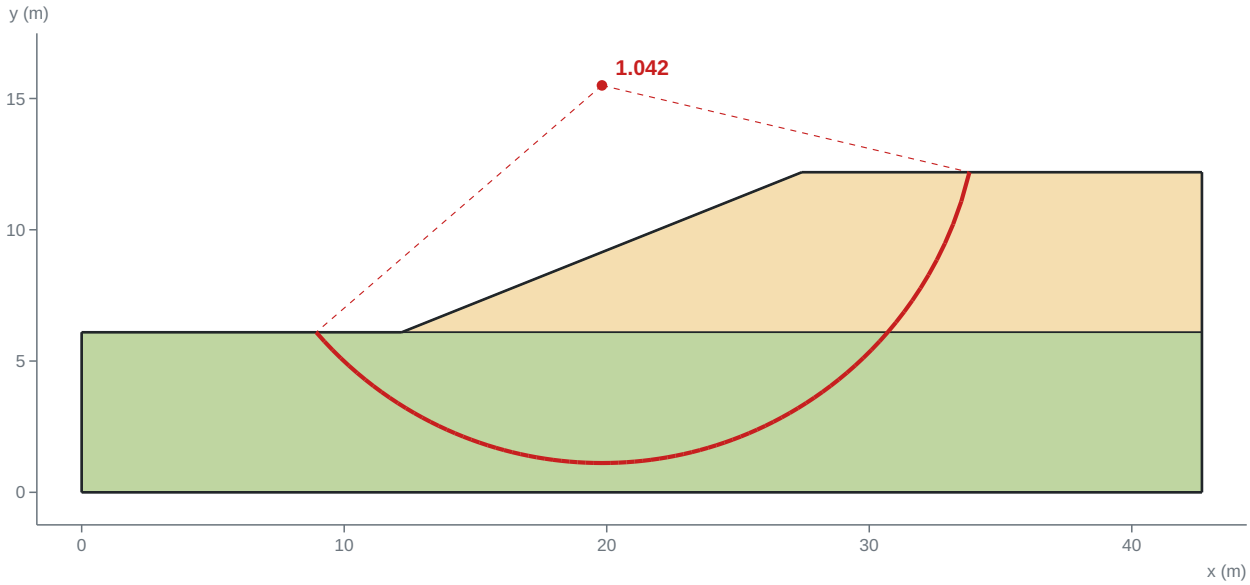


Figure 63. V036c: critical circle (Bishop simplified, FOS = 1.042).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Embankment	Mohr-Coulomb	19.64	c' = 0.00	35.00°
Foundation	Undrained (φ = 0)	15.71	Su = 14.36 at y = 6.10, +1.57 kPa/m below	0°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.045	1.042
Spencer (M-P, constant f(x))	Commercial program A	1.032	1.028
Spencer (M-P, constant f(x))	Duncan & Wright (reference)	1.030	1.028
Spencer (M-P, constant f(x))	Commercial program A (non-circular)	1.018	1.017

Notes

- Duncan & Wright's reference values: Spencer's method, read from fig. 15.9 to two digits.

3.64 V036d — Embankment on an undrained foundation, cz = 15 psf/ft

Duncan & Wright, Soil Strength and Slope Stability, fig. 15.9.

Critical circle search · Units: SI (m, kPa, kN/m³)

As V036a with cu increasing 15 psf/ft with depth.

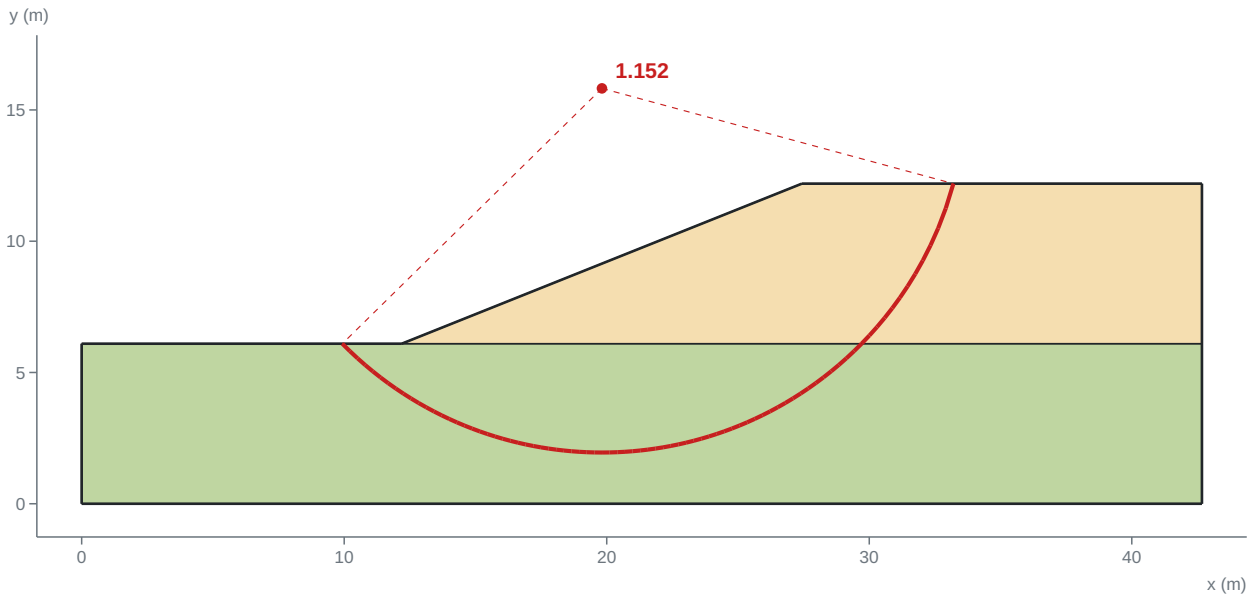


Figure 64. V036d: critical circle (Bishop simplified, FOS = 1.152).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Embankment	Mohr-Coulomb	19.64	c' = 0.00	35.00°
Foundation	Undrained (φ = 0)	15.71	Su = 14.36 at y = 6.10, +2.36 kPa/m below	0°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.154	1.152
Spencer (M-P, constant f(x))	Commercial program A	1.134	1.132
Spencer (M-P, constant f(x))	Duncan & Wright (reference)	1.130	1.132
Spencer (M-P, constant f(x))	Commercial program A (non-circular)	1.116	1.121

Notes

- Duncan & Wright's reference values: Spencer's method, read from fig. 15.9 to two digits.

3.65 V037 — Geogrid-reinforced fill on rock (D&W fig. 7.28)

Duncan & Wright, Soil Strength and Slope Stability, fig. 7.28.

Critical circle search · Units: SI (m, kPa, kN/m³)

24 ft fill slope (φ′ = 37°) on a strong rock foundation with five 800 lb/ft geogrids, 20 ft long, every 4 ft; circular search for surfaces through the toe. Converted to SI.

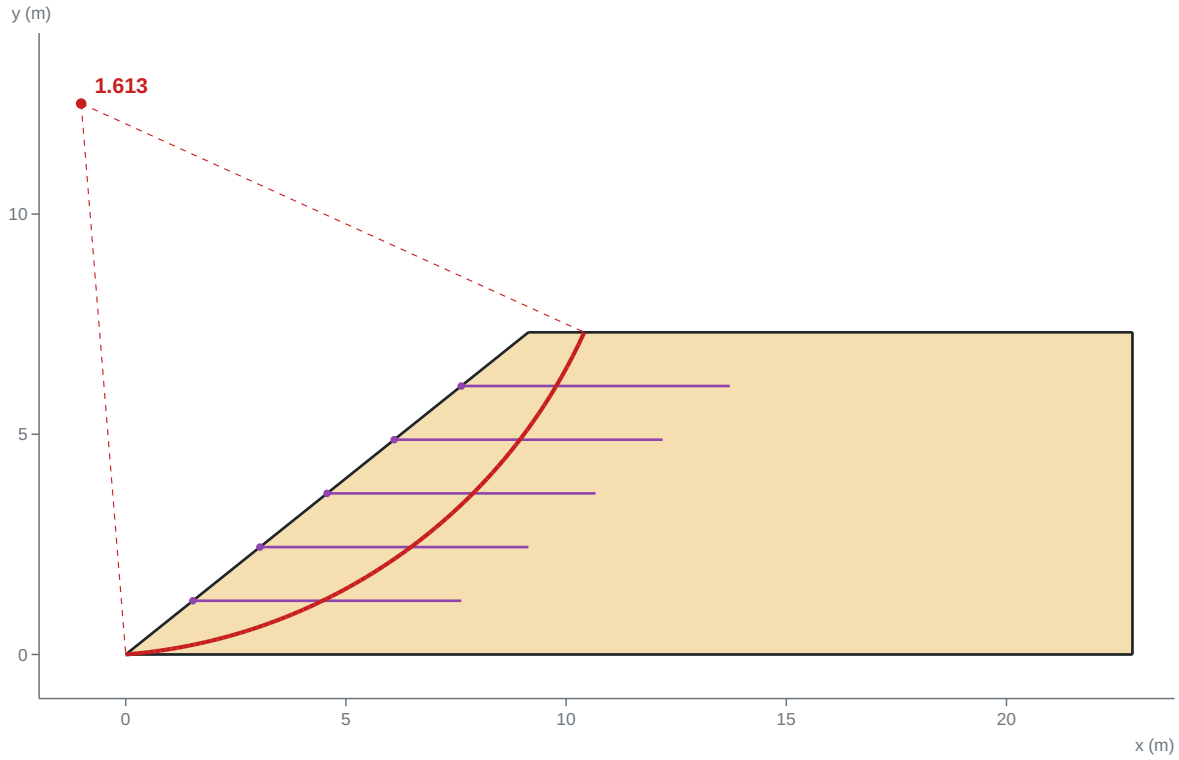


Figure 65. V037: critical circle (Bishop simplified, FOS = 1.613).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
fill	Mohr-Coulomb	20.42	c' = 0.00	37.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.629	1.613
Spencer (M-P, constant f(x))	Commercial program A	1.620	1.607

Notes

- Duncan & Wright 1.61. Without the exit limit the search finds a shallower surface between two geogrid layers (FOS ≈ 1.19), which the published analyses do not consider.

3.66 V038 — Homogeneous dam with a piezometric line (D&W fig. 7.19)

Duncan & Wright, Soil Strength and Slope Stability, fig. 7.19.

Critical circle search · Units: SI (m, kPa, kN/m³)

Homogeneous embankment (c′ = 100 psf, φ′ = 30°, γ = 100 pcf) with the pool at el. 40 ft and commercial program A's piezometric line; critical-circle search. Converted to SI.

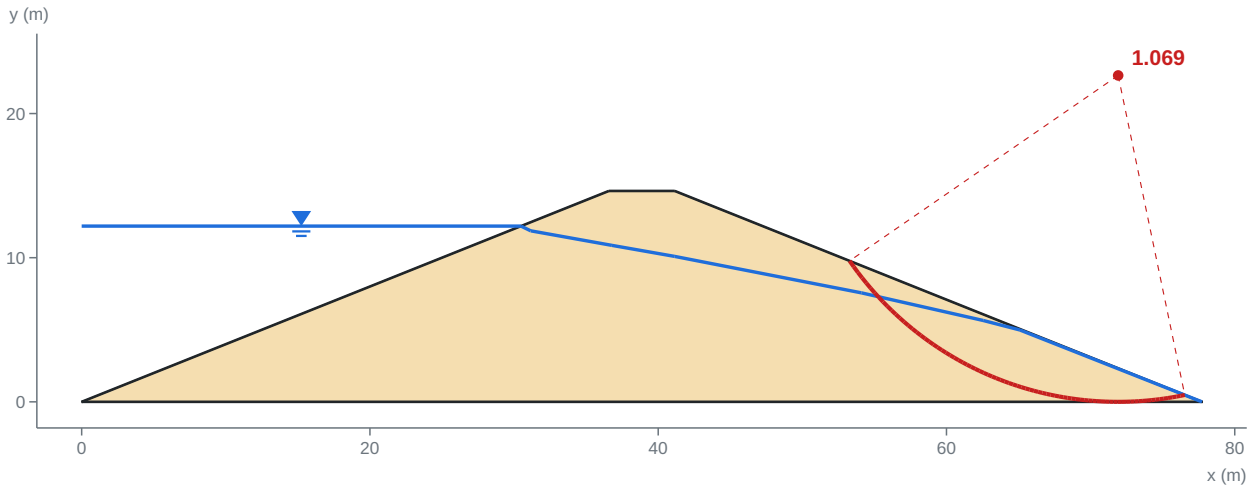


Figure 66. V038: critical circle (Bishop simplified, FOS = 1.069).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Material 1	Mohr-Coulomb	15.71	c' = 4.79	30.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.090	1.069
Spencer (M-P, constant f(x))	Commercial program A	1.100	1.077
Bishop simplified	Program D	1.070	1.069

Notes

- commercial program A 1.068/1.075 and program D 1.065/1.072 with FE seepage; D&W chart 1.16.

3.67 V039 — James Bay dyke (Duncan & Wright fig. 7.16)

Duncan & Wright, Soil Strength and Slope Stability, fig. 7.16.

Critical circle search · Units: SI (m, kPa, kN/m³)

Granular fill embankment with a wide berm (c′ = 0, φ′ = 30°) on a 4 m clay crust (Su = 41), 8 m marine clay (34.5) and 7 m lacustrine clay (31.2); circular search. The critical circle is deep and tangent to the base of the lacustrine clay.

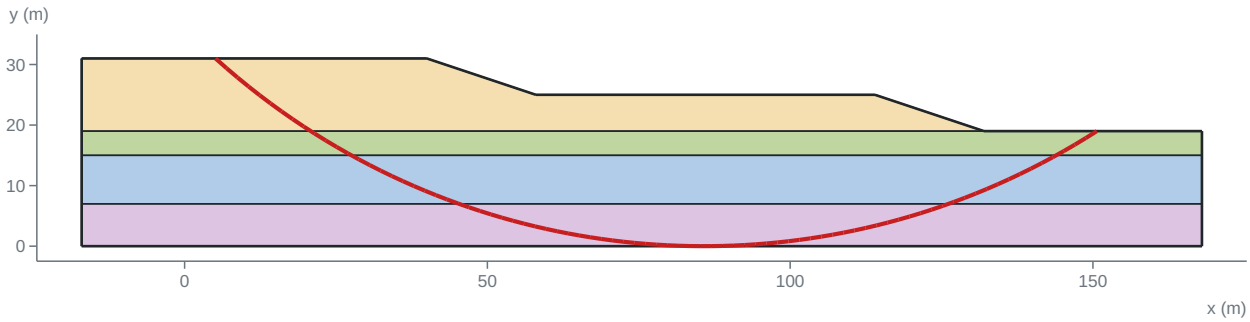


Figure 67. V039: critical circle (Bishop simplified, FOS = 1.424).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Fill	Mohr-Coulomb	20.00	c' = 0.00	30.00°
Clay crust	Undrained (φ = 0)	20.00	Su = 41.00	0°
Marine clay	Undrained (φ = 0)	18.80	Su = 34.50	0°
Lacustrine clay	Undrained (φ = 0)	20.30	Su = 31.20	0°

Method	Published by	Reference	cSlope
Bishop simplified	Duncan & Wright (reference)	1.450	1.424
Spencer (M-P, constant f(x))	Duncan & Wright (reference)	1.450	1.421
Spencer (M-P, constant f(x))	Commercial program A (non-circular)	1.167	1.155
Spencer (M-P, constant f(x))	Duncan & Wright (reference), non-circular	1.170	1.155

Notes

- program D's search gives 1.424 / 1.420; commercial program A reports 1.468 / 1.464 on its own (not the critical) circle. Non-circular: commercial program A Bishop 1.105, GLE 1.142.

3.68 V040 — James Bay dyke, reference model (deterministic)

El-Ramly et al. (2002); Duncan & Wright fig. 7.16.

Critical circle search · Units: SI (m, kPa, kN/m³)

The James Bay dyke as commercial program B models it (crest 36 m, berm at 30 m, clay crust Su = 43 kPa, till below y = 5); critical-circle search, deterministic.

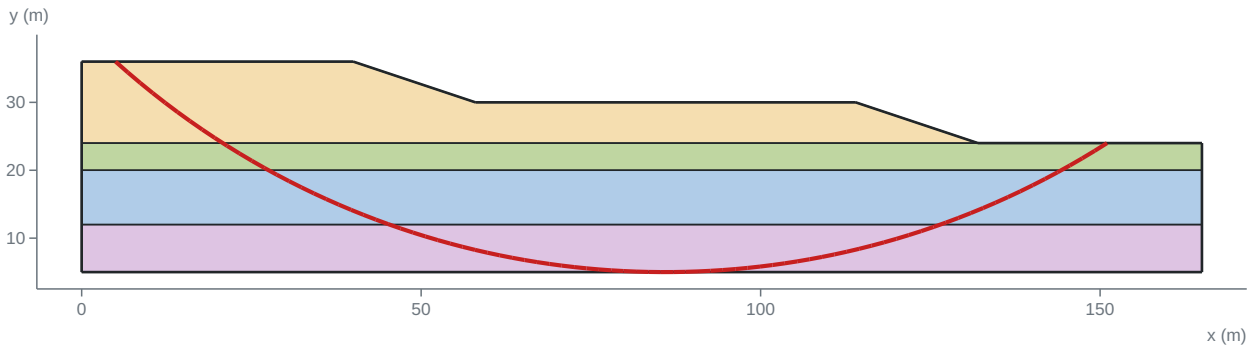


Figure 68. V040: critical circle (Bishop simplified, FOS = 1.433).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Embankment	Mohr-Coulomb	20.00	c' = 0.00	30.00°
Clay crust	Undrained (φ = 0)	18.80	Su = 43.00	0°
Marine clay	Undrained (φ = 0)	18.80	Su = 34.50	0°
Lacustrine clay	Undrained (φ = 0)	20.30	Su = 31.20	0°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program B	1.461	1.433

Notes

- Geometry read from Fig. 131/132 (axes, not labelled vertices): crest y = 36 for 40 m, 3:1 slopes, 56 m berm at y = 30, toe y = 24; layer tops 24/20/12, till (bedrock) below y = 5 — the model base here. Clay crust Su = 43, γ = 18.8 per Table 111 (V039 uses D&W's 41 kPa, γ = 20).
- cSlope's critical circle (centre 85.9, 126.1, R 121.1) grazes the till at y = 5.0 and gives 1.432; the commercial program B circle of Fig. 132 bottoms out at y ≈ 5.5 (fitted through (1, 36) and (150, 24): centre (83.8, 133.1), R 127.6), on which cSlope gives 1.462. The 2 % is the search (tangency to the till), not the model.
- Probabilistic (no spatial variability): mean 1.4611, σ 0.21295, PF 1.34 %, β 2.165 with σ(φ fill) = 1°, σ(γ fill) = 1, σ(c marine) = 8.14, σ(c lacustrine) = 8.65.

3.69 V041a — Submerged slope, pool 30 ft above the crest (D&W fig. 6.27)

Duncan & Wright, Soil Strength and Slope Stability, fig. 6.27.

Critical circle search · Units: SI (m, kPa, kN/m³)

Fully submerged homogeneous slope (c = 100 psf, φ = 20°, γ = 128 pcf) with the pool 30 ft above the crest: the extra water weight and pore pressure cancel. Converted to SI.

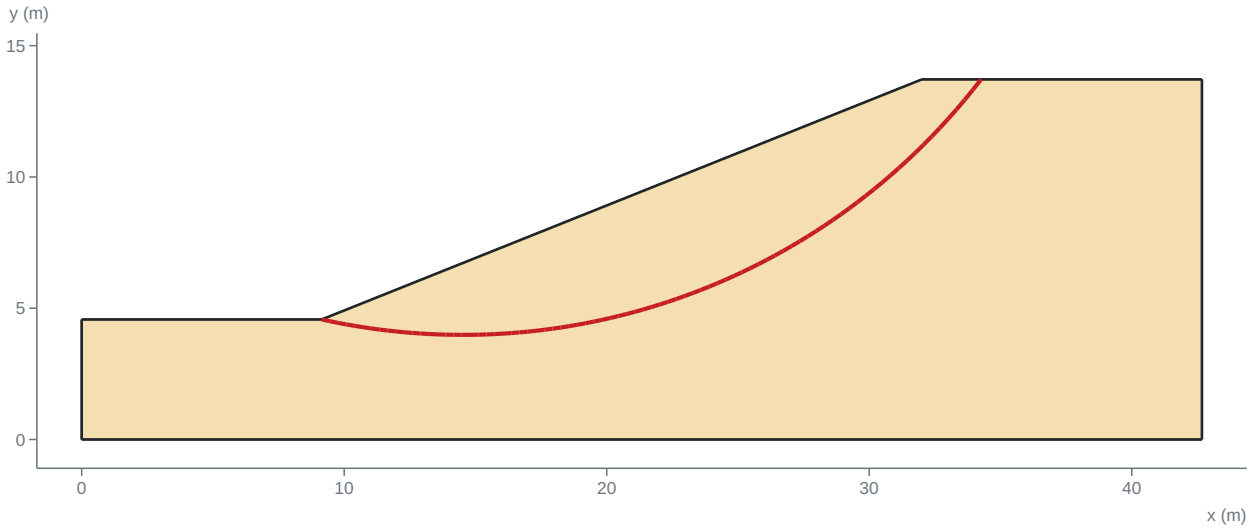


Figure 69. V041a: critical circle (Bishop simplified, FOS = 1.597).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Material 1	Mohr-Coulomb	20.11	c' = 4.79	20.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.603	1.597
Spencer (M-P, constant f(x))	Commercial program A	1.599	1.594
Spencer (M-P, constant f(x))	Duncan & Wright (reference)	1.600	1.594
Spencer (M-P, constant f(x))	Commercial program A (non-circular)	1.592	1.591

3.70 V041b — Submerged slope, pool 60 ft above the crest (D&W fig. 6.27)

Duncan & Wright, Soil Strength and Slope Stability, fig. 6.27.

Critical circle search · Units: SI (m, kPa, kN/m³)

As V041a with the pool 60 ft above the crest: the same factor of safety.

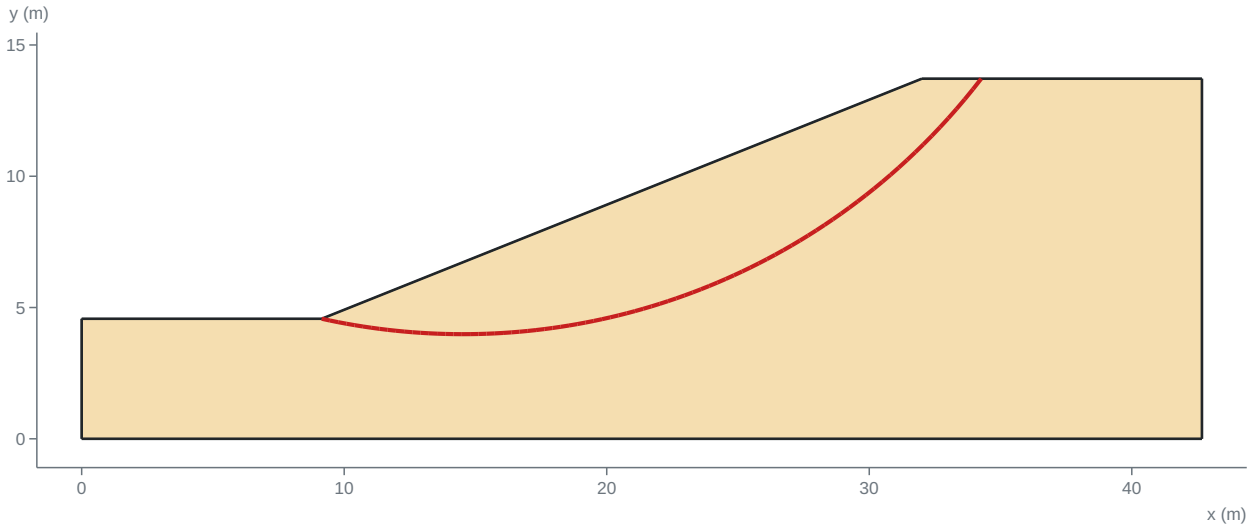


Figure 70. V041b: critical circle (Bishop simplified, FOS = 1.597).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Material 1	Mohr-Coulomb	20.11	c' = 4.79	20.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.603	1.597
Spencer (M-P, constant f(x))	Commercial program A	1.599	1.594
Spencer (M-P, constant f(x))	Duncan & Wright (reference)	1.600	1.594
Spencer (M-P, constant f(x))	Commercial program A (non-circular)	1.590	1.589

3.71 V042 — Thick-core dam with a piezometric line (D&W fig. 7.24)

Duncan & Wright, Soil Strength and Slope Stability, fig. 7.24.

Fixed circle · Units: SI (m, kPa, kN/m³)

Symmetric dam with a thick clay core (φ′ = 20°) in a shell (φ′ = 38°), pond at el. 315 ft, commercial program A's piezometric line; the deep circle. Converted to SI.

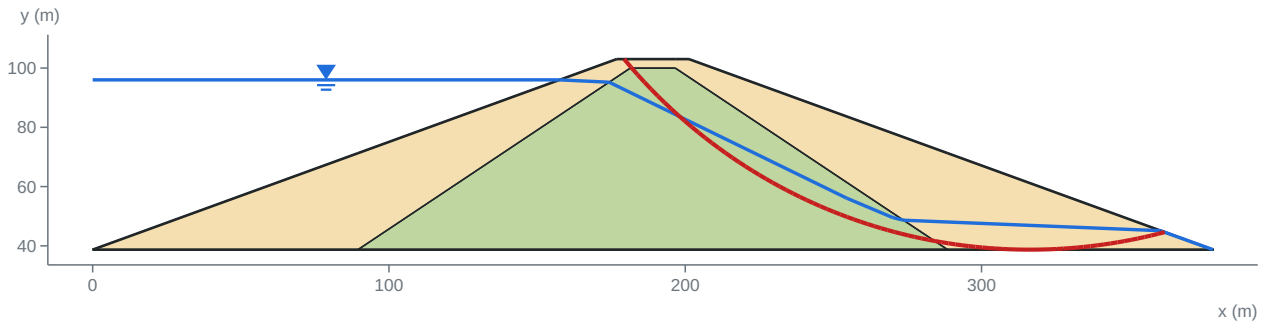


Figure 71. V042: analysed circle (Bishop simplified, FOS = 1.591).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Shell	Mohr-Coulomb	21.99	c' = 0.00	38.00°
Core	Mohr-Coulomb	18.85	c' = 0.00	20.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.584	1.591
Spencer (M-P, constant f(x))	Commercial program A	1.648	1.658
Bishop simplified	Program D	1.591	1.591
Morgenstern-Price (half-sine)	Program D	1.670	1.670

Notes

- With FE seepage: commercial program A 1.658/1.724.

3.72 V043 — Syncrude tailings dyke (El-Ramly et al.), composite circle

El-Ramly, Morgenstern & Cruden (2003), Can. Geotech. J. 40.

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

Cohesionless tailings sand, glacio-fluvial sands and tills (φ = 34°) on a presheared clay-shale (φ = 7.5°) with a water table; the reference circle (327.5, 394, R = 124) reaches below the model base, so the surface follows the base there (composite).

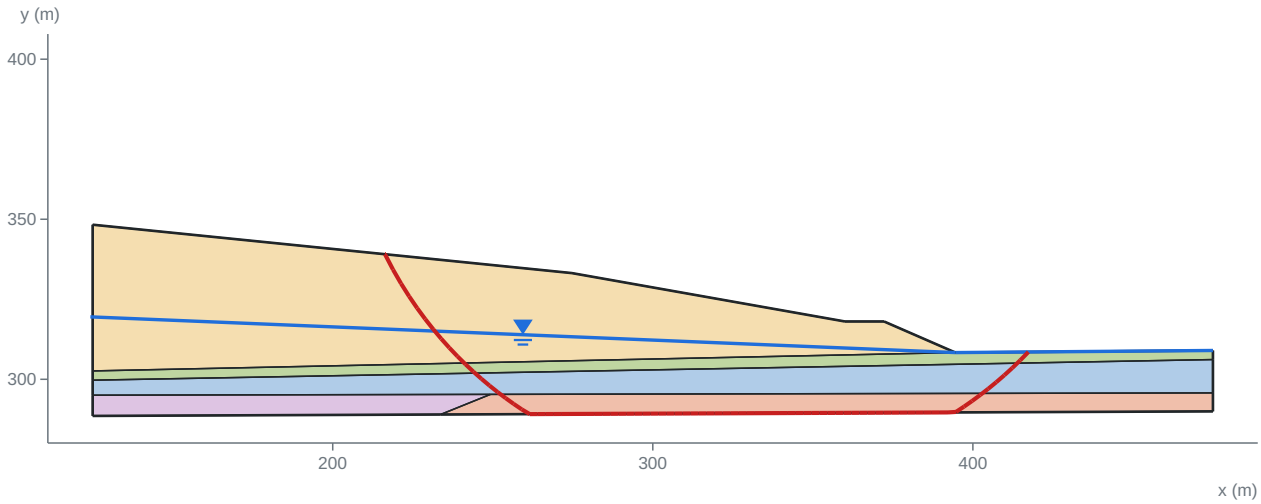


Figure 72. V043: analysed surface (Bishop simplified, FOS = 1.320).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Tailing sand (TS)	Mohr-Coulomb	20.00	c' = 0.00	34.00°
Glacio-fluvial sand (Pf4)	Mohr-Coulomb	17.00	c' = 0.00	34.00°
Sandy till (Pgs)	Mohr-Coulomb	17.00	c' = 0.00	34.00°
Clayey till (Pgc)	Mohr-Coulomb	17.00	c' = 0.00	7.50°
Disturbed clay-shale (Kca)	Mohr-Coulomb	17.00	c' = 0.00	7.50°

Method	Published by	Reference	cSlope
Bishop simplified	El-Ramly et al.	1.310	1.320
Bishop simplified	Program D	1.320	1.320

Notes

- commercial program A 1.305 is not used as a check: cSlope is 1.1 % above it and agrees with El-Ramly et al.'s own 1.31.

3.73 V044 — Fredlund & Krahn (1977), dry

Fredlund, D.G. & Krahn, J. (1977). Comparison of slope stability methods of analysis. Can. Geotech. J. 14(3).

Fixed circle · Units: imperial (ft, psf, pcf)

Homogeneous 2:1 slope, 40 ft high, on a firm base; the classic comparison problem of Fredlund & Krahn analysed with the published circle and no pore pressure.

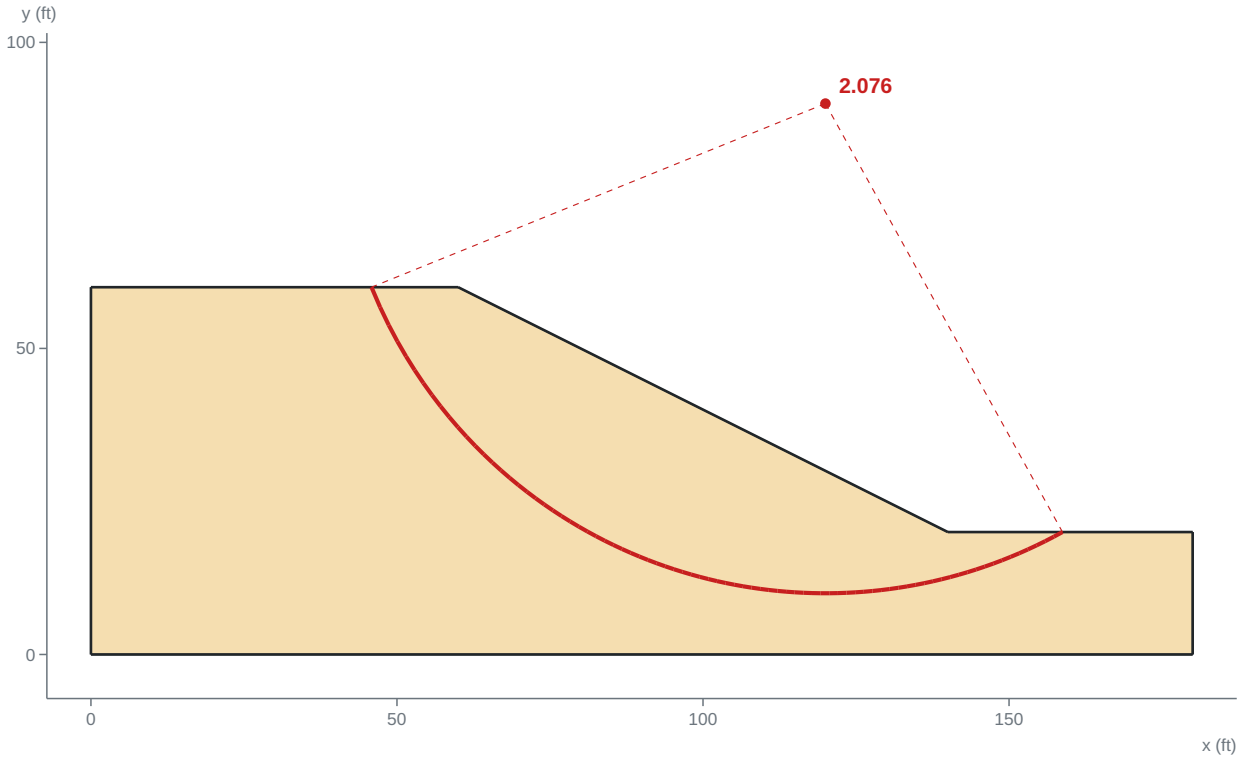


Figure 73. V044: analysed circle (Bishop simplified, FOS = 2.076).

Material	Model	γ (pcf)	Strength (psf)	φ
Soil	Mohr-Coulomb	120.0	c' = 600.0	20.00°

Method	Published by	Reference	cSlope
Bishop simplified	Fredlund & Krahn	2.080	2.076
Morgenstern-Price (half-sine)	Fredlund & Krahn	2.076	2.072

3.74 V045 — Fredlund & Krahn (1977), with water table

Fredlund, D.G. & Krahn, J. (1977). Can. Geotech. J. 14(3).

Fixed circle · Units: SI (m, kPa, kN/m³)

The Fredlund & Krahn slope with a piezometric line, analysed on the published circle. Converted to SI.

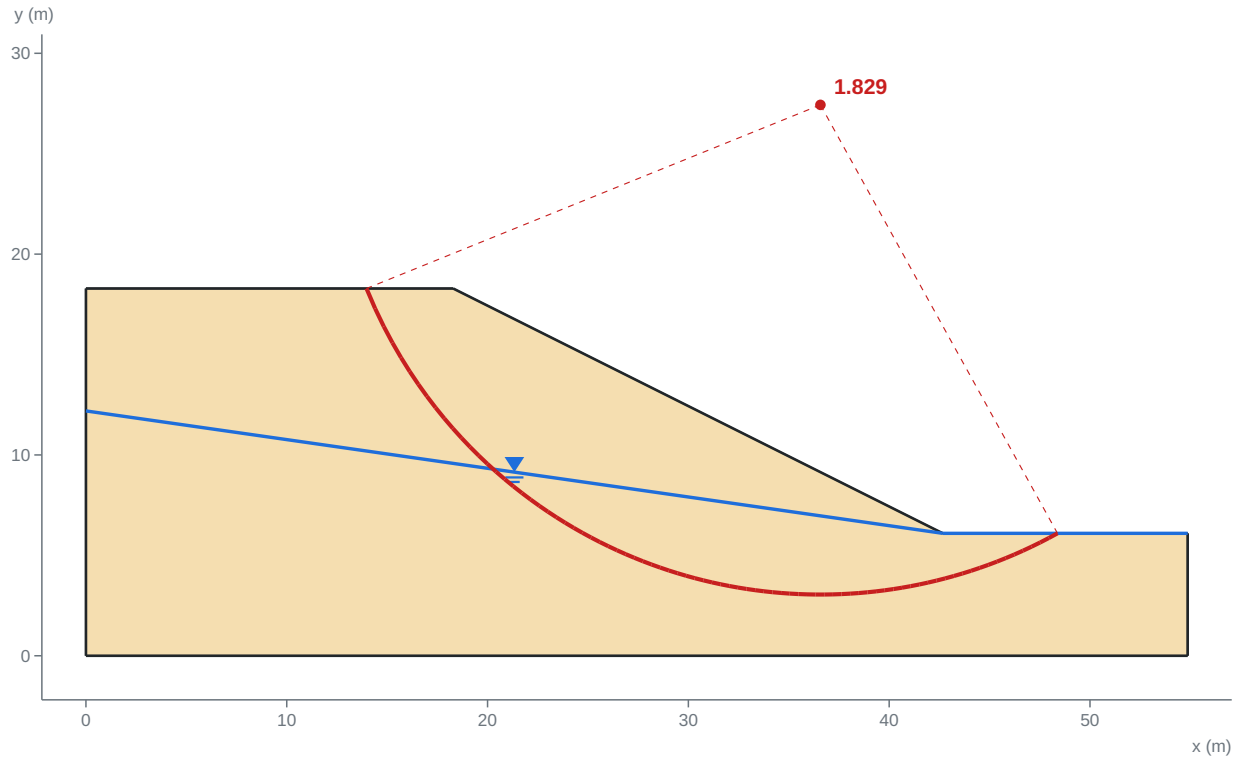


Figure 74. V045: analysed circle (Bishop simplified, FOS = 1.829).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	18.85	c' = 28.73	20.00°

Method	Published by	Reference	cSlope
Bishop simplified	Fredlund & Krahn	1.834	1.829
Morgenstern-Price (half-sine)	Fredlund & Krahn	1.832	1.827

3.75 V046 — Fredlund & Krahn slope on a weak seam, composite surface

Fredlund, D.G. & Krahn, J. (1977). Can. Geotech. J. 14(3).

Specified non-circular surface · Units: imperial (ft, psf, pcf)

The a related problem slope with a 1 ft weak seam (c′ = 0, φ′ = 10°) on the impenetrable base at el. 15; the published circle (120, 90, R = 80) reaches below the base, so the surface follows the arc down to the base, runs along the seam and climbs back on the arc. Dry.

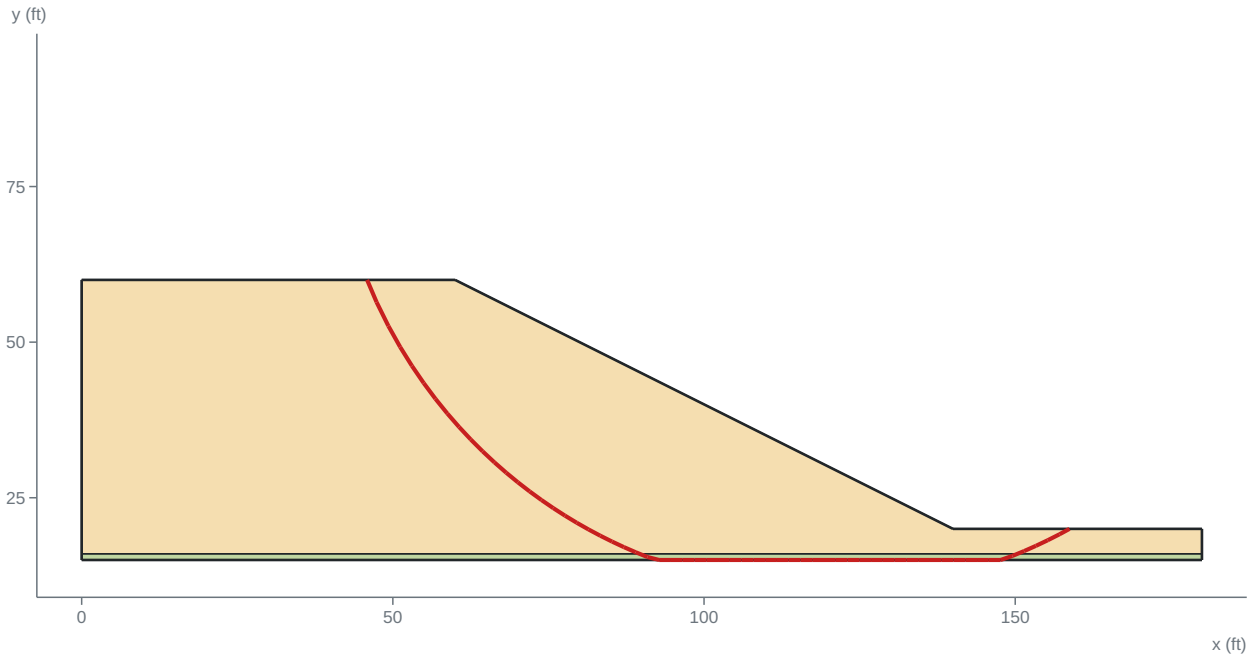


Figure 75. V046: analysed surface (Bishop simplified, FOS = 1.380).

Material	Model	γ (pcf)	Strength (psf)	φ
Upper soil	Mohr-Coulomb	120.0	c' = 600.0	20.00°
Weak seam	Mohr-Coulomb	120.0	c' = 0.00	10.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.382	1.380
Spencer (M-P, constant f(x))	Commercial program A	1.382	1.379
Morgenstern-Price (half-sine)	Fredlund & Krahn	1.370	1.370

3.76 V047a — ACADS 1(a), homogeneous slope, dry

Giam & Donald (1989).

Critical circle search · Units: SI (m, kPa, kN/m³)

ACADS 1(a): homogeneous 2:1 slope, 10 m high (c′ = 3 kPa, φ′ = 19.6°, γ = 20 kN/m³), dry; critical-circle search.

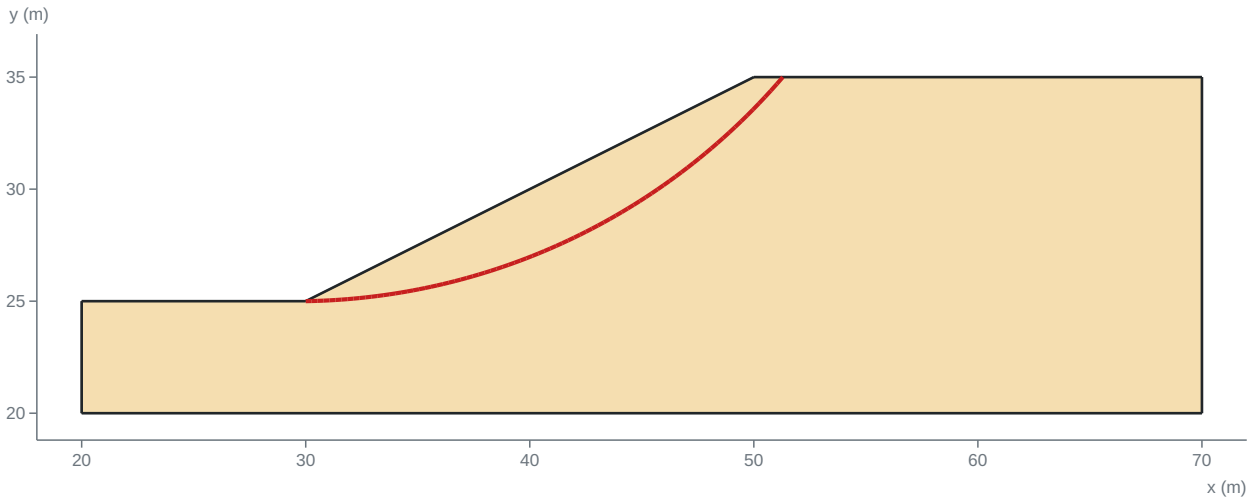


Figure 76. V047a: critical circle (Bishop simplified, FOS = 0.985).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	20.00	c' = 3.00	19.60°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	0.987	0.985
Spencer (M-P, constant f(x))	Commercial program A	0.986	0.984
Janbu corrected (f0)	Commercial program A	0.990	0.986
Morgenstern-Price (half-sine)	Commercial program B	0.991	0.984
Bishop simplified	Commercial program B (moment equilibrium, λ = 0)	0.992	0.985
Spencer (M-P, constant f(x))	ACADS	1.000	0.984

Notes

- Geometry (20, 20)–(20, 25)–(30, 25)–(50, 35)–(70, 35)–(70, 20) from Fig. 1.1, strength from Table 1.1. Giam & Donald's reference 1.00.
- Domain x = 20…70 as in Fig. 1 (the commercial program A/program D model is wider).
- Table 2 prints Bishop 0.963 / Janbu 0.992, but Fig. 3 (FS vs λ, slip 207) shows the moment FS at λ = 0 = 0.992 and the force FS at λ = 0 = 0.963: the two rows are swapped. commercial program B's Janbu (0.963) is the uncorrected simplified Janbu.

3.77 V047b — ACADS tension crack, reference crack depth 3.6 m (results correspond to a dry c

Giam & Donald (1989), ACADS problem 1(b).

Critical circle search · Units: SI (m, kPa, kN/m³)

ACADS 1(b) as commercial program B models it: c′ = 32 kPa, φ′ = 10°, a 3.6 m tension crack (dry, see the notes); critical-circle search.

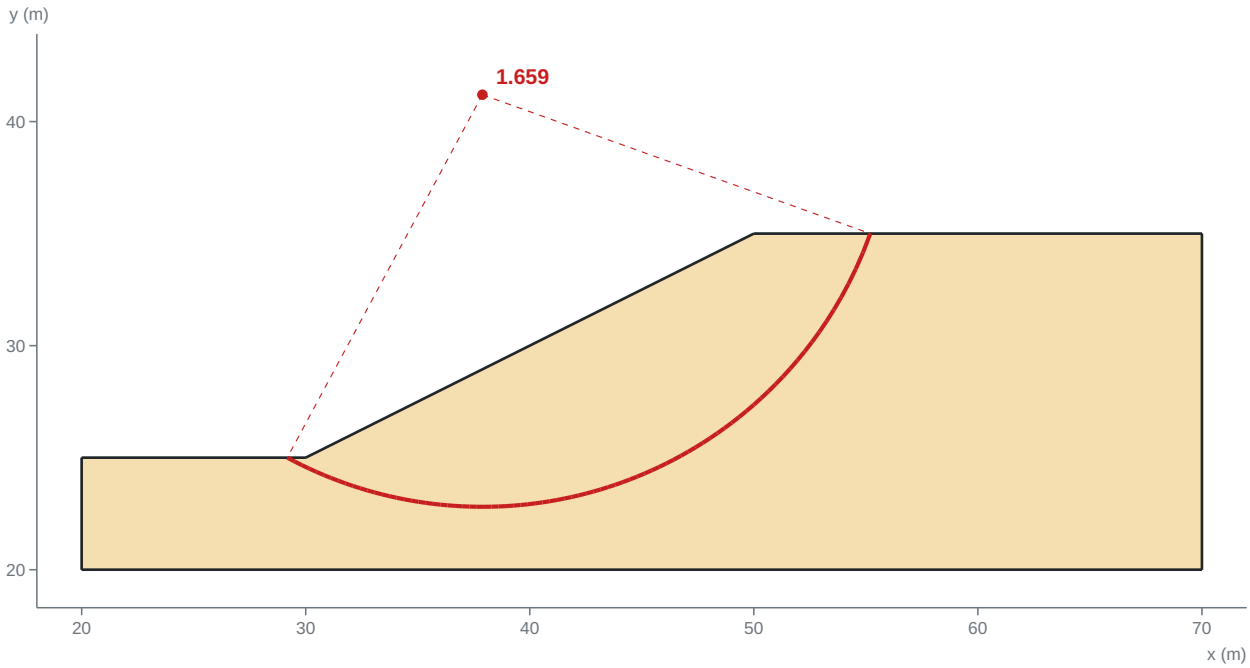


Figure 77. V047b: critical circle (Bishop simplified, FOS = 1.659).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	20.00	c' = 32.00	10.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program B	1.664	1.659
Morgenstern-Price (half-sine)	Commercial program B	1.660	1.656

Notes

- Entry on the crest (x = 49.9–55.2) and exit at the toe (x = 26.9–30.2), as in the source.
- The manual quotes 'about 3.6 m' for 2c/(γ√Ka); the formula gives 3.814 m (V048 uses 3.814). commercial program B Janbu (uncorrected) 1.471. ACADS band 1.65–1.70.
- The text says the crack is water-filled, but commercial program B's 1.664 / 1.660 are reproduced only with a DRY 3.6 m crack (cSlope 1.658 / 1.655). With the crack full of water cSlope gives 1.593 / 1.589 (3.6 m) and 1.589 / 1.586 (3.814 m, = V048; commercial program A 1.596 / 1.592). Without a crack 1.685 / 1.683.

3.78 V048 — ACADS 1(b), water-filled tension crack

Giam & Donald (1989), ACADS problem 1(b).

Critical circle search · Units: SI (m, kPa, kN/m³)

The ACADS 1(a) slope (10 m high, 2:1) with c' = 32 kPa, φ' = 10° and a water-filled tension crack 4 m deep, the crack zone of commercial program A's Fig. 2.1 (35 → 31 m; the formula 2c/(γ√Ka) gives 3.814 m); critical circle search.

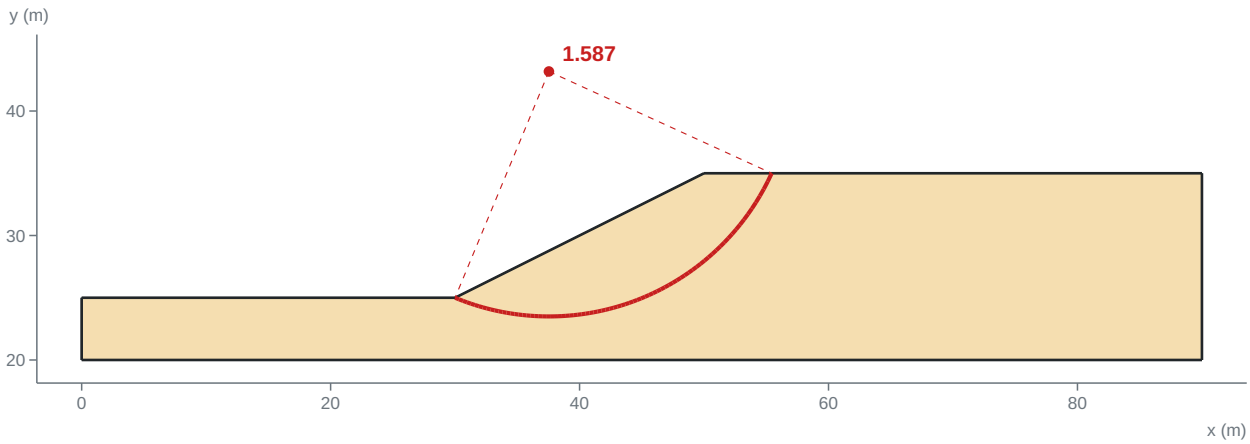


Figure 78. V048: critical circle (Bishop simplified, FOS = 1.587).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	20.00	c' = 32.00	10.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.596	1.587
Spencer (M-P, constant f(x))	Commercial program A	1.592	1.582
Janbu corrected (f0)	Commercial program A	1.489	1.475

Notes

- Search limited to x = 20–70, the model of the source.
- With the 3.814 m crack of the formula cSlope gives 1.590 / 1.585 / 1.481; the 4 m crack of the figure lowers them by 0.2–0.5 %.
- commercial program B's 1.664 (V047b) corresponds to a dry 3.6 m crack; the ACADS reference band is 1.65–1.70 (Giam 1.65). program D 1.589 with the 3.814 m crack.
- Janbu corrected: 1.5 % for a search whose critical circle (and so f0) differs from Bishop's.

3.79 V049 — ACADS 1(c), three-layer slope

Giam & Donald (1989), ACADS slope stability programs review.

Critical circle search · Units: SI (m, kPa, kN/m³)

Non-homogeneous three-layer slope (ACADS 1c); critical-circle search.

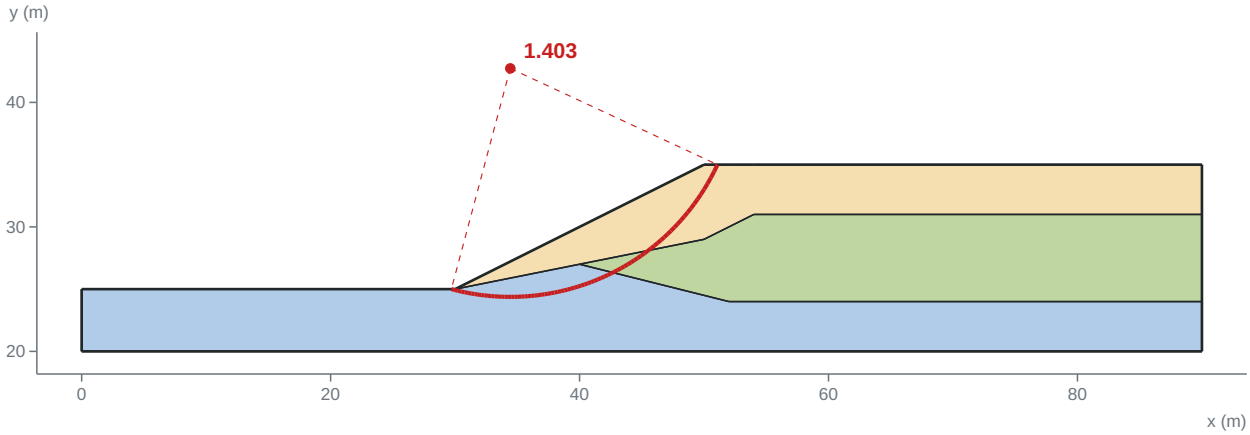


Figure 79. V049: critical circle (Bishop simplified, FOS = 1.403).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil #1	Mohr-Coulomb	19.50	c' = 0.00	38.00°
Soil #2	Mohr-Coulomb	19.50	c' = 5.30	23.00°
Soil #3	Mohr-Coulomb	19.50	c' = 7.20	20.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.405	1.403
Spencer (M-P, constant f(x))	Commercial program A	1.375	1.372
Janbu corrected (f0)	Commercial program A	1.357	1.355
Morgenstern-Price (half-sine)	Commercial program B	1.382	1.372
Bishop simplified	Program D	1.403	1.403

Notes

- Search limited to x = 20–70, the model of the source.
- ACADS 1.39; commercial program B Bishop 1.414.

3.80 V050a — ACADS 3(b), weak layer, specified non-circular surface

Giam & Donald (1989), ACADS problem 3(b).

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

Slope with a 0.5 m weak layer (c' = 0, φ' = 10°) and a water table at its base, analysed on the published four-point non-circular surface that runs along the weak layer.

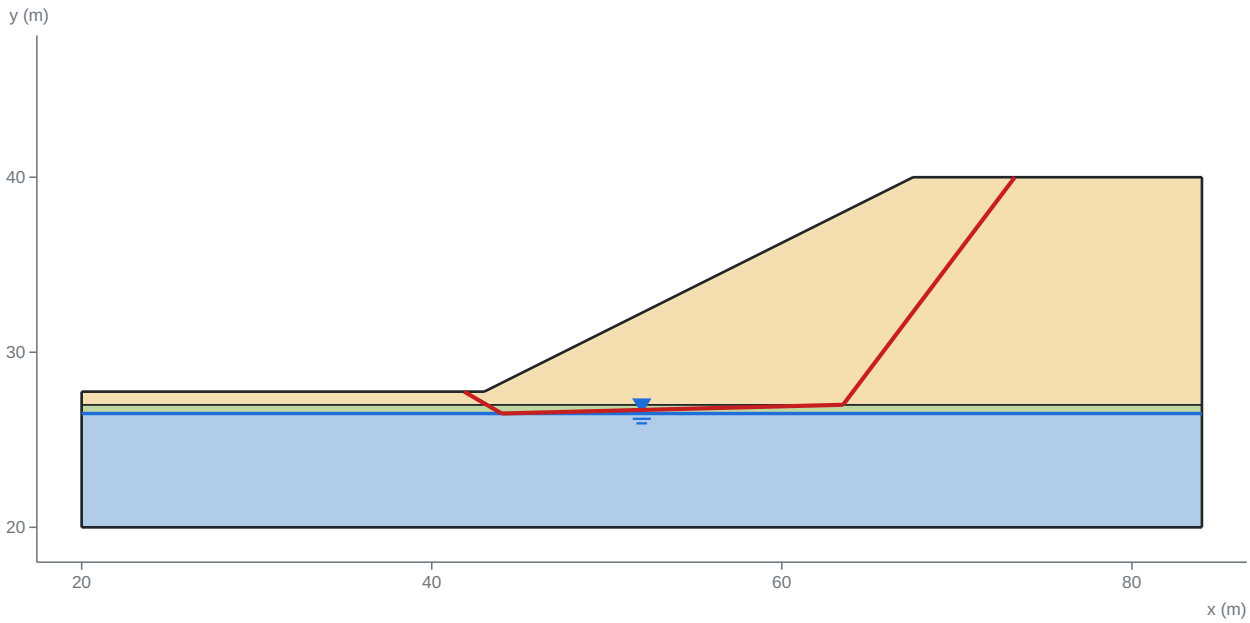


Figure 80. V050a: analysed surface (Janbu corrected (f0), FOS = 1.294).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil 1	Mohr-Coulomb	18.84	c' = 28.50	20.00°
Weak layer	Mohr-Coulomb	18.84	c' = 0.00	10.00°
Soil 1 below	Mohr-Coulomb	18.84	c' = 28.50	20.00°

Method	Published by	Reference	cSlope
Janbu corrected (f0)	Commercial program A	1.294	1.294
Spencer (M-P, constant f(x))	Commercial program A	1.277	1.276
Morgenstern-Price (half-sine)	Commercial program B	1.261	1.260

Notes

- Bishop on a non-circular surface depends on the moment axis; cSlope gives 1.250 about its default axis, commercial program B 1.259.

3.81 V050b — ACADS 3(a), weak layer, non-circular search

Giam & Donald (1989), ACADS problem 3(a).

Circle search with non-circular optimisation · Units: SI (m, kPa, kN/m³)

As V050a, with a circular search followed by non-circular optimisation; the critical surface slides along the weak layer with a back scarp to the crest.

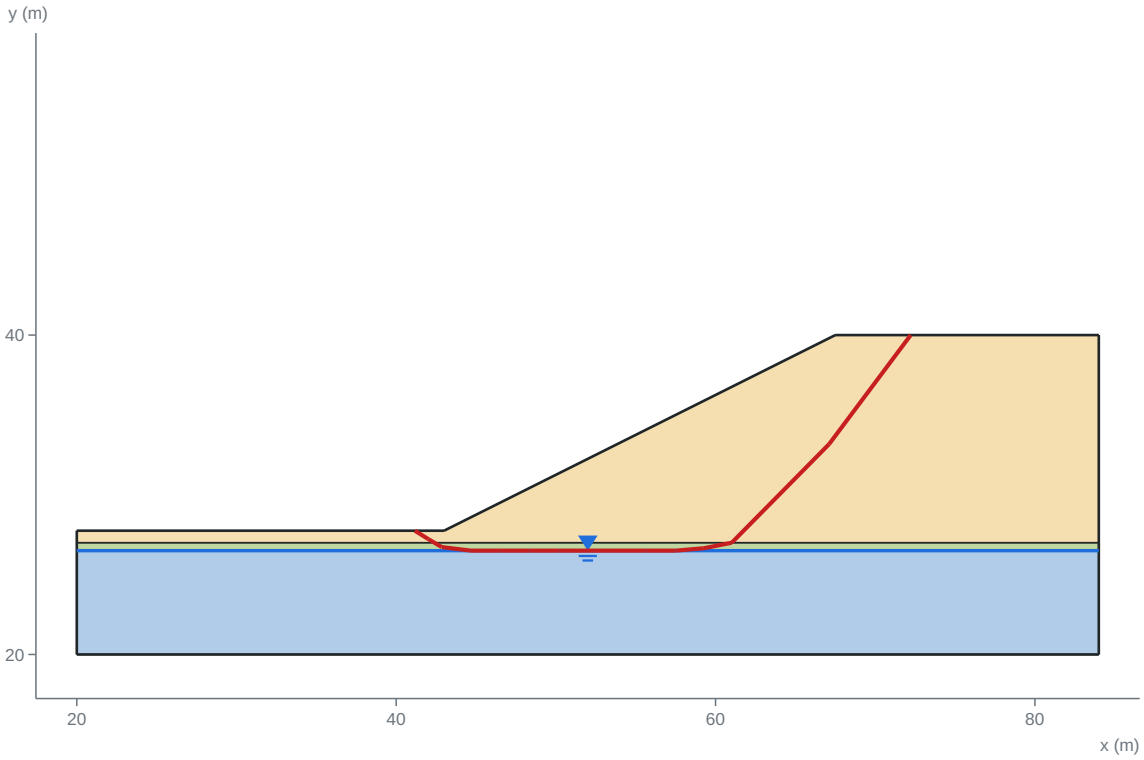


Figure 81. V050b: critical surface (Spencer (M-P, constant f(x)), FOS = 1.228).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil 1	Mohr-Coulomb	18.84	c' = 28.50	20.00°
Weak layer	Mohr-Coulomb	18.84	c' = 0.00	10.00°
Soil 1 below	Mohr-Coulomb	18.84	c' = 28.50	20.00°

Method	Published by	Reference	cSlope
Spencer (M-P, constant f(x))	Commercial program A	1.246	1.228

Notes

- The best circle gives 1.51; the optimised surface follows the weak layer with a wedge up to the crest. ACADS accepts about 1.24–1.27.
- Known gap: commercial program B's optimised Morgenstern-Price (half-sine) reports 1.261, the value of the specified surface of 3(b) (V050a), so its search found nothing lower; cSlope's search finds 1.218 (half-sine) and 1.228 (Spencer), a complete-equilibrium solution with one root that holds at 50, 100 and 200 slices. A search difference, not a model one: not a check.

3.82 V051 — ACADS 4, weak layer, water table and surcharges

Giam & Donald (1989), ACADS slope stability programs review.

Circle search with non-circular optimisation · Units: SI (m, kPa, kN/m³)

Weak-layer slope with a piezometric line and two surcharge strips (20 kPa on the bench, 20–40 kPa ramp on the crest); non-circular search. The published spread is wide: a search-difficulty benchmark.

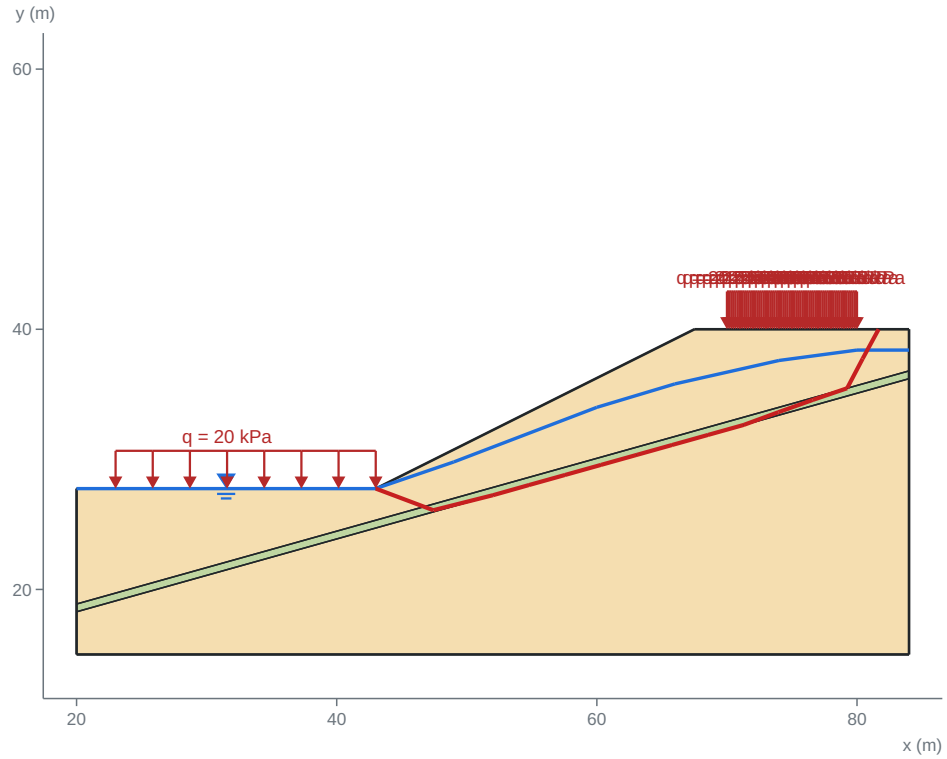


Figure 82. V051: critical surface (Spencer (M-P, constant f(x)), FOS = 0.698).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil 1	Mohr-Coulomb	18.84	c' = 28.50	20.00°
Weak Layer	Mohr-Coulomb	18.84	c' = 0.00	10.00°

Method	Published by	Reference	cSlope
Spencer (M-P, constant f(x))	Commercial program A (optimised)	0.707	0.698

Notes

- Known gap: commercial program A's optimised Janbu corrected is 0.699 (Table 9.5), cSlope's optimised Janbu corrected 0.754 (+7.8 %); the 0.734 of Table 9.4 is commercial program A's block search without optimisation and is not comparable with an optimised search. The search is sensitive to the method: Morgenstern-Price (half-sine) finds 0.718 against commercial program B's 0.689 and Spencer 0.718 against commercial program A's 0.707. commercial program A's optimised band 0.683–0.707; ACADS Slope 2000 (GLE) 0.688; commercial program A block search 0.760 (Spencer); program D's search 0.724 (Spencer) / 0.718 (Janbu); commercial program B Bishop 0.699; ACADS 20-program mean 0.808.

3.83 V052a — Talbingo Dam, end of construction

Giam & Donald (1989), ACADS slope stability programs review.

Critical circle search · Units: SI (m, kPa, kN/m³)

ACADS 2(a): Talbingo Dam, four zones, critical-circle search. The minimum is a shallow slide parallel to the steeper upstream face.

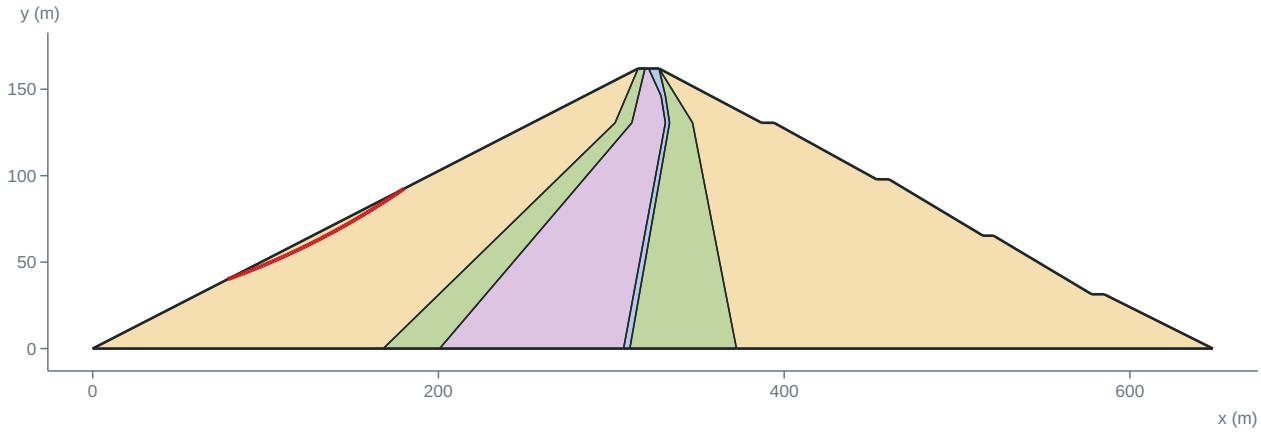


Figure 83. V052a: critical circle (Bishop simplified, FOS = 1.964).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Rockfill	Mohr-Coulomb	20.40	c' = 0.00	45.00°
Transitions	Mohr-Coulomb	20.40	c' = 0.00	45.00°
Filter	Mohr-Coulomb	20.40	c' = 0.00	45.00°
Core	Mohr-Coulomb	18.10	c' = 85.00	23.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.948	1.964
Spencer (M-P, constant f(x))	Commercial program A	1.948	1.964
Janbu corrected (f0)	Commercial program A	1.949	1.974
Bishop simplified	Commercial program B	1.951	1.964

Notes

- Giam 1.95; infinite-slope limit 1.9475; program D 1.955.

3.84 V052b — Talbingo Dam, specified circle

Giam & Donald (1989), ACADS slope stability programs review.

Fixed circle · Units: SI (m, kPa, kN/m³)

ACADS 2(b): Talbingo Dam on the circle (100.3, 291.0, R = 278.8).

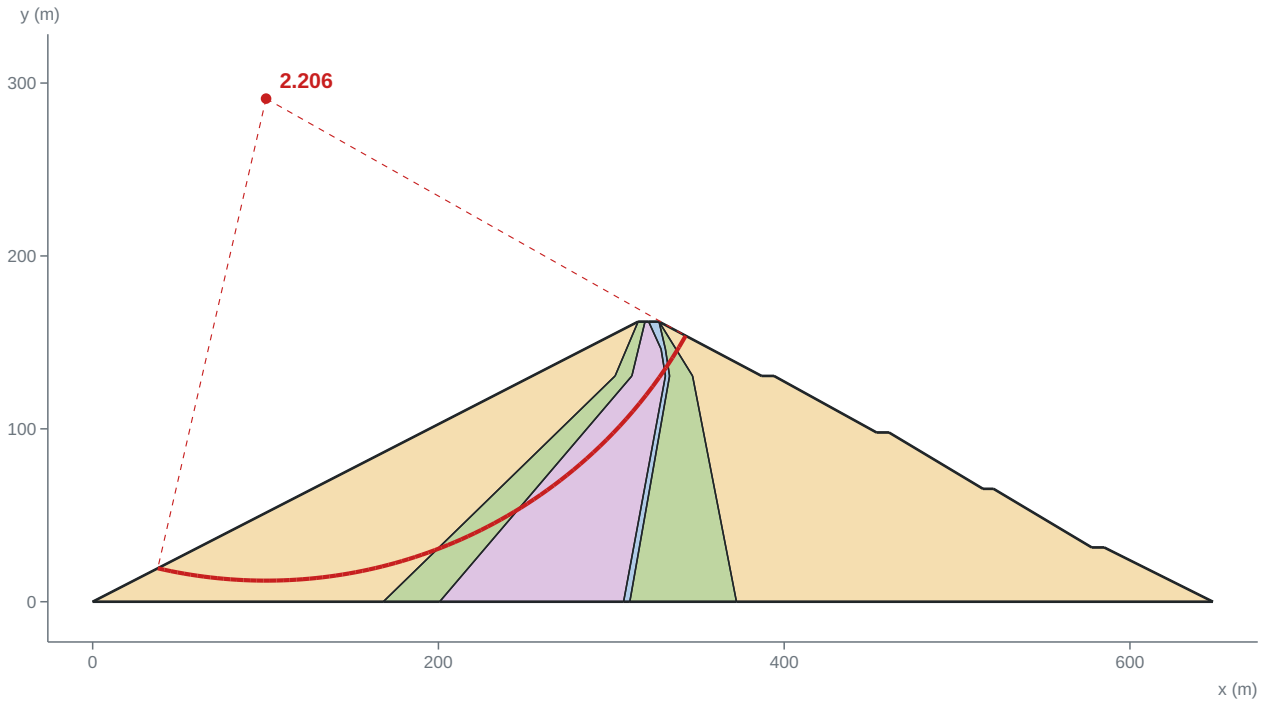


Figure 84. V052b: analysed circle (Bishop simplified, FOS = 2.206).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Rockfill	Mohr-Coulomb	20.40	c' = 0.00	45.00°
Transitions	Mohr-Coulomb	20.40	c' = 0.00	45.00°
Filter	Mohr-Coulomb	20.40	c' = 0.00	45.00°
Core	Mohr-Coulomb	18.10	c' = 85.00	23.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	2.208	2.206
Spencer (M-P, constant f(x))	Commercial program A	2.292	2.290
Janbu corrected (f0)	Commercial program A	2.073	2.073
Morgenstern-Price (half-sine)	Commercial program B	2.299	2.299

Notes

- Giam 2.29.

3.85 V053 — Greco (1996) ex. 5 / Chen & Shao weak-layer slope, reference geometry

Greco (1996) example 5; Chen & Shao (1988).

Circle search with non-circular optimisation · Units: SI (m, kPa, kN/m³)

Greco's weak-seam slope (V054) with commercial program B's geometry and water table; circular search with non-circular optimisation, vertical pore-pressure head.

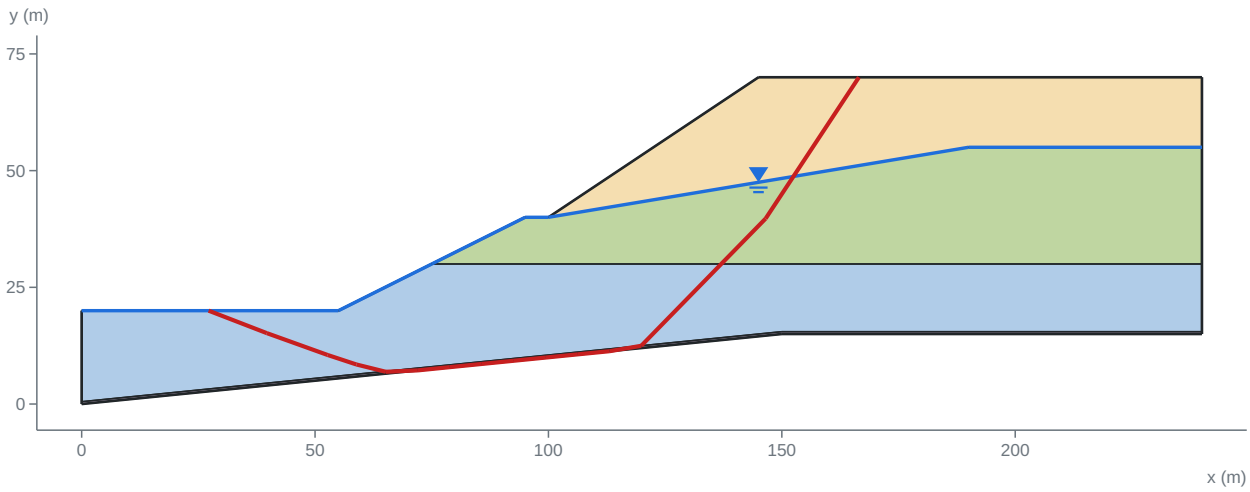


Figure 85. V053: critical surface (Spencer (M-P, constant f(x)), FOS = 1.009).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Layer 1	Mohr-Coulomb	20.00	c' = 9.80	35.00°
Layer 2	Mohr-Coulomb	19.00	c' = 58.80	25.00°
Layer 3	Mohr-Coulomb	21.50	c' = 19.80	30.00°
Layer 4	Mohr-Coulomb	21.50	c' = 9.80	16.00°

Method	Published by	Reference	cSlope
Spencer (M-P, constant f(x))	Commercial program A (block search)	1.010	1.009

Notes

- Entry on the crest (x = 145–240) and exit on the level ground (x = 0–55), as in the source.
- Fig. 40: base (0,0)–(150,15)–(240,15) under a 0.5 m Layer 4 whose top is (0,0.5)–(150,15.5)–(240,15.5); the thickness of Layer 4 is read from the label (0, 0.5) only partly legible. V054 (commercial program A) has the base rising to (240,16).
- commercial program B Janbu (uncorrected) 0.896, Bishop 1.040 (axis-dependent on a non-circular surface, not checked). Greco 1.08.
- The reference is commercial program A's block search (1.010, on its own geometry with the base rising to (240, 16)), the same kind of search as cSlope's: the critical surface slides along Layer 4 with a wedge up to the crest and a wedge out to the level ground, and Chen & Shao give 1.01–1.03. Commercial program B's optimised search on this geometry reports 1.054, which cSlope's search passes below (1.009): a search difference, not a model one.

3.86 V054 — Greco (1996) example 5 / Chen & Shao (1988)

Greco (1996), example 5; Chen & Shao (1988).

Critical circle search · Units: SI (m, kPa, kN/m³)

Four layers including a weak seam, with a water table; critical circle search, vertical pore-pressure head.

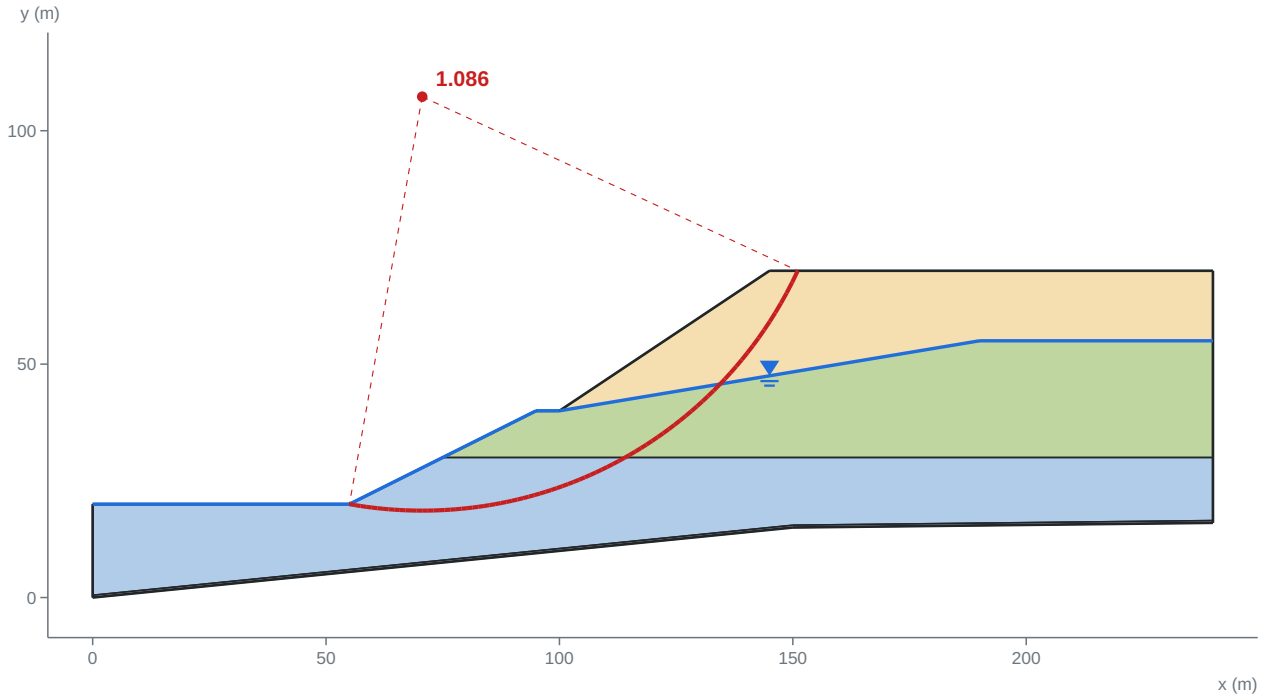


Figure 86. V054: critical circle (Bishop simplified (vertical head), FOS = 1.086).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Layer 1	Mohr-Coulomb	20.00	c' = 9.80	35.00°
Layer 2	Mohr-Coulomb	19.00	c' = 58.80	25.00°
Layer 3	Mohr-Coulomb	21.50	c' = 19.80	30.00°
Seam	Mohr-Coulomb	21.50	c' = 9.80	16.00°

Method	Published by	Reference	cSlope
Bishop simplified (vertical head)	Commercial program A	1.087	1.086
Spencer (vertical head)	Commercial program A	1.093	1.091

Notes

- Circles pass through the toe (55, 20), as in the source.
- With the phreatic correction (cos²θ) cSlope gives 1.141 (Bishop).

3.87 V055 — Griffiths & Lane example 2, slope with a foundation layer

Griffiths & Lane (1999), Géotechnique 49(3).

Critical circle search · Units: SI (m, kPa, kN/m³)

The 2:1 slope (c/γH = 0.05, φ = 20°) on a foundation of the same soil; unconstrained circular search finds the toe circle. Converted to SI.

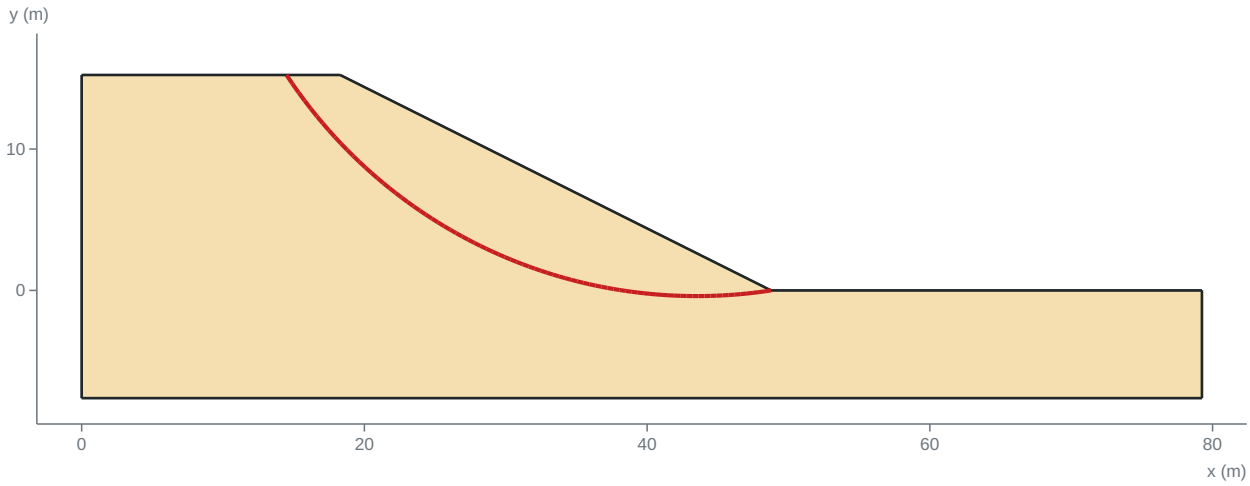


Figure 87. V055: critical circle (Spencer (M-P, constant f(x)), FOS = 1.366).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
soil	Mohr-Coulomb	19.64	c' = 14.96	20.00°

Method	Published by	Reference	cSlope
Spencer (M-P, constant f(x))	Griffiths & Lane (FE)	1.400	1.366
Spencer (M-P, constant f(x))	Program D	1.366	1.366

Notes

- Bishop & Morgenstern chart 1.380; a circle forced tangent to the foundation base gives 1.70.

3.88 V056 — Griffiths & Lane example 3, thin weak layer, cu2/cu1 = 0.2

Griffiths & Lane (1999), Géotechnique 49(3).

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

Undrained clay slope with a thin weak layer at a fifth of the strength, on the three-line wedge through the layer (the paper's Janbu mechanism). Converted to SI.

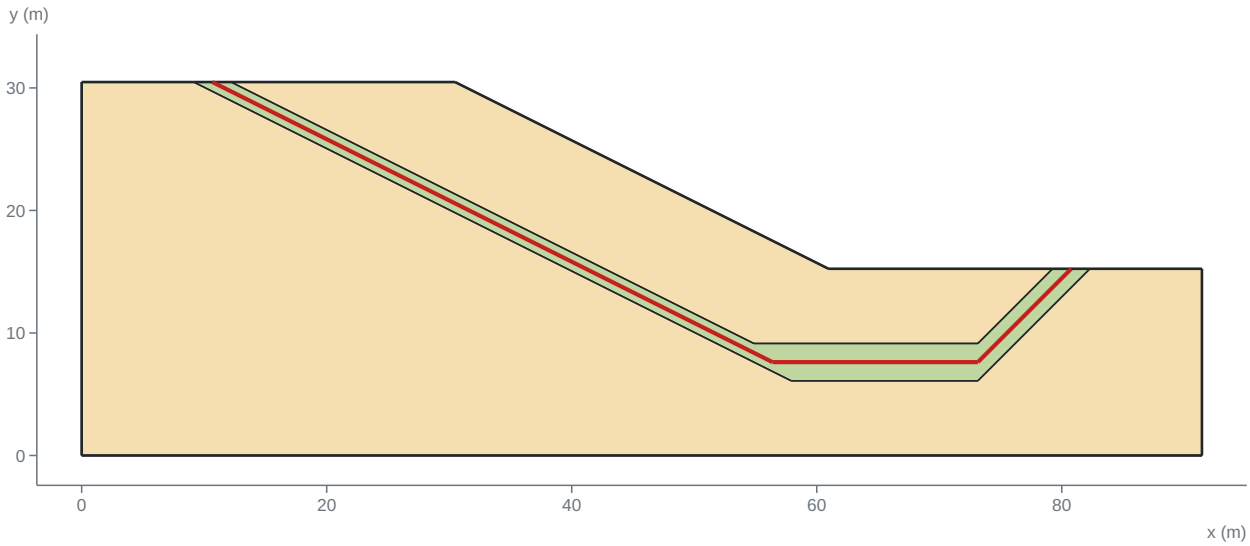


Figure 88. V056: analysed surface (Janbu corrected (f0), FOS = 0.499).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Surrounding clay	Undrained (φ = 0)	19.64	Su = 74.81	0°
Thin weak layer	Undrained (φ = 0)	19.64	Su = 14.96	0°

Method	Published by	Reference	cSlope
Janbu corrected (f0)	Griffiths & Lane (Janbu wedge, Fig. 7)	0.485	0.499

Notes

- Geometry as Fig. 6 (H = 50 ft, cu1/γH = 0.25, the layer 0.2H thick). The wedge runs along the middle of the layer; the paper shows its mechanism but gives no coordinates. Its factor of safety is read from Fig. 7 ("Slope program—wedges", Janbu's method) as 0.485; the paper does not say whether Janbu's correction factor f0 is applied: without it cSlope gives 0.458. FE 0.60.
- Spencer on the wedge 0.515; program D's non-circular search from this wedge reaches 0.462. cSlope's circular-seeded optimisation does not find the wedge (0.72); a circular search gives 1.24 (the paper: ≈1.3).

3.89 V057a — Griffiths & Lane example 4, cu2/cu1 = 1 (deep base circle)

Griffiths & Lane (1999), Géotechnique 49(3).

Critical circle search · Units: SI (m, kPa, kN/m³)

Undrained clay slope over a foundation of the same strength; the deep base circle. Converted to SI.

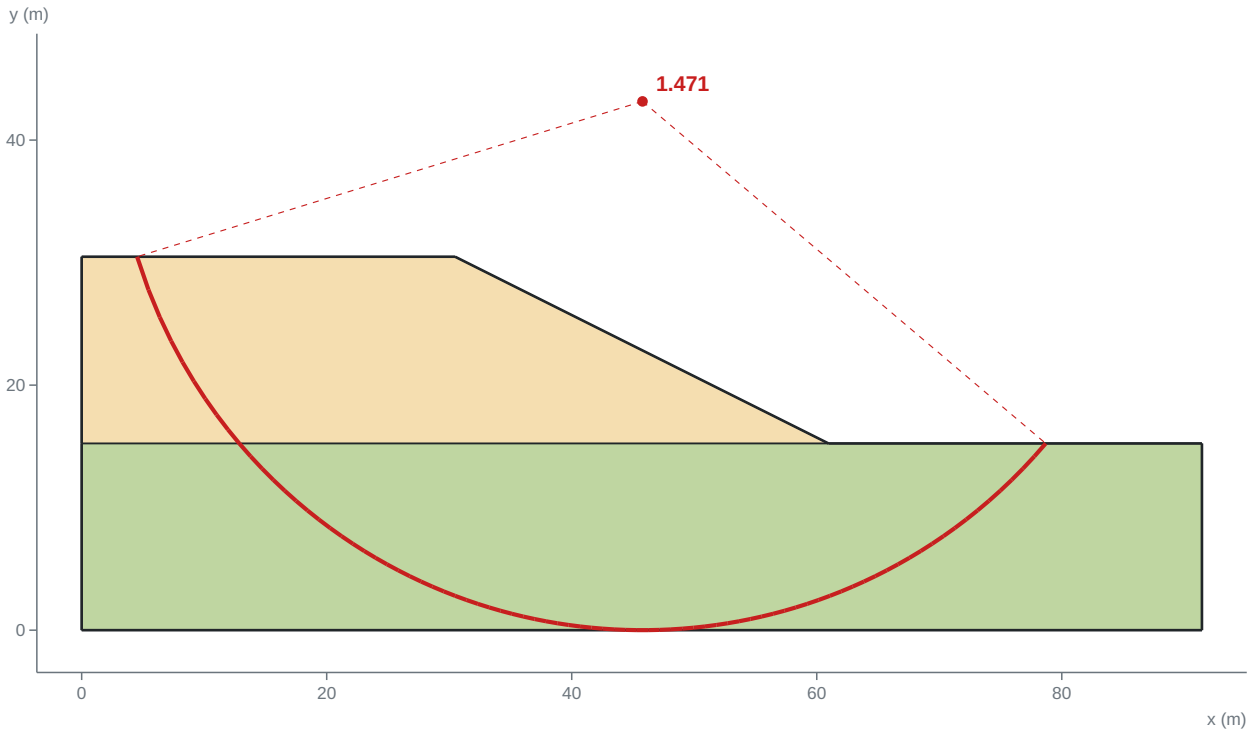


Figure 89. V057a: critical circle (Spencer (M-P, constant f(x)), FOS = 1.471).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Slope clay	Undrained (φ = 0)	19.64	Su = 74.81	0°
Foundation clay	Undrained (φ = 0)	19.64	Su = 74.81	0°

Method	Published by	Reference	cSlope
Spencer (M-P, constant f(x))	Griffiths & Lane (LE curve)	1.460	1.471
Spencer (M-P, constant f(x))	Program D	1.468	1.471

Notes

- FE 1.45.

3.90 V057b — Griffiths & Lane example 4, cu2/cu1 = 2 (toe circle)

Griffiths & Lane (1999), Géotechnique 49(3).

Critical circle search · Units: SI (m, kPa, kN/m³)

The same slope over a foundation twice as strong; the shallow toe circle. Converted to SI.

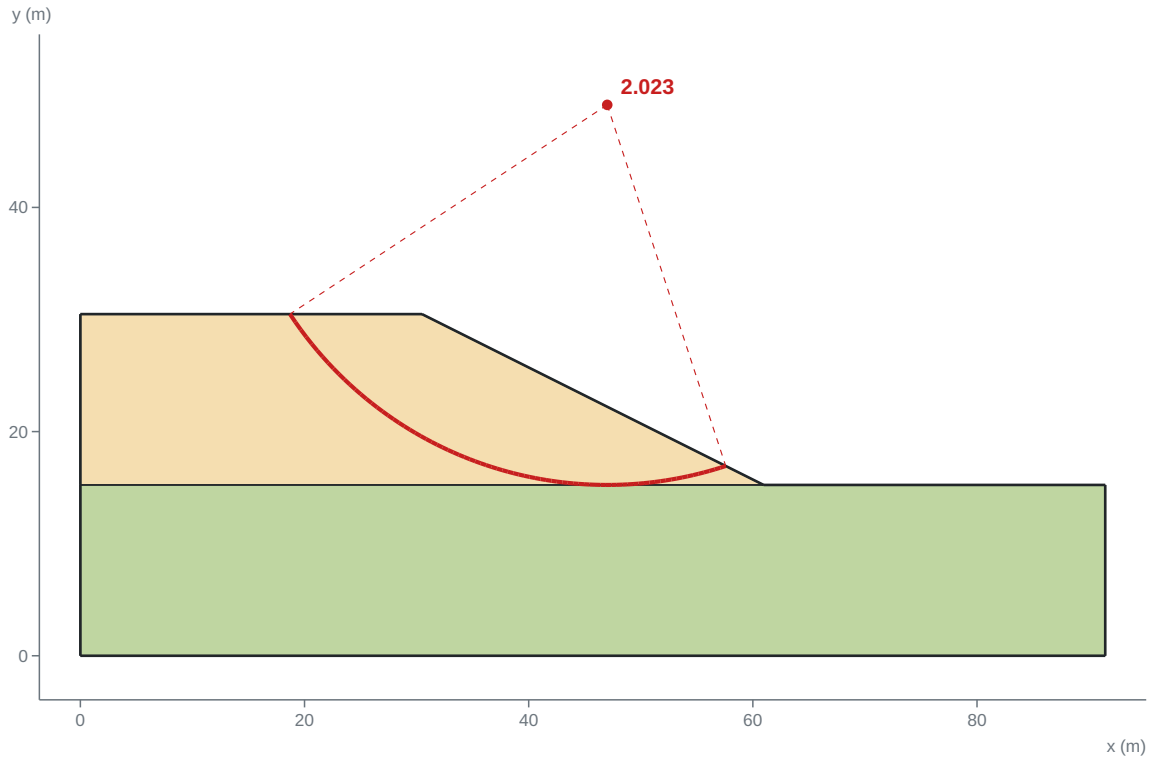


Figure 90. V057b: critical circle (Spencer (M-P, constant f(x)), FOS = 2.023).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Slope clay	Undrained (φ = 0)	19.64	Su = 74.81	0°
Foundation clay	Undrained (φ = 0)	19.64	Su = 149.6	0°

Method	Published by	Reference	cSlope
Spencer (M-P, constant f(x))	Griffiths & Lane (LE curve)	2.040	2.023
Spencer (M-P, constant f(x))	Program D	2.022	2.023

Notes

- FE 2.03.

3.91 V058 — Griffiths & Lane example 6, earth dam with full reservoir

Griffiths & Lane (1999), Géotechnique 49(3).

Critical circle search · Units: SI (m, kPa, kN/m³)

Two-sided earth dam with the reservoir full and a piezometric line; circular search. Converted to SI.

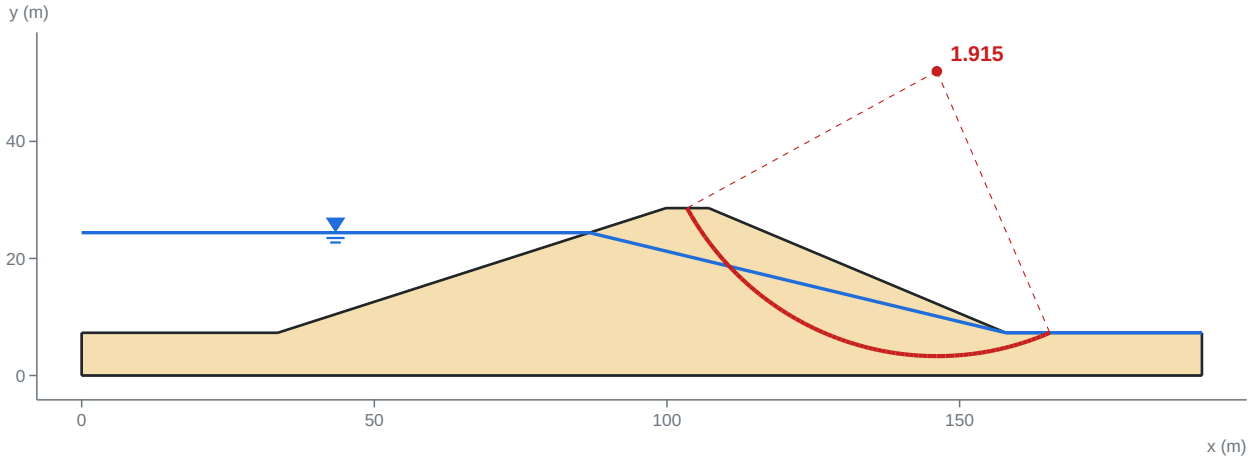


Figure 91. V058: critical circle (Spencer (M-P, constant f(x)), FOS = 1.915).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Embankment	Mohr-Coulomb	18.20	c' = 13.80	37.00°

Method	Published by	Reference	cSlope
Spencer (M-P, constant f(x))	Griffiths & Lane (FE)	1.900	1.915
Spencer (M-P, constant f(x))	Program D	1.915	1.915

3.92 V059a — Guo & Griffiths two-layer undrained slope, cu2/cu1 = 1.4

Guo & Griffiths (2020), Can. Geotech. J. 57.

Circle search with non-circular optimisation · Units: SI (m, kPa, kN/m³)

18 m embankment at 1V:2H (Su = 60 kPa) on an 18 m undrained foundation (Su = 84 kPa), γ = 20, firm base; optimised non-circular search. The deep mechanism governs.

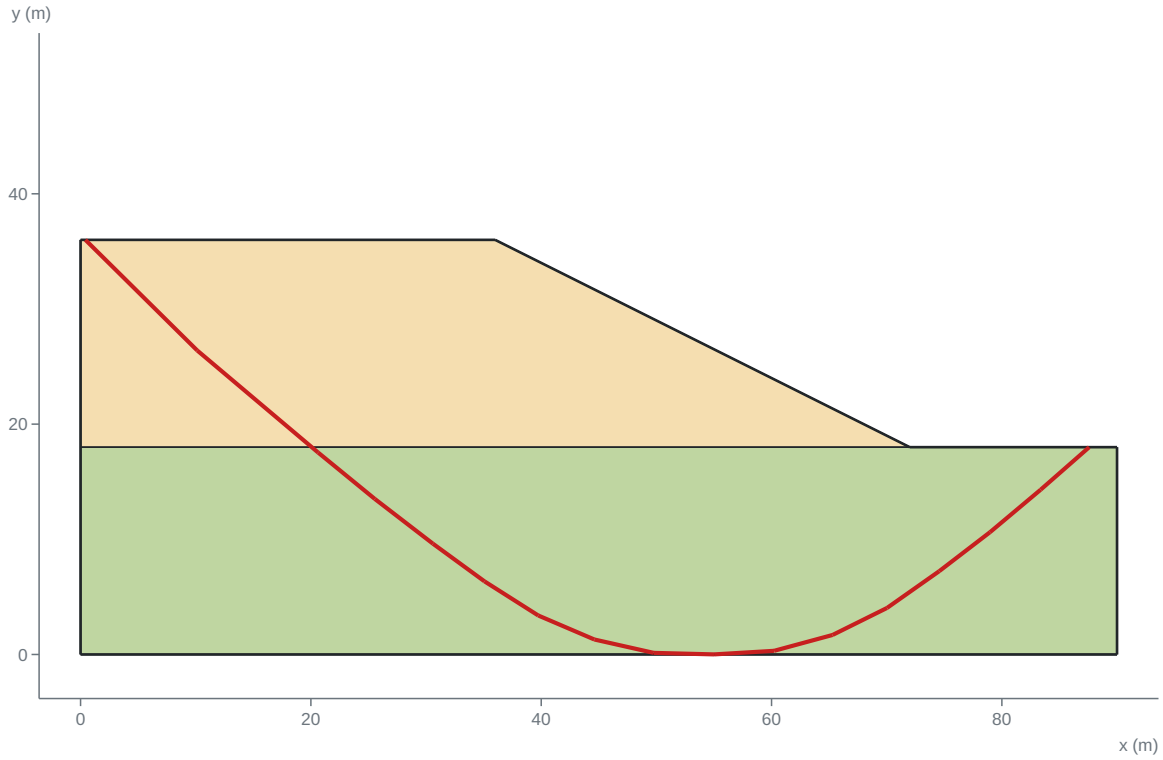


Figure 92. V059a: critical surface (Spencer (M-P, constant f(x)), FOS = 1.216).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Embankment	Undrained (φ = 0)	20.00	Su = 60.00	0°
Foundation	Undrained (φ = 0)	20.00	Su = 84.00	0°

Method	Published by	Reference	cSlope
Spencer (M-P, constant f(x))	Commercial program A (optimised)	1.215	1.216

3.93 V059b — Guo & Griffiths two-layer undrained slope, cu2/cu1 = 1.5

Guo & Griffiths (2020), Can. Geotech. J. 57.

Circle search with non-circular optimisation · Units: SI (m, kPa, kN/m³)

18 m embankment at 1V:2H (Su = 60 kPa) on an 18 m undrained foundation (Su = 90 kPa), γ = 20, firm base; optimised non-circular search. The deep mechanism governs.

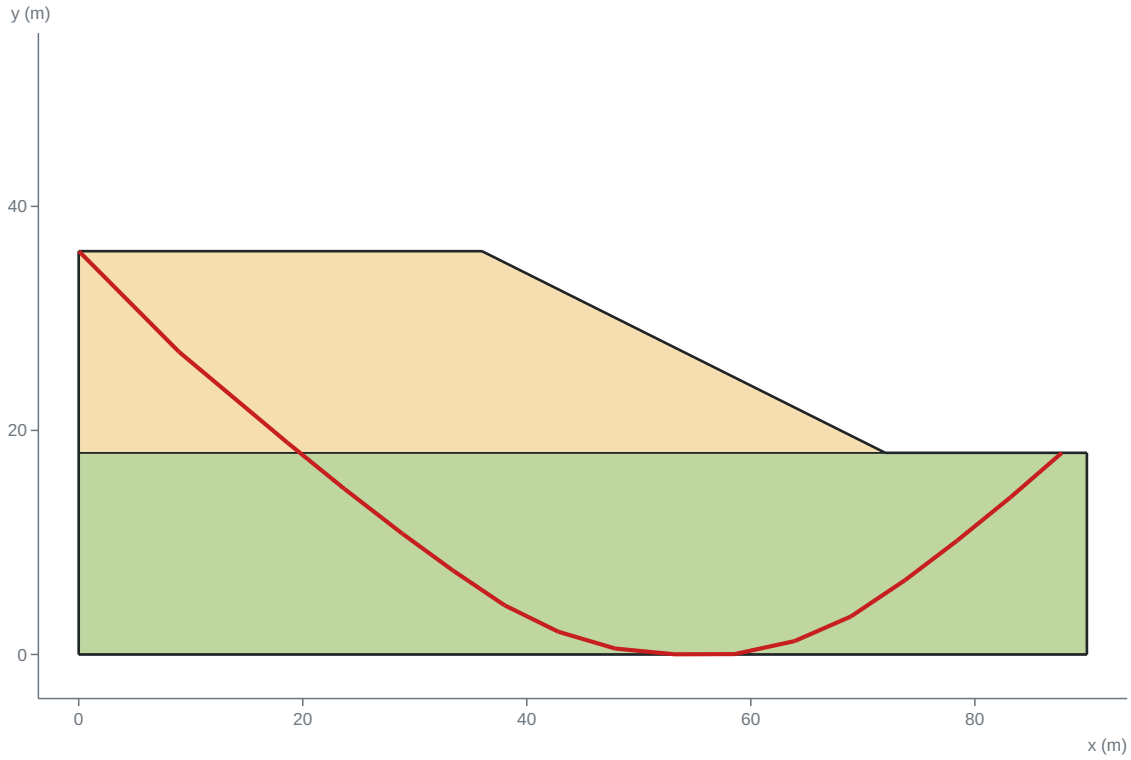


Figure 93. V059b: critical surface (Spencer (M-P, constant f(x)), FOS = 1.292).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Embankment	Undrained (φ = 0)	20.00	Su = 60.00	0°
Foundation	Undrained (φ = 0)	20.00	Su = 90.00	0°

Method	Published by	Reference	cSlope
Spencer (M-P, constant f(x))	Commercial program A (optimised)	1.290	1.292

3.94 V059c — Guo & Griffiths two-layer undrained slope, cu2/cu1 = 1.6

Guo & Griffiths (2020), Can. Geotech. J. 57.

Circle search with non-circular optimisation · Units: SI (m, kPa, kN/m³)

18 m embankment at 1V:2H (Su = 60 kPa) on an 18 m undrained foundation (Su = 96 kPa), γ = 20, firm base; optimised non-circular search. The shallow mechanism governs.

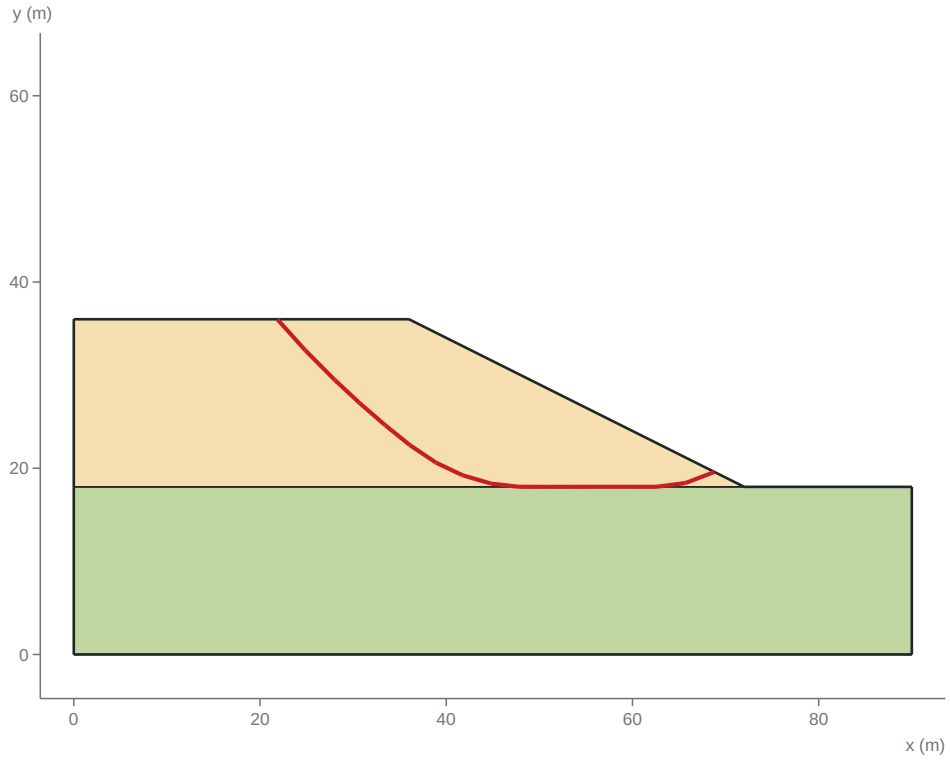


Figure 94. V059c: critical surface (Spencer (M-P, constant f(x)), FOS = 1.314).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Embankment	Undrained (φ = 0)	20.00	Su = 60.00	0°
Foundation	Undrained (φ = 0)	20.00	Su = 96.00	0°

Method	Published by	Reference	cSlope
Spencer (M-P, constant f(x))	Commercial program A (optimised)	1.315	1.314

3.95 V060a — Cannon Dam end of construction (Hassan & Wolff), critical circle

Hassan & Wolff (1999), J. Geotech. Geoenviron. Eng. 125(4).

Critical circle search · Units: SI (m, kPa, kN/m³)

Two clay fills with a sand-filter strip under the crest and a spoil-covered downstream toe; Bishop search for the minimum factor of safety at the mean strengths. The reliability index on the minimum-β surface is V060b.

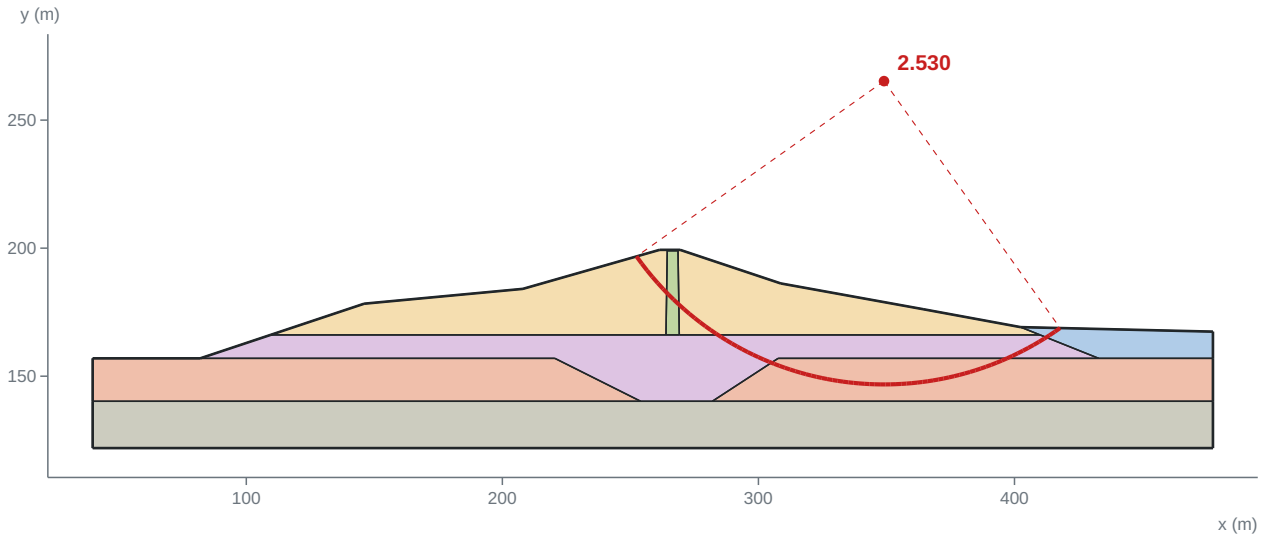


Figure 95. V060a: critical circle (Bishop simplified, FOS = 2.530).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Phase II clay fill	Mohr-Coulomb	22.00	c' = 143.6	15.00°
Sand filter	Mohr-Coulomb	22.00	c' = 0.00	35.00°
Spoil fill	Mohr-Coulomb	25.00	c' = 5.00	35.00°
Phase I clay fill	Mohr-Coulomb	22.00	c' = 117.8	8.50°
Foundation sand	Mohr-Coulomb	20.00	c' = 5.00	18.00°
Limestone	Mohr-Coulomb	23.00	c' = 1000.0	45.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	2.551	2.530
Bishop simplified	Program D	2.529	2.530

Notes

- 2.551 is commercial program A's Bishop value on Hassan & Wolff's surface 7A (Table 35.2), the lowest of the published circles, not a search result; cSlope's search finds 2.530 (V062a checks the circle itself). Hassan & Wolff 2.753 on 7A (their own LEM).

3.96 V060b — Cannon Dam end of construction (Hassan & Wolff), reliability on the minimum-β c

Hassan & Wolff (1999), J. Geotech. Geoenviron. Eng. 125(4).

Fixed circle · Units: SI (m, kPa, kN/m³)

The V060a dam on the circle of minimum reliability index: Phase I fill φ = 8.5° ± 8.5°, c = 117.8 ± 58.9 kPa; Phase II fill c = 143.6 ± 79 kPa, φ = 15° ± 9°, treated as independent normal variables (program D's Taylor series has no correlation term); Bishop.

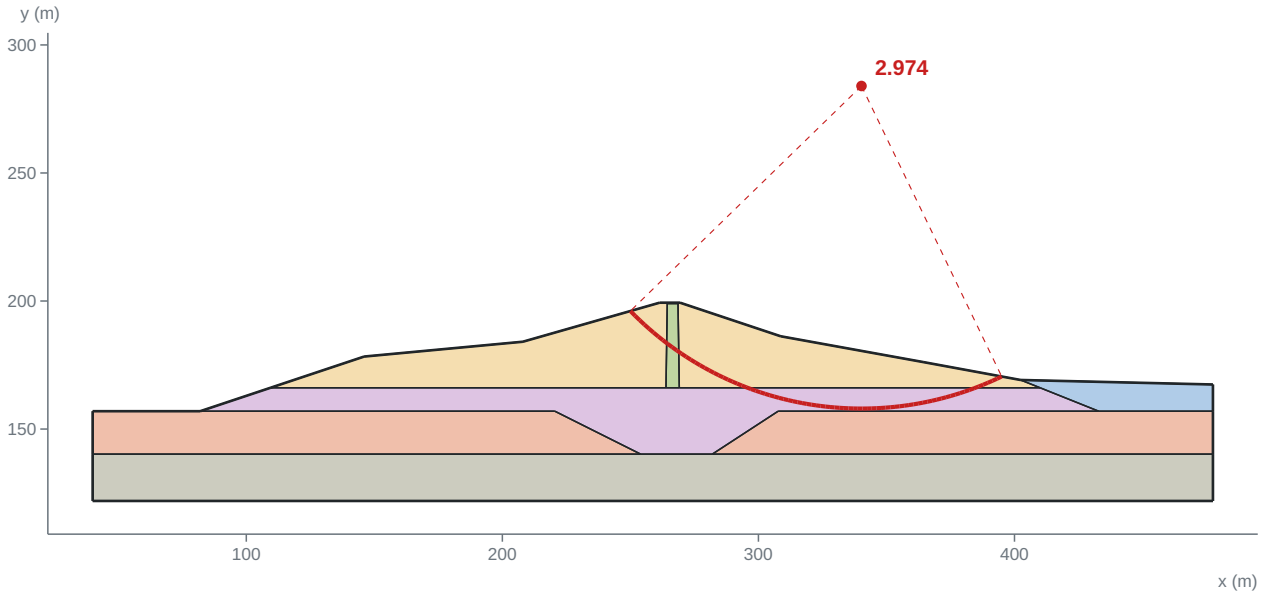


Figure 96. V060b: analysed circle (Bishop simplified, FOS = 2.974).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Phase II clay fill	Mohr-Coulomb	22.00	c' = 143.6	15.00°
Sand filter	Mohr-Coulomb	22.00	c' = 0.00	35.00°
Spoil fill	Mohr-Coulomb	25.00	c' = 5.00	35.00°
Phase I clay fill	Mohr-Coulomb	22.00	c' = 117.8	8.50°
Foundation sand	Mohr-Coulomb	20.00	c' = 5.00	18.00°
Limestone	Mohr-Coulomb	23.00	c' = 1000.0	45.00°

Method	Published by	Reference	cSlope
Bishop simplified	Program D	2.973	2.974

Probabilistic analysis on the same surface: 4 normal variables (independent); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1). Differences in β and Pf are absolute.

Quantity	Published by	Reference	cSlope
β lognormal (Taylor series)	Program D	3.353	3.354

Notes

- commercial program A's reliability indices on Hassan & Wolff's nine circles, with the correlations of Table 35.1, are checked in V062a … V062i; the critical factor of safety at the means is V060a.

3.97 V061 — Clarence Cannon Dam (Wolff & Harr), specified surface

Wolff & Harr (1987).

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

Phase I and II fills with a chimney drain and a cutoff trench to rock, flat water table at el. 557 ft; the printed non-circular surface. Converted to SI.

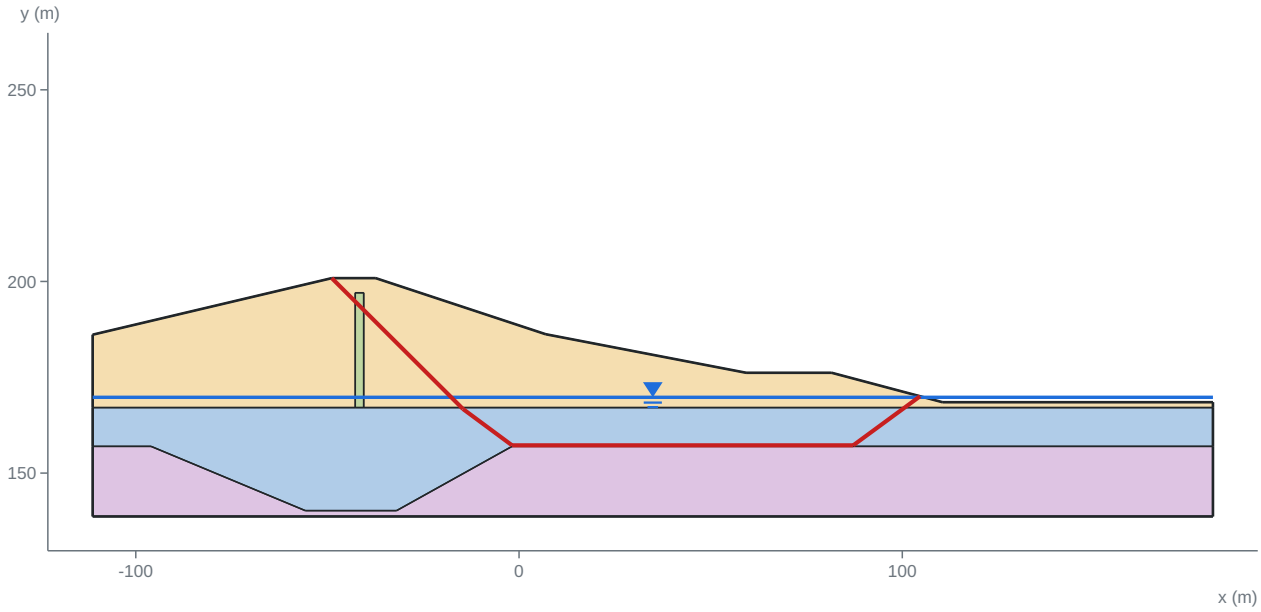


Figure 97. V061: analysed surface (Morgenstern-Price (half-sine), FOS = 2.384).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Phase II fill	Mohr-Coulomb	23.56	c' = 138.9	14.80°
Sand drain	Mohr-Coulomb	18.85	c' = 0.00	30.00°
Phase I fill	Mohr-Coulomb	23.56	c' = 106.8	6.34°
Foundation sand	Mohr-Coulomb	18.85	c' = 0.00	30.00°

Method	Published by	Reference	cSlope
Morgenstern-Price (half-sine)	Wolff & Harr	2.360	2.384
Spencer (M-P, constant f(x))	Program D	2.423	2.423
Morgenstern-Price (half-sine)	Program D	2.384	2.384

Notes

- commercial program A Spencer 2.383 and GLE 2.333 are not used as checks: cSlope agrees with Wolff & Harr's published 2.36 and with program D on the same input.

3.98 V062a — Clarence Cannon Dam, surface 7A

Hassan & Wolff (1999).

Fixed circle · Units: SI (m, kPa, kN/m³)

Six-zone embankment dam on sand over limestone, analysed on the published circle 7A (centre 320.175, 313.336; tangent elevation 145.347), Bishop.

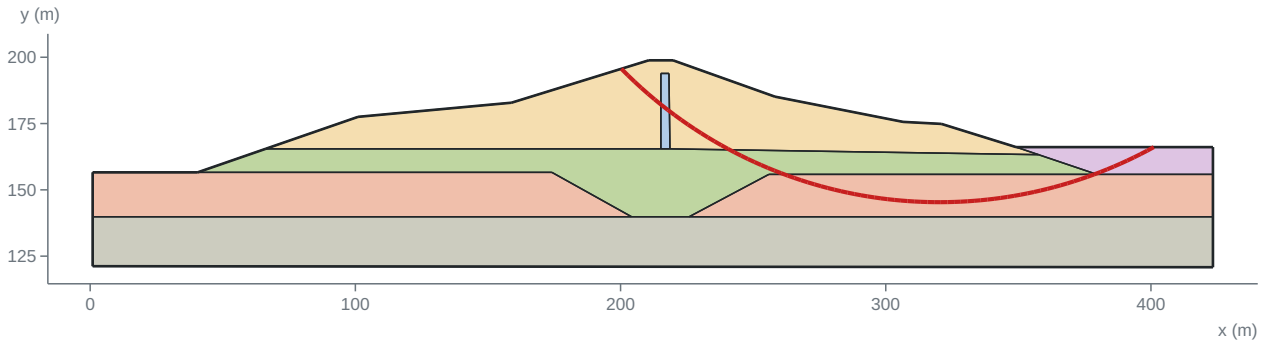


Figure 98. V062a: analysed circle (Bishop simplified, FOS = 2.549).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Phase II fill	Mohr-Coulomb	22.00	c' = 143.6	15.00°
Phase I fill	Mohr-Coulomb	22.00	c' = 117.8	8.50°
Sand filter	Mohr-Coulomb	22.00	c' = 0.00	35.00°
Spoil fill	Mohr-Coulomb	25.00	c' = 5.00	35.00°
Foundation sand	Mohr-Coulomb	20.00	c' = 5.00	18.00°
Limestone	Mohr-Coulomb	20.00	c' = 100.0	35.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program B	2.560	2.549
Bishop simplified	Commercial program A	2.551	2.549

Probabilistic analysis on the same surface: 4 normal variables (ρ = -0.55, ρ = 0.1 between pairs); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1). Differences in β and Pf are absolute.

Quantity	Published by	Reference	cSlope
β lognormal (Monte Carlo)	Commercial program A (lognormal)	10.953	11.034

Notes

- Both programs use Bishop's method here (commercial program B §2.22, commercial program A Table 35.2). commercial program B's values come from its own model of the dam, with the circles placed by its own digitisation, and differ from commercial program A's by up to 0.5 % on the same nominal circle. commercial program B's manual prints commercial program A's values under the heading "Hassan and Wolff"; Hassan & Wolff's own factors of safety differ (e.g. 2.753 on 7A).

3.99 V062b — Clarence Cannon Dam, surface 7B

Hassan & Wolff (1999).

Fixed circle · Units: SI (m, kPa, kN/m³)

Six-zone embankment dam on sand over limestone, analysed on the published circle 7B (centre 295.235, 277.589; tangent elevation 156.298), Bishop.

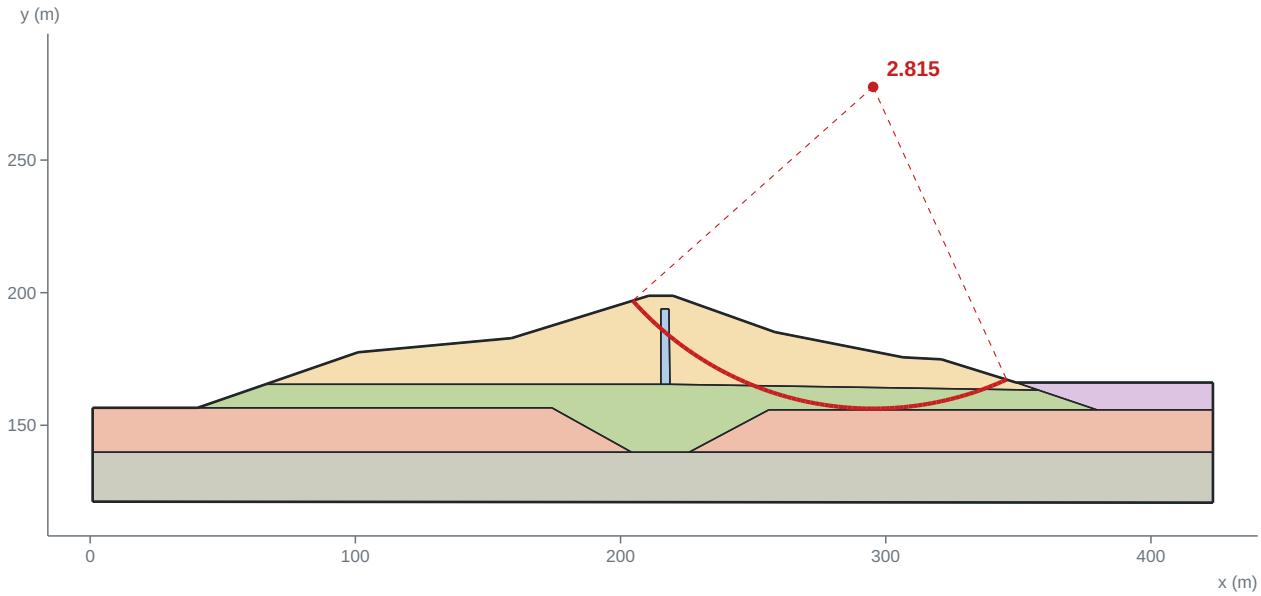


Figure 99. V062b: analysed circle (Bishop simplified, FOS = 2.815).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Phase II fill	Mohr-Coulomb	22.00	c' = 143.6	15.00°
Phase I fill	Mohr-Coulomb	22.00	c' = 117.8	8.50°
Sand filter	Mohr-Coulomb	22.00	c' = 0.00	35.00°
Spoil fill	Mohr-Coulomb	25.00	c' = 5.00	35.00°
Foundation sand	Mohr-Coulomb	20.00	c' = 5.00	18.00°
Limestone	Mohr-Coulomb	20.00	c' = 100.0	35.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program B	2.806	2.815
Bishop simplified	Commercial program A	2.820	2.815

Probabilistic analysis on the same surface: 4 normal variables (ρ = -0.55, ρ = 0.1 between pairs); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1). Differences in β and Pf are absolute.

Quantity	Published by	Reference	cSlope
β lognormal (Monte Carlo)	Commercial program A (lognormal)	4.351	4.402

Notes

- Both programs use Bishop's method here (commercial program B §2.22, commercial program A Table 35.2). commercial program B's values come from its own model of the dam, with the circles placed by its own digitisation, and differ from commercial program A's by up to 0.5 % on the same nominal circle. commercial program B's manual prints commercial program A's values under the heading "Hassan and Wolff"; Hassan & Wolff's own factors of safety differ (e.g. 2.753 on 7A).

3.100 V062c — Clarence Cannon Dam, surface 7C

Hassan & Wolff (1999).

Fixed circle · Units: SI (m, kPa, kN/m³)

Six-zone embankment dam on sand over limestone, analysed on the published circle 7C (centre 303.683, 310.206; tangent elevation 155.928), Bishop.

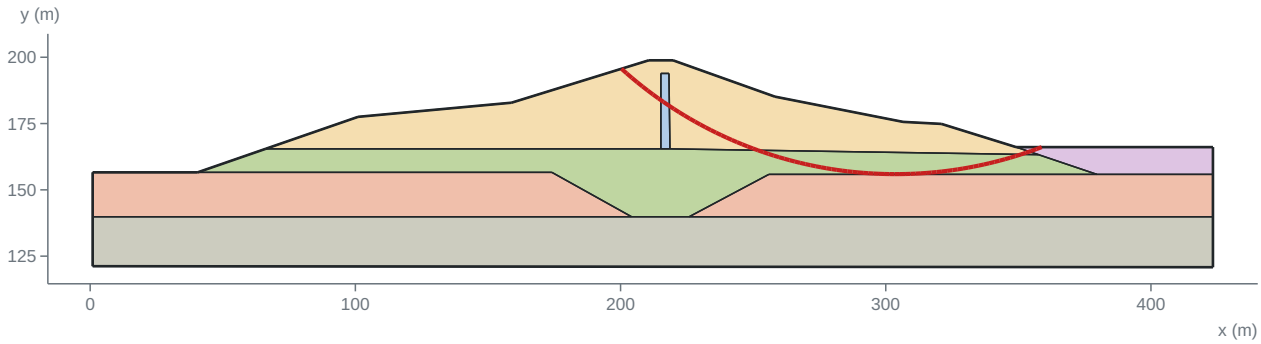


Figure 100. V062c: analysed circle (Bishop simplified, FOS = 2.779).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Phase II fill	Mohr-Coulomb	22.00	c' = 143.6	15.00°
Phase I fill	Mohr-Coulomb	22.00	c' = 117.8	8.50°
Sand filter	Mohr-Coulomb	22.00	c' = 0.00	35.00°
Spoil fill	Mohr-Coulomb	25.00	c' = 5.00	35.00°
Foundation sand	Mohr-Coulomb	20.00	c' = 5.00	18.00°
Limestone	Mohr-Coulomb	20.00	c' = 100.0	35.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program B	2.771	2.779
Bishop simplified	Commercial program A	2.777	2.779

Probabilistic analysis on the same surface: 4 normal variables (ρ = -0.55, ρ = 0.1 between pairs); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1). Differences in β and Pf are absolute.

Quantity	Published by	Reference	cSlope
β lognormal (Monte Carlo)	Commercial program A (lognormal)	4.263	4.325

Notes

- Both programs use Bishop's method here (commercial program B §2.22, commercial program A Table 35.2). commercial program B's values come from its own model of the dam, with the circles placed by its own digitisation, and differ from commercial program A's by up to 0.5 % on the same nominal circle. commercial program B's manual prints commercial program A's values under the heading "Hassan and Wolff"; Hassan & Wolff's own factors of safety differ (e.g. 2.753 on 7A).

3.101 V062d — Clarence Cannon Dam, surface 7D

Hassan & Wolff (1999).

Fixed circle · Units: SI (m, kPa, kN/m³)

Six-zone embankment dam on sand over limestone, analysed on the published circle 7D (centre 322.152, 310.206; tangent elevation 145.36), Bishop.

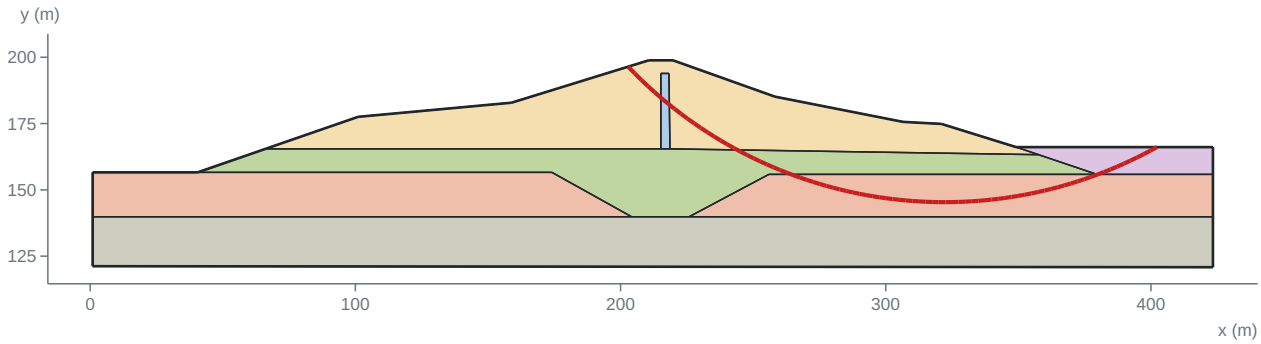


Figure 101. V062d: analysed circle (Bishop simplified, FOS = 2.577).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Phase II fill	Mohr-Coulomb	22.00	c' = 143.6	15.00°
Phase I fill	Mohr-Coulomb	22.00	c' = 117.8	8.50°
Sand filter	Mohr-Coulomb	22.00	c' = 0.00	35.00°
Spoil fill	Mohr-Coulomb	25.00	c' = 5.00	35.00°
Foundation sand	Mohr-Coulomb	20.00	c' = 5.00	18.00°
Limestone	Mohr-Coulomb	20.00	c' = 100.0	35.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program B	2.589	2.577
Bishop simplified	Commercial program A	2.583	2.577

Probabilistic analysis on the same surface: 4 normal variables (ρ = -0.55, ρ = 0.1 between pairs); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1). Differences in β and Pf are absolute.

Quantity	Published by	Reference	cSlope
β lognormal (Monte Carlo)	Commercial program A (lognormal)	11.092	11.076

Notes

- Both programs use Bishop's method here (commercial program B §2.22, commercial program A Table 35.2). commercial program B's values come from its own model of the dam, with the circles placed by its own digitisation, and differ from commercial program A's by up to 0.5 % on the same nominal circle. commercial program B's manual prints commercial program A's values under the heading "Hassan and Wolff"; Hassan & Wolff's own factors of safety differ (e.g. 2.753 on 7A).

3.102 V062e — Clarence Cannon Dam, surface 7E

Hassan & Wolff (1999).

Fixed circle · Units: SI (m, kPa, kN/m³)

Six-zone embankment dam on sand over limestone, analysed on the published circle 7E (centre 327.693, 324.018; tangent elevation 147.163), Bishop.

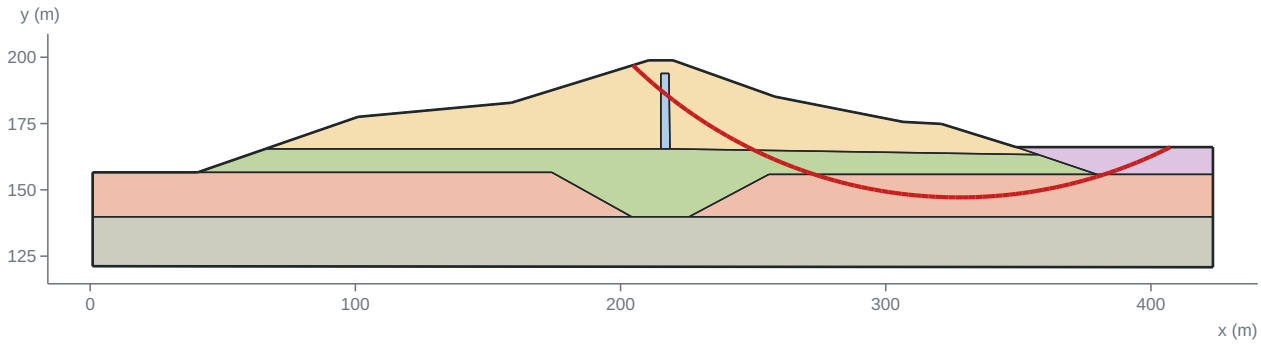


Figure 102. V062e: analysed circle (Bishop simplified, FOS = 2.690).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Phase II fill	Mohr-Coulomb	22.00	c' = 143.6	15.00°
Phase I fill	Mohr-Coulomb	22.00	c' = 117.8	8.50°
Sand filter	Mohr-Coulomb	22.00	c' = 0.00	35.00°
Spoil fill	Mohr-Coulomb	25.00	c' = 5.00	35.00°
Foundation sand	Mohr-Coulomb	20.00	c' = 5.00	18.00°
Limestone	Mohr-Coulomb	20.00	c' = 100.0	35.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program B	2.703	2.690
Bishop simplified	Commercial program A	2.692	2.690

Probabilistic analysis on the same surface: 4 normal variables (ρ = -0.55, ρ = 0.1 between pairs); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1). Differences in β and Pf are absolute.

Quantity	Published by	Reference	cSlope
β lognormal (Monte Carlo)	Commercial program A (lognormal)	10.281	10.329

Notes

- Both programs use Bishop's method here (commercial program B §2.22, commercial program A Table 35.2). commercial program B's values come from its own model of the dam, with the circles placed by its own digitisation, and differ from commercial program A's by up to 0.5 % on the same nominal circle. commercial program B's manual prints commercial program A's values under the heading "Hassan and Wolff"; Hassan & Wolff's own factors of safety differ (e.g. 2.753 on 7A).

3.103 V062f — Clarence Cannon Dam, surface 8B

Hassan & Wolff (1999).

Fixed circle · Units: SI (m, kPa, kN/m³)

Six-zone embankment dam on sand over limestone, analysed on the published circle 8B (centre 304.239, 307.858; tangent elevation 155.311), Bishop.

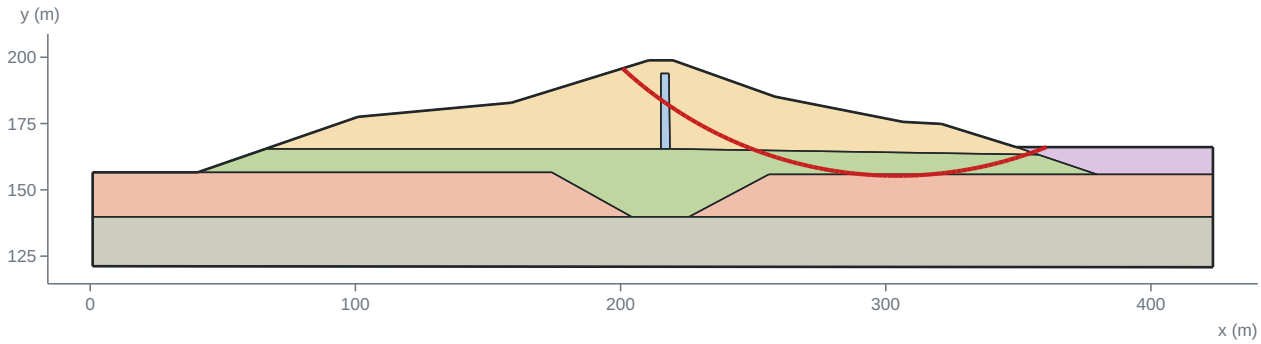


Figure 103. V062f: analysed circle (Bishop simplified, FOS = 2.673).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Phase II fill	Mohr-Coulomb	22.00	c' = 143.6	15.00°
Phase I fill	Mohr-Coulomb	22.00	c' = 117.8	8.50°
Sand filter	Mohr-Coulomb	22.00	c' = 0.00	35.00°
Spoil fill	Mohr-Coulomb	25.00	c' = 5.00	35.00°
Foundation sand	Mohr-Coulomb	20.00	c' = 5.00	18.00°
Limestone	Mohr-Coulomb	20.00	c' = 100.0	35.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program B	2.673	2.673
Bishop simplified	Commercial program A	2.672	2.673

Probabilistic analysis on the same surface: 4 normal variables (ρ = -0.55, ρ = 0.1 between pairs); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1). Differences in β and Pf are absolute.

Quantity	Published by	Reference	cSlope
β lognormal (Monte Carlo)	Commercial program A (lognormal)	4.858	4.933

Notes

- Both programs use Bishop's method here (commercial program B §2.22, commercial program A Table 35.2). commercial program B's values come from its own model of the dam, with the circles placed by its own digitisation, and differ from commercial program A's by up to 0.5 % on the same nominal circle. commercial program B's manual prints commercial program A's values under the heading "Hassan and Wolff"; Hassan & Wolff's own factors of safety differ (e.g. 2.753 on 7A).

3.104 V062g — Clarence Cannon Dam, surface 8F

Hassan & Wolff (1999).

Fixed circle · Units: SI (m, kPa, kN/m³)

Six-zone embankment dam on sand over limestone, analysed on the published circle 8F (centre 313.114, 304.784; tangent elevation 160.368), Bishop.

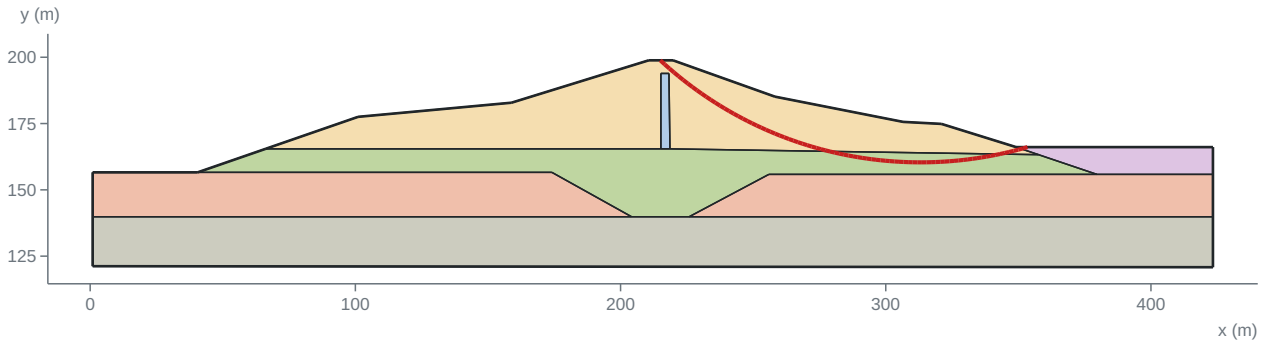


Figure 104. V062g: analysed circle (Bishop simplified, FOS = 3.592).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Phase II fill	Mohr-Coulomb	22.00	c' = 143.6	15.00°
Phase I fill	Mohr-Coulomb	22.00	c' = 117.8	8.50°
Sand filter	Mohr-Coulomb	22.00	c' = 0.00	35.00°
Spoil fill	Mohr-Coulomb	25.00	c' = 5.00	35.00°
Foundation sand	Mohr-Coulomb	20.00	c' = 5.00	18.00°
Limestone	Mohr-Coulomb	20.00	c' = 100.0	35.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program B	3.586	3.592
Bishop simplified	Commercial program A	3.598	3.592

Probabilistic analysis on the same surface: 4 normal variables (ρ = -0.55, ρ = 0.1 between pairs); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1). Differences in β and Pf are absolute.

Quantity	Published by	Reference	cSlope
β lognormal (Monte Carlo)	Commercial program A (lognormal)	5.485	5.528

Notes

- Both programs use Bishop's method here (commercial program B §2.22, commercial program A Table 35.2). commercial program B's values come from its own model of the dam, with the circles placed by its own digitisation, and differ from commercial program A's by up to 0.5 % on the same nominal circle. commercial program B's manual prints commercial program A's values under the heading "Hassan and Wolff"; Hassan & Wolff's own factors of safety differ (e.g. 2.753 on 7A).

3.105 V062h — Clarence Cannon Dam, surface 8G

Hassan & Wolff (1999).

Fixed circle · Units: SI (m, kPa, kN/m³)

Six-zone embankment dam on sand over limestone, analysed on the published circle 8G (centre 314.856, 305.604; tangent elevation 165.756), Bishop.

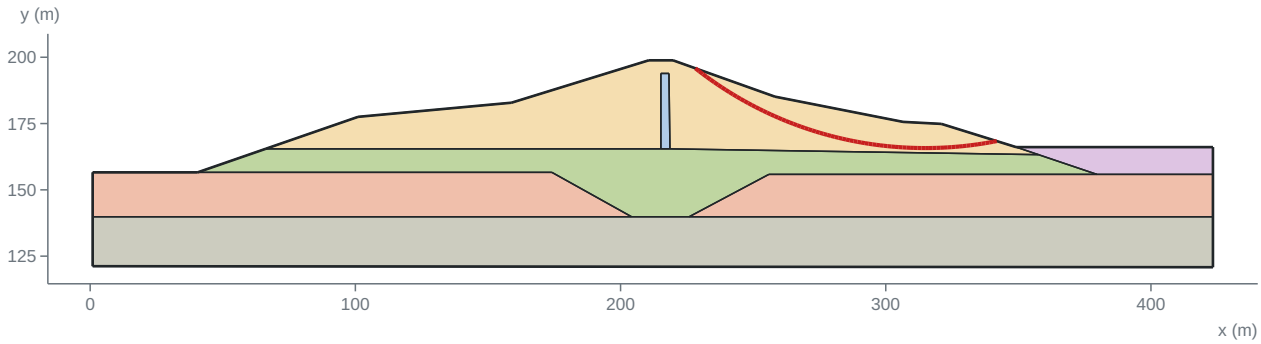


Figure 105. V062h: analysed circle (Bishop simplified, FOS = 6.063).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Phase II fill	Mohr-Coulomb	22.00	c' = 143.6	15.00°
Phase I fill	Mohr-Coulomb	22.00	c' = 117.8	8.50°
Sand filter	Mohr-Coulomb	22.00	c' = 0.00	35.00°
Spoil fill	Mohr-Coulomb	25.00	c' = 5.00	35.00°
Foundation sand	Mohr-Coulomb	20.00	c' = 5.00	18.00°
Limestone	Mohr-Coulomb	20.00	c' = 100.0	35.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program B	6.069	6.063
Bishop simplified	Commercial program A	6.074	6.063

Probabilistic analysis on the same surface: 4 normal variables (ρ = -0.55, ρ = 0.1 between pairs); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1). Differences in β and Pf are absolute.

Quantity	Published by	Reference	cSlope
β lognormal (Monte Carlo)	Commercial program A (lognormal)	5.563	5.490

Notes

- Both programs use Bishop's method here (commercial program B §2.22, commercial program A Table 35.2). commercial program B's values come from its own model of the dam, with the circles placed by its own digitisation, and differ from commercial program A's by up to 0.5 % on the same nominal circle. commercial program B's manual prints commercial program A's values under the heading "Hassan and Wolff"; Hassan & Wolff's own factors of safety differ (e.g. 2.753 on 7A).

3.106 V062i — Clarence Cannon Dam, surface 8H

Hassan & Wolff (1999).

Fixed circle · Units: SI (m, kPa, kN/m³)

Six-zone embankment dam on sand over limestone, analysed on the published circle 8H (centre 316.198, 305.604; tangent elevation 170.537), Bishop.

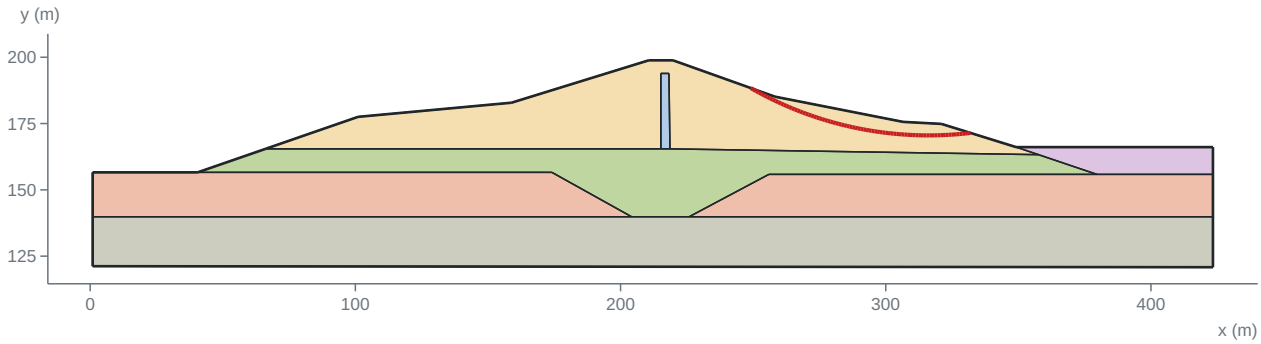


Figure 106. V062i: analysed circle (Bishop simplified, FOS = 11.567).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Phase II fill	Mohr-Coulomb	22.00	c' = 143.6	15.00°
Phase I fill	Mohr-Coulomb	22.00	c' = 117.8	8.50°
Sand filter	Mohr-Coulomb	22.00	c' = 0.00	35.00°
Spoil fill	Mohr-Coulomb	25.00	c' = 5.00	35.00°
Foundation sand	Mohr-Coulomb	20.00	c' = 5.00	18.00°
Limestone	Mohr-Coulomb	20.00	c' = 100.0	35.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program B	11.583	11.567

Probabilistic analysis on the same surface: 4 normal variables (ρ = -0.55, ρ = 0.1 between pairs); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1). Differences in β and Pf are absolute.

Quantity	Published by	Reference	cSlope
β lognormal (Monte Carlo)	Commercial program A (lognormal)	6.394	6.431

Notes

- Both programs use Bishop's method here (commercial program B §2.22, commercial program A Table 35.2). commercial program B's values come from its own model of the dam, with the circles placed by its own digitisation, and differ from commercial program A's by up to 0.5 % on the same nominal circle. commercial program B's manual prints commercial program A's values under the heading "Hassan and Wolff"; Hassan & Wolff's own factors of safety differ (e.g. 2.753 on 7A).
- commercial program A's Bishop value for 8H (11.23) disagrees with commercial program B (11.583) and two other codes and is not used.

3.107 V063 — Huang & Jia earth dam, dry

Huang & Jia (2008).

Critical circle search · Units: SI (m, kPa, kN/m³)

Homogeneous earth dam (c′ = 13.8 kPa, φ′ = 37°, γ = 18.2 kN/m³) before the reservoir is filled; circular search.

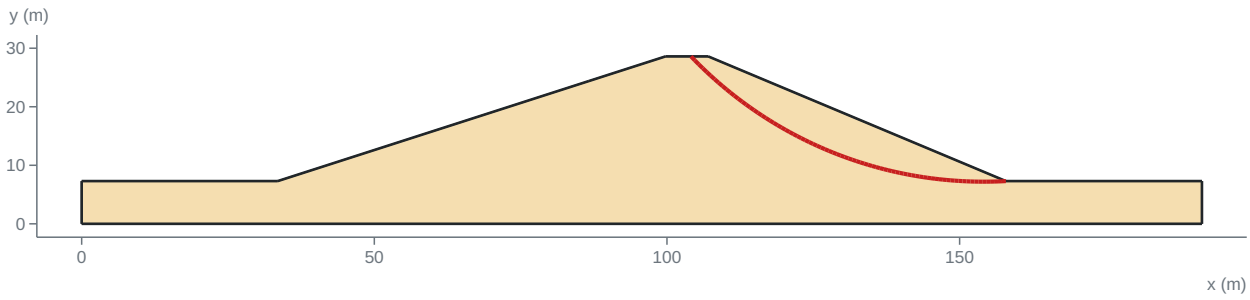


Figure 107. V063: critical circle (Spencer (M-P, constant f(x)), FOS = 2.451).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Material 1	Mohr-Coulomb	18.20	c' = 13.80	37.00°

Method	Published by	Reference	cSlope
Spencer (M-P, constant f(x))	Commercial program A	2.455	2.451
Spencer (M-P, constant f(x))	Huang & Jia (FE strength reduction)	2.430	2.451

Notes

- Search limited to x = 101.7–185.6 (crest to the downstream berm), as in the source.
- commercial program A reports the Spencer value only. The steady-seepage and drawdown stages need a seepage analysis.

3.108 V064 — Leshchinsky & Han MSE wall, five tiers

Leshchinsky & Han (2004), J. Geotech. Geoenviron. Eng. 130(12).

Fixed circle · Units: SI (m, kPa, kN/m³)

Five 1.8 m tiers (Ta = 10.1 kN/m).

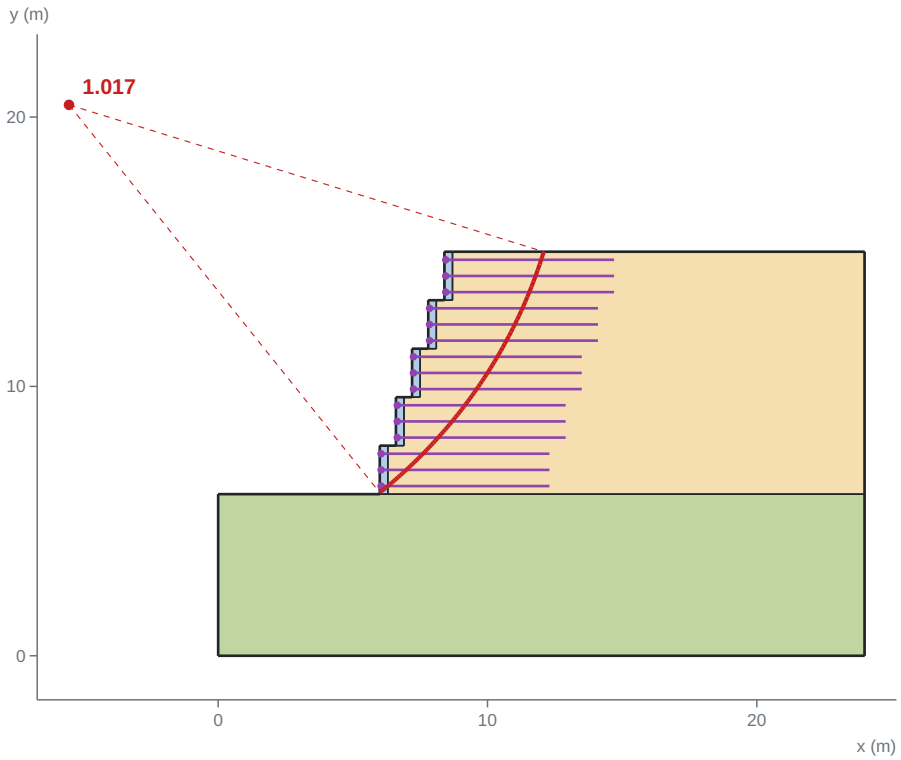


Figure 108. V064: analysed circle (Bishop simplified, FOS = 1.017).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Reinforced and retained fill	Mohr-Coulomb	18.00	c' = 0.00	34.00°
Foundation soil	Mohr-Coulomb	18.00	c' = 10.00	34.00°
Blocks	Mohr-Coulomb	18.00	c' = 2.50	34.00°

Method	Published by	Reference	cSlope
Bishop simplified	Program D	1.020	1.017

3.109 V065 — Leshchinsky & Han MSE wall, weak foundation

Leshchinsky & Han (2004), J. Geotech. Geoenviron. Eng. 130(12).

Fixed circle · Units: SI (m, kPa, kN/m³)

As V066a on a c = 0, φ = 18° foundation.

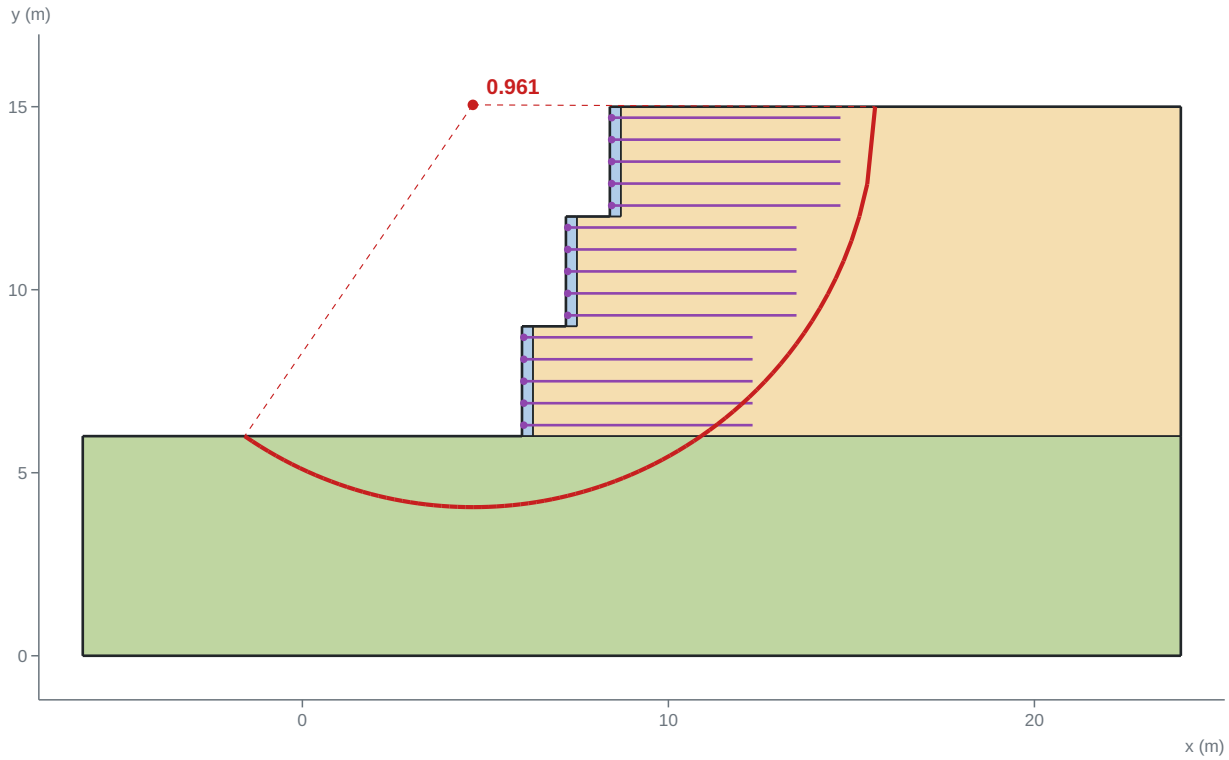


Figure 109. V065: analysed circle (Spencer (M-P, constant f(x)), FOS = 0.961).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Reinforced and retained fill	Mohr-Coulomb	18.00	c' = 0.00	34.00°
Foundation soil	Mohr-Coulomb	18.00	c' = 0.00	18.00°
Blocks	Mohr-Coulomb	18.00	c' = 2.50	34.00°

Method	Published by	Reference	cSlope
Spencer (M-P, constant f(x))	Commercial program A	0.964	0.961

3.110 V066a — Leshchinsky & Han three-tier MSE wall, baseline

Leshchinsky & Han (2004), J. Geotech. Geoenviron. Eng. 130(12).

Fixed circle · Units: SI (m, kPa, kN/m³)

Three 3 m tiers offset 1.2 m, block facing, geotextiles every 0.6 m (L = 6.3 m, Ta = 10 kN/m, applied horizontally, passive), commercial program A's critical circle.

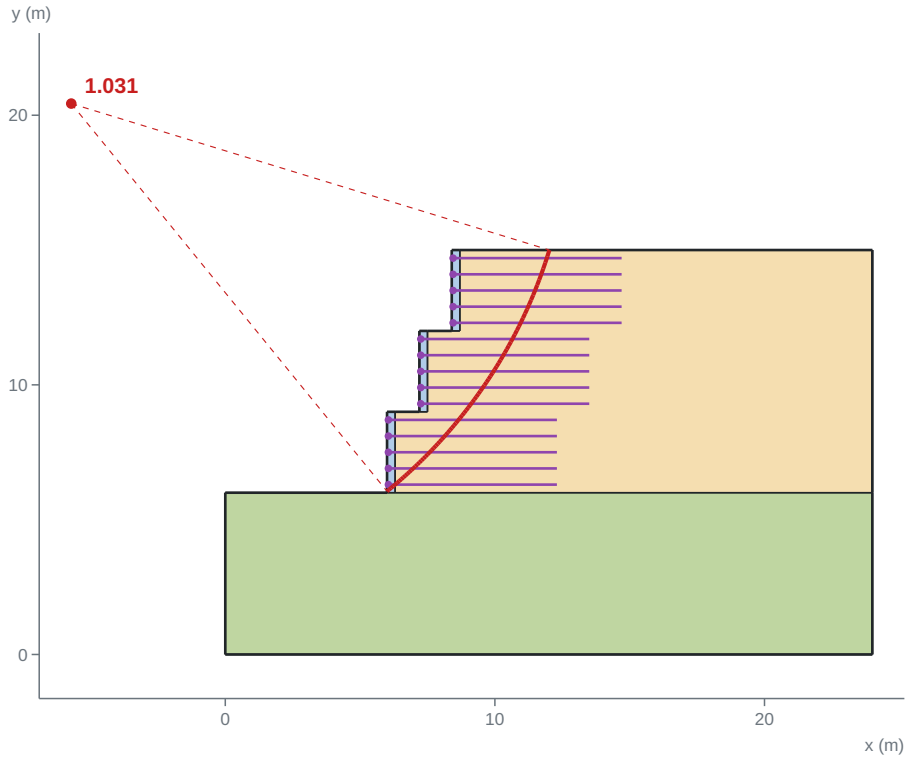


Figure 110. V066a: analysed circle (Bishop simplified, FOS = 1.031).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Reinforced and retained fill	Mohr-Coulomb	18.00	c' = 0.00	34.00°
Foundation soil	Mohr-Coulomb	18.00	c' = 10.00	34.00°
Blocks	Mohr-Coulomb	18.00	c' = 2.50	34.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.040	1.031

Notes

- L&H finite difference 0.99.

3.111 V066b — Leshchinsky & Han MSE wall, L = 4.2 m

Leshchinsky & Han (2004), J. Geotech. Geoenviron. Eng. 130(12).

Fixed circle · Units: SI (m, kPa, kN/m³)

As V066a with 4.2 m geotextiles (Ta = 11.4 kN/m).

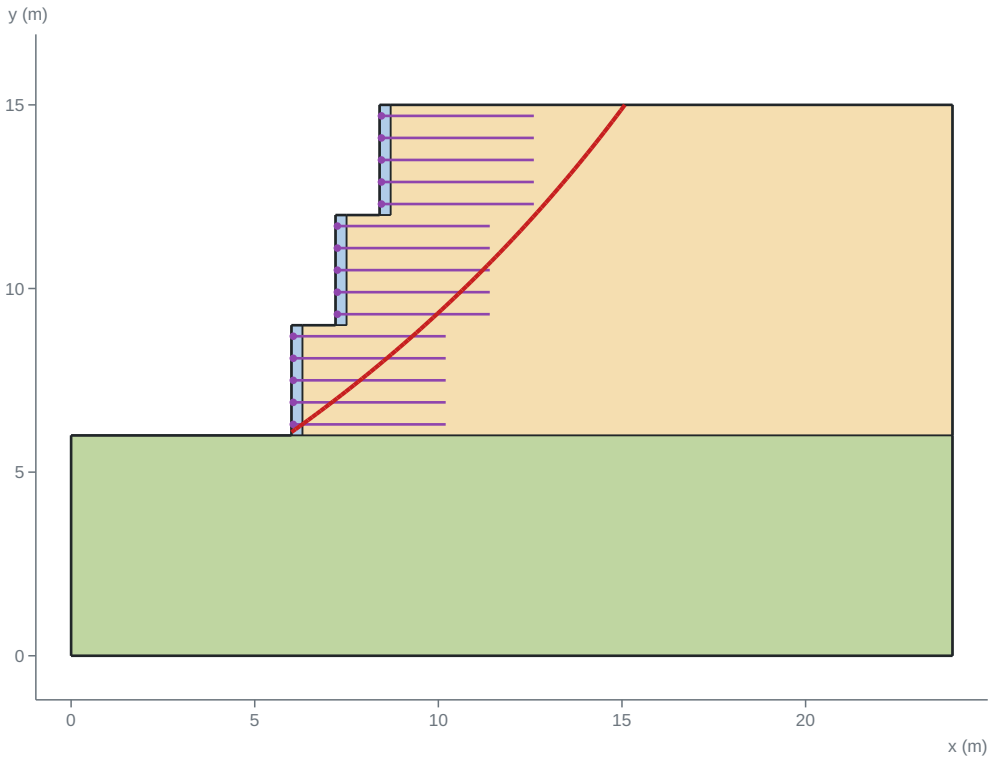


Figure 111. V066b: analysed circle (Spencer (M-P, constant f(x)), FOS = 1.011).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Reinforced and retained fill	Mohr-Coulomb	18.00	c' = 0.00	34.00°
Foundation soil	Mohr-Coulomb	18.00	c' = 10.00	34.00°
Blocks	Mohr-Coulomb	18.00	c' = 2.50	34.00°

Method	Published by	Reference	cSlope
Spencer (M-P, constant f(x))	Program D	1.011	1.011

Notes

- commercial program A reports 0.971 (Spencer) and 0.976 (Bishop) with the same Ta = 11.4 kN/m (Table 89.2); cSlope and program D are 4 % higher on the reference circle. The difference is not explained (the reinforcement model of the tiers is the likely cause), so the commercial program A value is not used as a check. L&H 0.98 (finite difference).

3.112 V066c — Leshchinsky & Han MSE wall, two reinforcement types

Leshchinsky & Han (2004), J. Geotech. Geoenviron. Eng. 130(12).

Fixed circle · Units: SI (m, kPa, kN/m³)

As V066a with two reinforcement strengths (7.5 and 11.0 kN/m).

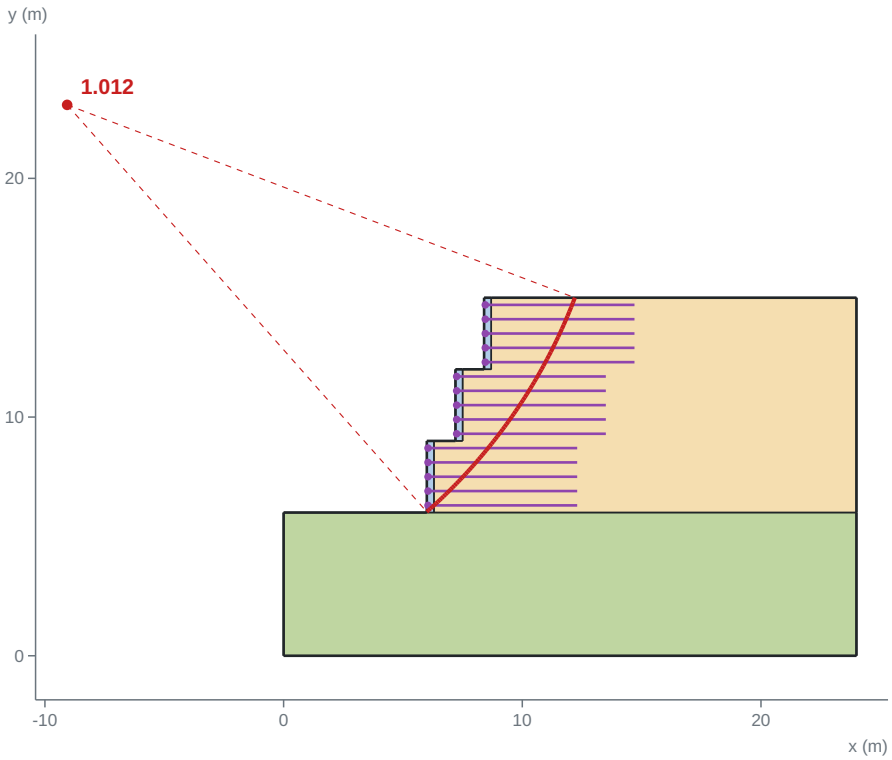


Figure 112. V066c: analysed circle (Bishop simplified, FOS = 1.012).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Reinforced and retained fill	Mohr-Coulomb	18.00	c' = 0.00	34.00°
Foundation soil	Mohr-Coulomb	18.00	c' = 10.00	34.00°
Blocks	Mohr-Coulomb	18.00	c' = 2.50	34.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.004	1.012

3.113 V066d — Leshchinsky & Han MSE wall, water hw = 3 m

Leshchinsky & Han (2004), J. Geotech. Geoenviron. Eng. 130(12).

Fixed circle · Units: SI (m, kPa, kN/m³)

As V066a with the water table 3 m above the toe: pore pressure in the foundation only (drained fill); Ta = 9.25 kN/m. The pond in front of the wall is left out of this comparison: program D's water loads apply no pressure to the vertical facing, whereas cSlope does.

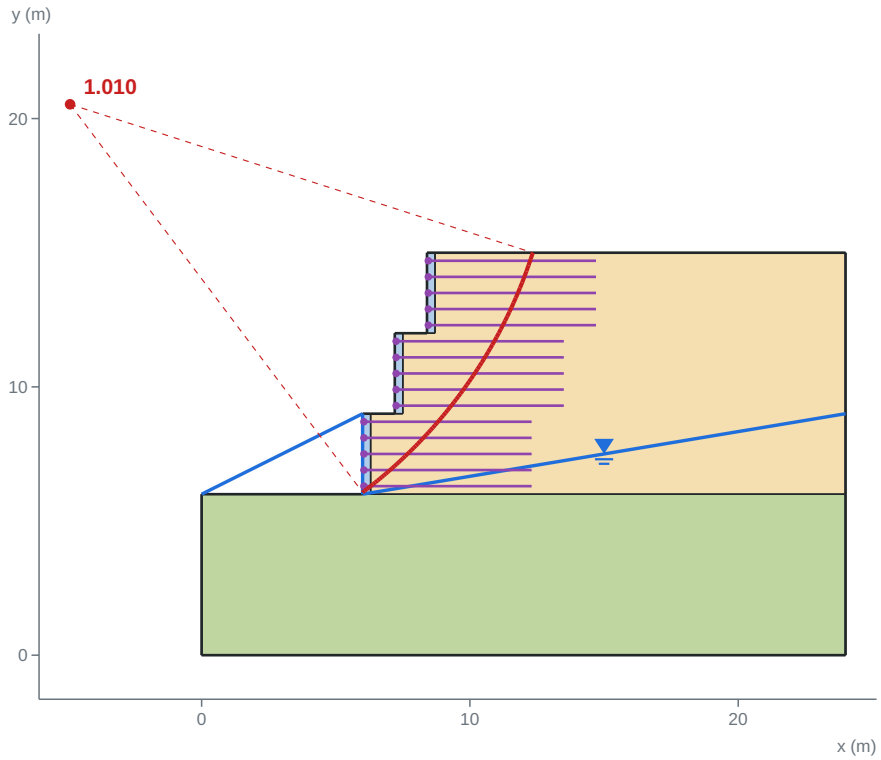


Figure 113. V066d: analysed circle (Bishop simplified, FOS = 1.010).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Reinforced and retained fill	Mohr-Coulomb	18.00	c' = 0.00	34.00°
Foundation soil	Mohr-Coulomb	18.00	c' = 10.00	34.00°
Blocks	Mohr-Coulomb	18.00	c' = 2.50	34.00°

Method	Published by	Reference	cSlope
Bishop simplified	Program D	1.010	1.010

Notes

- commercial program A reports 1.037 (Bishop) with the same Ta = 9.25 kN/m (Table 92.2) and the pond in front of the wall; the difference is not explained, so it is not used as a check. L&H 1.01 (finite difference).
- With the pond against the lower tier (water at el. 9 in front of the wall) the hydrostatic thrust on the facing resists the sliding and cSlope gives 1.18.

3.114 V066e — Leshchinsky & Han MSE wall, surcharge

Leshchinsky & Han (2004), J. Geotech. Geoenviron. Eng. 130(12).

Fixed circle · Units: SI (m, kPa, kN/m³)

As V066a with a 20 kPa surcharge.

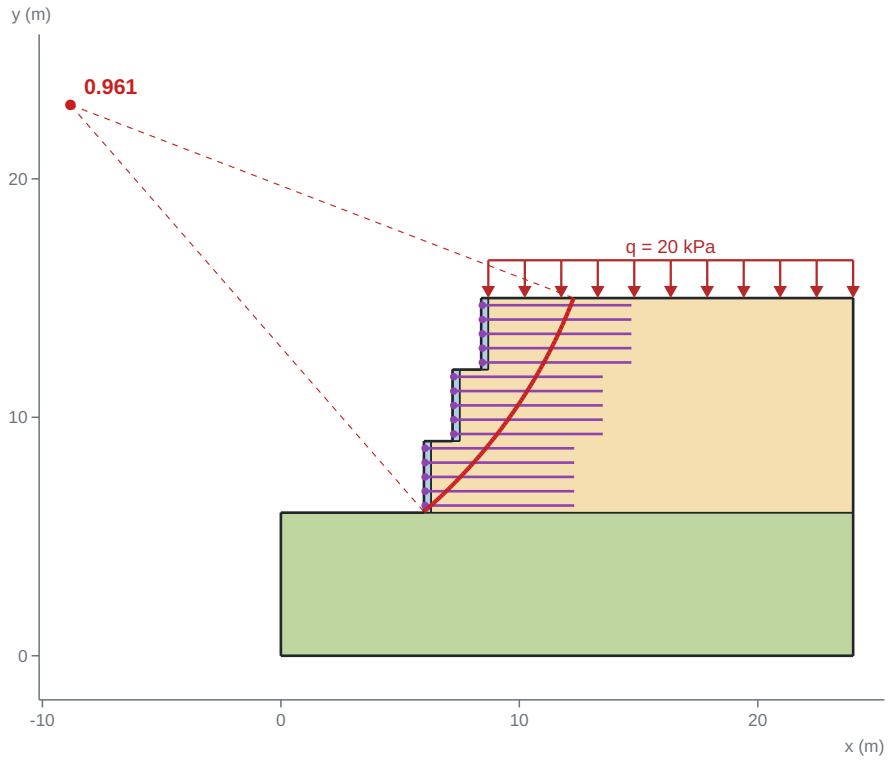


Figure 114. V066e: analysed circle (Bishop simplified, FOS = 0.961).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Reinforced and retained fill	Mohr-Coulomb	18.00	c' = 0.00	34.00°
Foundation soil	Mohr-Coulomb	18.00	c' = 10.00	34.00°
Blocks	Mohr-Coulomb	18.00	c' = 2.50	34.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	0.958	0.961

3.115 V067 — Low (1989), layered undrained clay

Low (1989).

Critical circle search · Units: SI (m, kPa, kN/m³)

Embankment on three undrained clay layers of different strength; critical circle search.

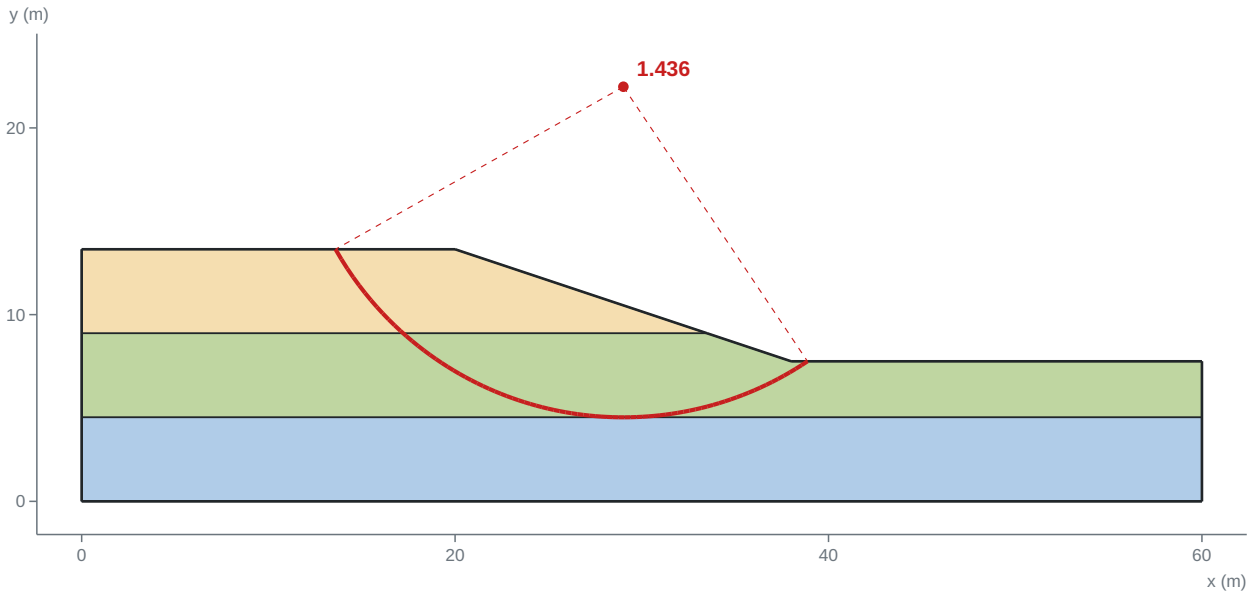


Figure 115. V067: critical circle (Bishop simplified, FOS = 1.436).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
A	Undrained (φ = 0)	18.00	Su = 30.00	0°
B	Undrained (φ = 0)	18.00	Su = 20.00	0°
C	Undrained (φ = 0)	18.00	Su = 150.0	0°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.439	1.436

3.116 V068 — Low (1989), stiff crust over soft clays, published circle

Low (1989).

Fixed circle · Units: SI (m, kPa, kN/m³)

Stiff upper slope on two undrained clays, the lowest with Su increasing linearly with depth (S = f(datum)), on the circle that carries commercial program A's published values.

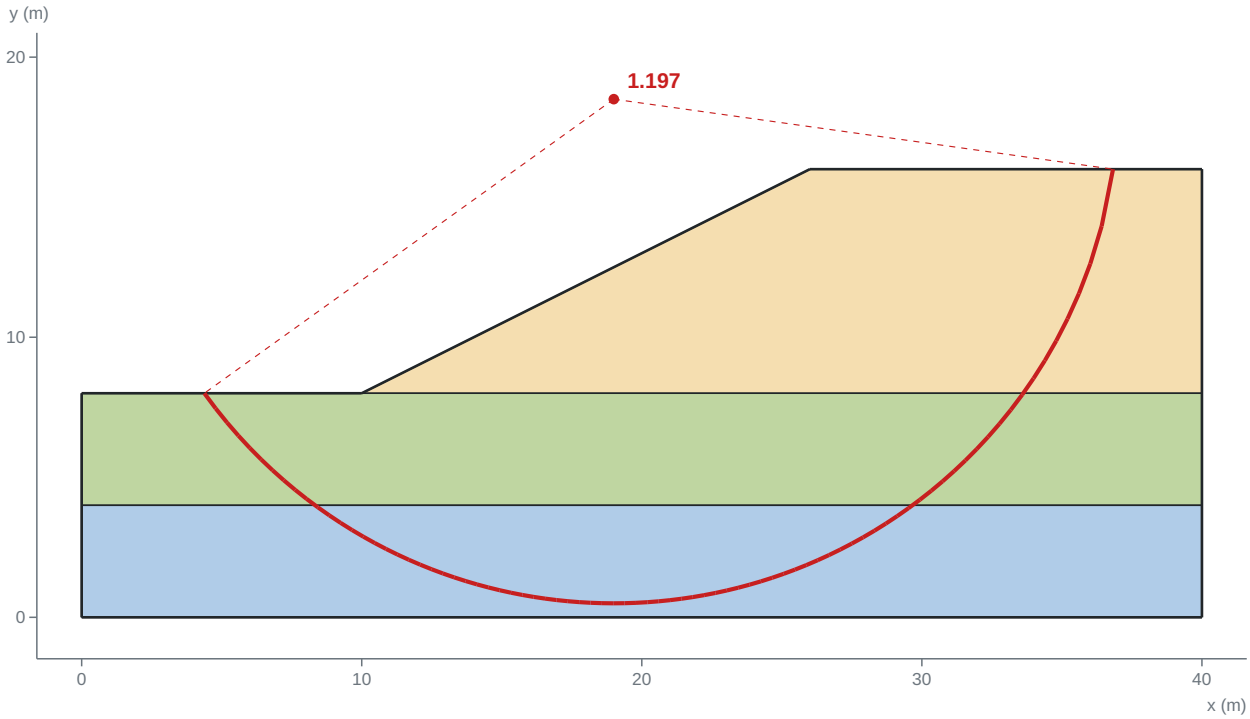


Figure 116. V068: analysed circle (Bishop simplified, FOS = 1.197).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Upper	Mohr-Coulomb	20.00	c' = 95.00	15.00°
Middle	Mohr-Coulomb	20.00	c' = 15.00	0.00°
Lower	S = f(datum)	20.00	Su = 15.00 at y = 4.00, 3.75 kPa/m below	0°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.192	1.197

3.117 V069a — Morgenstern (1963) complete drawdown, 3:1

Morgenstern, N. (1963). Stability charts for earth slopes during rapid drawdown. Géotechnique 13(2).

Critical circle search · Units: SI (m, kPa, kN/m³)

3:1 slope after complete drawdown: piezometric line on the slope face, base at the toe (D = 1). Pore pressure from the vertical head. Converted to SI.

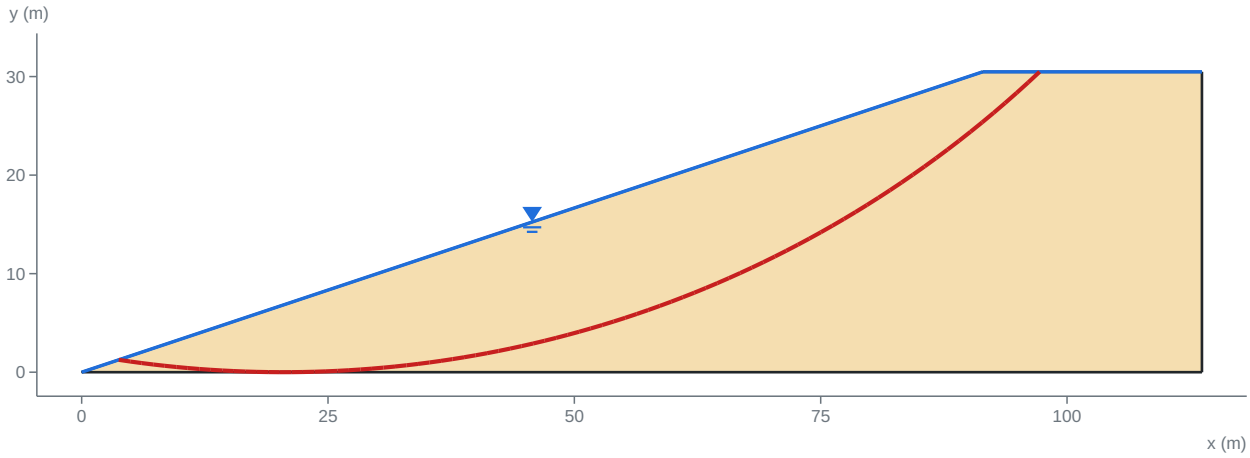


Figure 117. V069a: critical circle (Bishop simplified (vertical head), FOS = 1.200).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	19.60	c' = 14.94	30.00°

Method	Published by	Reference	cSlope
Bishop simplified (vertical head)	Morgenstern (chart)	1.200	1.200
Spencer (vertical head)	Morgenstern (chart)	1.200	1.205

Notes

- commercial program A reports 1.212 (Bishop) on its critical circle, checked in V069b. With the phreatic correction ($\cos^2\theta$) cSlope gives 1.303.

3.118 V069b — Morgenstern complete drawdown, the reference circle

Morgenstern, N. (1963). Stability charts for earth slopes during rapid drawdown. Géotechnique 13(2). Verification manual of a commercial slope stability program.

Fixed circle · Units: SI (m, kPa, kN/m³)

The V069a slope after complete drawdown on the reference critical circle (Fig. 100.2); pore pressure from the vertical head. Converted to SI.

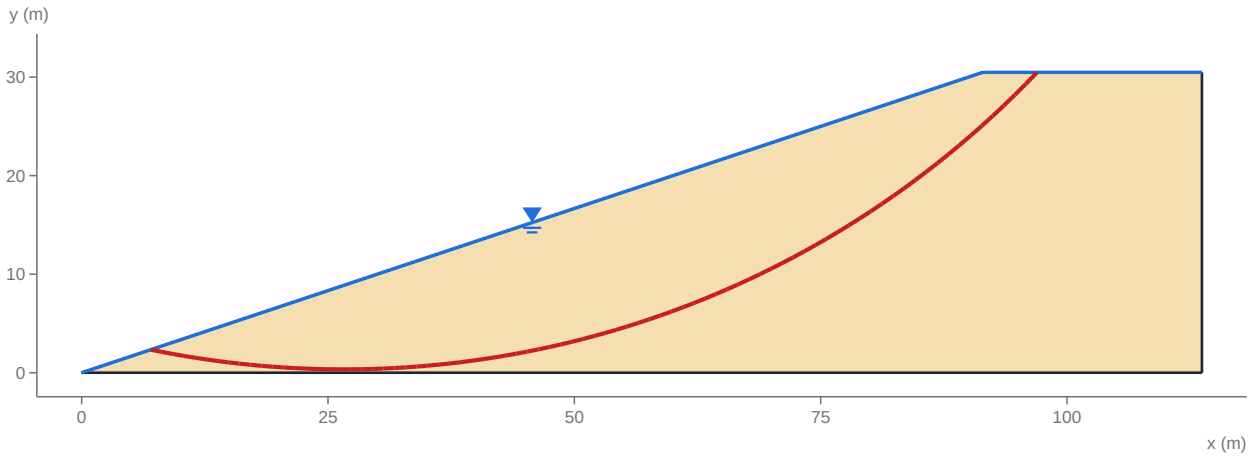


Figure 118. V069b: analysed circle (Bishop simplified, FOS = 1.209).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	19.60	c' = 14.94	30.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.212	1.209

3.119 V070 — Partial drawdown, $\bar{B} = 1$ (Morgenstern chart)

Morgenstern (1963).

Critical circle search · Units: SI (m, kPa, kN/m³)

Homogeneous embankment after a partial drawdown from 100 to 50 ft with $\bar{B} = 1$: the piezometric line follows the face above the pool and stays at the pool below it, which loads the submerged face. Converted to SI.

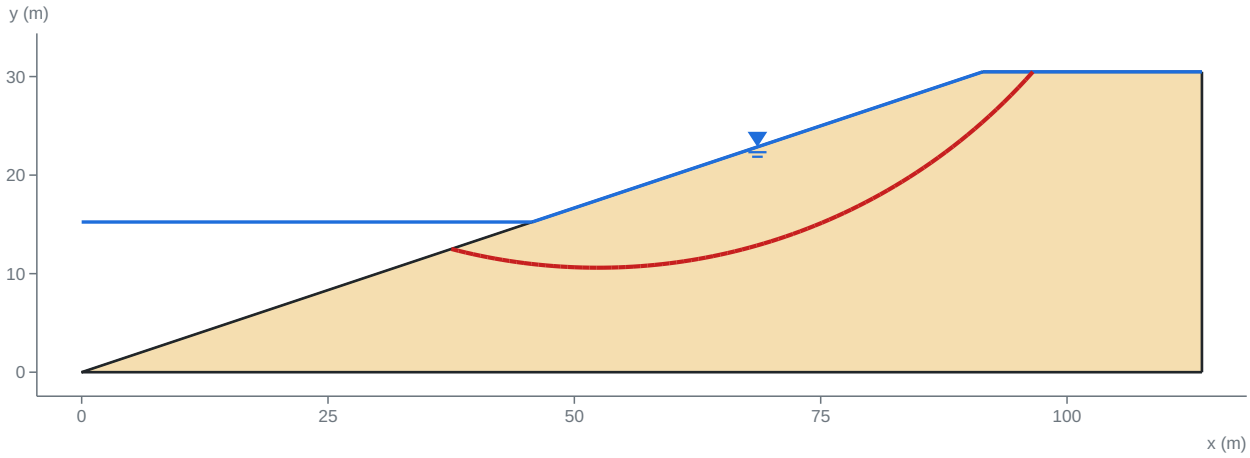


Figure 119. V070: critical circle (Bishop simplified, FOS = 1.415).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Morgenstern soil	Mohr-Coulomb	19.60	c' = 14.94	30.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.417	1.415
Bishop simplified	Morgenstern chart	1.410	1.415
Spencer (M-P, constant f(x))	Program D	1.422	1.421

3.120 V071a — Pockoski & Duncan (2000) slope 1

Pockoski & Duncan (2000), slope 1.

Fixed circle · Units: SI (m, kPa, kN/m³)

2:1 slope with a water table, on the circle published by commercial program A; pore pressure from the vertical head below the piezometric line. Converted to SI.

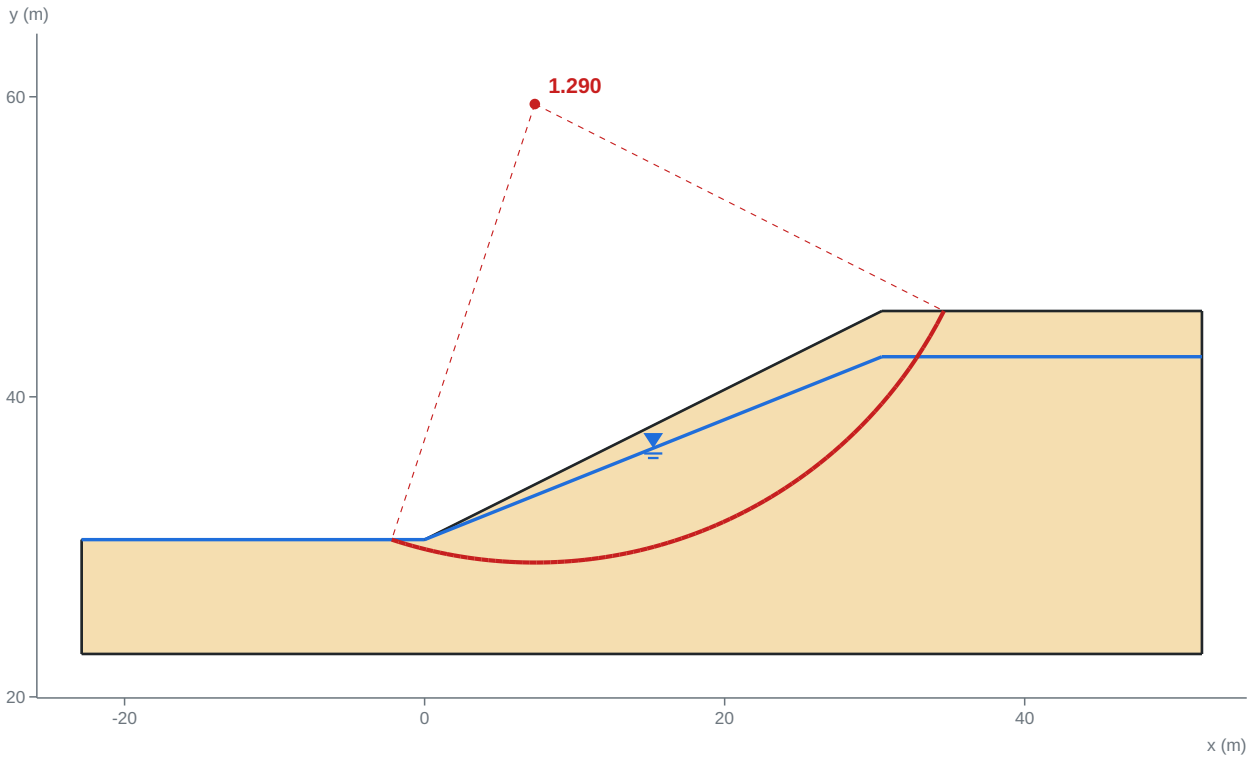


Figure 120. V071a: analysed circle (Bishop simplified (vertical head), FOS = 1.290).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Sandy clay	Mohr-Coulomb	18.85	c' = 14.36	30.00°

Method	Published by	Reference	cSlope
Bishop simplified (vertical head)	Commercial program A	1.293	1.290
Spencer (vertical head)	Commercial program A	1.300	1.296

Notes

- With the phreatic correction (cos²θ) cSlope gives 1.381 (+6.8 %).

3.121 V071b — Pockoski & Duncan slope 1, circular search

Pockoski & Duncan (2000), slope 1. Verification manual of a commercial slope stability program.

Critical circle search · Units: SI (m, kPa, kN/m³)

The V071a slope (Pockoski & Duncan slope 1) with a circular search instead of the given circle; pore pressure from the vertical head. Converted to SI.

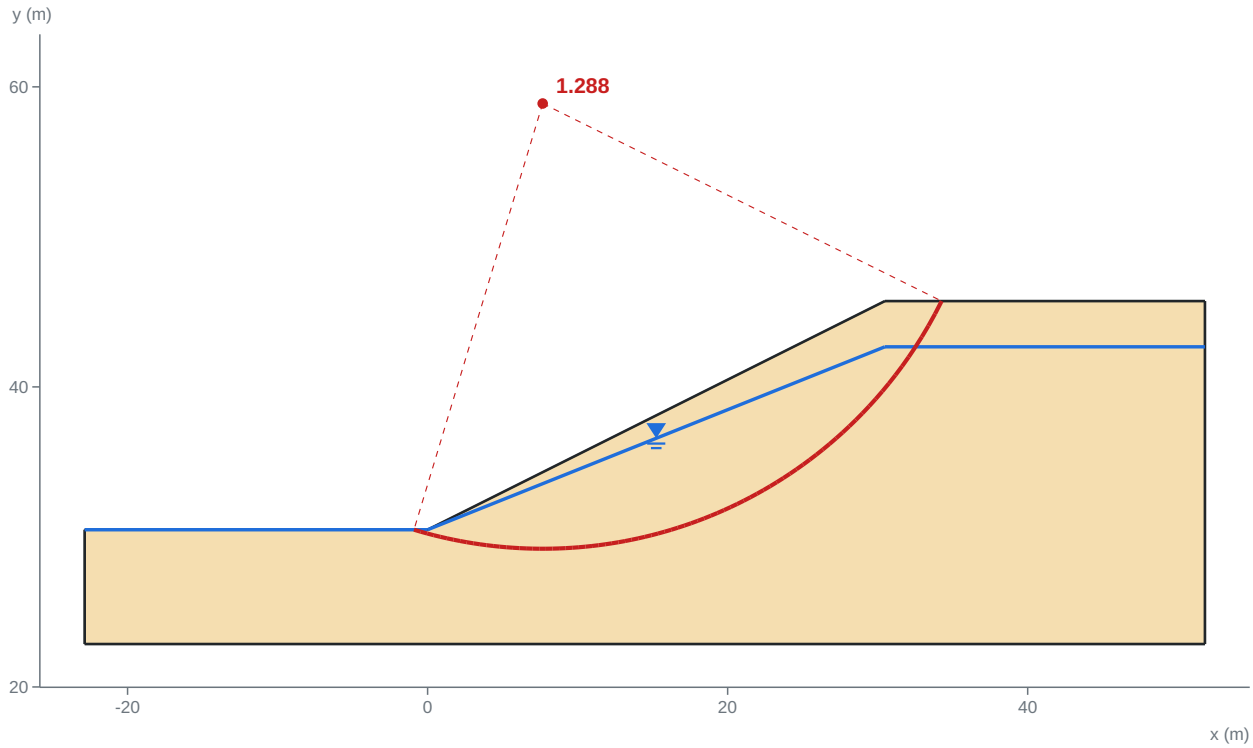


Figure 121. V071b: critical circle (Bishop simplified, FOS = 1.288).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Sandy clay	Mohr-Coulomb	18.85	c' = 14.36	30.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.293	1.288
Spencer (M-P, constant f(x))	Commercial program A	1.300	1.294
Janbu simplified (uncorrected)	Commercial program A	1.151	1.147

Notes

- program G and commercial program B (Pockoski & Duncan 2000): 1.30 / 1.29 / 1.15 (Spencer / Bishop / Janbu simplified).

3.122 V072a — Pockoski & Duncan test slope 2, dry tension crack

Pockoski & Duncan (2000).

Fixed circle · Units: SI (m, kPa, kN/m³)

The V071a slope with a dry 5.5 ft tension crack, on the reference critical circle. Converted to SI.

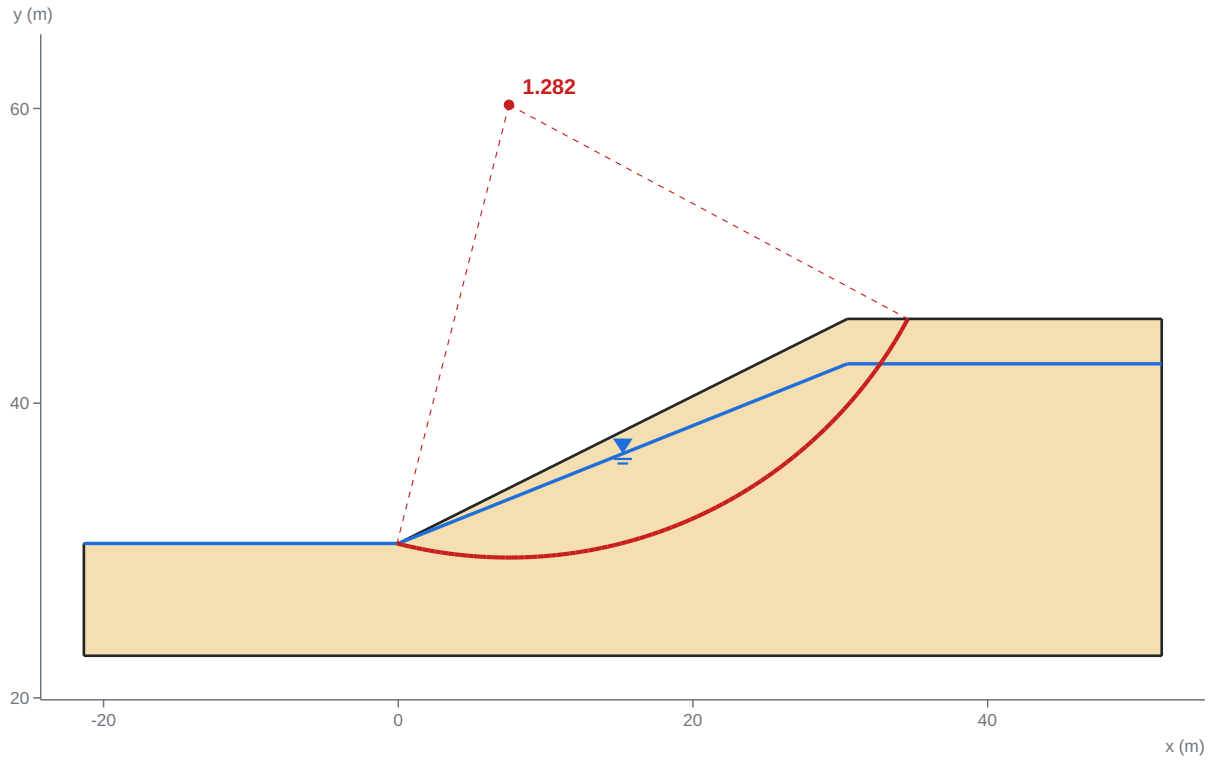


Figure 122. V072a: analysed circle (Bishop simplified, FOS = 1.282).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Sandy clay	Mohr-Coulomb	18.85	c' = 14.36	30.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.285	1.282
Spencer (M-P, constant f(x))	Commercial program A	1.290	1.287

3.123 V072b — Pockoski & Duncan slope 2, circular search

Pockoski & Duncan (2000). Verification manual of a commercial slope stability program.

Critical circle search · Units: SI (m, kPa, kN/m³)

The V072a slope (dry 5.5 ft tension crack) with a circular search. Converted to SI.

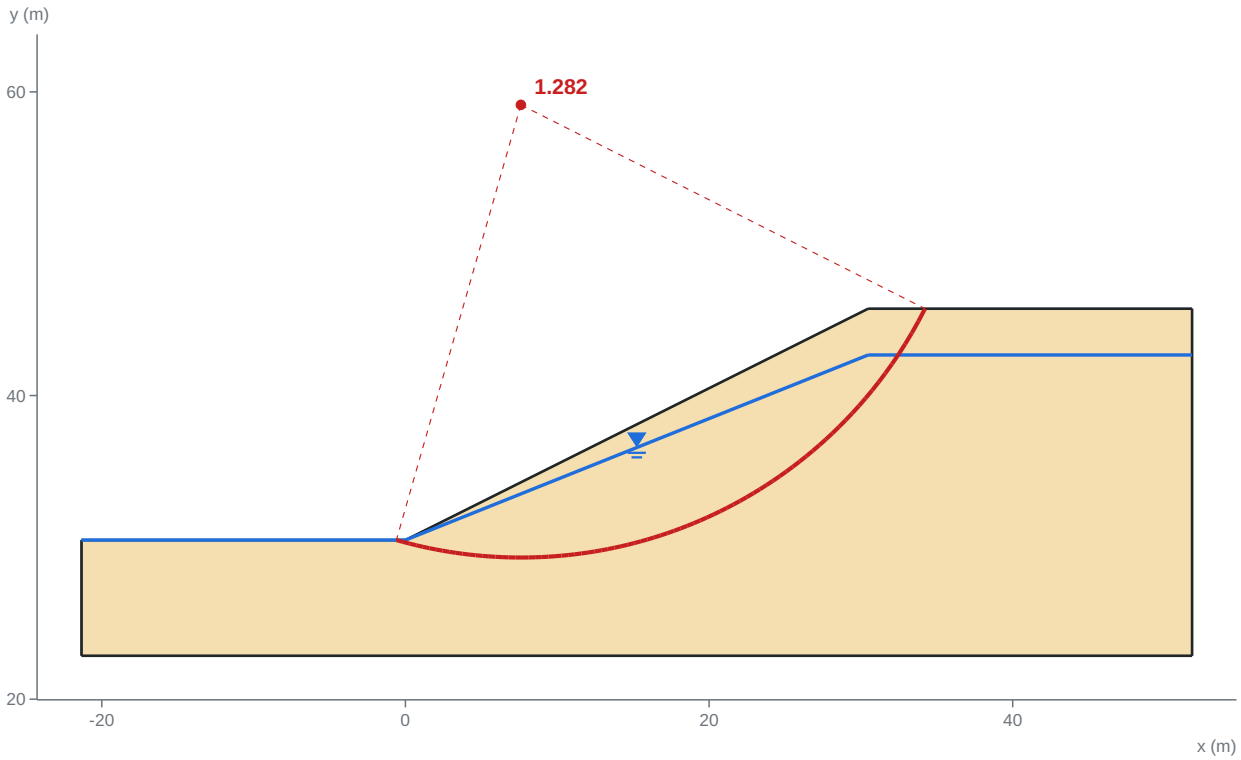


Figure 123. V072b: critical circle (Bishop simplified, FOS = 1.282).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Sandy clay	Mohr-Coulomb	18.85	c' = 14.36	30.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.285	1.282
Spencer (M-P, constant f(x))	Commercial program A	1.290	1.287
Janbu simplified (uncorrected)	Commercial program A	1.141	1.135

3.124 V073 — Pockoski & Duncan test slope 3, clay seam and dry crack

Pockoski & Duncan (2000).

Critical circle search · Units: SI (m, kPa, kN/m³)

Sandy clay over a 5 ft highly plastic clay seam on the model base, water table and a dry 6 ft tension crack; circular search. Converted to SI.

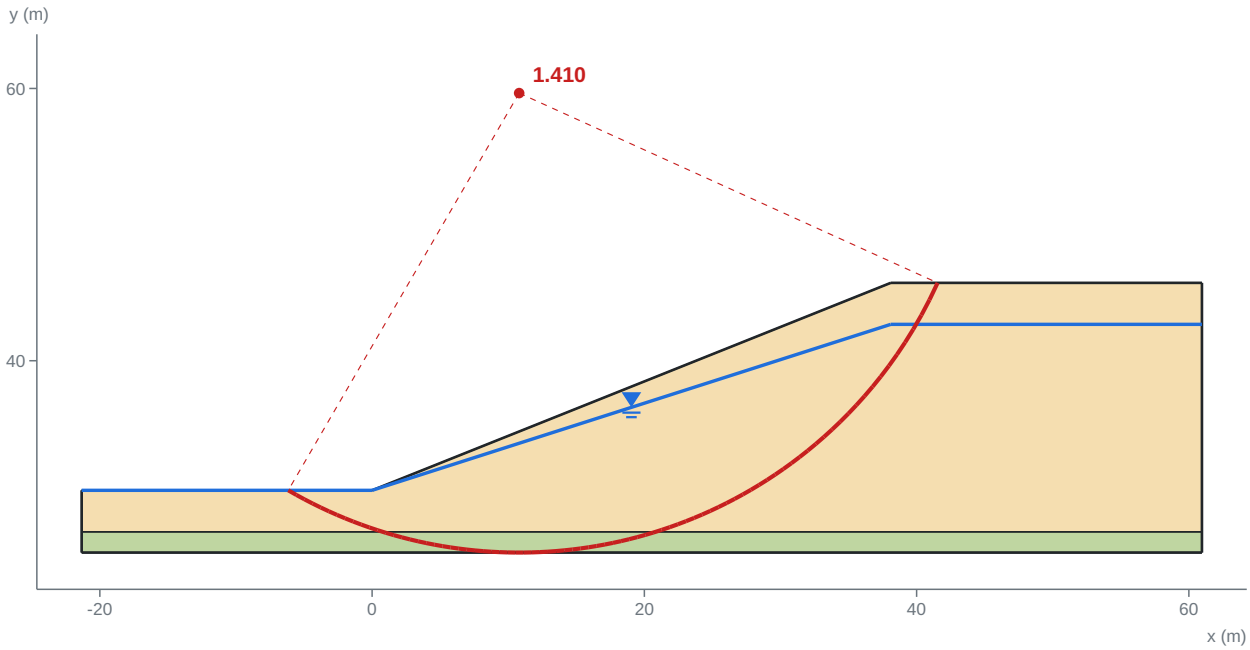


Figure 124. V073: critical circle (Bishop simplified, FOS = 1.410).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Sandy clay	Mohr-Coulomb	20.42	c' = 14.36	35.00°
Plastic clay seam	Mohr-Coulomb	20.42	c' = 0.00	25.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A (circles)	1.417	1.410
Spencer (M-P, constant f(x))	Commercial program A (circles)	1.422	1.415

Notes

- Search limited to x = −15.24 to 50 m (−50 to 164 ft: the toe bench to the crest), as in the source.
- commercial program A's composite surfaces (circles cut off at the base) give 1.392 / 1.400; cSlope's non-circular optimisation, which is not restricted to that shape, finds 1.353 / 1.369 along the seam.

3.125 V074 — Soil-nailed wall with tension crack and surcharges

Pockoski & Duncan (2000), Comparison of computer programs for analysis of reinforced slopes. Virginia Tech.

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

25 ft nailed wall in undrained clay (800 psf) with crest surcharges, a dry 7 ft crack and five passive nail rows at 15°; the reference circle truncated by the crack. Converted to SI.

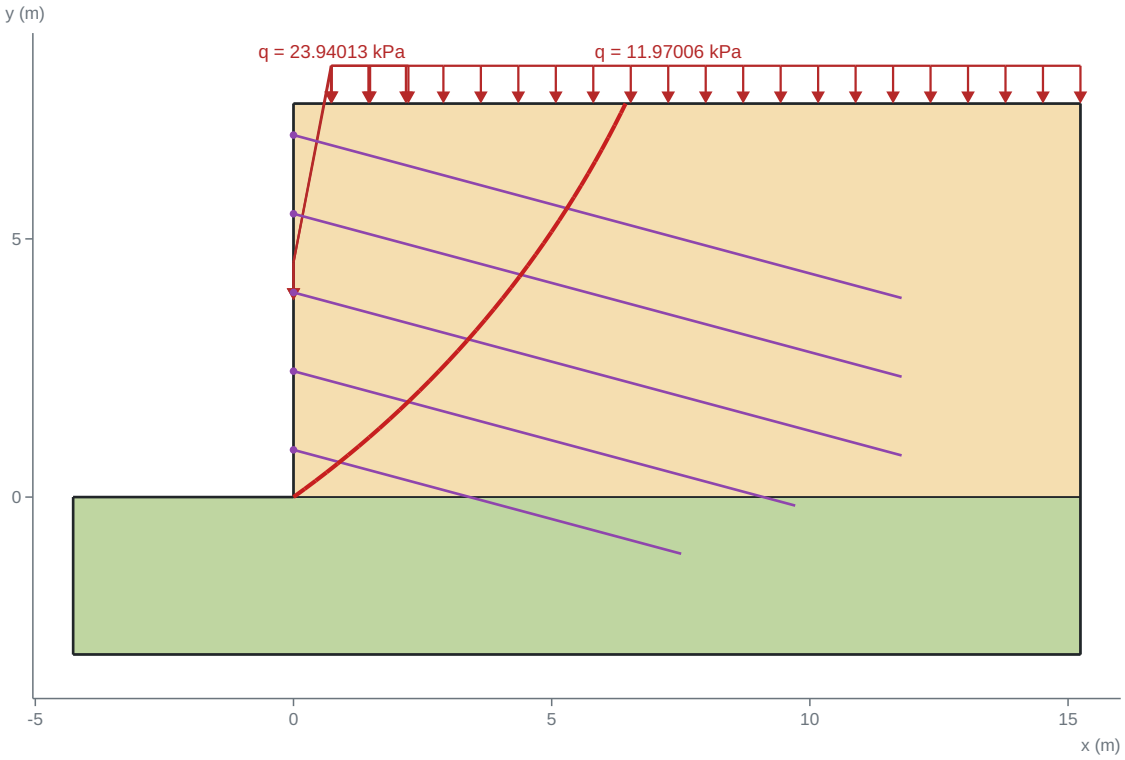


Figure 125. V074: analysed surface (Spencer (M-P, constant f(x)), FOS = 1.010).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Sandy clay	Undrained (φ = 0)	18.85	Su = 38.30	0°
Firm soil	Mohr-Coulomb	20.42	c' = 239.4	35.00°

Method	Published by	Reference	cSlope
Spencer (M-P, constant f(x))	Commercial program A	1.009	1.010
Janbu corrected (f0)	Program D	1.073	1.075

Notes

- commercial program A (Table 60.3): Spencer 1.009, Bishop 0.997, Janbu simplified 1.041. Pockoski & Duncan (2000), as quoted there: program G 1.02 / 1.00 and commercial program B 1.02 / 1.01 (Spencer / Bishop); the commercial program B 2022 verification manual (§2.37) prints commercial program B 1.00 / 0.995.

3.126 V075 — Tieback wall in sand, drawdown water table

Pockoski & Duncan (2000), Comparison of computer programs for analysis of reinforced slopes. Virginia Tech.

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

Single-row tieback wall in sand (φ′ = 30°) with a water table drawn down to the wall face (a vertical step there) and the phreatic correction; the printed surface from the wall toe. Converted to SI.

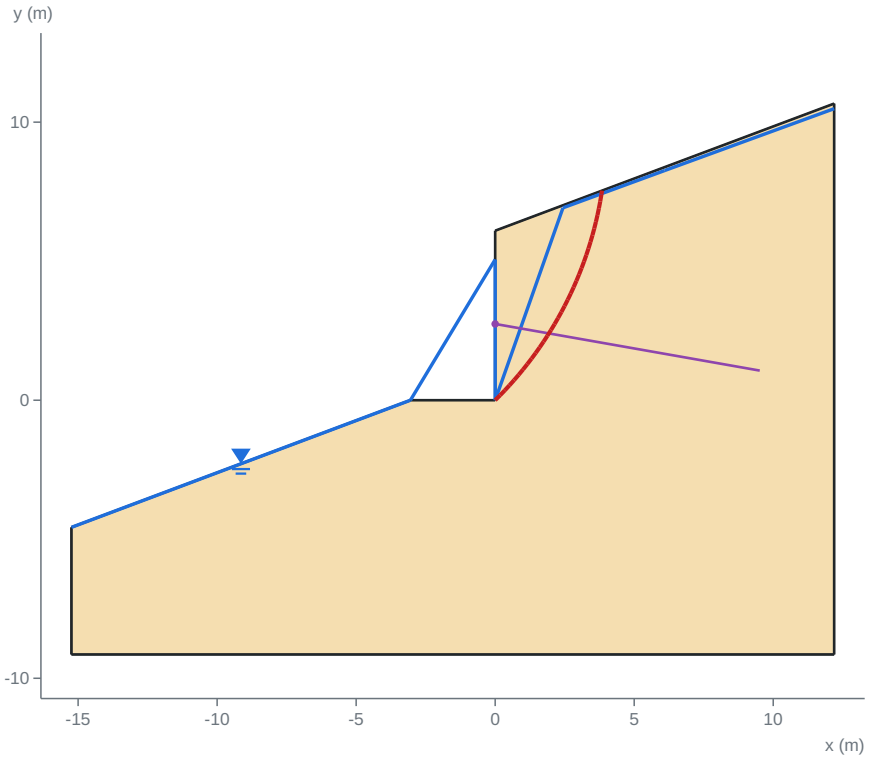


Figure 126. V075: analysed surface (Janbu corrected (f0), FOS = 0.578).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Sand	Mohr-Coulomb	18.85	c' = 0.00	30.00°

Method	Published by	Reference	cSlope
Janbu corrected (f0)	Program D	0.579	0.578
Spencer (M-P, constant f(x))	Commercial program A	0.596	0.599

Notes

- commercial program A (Table 59.3): Spencer 0.596, Bishop 0.582, Janbu simplified 0.583, each on its own critical surface (V075 checks the Spencer circle). Pockoski & Duncan (2000), as quoted there: program G 0.65 / 0.56 / 0.64 and commercial program B 0.60 / 0.60 / 0.61 (Spencer / Bishop / Janbu); the commercial program B 2022 verification manual (§2.36) prints commercial program B 0.57 / 0.53 / 0.58.
- Surface = commercial program A's Spencer circle (-30.872, 31.315, R 43.975 ft). commercial program A's Bishop 0.582 / Janbu 0.583 belong to their own critical circles.
- No complete-equilibrium solution exists on this surface: moment equilibrium needs F >= 0.600 for every interslice inclination, force equilibrium gives at most 0.583 (at 45 deg). The base is inclined 45-81 deg, above the active limit 45 + phi/2 = 60 deg over the upper 29 slices, with an anchor force of 198 kN against a weight of 294 kN. The published 0.596 is a value accepted with a gap; cSlope reports the closest pair (gap 0.020 at lambda 1.6) and flags it (equilibriumGap).

3.127 V076 — Tied-back wall in layered soil

Pockoski & Duncan (2000), Comparison of computer programs for analysis of reinforced slopes. Virginia Tech.

Fixed circle · Units: SI (m, kPa, kN/m³)

44 ft tied-back excavation in eight layers, water table at grade in front of the wall and at el. 102.5 behind it (a vertical step at the wall), three tieback rows at 20° (88 ft, 40 ft bond). commercial program A's critical circle. Converted to SI.

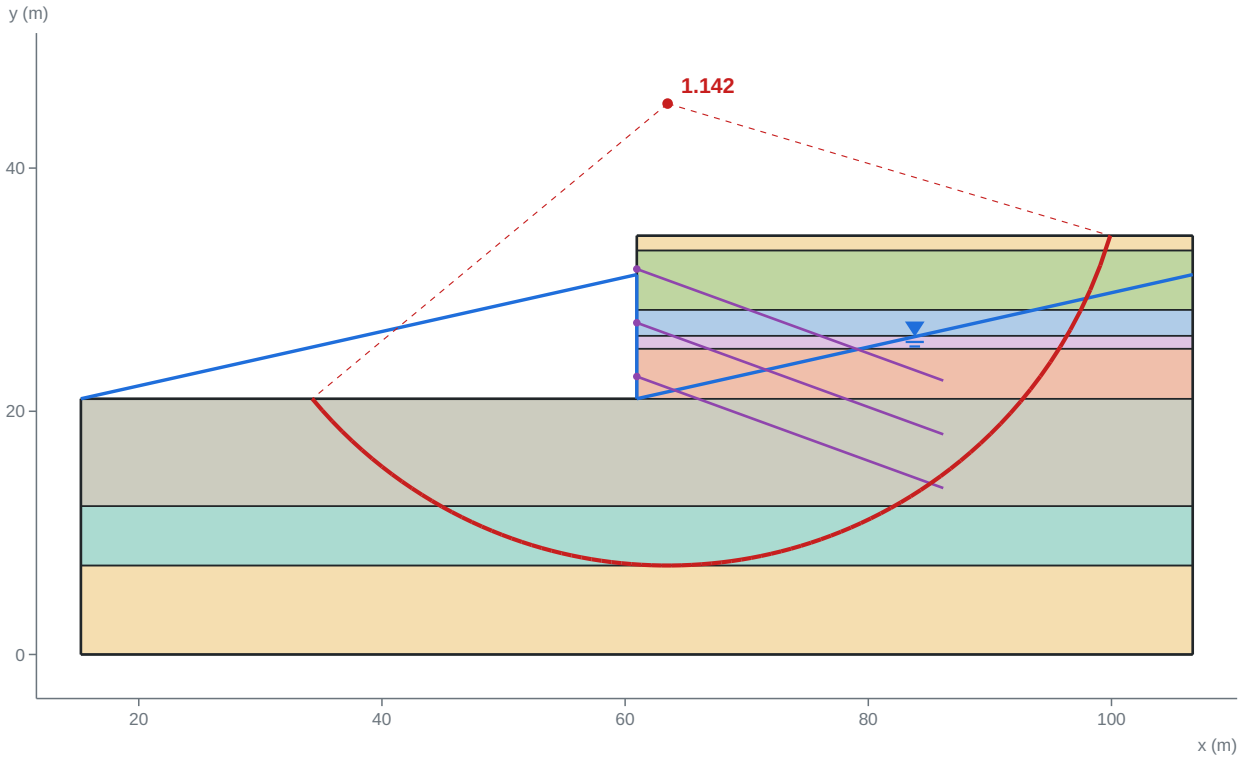


Figure 127. V076: analysed circle (Bishop simplified, FOS = 1.142).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Granular Fill	Mohr-Coulomb	18.91	c' = 0.00	30.00°
Cohesive Fill	Mohr-Coulomb	18.02	c' = 0.00	30.00°
Organic Silt	Undrained (φ = 0)	17.31	Su = 43.09	0°
OC Crust	Undrained (φ = 0)	18.50	Su = 119.0	0°
Upper Marine Clay	Undrained (φ = 0)	18.50	Su = 79.96	0°
Middle Marine Clay	Undrained (φ = 0)	18.50	Su = 45.97	0°
Lower Marine Clay	Undrained (φ = 0)	18.50	Su = 51.95	0°
Glaciomarine Deposits	Undrained (φ = 0)	23.11	Su = 71.82	0°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.147	1.142
Spencer (M-P, constant f(x))	Commercial program A	1.145	1.140

Notes

- program G and commercial program B 1.14.

3.128 V077 — Prandtl bearing capacity on level ground

Prandtl (1920)

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

Weightless (γ = 0.01 kN/m³) c = 20 kPa soil (φ = 0) loaded by a strip load of 102.83 kPa = c·(2 + π) over 10 m, the exact bearing capacity. Analysed on the Prandtl mechanism of Fig. 26.2: active wedge under the load, a 90° fan arc of radius 7.071 m about the load's edge (5, 10), lowest point 2.929, and the passive wedge.

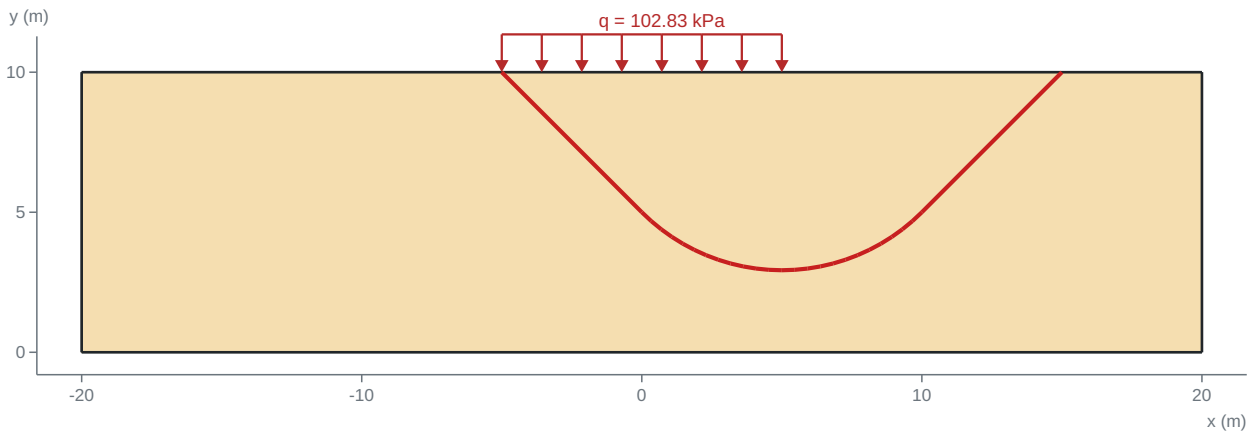


Figure 128. V077: analysed surface (Spencer (M-P, constant f(x)), FOS = 0.943).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
soil	Undrained (φ = 0)	0.01	Su = 20.00	0°

Method	Published by	Reference	cSlope
Spencer (M-P, constant f(x))	Commercial program A	0.940	0.943

Notes

- Theoretical factor of safety 1.0; limit equilibrium with slices gives 6 % less on this mechanism (commercial program A and cSlope alike).
- program D 1.043 on an earlier transcription whose fan was a single kink at (5, 3.9645).

3.129 V078 — Prandtl bearing capacity, fully specified Prandtl surface (reference values)

Prandtl (1921).

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

Prandtl bearing capacity as commercial program B specifies it: weightless c = 20 kPa soil, 102.83 kPa over 10 m, the fully specified Prandtl surface (active wedge, 90° fan about the load edge, passive wedge).

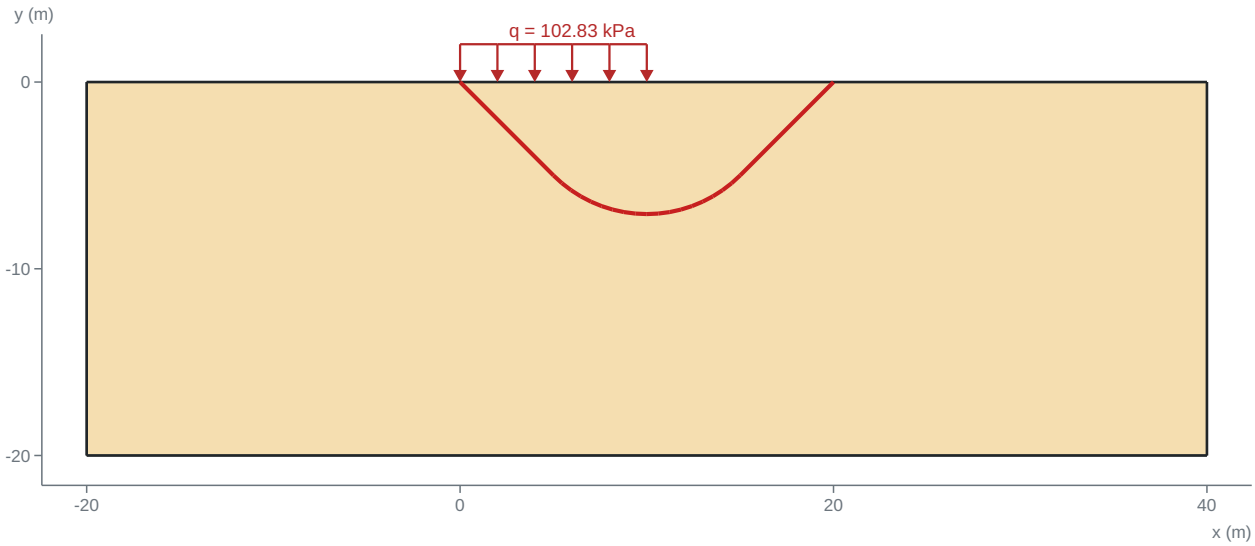


Figure 129. V078: analysed surface (Morgenstern-Price (half-sine), FOS = 0.957).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Undrained (φ = 0)	0.00	Su = 20.00	0°

Method	Published by	Reference	cSlope
Morgenstern-Price (half-sine)	Commercial program B	0.960	0.957

Notes

- Strip load 102.83 kPa = c(π + 2) over 10 m; the domain size is not given in the manual (irrelevant for a weightless soil on a fixed surface).
- Surface: active wedge under the load (apex at x = 5, 5 m deep), fan centred on the load's right edge (R = 7.071 m), passive wedge to x = 20 — the mechanism drawn in Fig. 49. commercial program B Janbu (uncorrected) 0.895; entry-exit search: Bishop 1.056, Janbu 1.031, M-P 1.08. Theory 1.0.
- commercial program B Bishop 0.924 is not checked: Bishop on a non-circular surface depends on the moment axis (cSlope 0.909).

3.130 V079 — Priest (1993) rigid block on a plane, water-filled crack

Priest (1993).

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

Block on a 30° plane from the toe, a 15 m tension crack holding 3.75 m of water and a water table along the plane up to the crack.

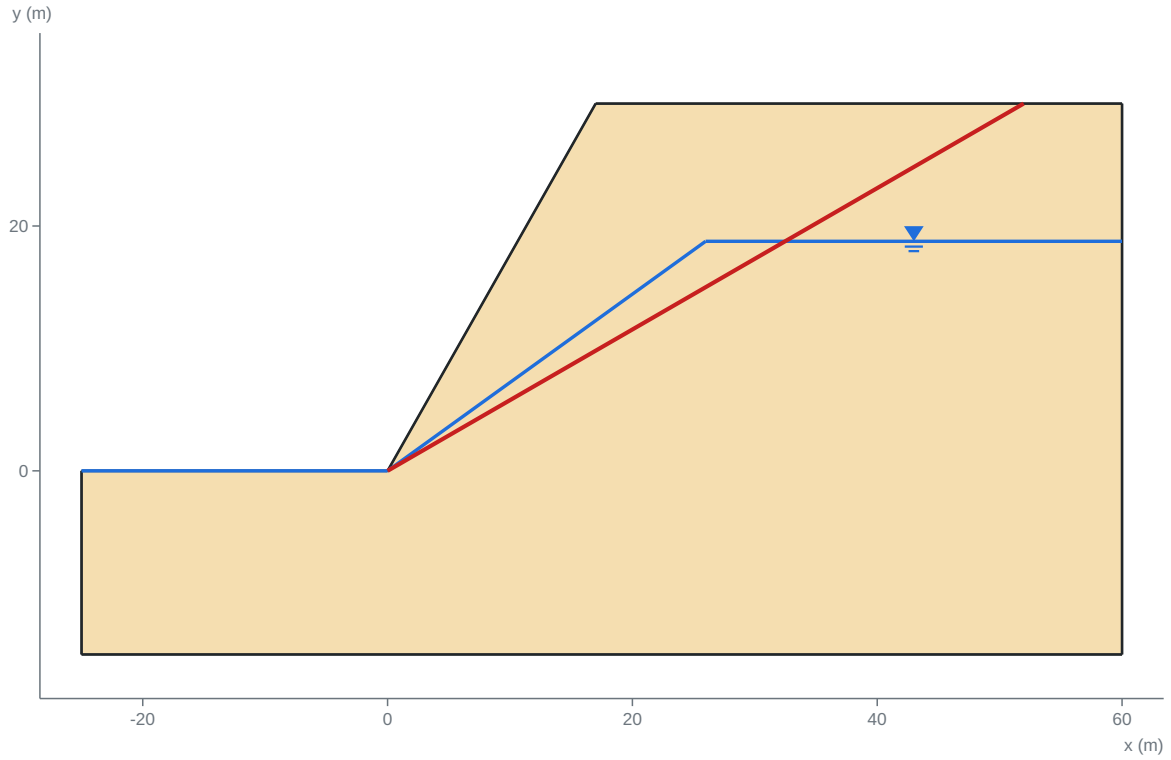


Figure 130. V079: analysed surface (Janbu simplified, FOS = 1.048).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Block	Mohr-Coulomb	25.00	c' = 20.00	30.00°

Method	Published by	Reference	cSlope
Janbu simplified	Commercial program A, commercial program C, Priest	1.049	1.048
Spencer (M-P, constant f(x))	Commercial program A, commercial program C, Priest	1.049	1.048

3.131 V080 — Priest (1993) rigid block, reference geometry (60° face)

Priest (1993).

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

Priest's rigid block on a 30° plane with commercial program B's 60° face (crest at x = 17.321): 15 m tension crack with 3.75 m of water and a water table along the plane.

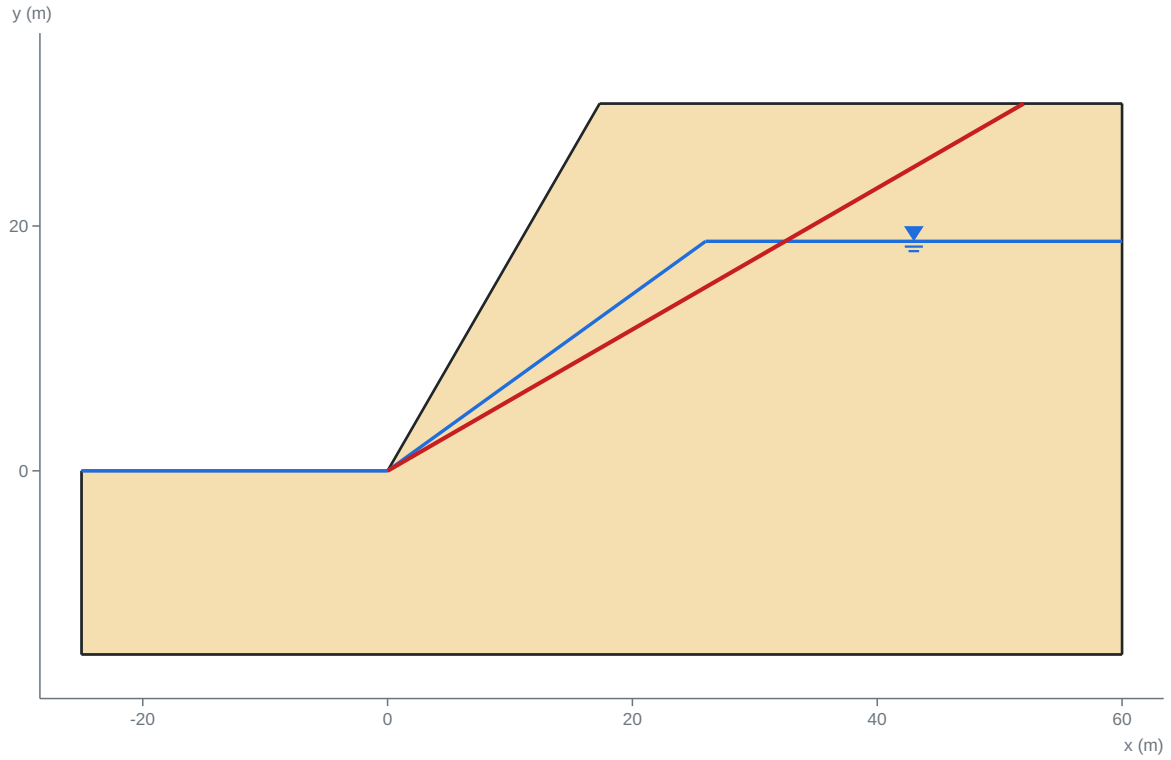


Figure 131. V080: analysed surface (Morgenstern-Price (half-sine), FOS = 1.049).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Block	Mohr-Coulomb	25.00	c' = 20.00	30.00°

Method	Published by	Reference	cSlope
Morgenstern-Price (half-sine)	Commercial program B	1.049	1.049
Spencer (M-P, constant f(x))	Priest	1.049	1.049

Notes

- Crest at (17.321, 30) per Fig. 106 (V079 uses 17.0). commercial program B Janbu 1.049.

3.132 V081a — Sharma (1996) reinforcement example, unreinforced

Sharma (1996).

Fixed circle · Units: SI (m, kPa, kN/m³)

45° cohesionless sand slope with a 40 kPa surcharge behind the crest, on the published toe circle.

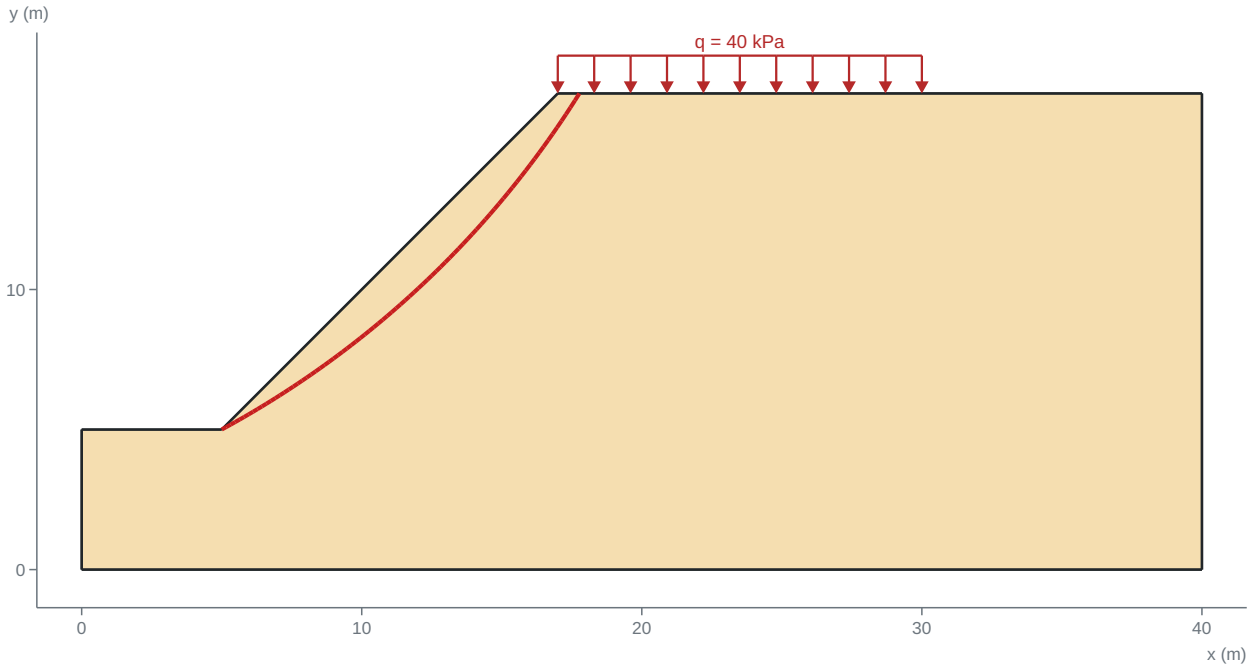


Figure 132. V081a: analysed circle (Bishop simplified, FOS = 0.764).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Sand	Mohr-Coulomb	20.00	c' = 0.00	36.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	0.764	0.764

Notes

- commercial program A reports the Bishop value only (Fig. 37.3); cSlope's Spencer is 0.764 as well. V081b is the slope with the reinforced zone.

3.133 V081b — Sharma (1996) slope with a reinforced zone (φ = 54.93°)

Sharma (1996). Verification manual of a commercial slope stability program.

Fixed circle · Units: SI (m, kPa, kN/m³)

The V081a sand slope (φ′ = 36°, 40 kPa surcharge) with a 7.6 m reinforced zone of φ = 54.93° (tan φ = 1.96 tan 36°) behind the face, on the reference critical toe circle (8.707, 18.307, R = 13.813).

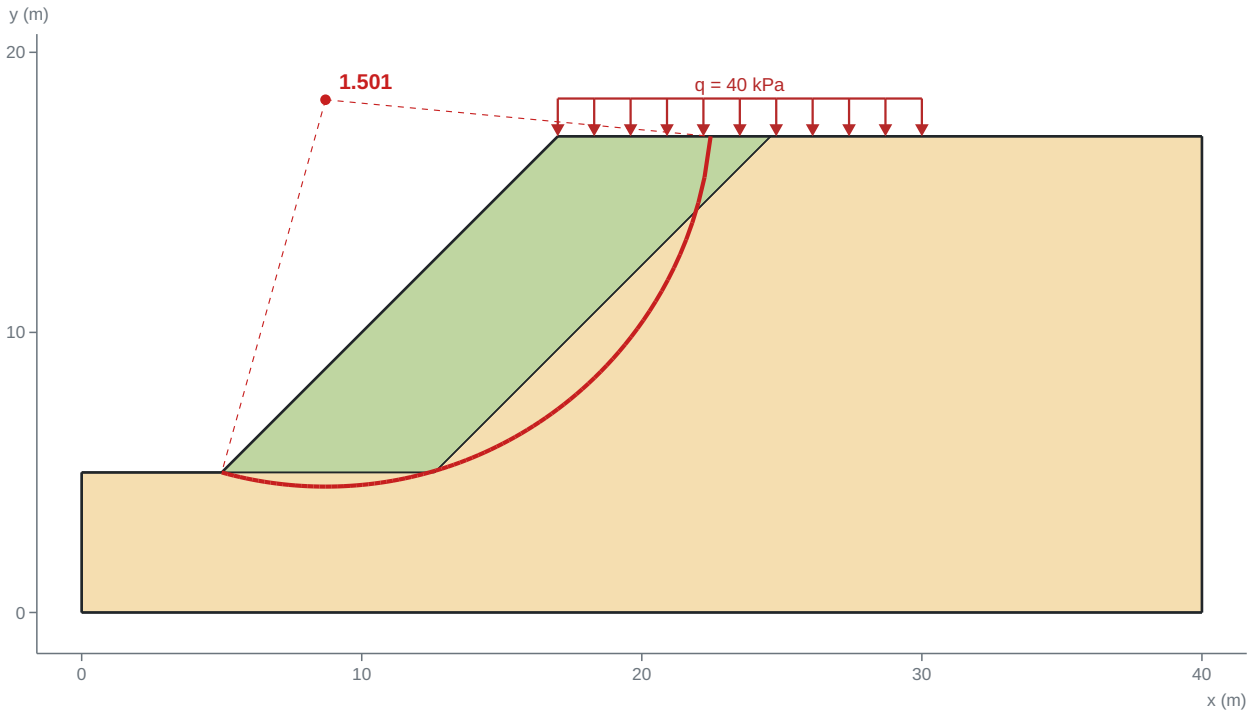


Figure 133. V081b: analysed circle (Bishop simplified, FOS = 1.501).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Material 1	Mohr-Coulomb	20.00	c' = 0.00	36.00°
Reinforced zone	Mohr-Coulomb	20.00	c' = 0.00	54.93°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.504	1.501

Notes

- φ = 36° and γ = 20 kN/m³ of the slope are not printed in the manual (tan 54.93° = 1.96 tan 36°; γ as V081a). Zone base (5, 5)–(12.6, 5), back face parallel to the slope face; 40 kPa from x = 17 to 30.
- commercial program A's required force 351 kN and zone length 7.6 m (program F 345 kN, 7.5 m) give FOS 1.504 against the target 1.5.

3.134 V082a — Sharma (1996) slope, 11 ft dry tension crack instead of the zero-strength cap

Sharma (1996); Malkawi et al. (2001). Verification manual of a commercial slope stability program.

Fixed circle · Units: SI (m, kPa, kN/m³)

The V082c slope with the zero-strength cap replaced by an 11 ft dry tension crack (Table 27.4, analysis 3), on the V082c circle; phreatic correction. Converted to SI.

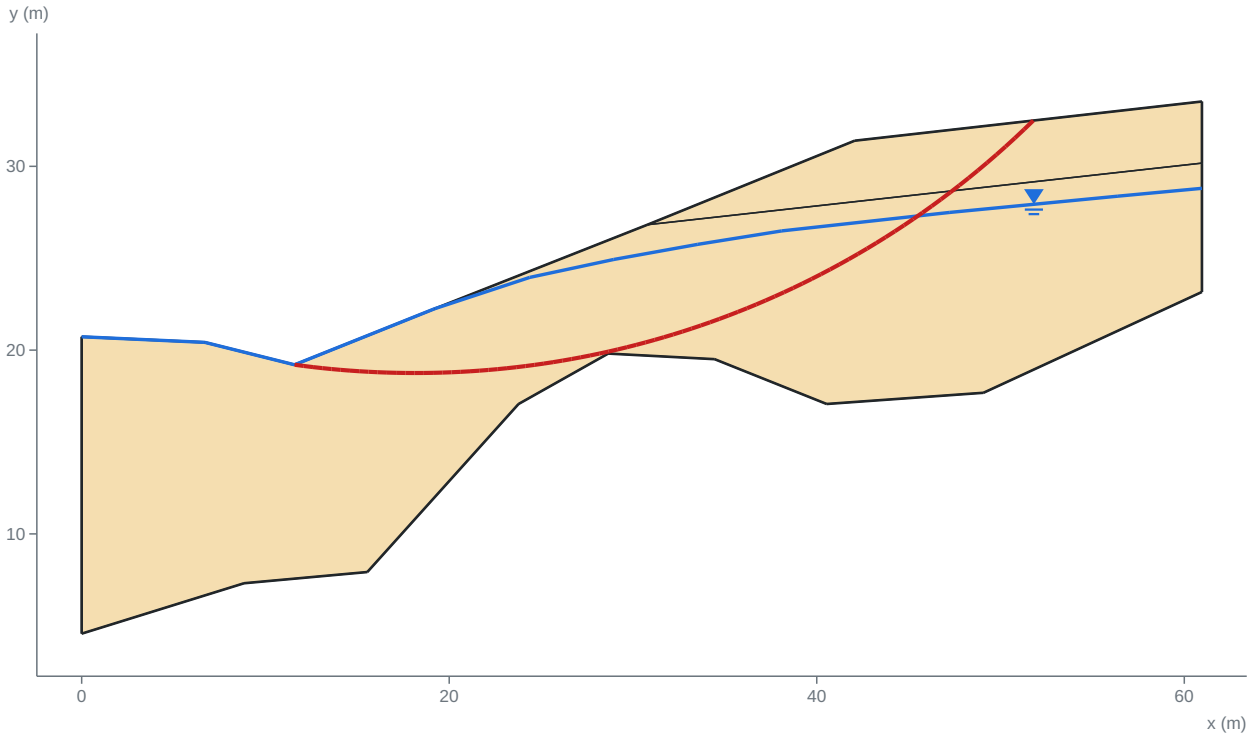


Figure 134. V082a: analysed circle (Bishop simplified, FOS = 1.504).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil 1	Mohr-Coulomb	18.28	c' = 23.94	14.00°
Soil 2 (zero strength)	Undrained (φ = 0)	18.28	Su = 0.00	0°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.532	1.504
Spencer (M-P, constant f(x))	Commercial program A	1.532	1.503
Morgenstern-Price (half-sine)	Commercial program A	1.532	1.503

Notes

- Same 1.8–1.9 % offset as V082c (the water table is traced from the figure).

3.135 V082b — Sharma (1996) slope, 11 ft tension crack with 6 ft of water

Sharma (1996); Malkawi et al. (2001). Verification manual of a commercial slope stability program.

Fixed circle · Units: SI (m, kPa, kN/m³)

As V082a with 6 ft of water in the 11 ft crack (Table 27.4, analysis 4).

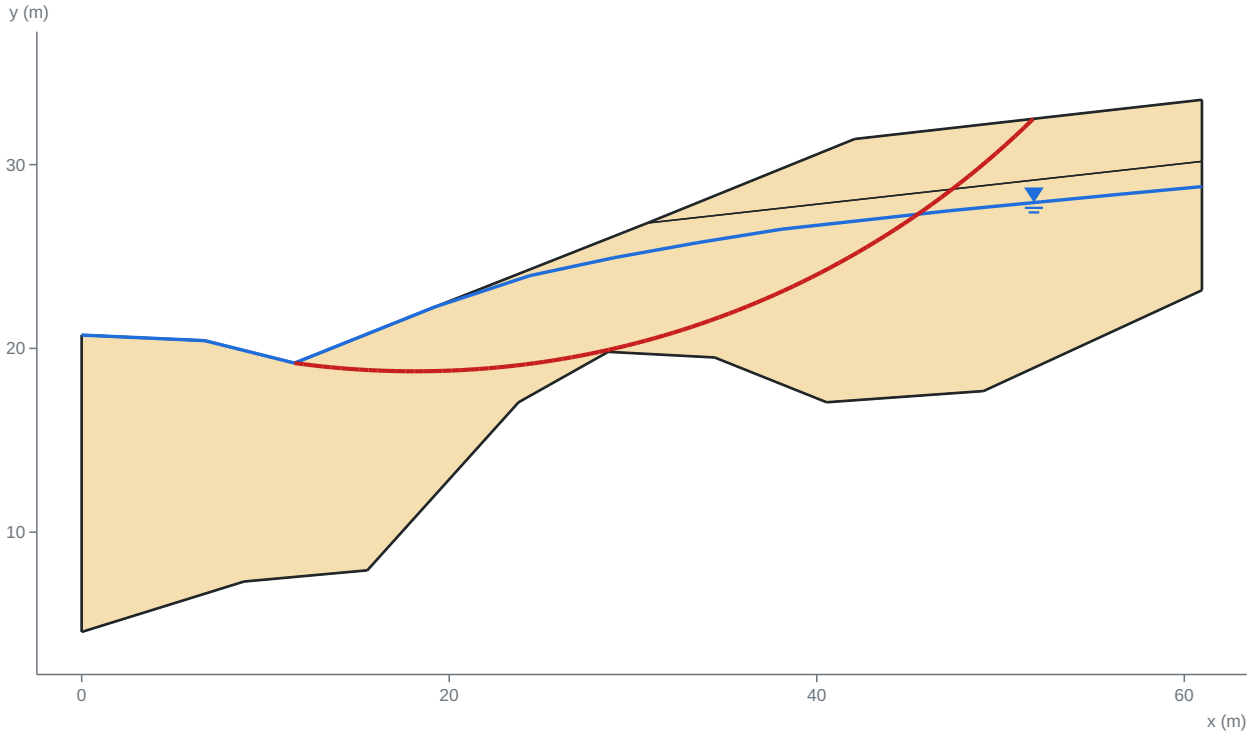


Figure 135. V082b: analysed circle (Bishop simplified, FOS = 1.483).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil 1	Mohr-Coulomb	18.28	c' = 23.94	14.00°
Soil 2 (zero strength)	Undrained (φ = 0)	18.28	Su = 0.00	0°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.511	1.483
Spencer (M-P, constant f(x))	Commercial program A	1.510	1.482

Notes

- Same 1.8 % offset as V082c and V082a (water table traced from the figure).

3.136 V082c — Sharma (1996) slope over undulating bedrock, zero-strength cap

Sharma (1996); Malkawi et al. (2001).

Fixed circle · Units: SI (m, kPa, kN/m³)

Two-material slope over undulating bedrock with a zero-strength cap layer and a water table (phreatic correction); soil 1 has moist/saturated unit weights 116.4/124.2 pcf. Specified circle (59.52, 219.21, R = 157.68). Converted to SI.

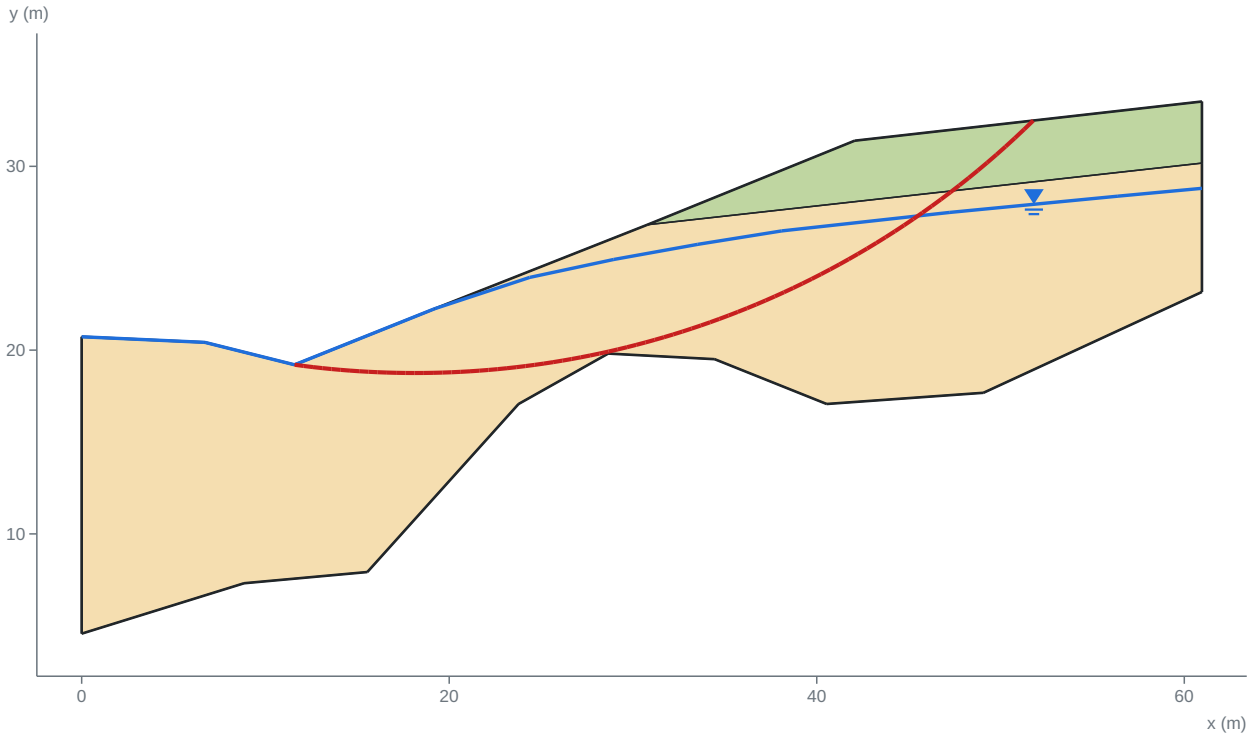


Figure 136. V082c: analysed circle (Bishop simplified, FOS = 1.369).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil 1	Mohr-Coulomb	18.28	c' = 23.94	14.00°
Soil 2 (zero strength)	Undrained (φ = 0)	18.28	Su = 0.00	0°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.396	1.369
Spencer (M-P, constant f(x))	Commercial program A	1.402	1.375
Bishop simplified	Program D	1.369	1.369
Spencer (M-P, constant f(x))	Program D	1.375	1.375

Notes

- The water table is traced from the manual's figure to about ±2 ft; program D reads 1.9 % below commercial program A for every method on the same input, as cSlope does. V082a and V082b are the tension-crack variants of Table 27.4.
- commercial program A's circular search with exit limits (Table 27.3: Bishop 1.376, Spencer 1.382, Janbu corrected 1.345) is not checked: cSlope's search on this input is 4.2 % lower, beyond the offset explained by the traced water table.

3.137 V083 — Soil-nailed wall in clay (Amherst test wall)

Sheahan & Ho (2003).

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

6 m vertical cut in undrained clay (Su = 25 kPa) with two nail rows at 20° (tensile 118 kN, plate 86 kN, bond 15 kN/m, spacing 1.5 m; axial, passive) and the facing weight as a line load; critical plane at 44.5°.

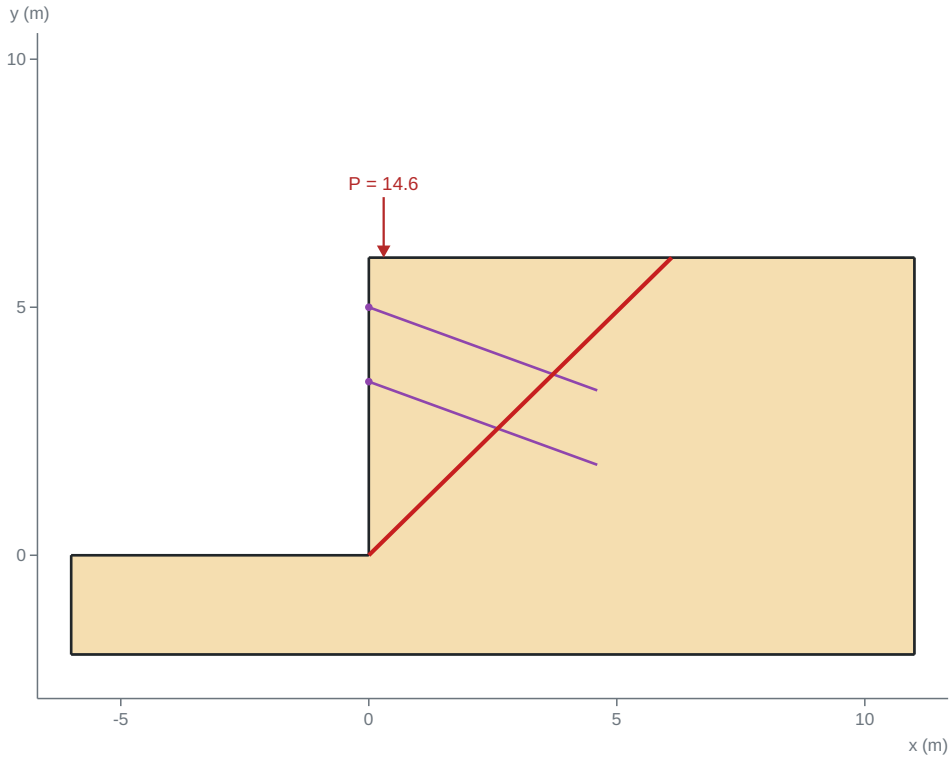


Figure 137. V083: analysed surface (Janbu corrected (f0), FOS = 0.899).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Amherst Clay	Undrained (φ = 0)	18.90	Su = 25.00	0°

Method	Published by	Reference	cSlope
Janbu corrected (f0)	Commercial program A	0.890	0.899

Notes

- Sheahan (trial wedge) 0.887; program D 0.899.

3.138 V084a — Soil-nailed wall in sand (Clouterre test wall)

Sheahan & Ho (2003).

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

7 m cut in sand (c′ = 3 kPa, φ′ = 38°) with seven passive nail rows at 10° (tensile 15 kN, plate 59 kN, bond 7.5 kN/m, i.e. the tensile capacity develops over 2 m; spacing 1.15 m) and the facing weight; 55° plane through the toe. V084b … V084f are the other planes of Table 48.3.

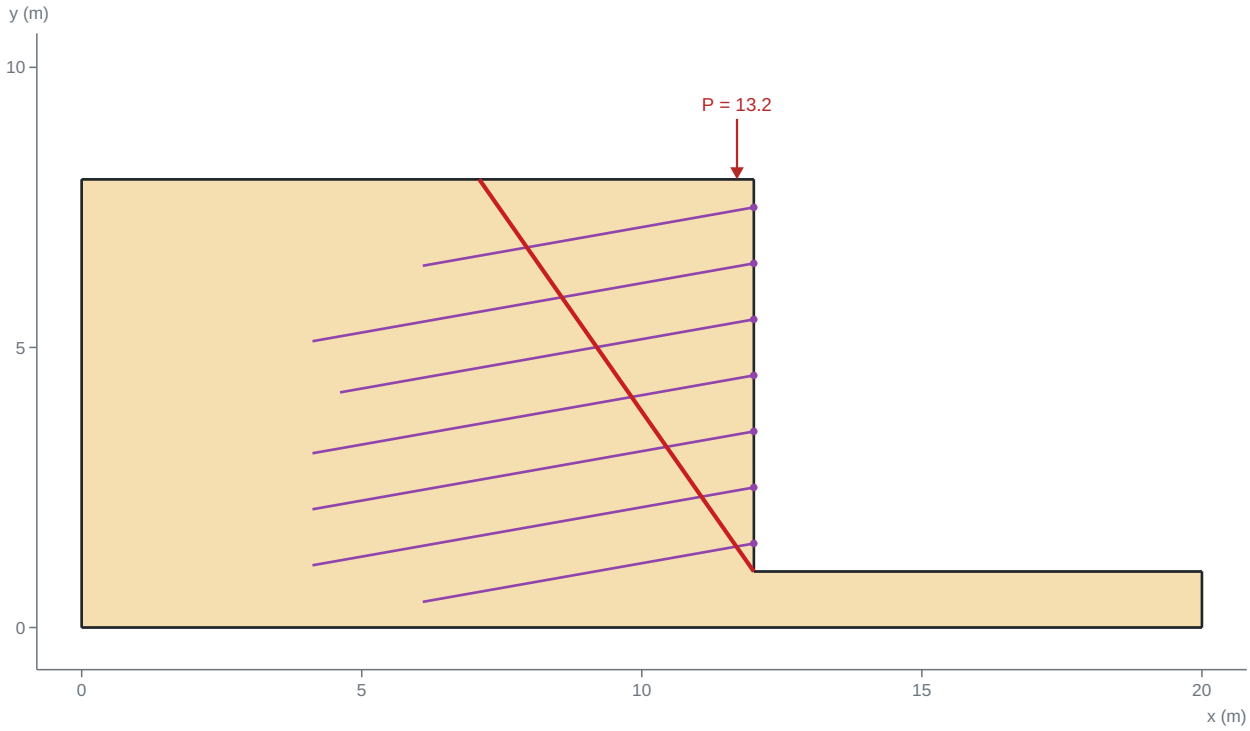


Figure 138. V084a: analysed surface (Janbu corrected (f0), FOS = 0.988).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Fontainebleau Sand	Mohr-Coulomb	20.00	c' = 3.00	38.00°

Method	Published by	Reference	cSlope
Janbu corrected (f0)	Commercial program A	0.989	0.988

Notes

- Table 48.2 lists an out-of-plane spacing of 1.5 m; 1.15 m (the Clouterre wall) reproduces commercial program A's results at all six plane angles and is used here.
- Sheahan & Ho 0.989 at 55° (force-equilibrium trial wedge).

3.139 V084b — Soil-nailed wall in sand (Clouterre test wall), 45° plane

Sheahan & Ho (2003).

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

The V084a wall on the 45° plane through the toe.

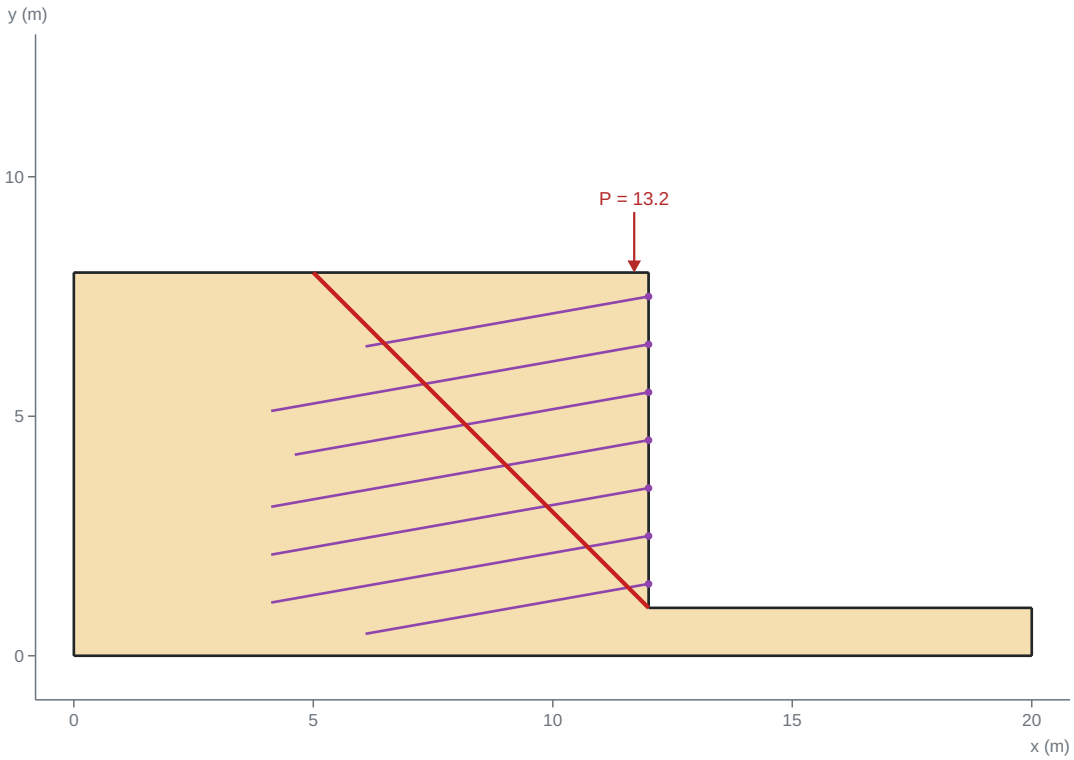


Figure 139. V084b: analysed surface (Janbu corrected (f0), FOS = 1.124).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Fontainebleau Sand	Mohr-Coulomb	20.00	c' = 3.00	38.00°

Method	Published by	Reference	cSlope
Janbu corrected (f0)	Commercial program A	1.123	1.124

Notes

- Sheahan & Ho 1.176 (force-equilibrium trial wedge).

3.140 V084c — Soil-nailed wall in sand (Clouterre test wall), 50° plane

Sheahan & Ho (2003).

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

The V084a wall on the 50° plane through the toe.

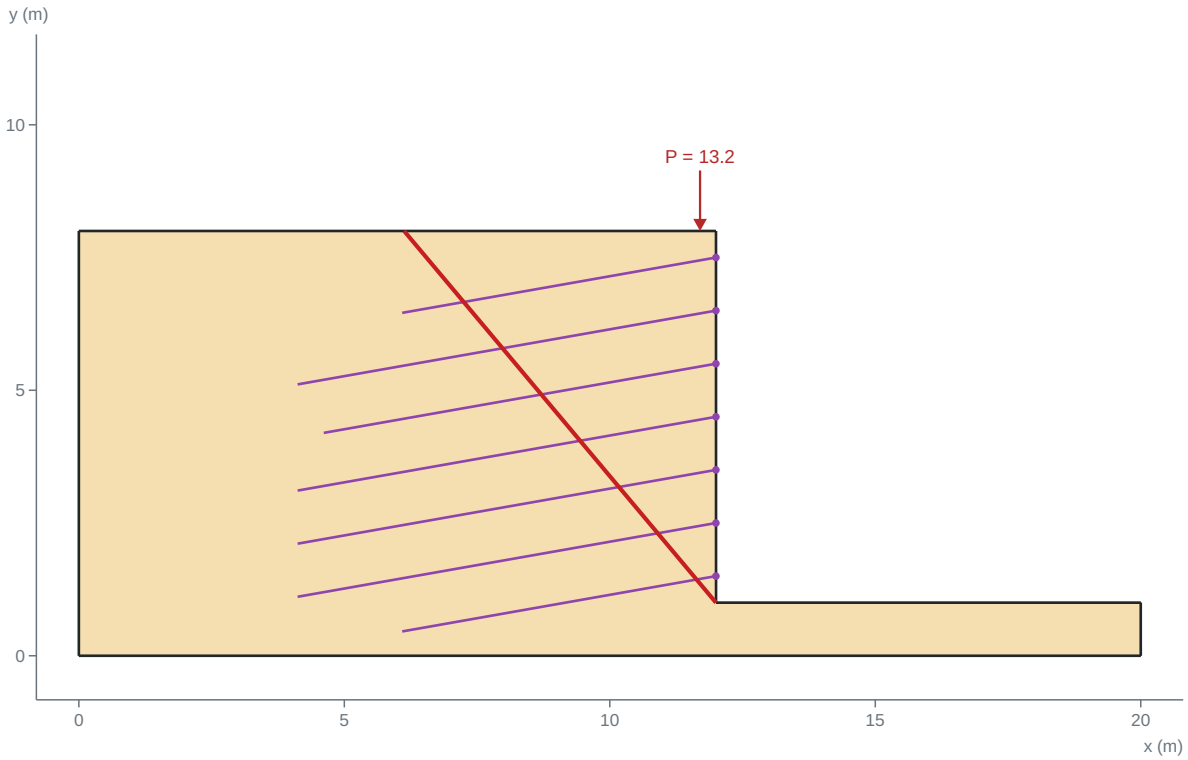


Figure 140. V084c: analysed surface (Janbu corrected (f0), FOS = 1.043).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Fontainebleau Sand	Mohr-Coulomb	20.00	c' = 3.00	38.00°

Method	Published by	Reference	cSlope
Janbu corrected (f0)	Commercial program A	1.043	1.043

Notes

- Sheahan & Ho 1.070 (force-equilibrium trial wedge).

3.141 V084d — Soil-nailed wall in sand (Clouterre test wall), 60° plane

Sheahan & Ho (2003).

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

The V084a wall on the 60° plane through the toe.

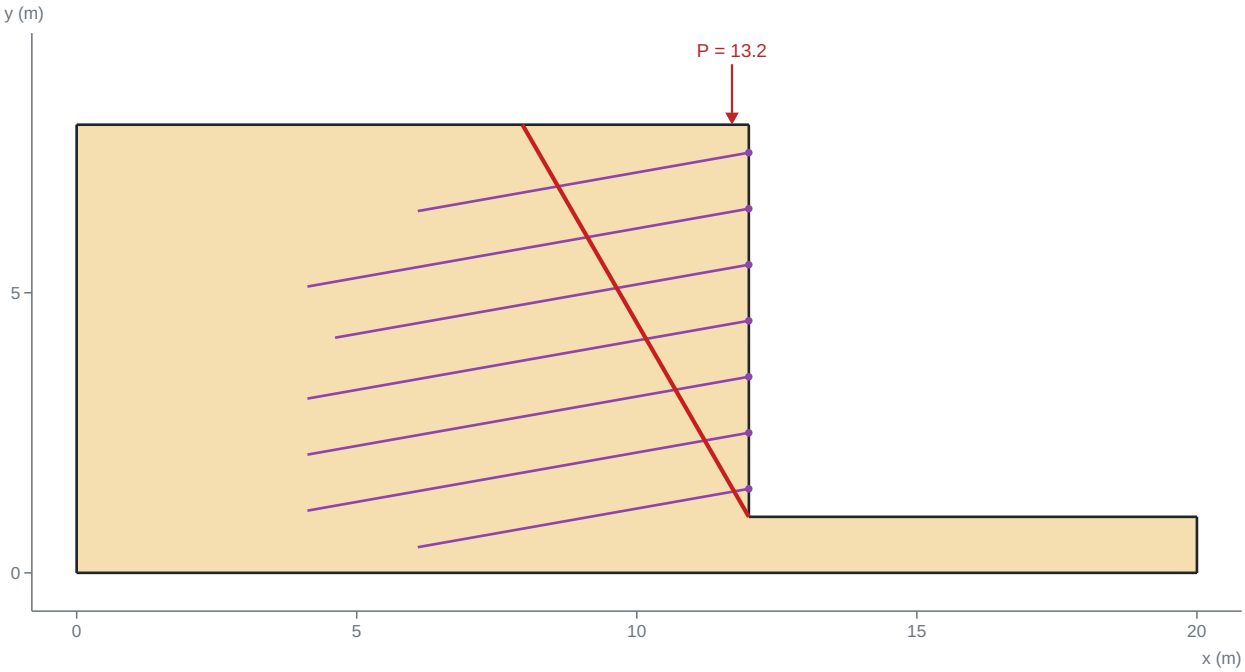


Figure 141. V084d: analysed surface (Janbu corrected (f0), FOS = 0.944).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Fontainebleau Sand	Mohr-Coulomb	20.00	c' = 3.00	38.00°

Method	Published by	Reference	cSlope
Janbu corrected (f0)	Commercial program A	0.945	0.944

Notes

- Sheahan & Ho 0.929 (force-equilibrium trial wedge).

3.142 V084e — Soil-nailed wall in sand (Clouterre test wall), 65° plane

Sheahan & Ho (2003).

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

The V084a wall on the 65° plane through the toe.

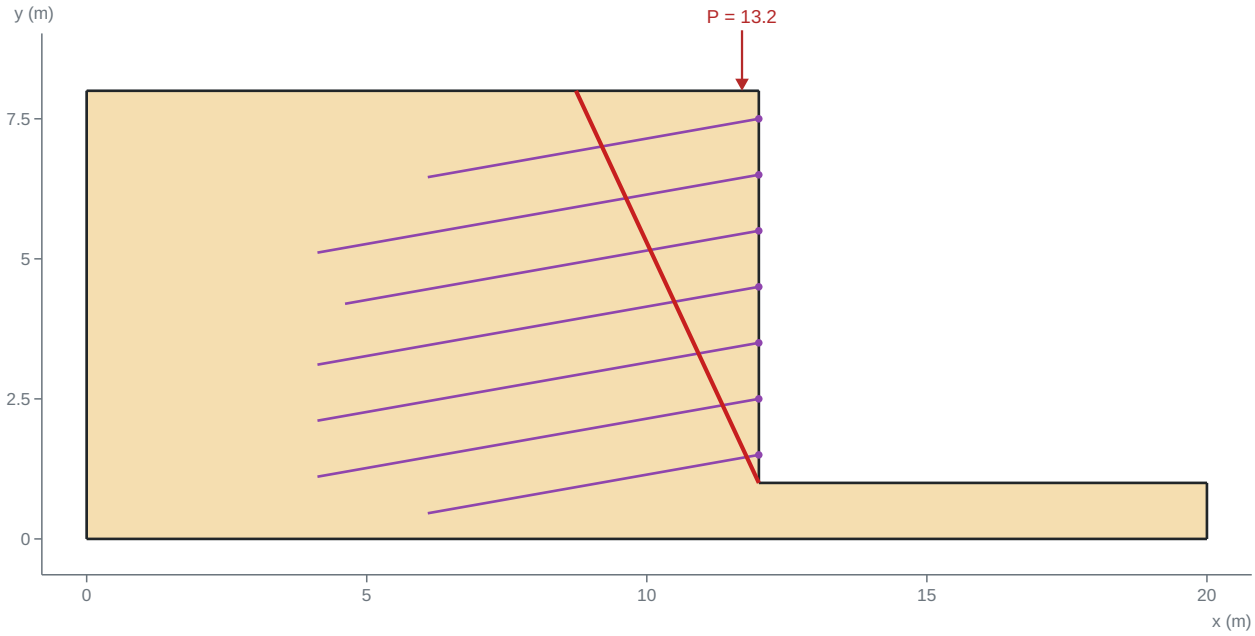


Figure 142. V084e: analysed surface (Janbu corrected (f0), FOS = 0.920).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Fontainebleau Sand	Mohr-Coulomb	20.00	c' = 3.00	38.00°

Method	Published by	Reference	cSlope
Janbu corrected (f0)	Commercial program A	0.922	0.920

Notes

- Sheahan & Ho 0.893 (force-equilibrium trial wedge).

3.143 V084f — Soil-nailed wall in sand (Clouterre test wall), 70° plane

Sheahan & Ho (2003).

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

The V084a wall on the 70° plane through the toe.

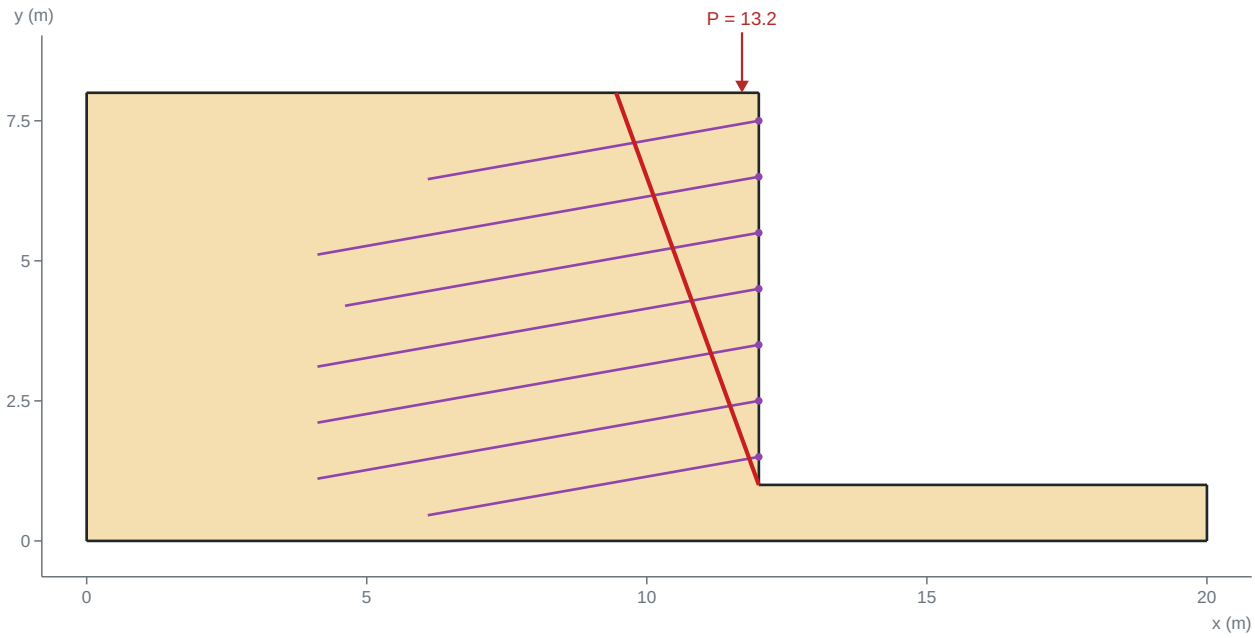


Figure 143. V084f: analysed surface (Janbu corrected (f0), FOS = 0.920).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Fontainebleau Sand	Mohr-Coulomb	20.00	c' = 3.00	38.00°

Method	Published by	Reference	cSlope
Janbu corrected (f0)	Commercial program A	0.923	0.920

Notes

- Sheahan & Ho 0.887 (force-equilibrium trial wedge).

3.144 V085 — Bradwell excavated slope, stiff-fissured London clay

Skempton & LaRochelle (1965); Duncan & Wright, Soil Strength and Slope Stability.

Critical circle search · Units: imperial (ft, psf, pcf)

Excavation in London clay (six sublayers with Su increasing with depth, S = f(datum)) under Marsh clay and a clay fill cracked to its full depth (11.4 ft); search for the deep failure, entering on the crest and leaving at the excavation toe.

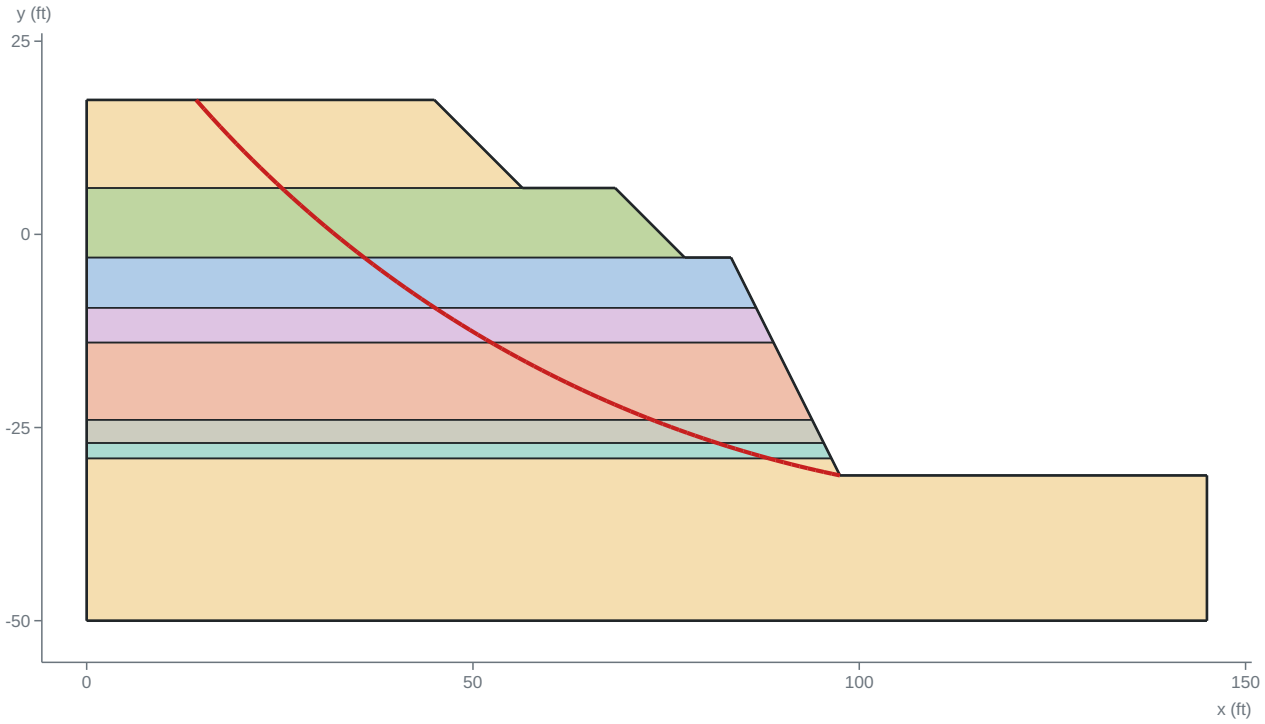


Figure 144. V085: critical circle (Bishop simplified, FOS = 1.759).

Material	Model	γ (pcf)	Strength (psf)	φ
Clay fill	Mohr-Coulomb	110.0	c' = 1.00	35.00°
Marsh clay	Mohr-Coulomb	105.0	c' = 300.0	0.00°
Brown London clay 1	S = f(datum)	120.0	Su = 750.0 at y = -3.00, 90.00 psf/ft below	0°
Brown London clay 2	S = f(datum)	120.0	Su = 1335.0 at y = -9.50, 82.00 psf/ft below	0°
Brown London clay 3	S = f(datum)	120.0	Su = 1704.0 at y = -14.00, 53.00 psf/ft below	0°
Brown London clay 4	S = f(datum)	120.0	Su = 2234.0 at y = -24.00, 47.00 psf/ft below	0°
Blue London clay 1	S = f(datum)	120.0	Su = 2375.0 at y = -27.00, 47.00 psf/ft below	0°
Blue London clay 2	S = f(datum)	120.0	Su = 2469.0 at y = -29.00, 39.00 psf/ft below	0°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.762	1.759
Spencer (M-P, constant f(x))	Commercial program A	1.758	1.759
Janbu corrected (f0)	Commercial program A	1.736	1.735

Notes

- Without limits the search finds shallower mechanisms: a sliver in the cohesionless 1:1 fill (0.74) and, with the crack, a surface through the fill and Marsh clay (1.15). The published value is the deep failure.
- The source searches the whole model and models the fill as a tension-crack zone, so its circles start at the base of the fill; these limits stand in for that.

3.145 V086a — Tandjiria embankment, clay fill, geosynthetic T = 169 kN/m

Tandjiria (2002).

Fixed circle · Units: SI (m, kPa, kN/m³)

The V086b clay-fill embankment with a 169 kN/m geosynthetic at its base (axial, active, fully anchored) and the water-filled 2 m crack, on the critical circle.

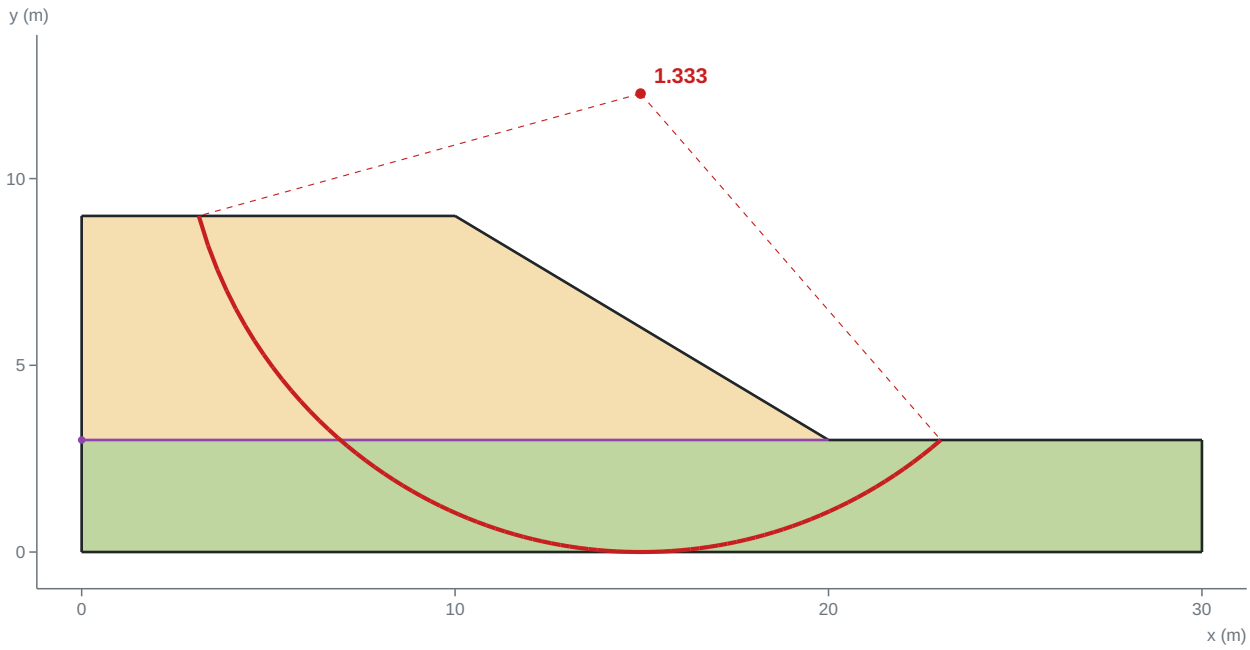


Figure 145. V086a: analysed circle (Spencer (M-P, constant f(x)), FOS = 1.333).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Fill	Undrained (φ = 0)	19.40	Su = 20.00	0°
Soft Clay	Undrained (φ = 0)	19.40	Su = 20.00	0°

Method	Published by	Reference	cSlope
Spencer (M-P, constant f(x))	Commercial program A	1.350	1.333
Spencer (M-P, constant f(x))	Program D	1.332	1.333

3.146 V086b — Tandjiria embankment, clay fill, water-filled crack

Tandjiria (2002).

Critical circle search · Units: SI (m, kPa, kN/m³)

Half-embankment (9 m crest, 10 m wide crest, 1V:1.67H face) of clay fill (Su = 20 kPa) on soft clay (Su = 20 kPa), γ = 19.4; 2 m water-filled tension crack, unreinforced; circular search. Also run with the automatic (Rankine) crack depth.

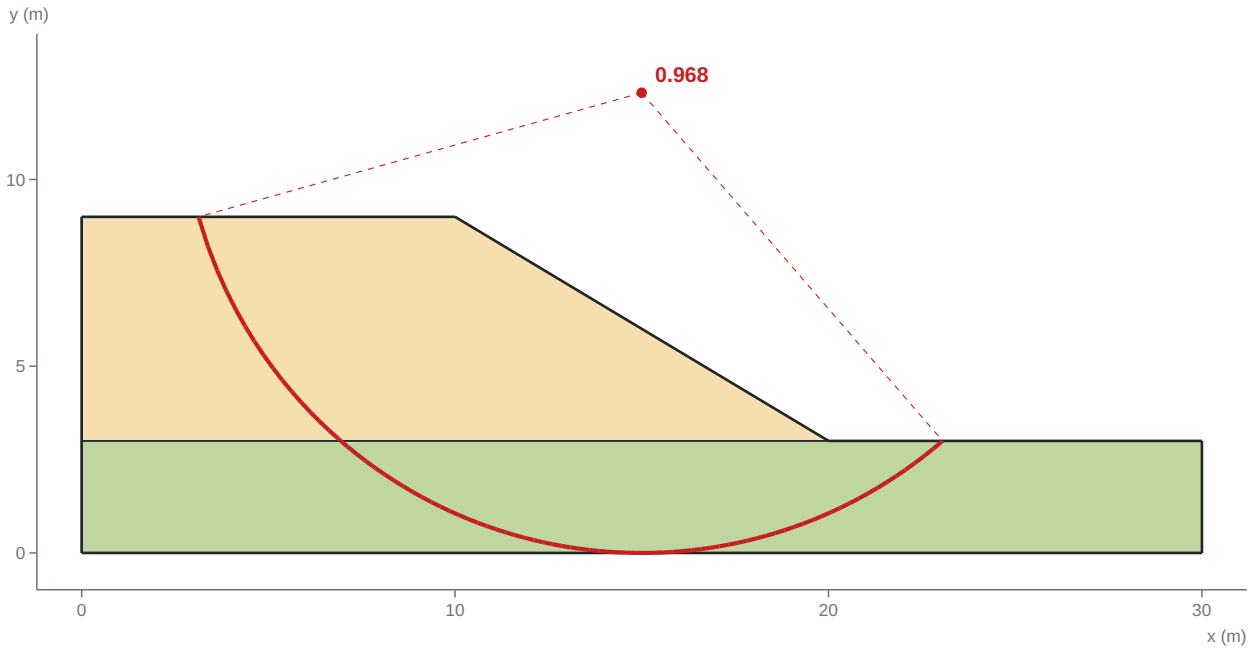


Figure 146. V086b: critical circle (Spencer (M-P, constant f(x)), FOS = 0.968).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Fill	Undrained (φ = 0)	19.40	Su = 20.00	0°
Soft clay	Undrained (φ = 0)	19.40	Su = 20.00	0°

Method	Published by	Reference	cSlope
Spencer (M-P, constant f(x))	Commercial program A	0.975	0.968
Spencer (M-P, constant f(x))	Commercial program A	0.975	0.966

Notes

- program D gives 0.968 on the reference circle, as cSlope does; Tandjiria 0.981.
- Automatic crack depth 2c/γ = 2.06 m instead of the given 2 m.

3.147 V086c — Tandjiria embankment, sand fill, dry crack

Tandjiria (2002).

Critical circle search · Units: SI (m, kPa, kN/m³)

The V086 embankment as a sand fill (φ′ = 37°, γ = 17) on soft clay (Su = 20 kPa, γ = 20) with a dry 2 m tension crack, unreinforced; circular search.

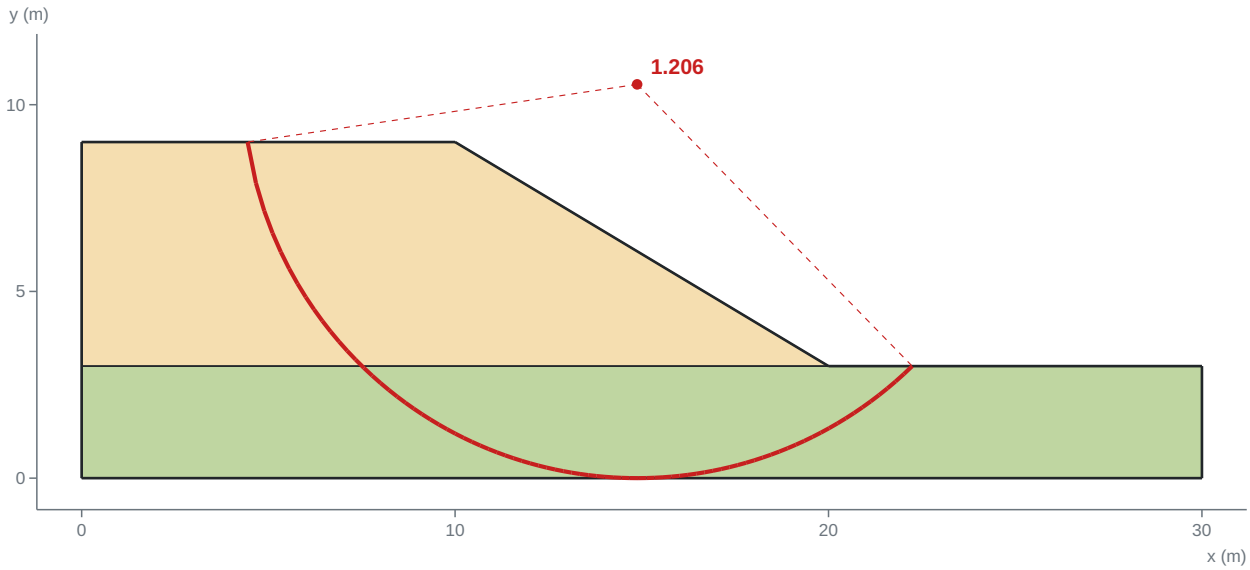


Figure 147. V086c: critical circle (Spencer (M-P, constant f(x)), FOS = 1.206).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Fill	Mohr-Coulomb	17.00	c' = 0.00	37.00°
Soft clay	Undrained (φ = 0)	20.00	Su = 20.00	0°

Method	Published by	Reference	cSlope
Spencer (M-P, constant f(x))	Commercial program A	1.209	1.206

Notes

- program D gives 1.200 on the reference circle; Tandjiria 1.219.

3.148 V087 — USACE dam, chart-check properties (EM 1110-2-1902 fig. 4-3)

USACE EM 1110-2-1902 (2003), fig. 4-3.

Fixed circle · Units: SI (m, kPa, kN/m³)

The V088 dam family with single unit weights (embankment c = 200 psf, φ = 25°, γ = 115; sand 0/35/130; clay 0/27/115), slightly flatter faces and ponded water at el. 20 on both faces. the reference circle (−135, 169, R = 169.1), tangent to the sand top. Converted to SI.

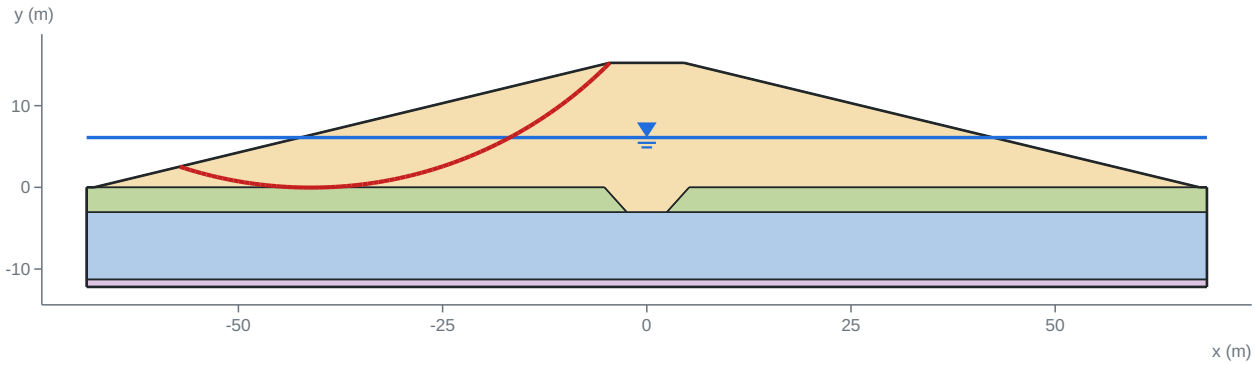


Figure 148. V087: analysed circle (Spencer (M-P, constant f(x)), FOS = 2.258).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Embankment	Mohr-Coulomb	18.07	c' = 9.58	25.00°
Sand	Mohr-Coulomb	20.42	c' = 0.00	35.00°
Foundation clay	Mohr-Coulomb	18.07	c' = 0.00	27.00°
Rock	Mohr-Coulomb	25.13	c' = 0.00	45.00°

Method	Published by	Reference	cSlope
Spencer (M-P, constant f(x))	Program D	2.258	2.258
Bishop simplified	Program D	2.254	2.255
Janbu corrected (f0)	Commercial program A	2.290	2.258

Notes

- The three commercial program A models carry slightly different digitisations of the same dam; program D reads about 2 % below commercial program A here on the same input, as cSlope does. USACE 2.30; commercial program A GLE 2.309.
- An independent Bishop calculation on commercial program A's circle (R = 169.0 ft) gives 2.243, as cSlope does; commercial program A's 2.307 (Bishop and Spencer) is not explained and is not used as a check.

3.149 V088 — USACE dam, end of construction (EM 1110-2-1902 fig. 4-1)

USACE EM 1110-2-1902 (2003), fig. 4-1.

Fixed circle · Units: SI (m, kPa, kN/m³)

Symmetric 50 ft embankment (crest ±12 ft, toes ±206.18 ft) with c = 1000 psf, φ = 5°, a core trench through a 10 ft sand blanket over foundation clay (Su = 3000 psf) and rock; groundwater at the sand top, 7 ft dry crest crack; moist/saturated unit weights. Specified circle (102, 163, R = 163). Converted to SI.

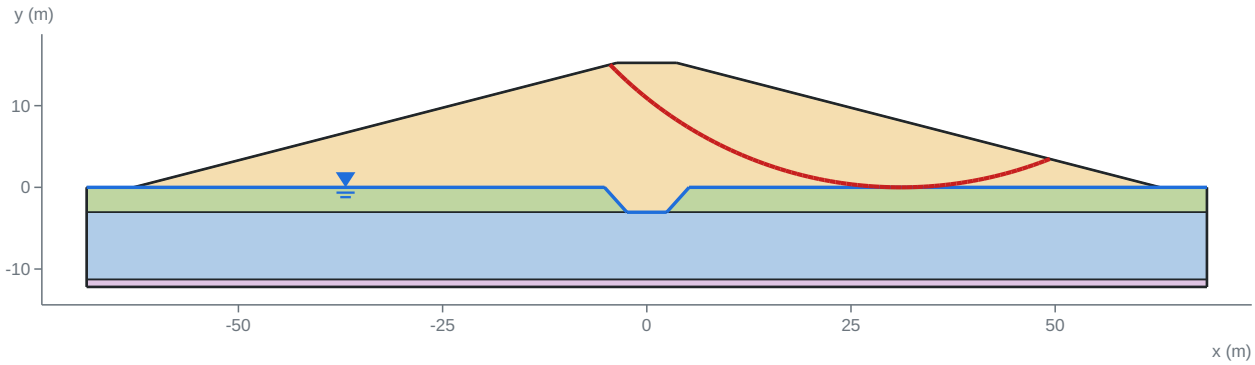


Figure 149. V088: analysed circle (Spencer (M-P, constant f(x)), FOS = 2.442).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Embankment	Mohr-Coulomb	18.07	c' = 47.88	5.00°
Sand	Mohr-Coulomb	19.64	c' = 0.00	35.00°
Foundation clay	Undrained (φ = 0)	17.28	Su = 143.6	0°
Rock	Mohr-Coulomb	25.13	c' = 0.00	45.00°

Method	Published by	Reference	cSlope
Spencer (M-P, constant f(x))	Commercial program A	2.445	2.442
Bishop simplified	Commercial program A	2.447	2.442
Janbu corrected (f0)	Commercial program A	2.430	2.444

Notes

- Neither figure labels its vertices. The crest (±12 ft) and toes (±206.18 ft) are inferred from commercial program A's printed exit point (161.895, 11.403 ft), which lies on this face; commercial program A GLE 2.447, USACE 2.44.
- The earlier outline after program D (crest ±17 ft, toes ±217 ft, from USACE's slice table) misses that exit point by 2.4 ft and gave 2.489 / 2.488 (program D and cSlope alike, 1.8 % above commercial program A).

3.150 V089 — USACE dam, upstream low pool (EM 1110-2-1902 fig. 4-2)

USACE EM 1110-2-1902 (2003), fig. 4-2.

Fixed circle · Units: SI (m, kPa, kN/m³)

The V088 dam family, 45 ft high (crest ±15 ft, toes ±200 ft), with drained strengths (embankment c = 100 psf, φ = 25°; sand 0/35; clay 0/28; rock 0/45; moist/saturated unit weights) and the pool at el. 20 standing on the upstream face. USACE's circle (−102, 163, R = 173), tangent to the clay top. Converted to SI.

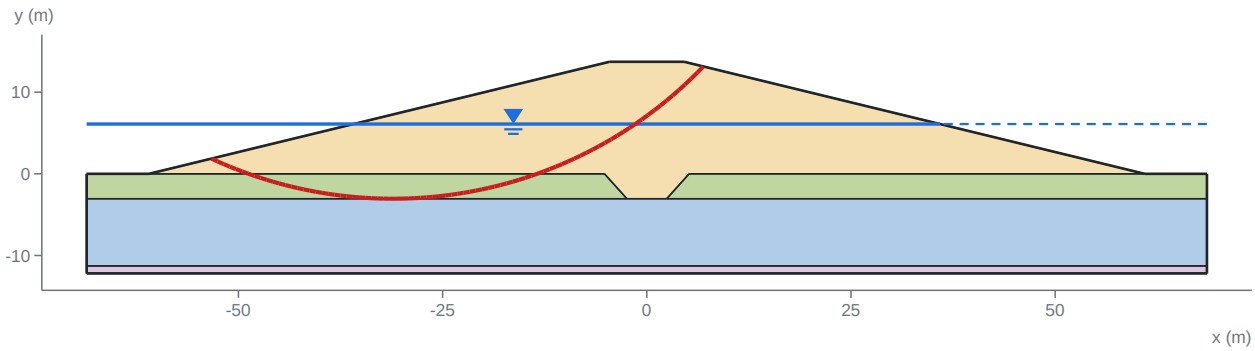


Figure 150. V089: analysed circle (Bishop simplified, FOS = 2.698).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Embankment	Mohr-Coulomb	18.07	c' = 4.79	25.00°
Sand	Mohr-Coulomb	19.64	c' = 0.00	35.00°
Foundation clay	Mohr-Coulomb	17.28	c' = 0.00	28.00°
Rock	Mohr-Coulomb	25.13	c' = 0.00	45.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	2.716	2.698
Spencer (M-P, constant f(x))	Commercial program A	2.736	2.721
Janbu corrected (f0)	Commercial program A	2.650	2.648

Notes

- Outline from commercial program A's printed slip-surface endpoints (Fig. 65.2): 45 ft high, crest ±15 ft, toes ±200 ft; the pool line ends on the upstream face at x = 117.778 ft. commercial program A GLE 2.744, USACE 2.71 (Bishop).
- Table 65.1 prints φ = 35° for the embankment, which gives 3.10–3.14; φ = 25° (USACE fig. 4-2) reproduces commercial program A's results and is used.
- program D 2.725 / 2.748 on the earlier 50 ft outline (crest ±17 ft, toes ±217 ft), where cSlope gave 2.724 / 2.747.

3.151 V090 — USACE EM 1110-2-1902, example F-5

US Army Corps of Engineers (2003). Slope Stability, EM 1110-2-1902, Appendix F.

Fixed circle · Units: imperial (ft, psf, pcf)

Embankment (c = 1780 psf, φ = 5°) on a foundation (c = 1600 psf, φ = 2°), analysed with the circle given in the Corps of Engineers example.

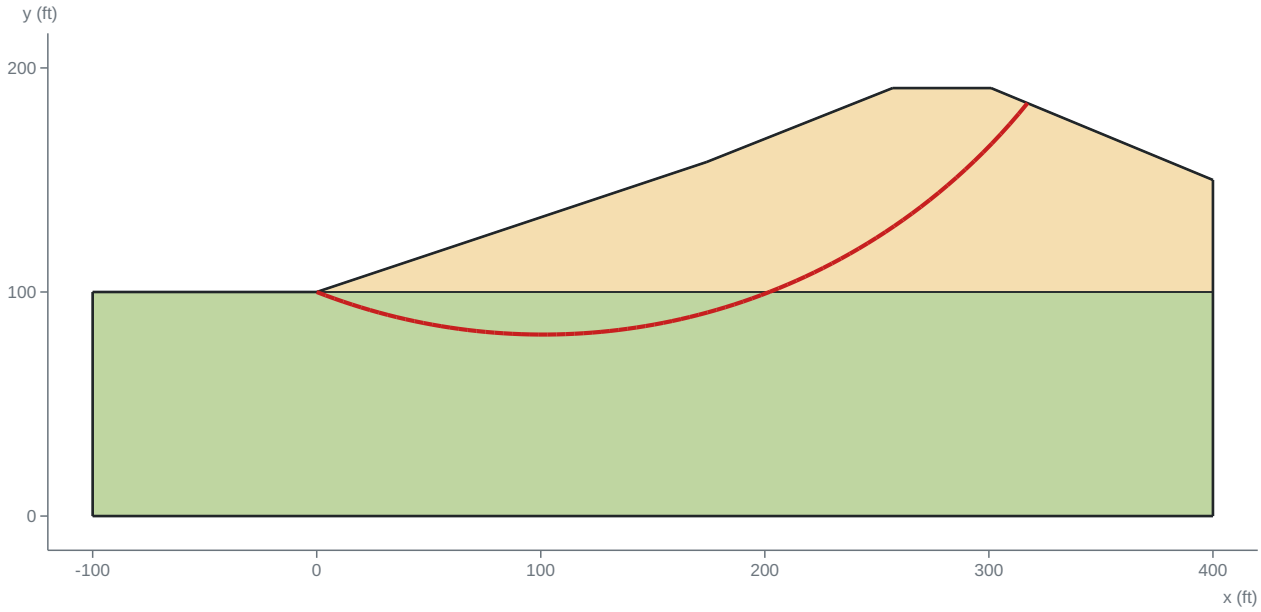


Figure 151. V090: analysed circle (Bishop simplified, FOS = 1.320).

Material	Model	γ (pcf)	Strength (psf)	φ
Embankment	Mohr-Coulomb	135.0	c' = 1780.0	5.00°
Foundation	Mohr-Coulomb	127.0	c' = 1600.0	2.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.332	1.320
Spencer (M-P, constant f(x))	Commercial program A	1.328	1.317
Janbu corrected (f0)	Commercial program A	1.345	1.341

Notes

- USACE's reference value is 1.33 (two digits); commercial program A GLE 1.327.

3.152 V091 — USACE example E-10, ponded water

USACE EM 1110-2-1902, example E-10.

Fixed circle · Units: SI (m, kPa, kN/m³)

Undrained three-layer slope with 8 ft of water ponded against its toe (pool el. 0), on commercial program A's circle (48.262, 28, R = 47.683 ft; Fig. 68.2). Converted to SI.

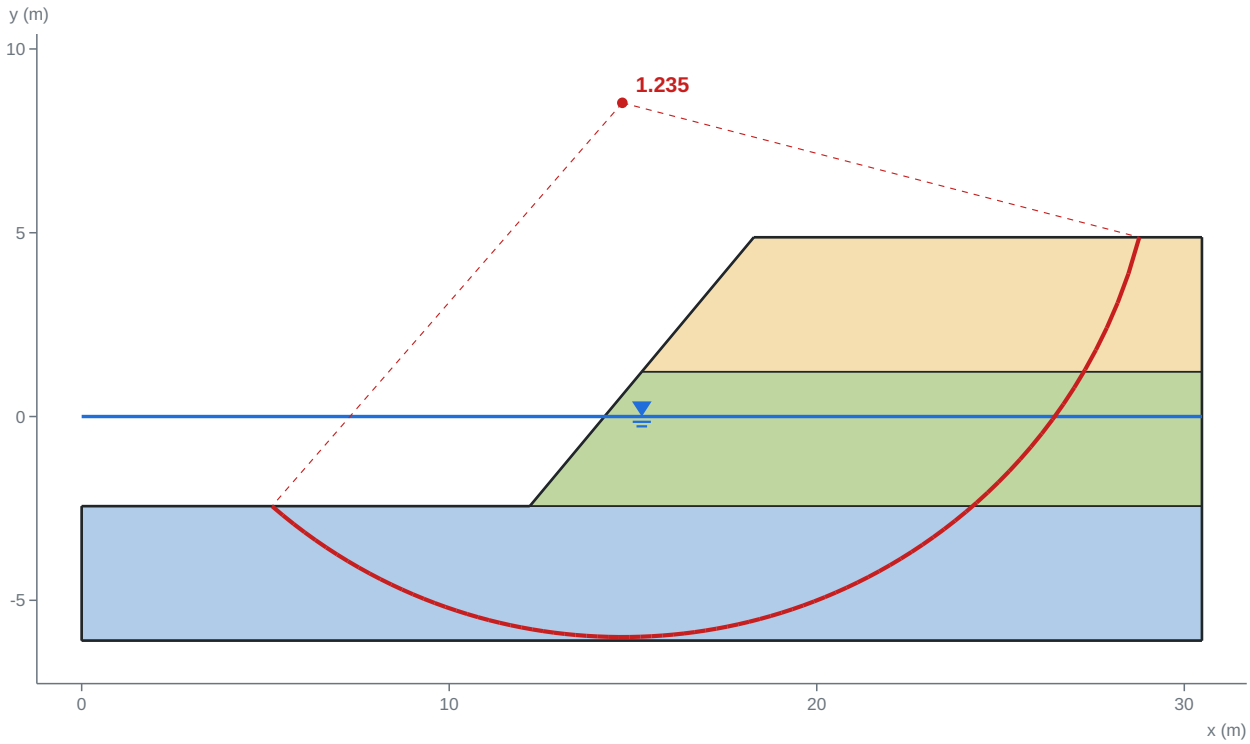


Figure 152. V091: analysed circle (Bishop simplified, FOS = 1.235).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil 1	Undrained (φ = 0)	18.85	Su = 28.73	0°
Soil 2	Undrained (φ = 0)	15.71	Su = 19.15	0°
Soil 3	Undrained (φ = 0)	16.49	Su = 23.94	0°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.241	1.235
Spencer (M-P, constant f(x))	Commercial program A	1.241	1.235
Morgenstern-Price (half-sine)	Commercial program A (GLE)	1.244	1.235
Janbu corrected (f0)	Commercial program A	1.385	1.380

Notes

- The text describes USACE's circle (centre 8.4 ft right of and 36 ft above the toe, tangent to the base of soil 3: 48.4, 28, R = 48); commercial program A's printed circle is slightly smaller. cSlope gives the same FOS on both (1.235).
- The water above the toe acts as a load on the ground surface; the published USACE chart value is 1.33.

3.153 V092 — USACE example F-6, steady-seepage dam

USACE EM 1110-2-1902, example F-6.

Fixed circle · Units: SI (m, kPa, kN/m³)

112 ft embankment on a granular foundation under steady seepage: piezometric line from the pool through the chimney drain to the tailwater, which stands on the downstream toe (free water). USACE's circle. Converted to SI.

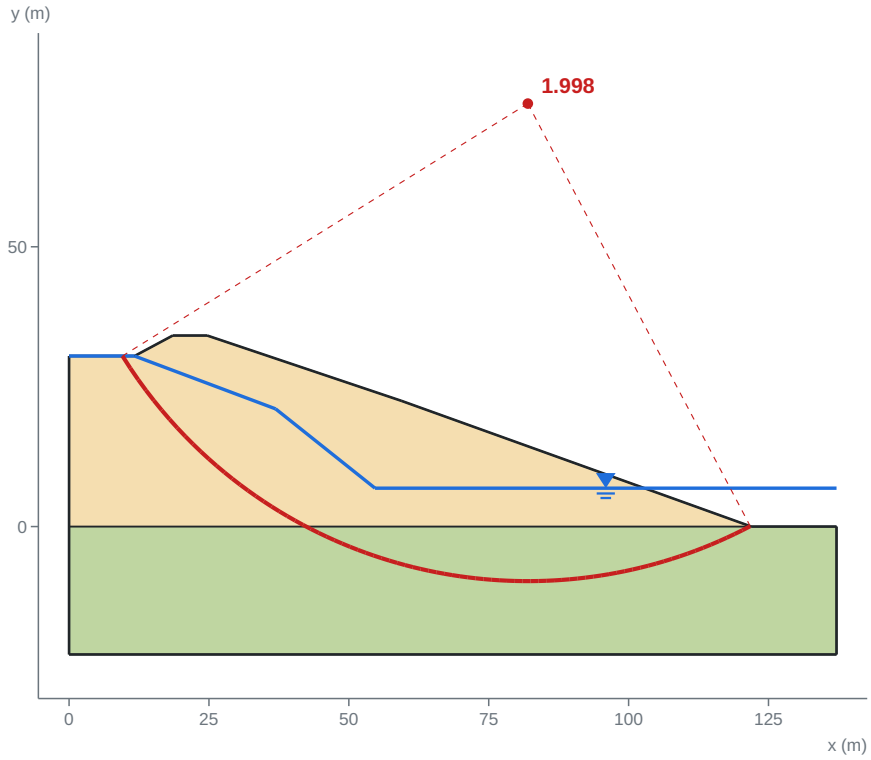


Figure 153. V092: analysed circle (Bishop simplified, FOS = 1.998).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Embankment	Mohr-Coulomb	20.42	c' = 0.00	34.00°
Foundation	Mohr-Coulomb	19.64	c' = 0.00	35.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	2.011	1.998
Spencer (M-P, constant f(x))	Commercial program A	2.026	2.012
Janbu corrected (f0)	Commercial program A	1.830	1.832

Notes

- The kink of the piezometric line is located from the figure to about a foot; program D also reads 0.6 % low.

3.154 V093 — Yamagami & Ueta (1988)

Yamagami & Ueta (1988).

Critical circle search · Units: SI (m, kPa, kN/m³)

Homogeneous 2:1 slope, 5 m high; critical circle search.

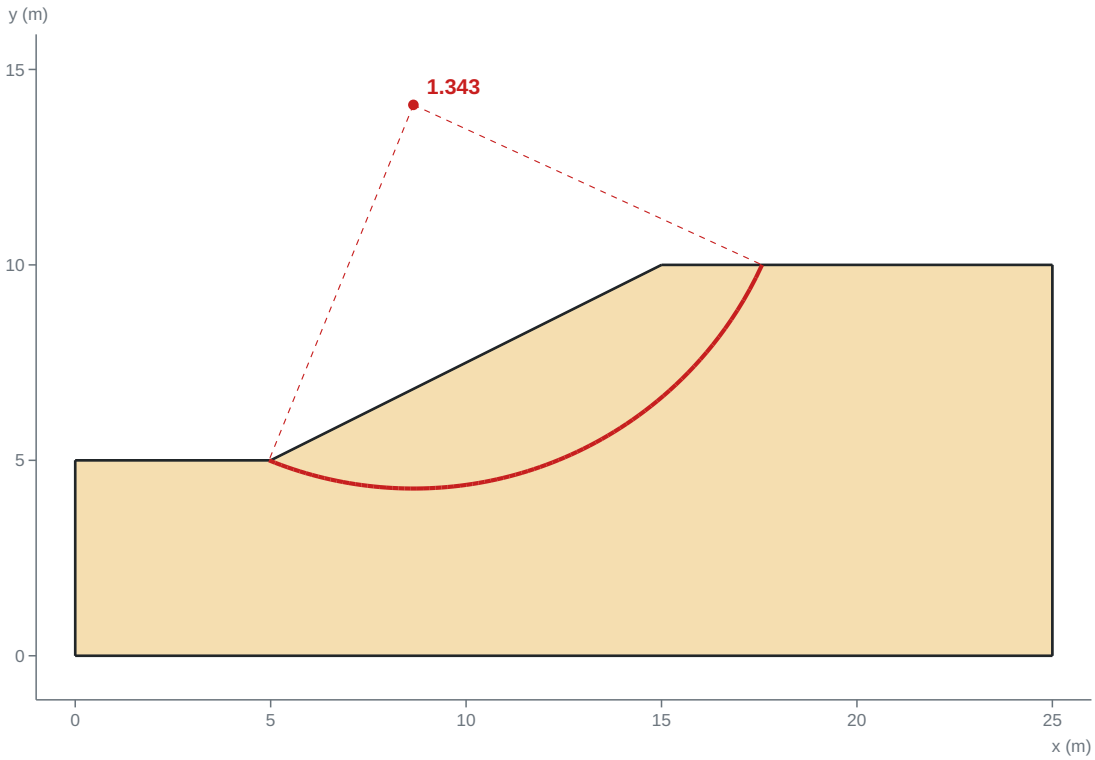


Figure 154. V093: critical circle (Bishop simplified, FOS = 1.343).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	17.64	c' = 9.80	10.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.344	1.343
Spencer (M-P, constant f(x))	Commercial program A (non-circular)	1.325	1.324

Notes

- Yamagami & Ueta: Bishop 1.348, Spencer 1.339 (non-circular); Greco 1.33.

3.155 V094 — Yamagami & Ueta (1988) four-layer slope, non-circular

Greco (1996) example 4; Yamagami & Ueta (1988).

Circle search with non-circular optimisation · Units: SI (m, kPa, kN/m³)

Four-layer slope without water whose critical surface is non-circular; circular search followed by non-circular optimisation.

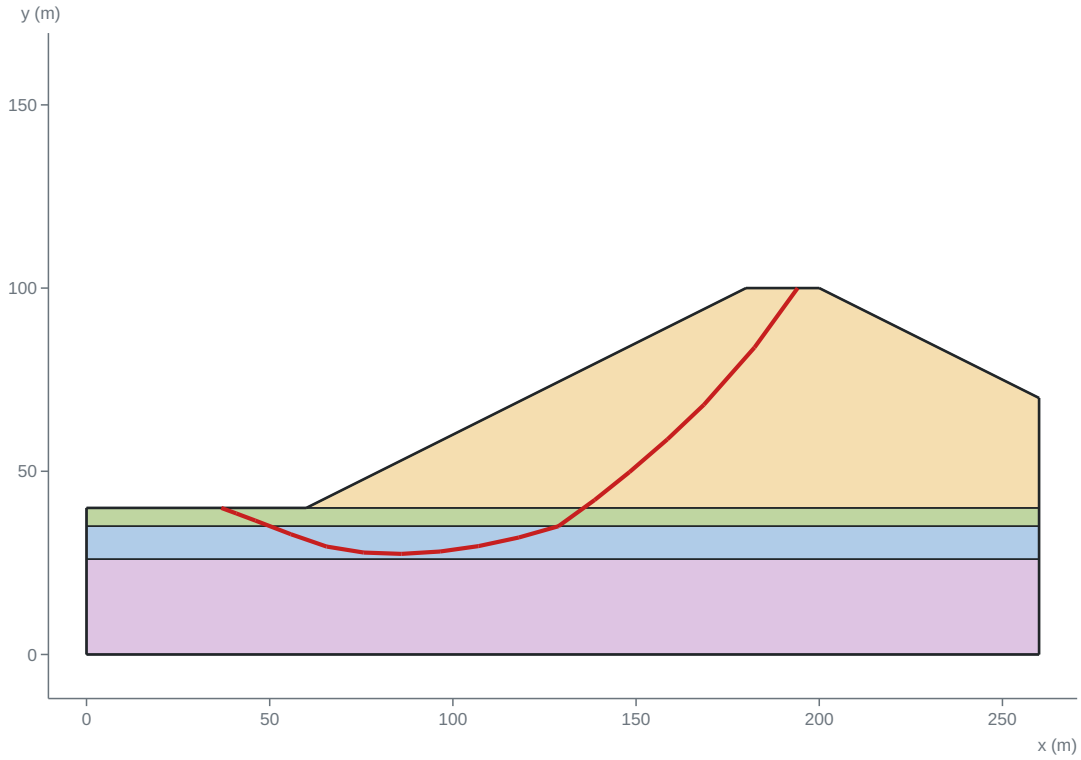


Figure 155. V094: critical surface (Spencer (M-P, constant f(x)), FOS = 1.401).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Upper layer	Mohr-Coulomb	20.38	c' = 49.00	29.00°
Layer 2	Mohr-Coulomb	17.64	c' = 0.00	30.00°
Layer 3	Mohr-Coulomb	20.38	c' = 7.84	20.00°
Bottom layer	Mohr-Coulomb	17.64	c' = 0.00	30.00°

Method	Published by	Reference	cSlope
Spencer (M-P, constant f(x))	Commercial program A (Monte Carlo)	1.398	1.401
Morgenstern-Price (half-sine)	Commercial program B	1.389	1.395

Notes

- Search limited to x = 0–200; the down-slope beyond the crest is not searched, as in the source.
- Greco publishes 1.40–1.42 for the non-circular surface; the best circle gives 1.430.

3.156 V095a — Yamagami micro-pile slope, no pile

Yamagami micro-pile slope example.

Fixed circle · Units: SI (m, kPa, kN/m³)

Homogeneous 2:1 slope, 4 m high, without piles, on the published circle.

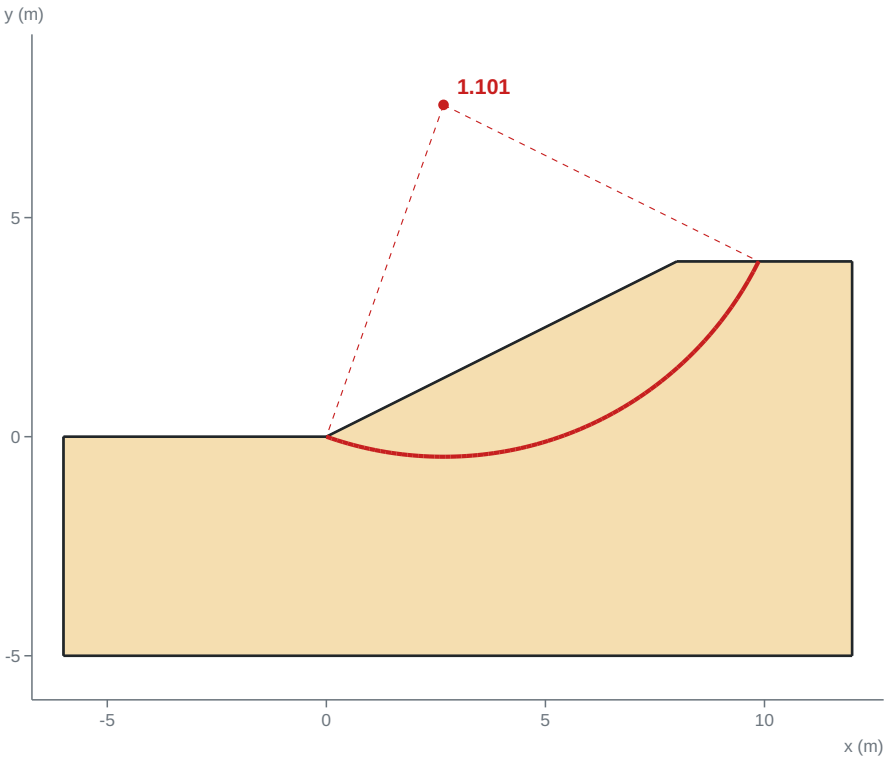


Figure 156. V095a: analysed circle (Bishop simplified, FOS = 1.101).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	15.68	c' = 4.90	10.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.102	1.101

3.157 V095b — Homogeneous slope, no micro piles

Yamagami et al. (2000).

Fixed circle · Units: SI (m, kPa, kN/m³)

Homogeneous slope on the printed critical circle, unreinforced.

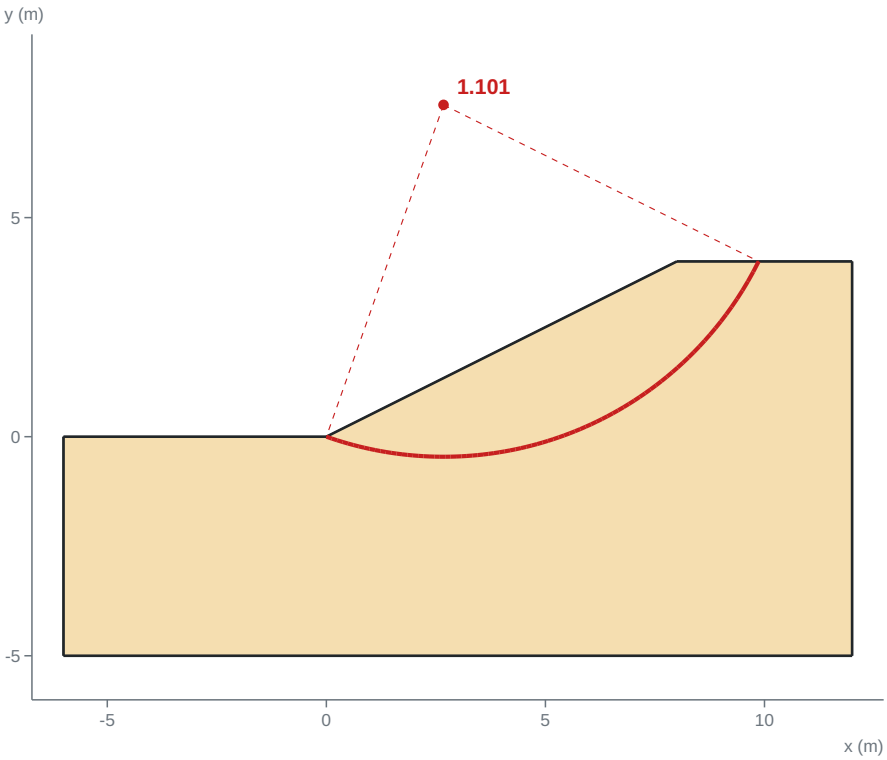


Figure 157. V095b: analysed circle (Bishop simplified, FOS = 1.101).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Material 1	Mohr-Coulomb	15.68	c' = 4.90	10.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.102	1.101

Notes

- Yamagami 1.10, commercial program B 1.102.

3.158 V095c — Homogeneous slope with a micro-pile row

Yamagami et al. (2000).

Fixed circle · Units: SI (m, kPa, kN/m³)

As V095b with a micro-pile row (1 m spacing) whose shear strength of 10.7 kN acts as a passive force tangent to the slip surface, on the reference with-pile circle (2.674, 7.376, R = 8.102).

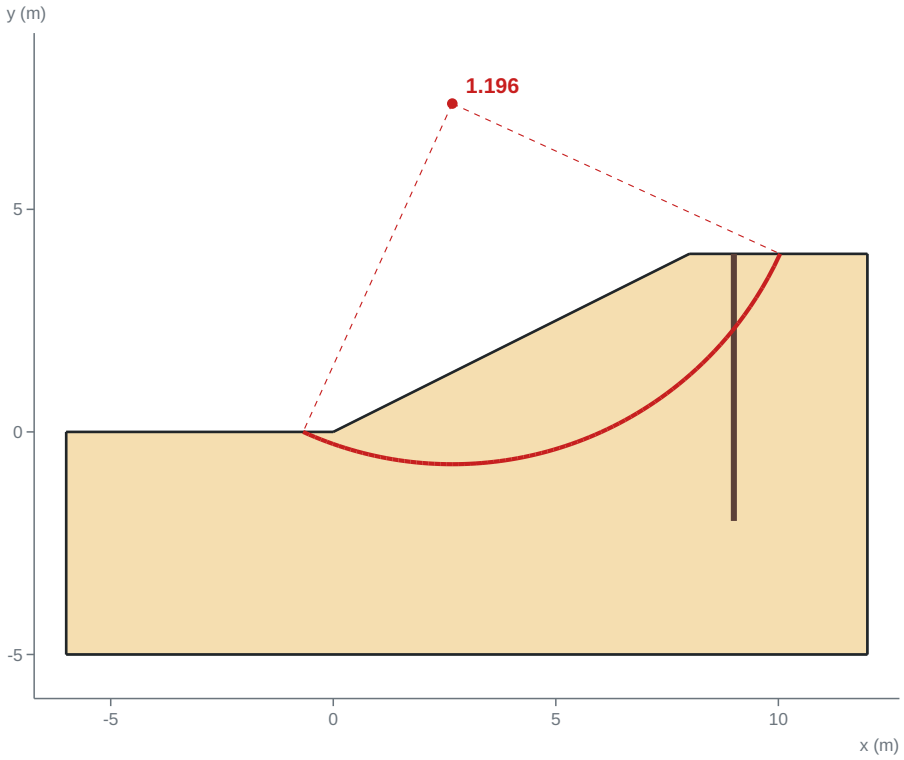


Figure 158. V095c: analysed circle (Bishop simplified, FOS = 1.196).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Material 1	Mohr-Coulomb	15.68	c' = 4.90	10.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.193	1.196

Notes

- Yamagami 1.20, commercial program B 1.223; program D 1.185 (Ito & Matsui force, active, on the no-pile circle).
- On this circle the 10.7 kN passive force gives 1.172 horizontal and 1.195 tilted 51.3° (the base inclination where the circle crosses the pile).

3.159 V096a — Zhu & Lee heterogeneous slope, dry, deep circle

Zhu & Lee (2002).

Critical circle search · Units: SI (m, kPa, kN/m³)

Four horizontal layers under a two-bench slope with a dry tension crack 1.015 m deep at the crest (Fig. 52.1; the depth of commercial program A's surface end below the crest), dry; the circular search lands in the deep family (surface 3).

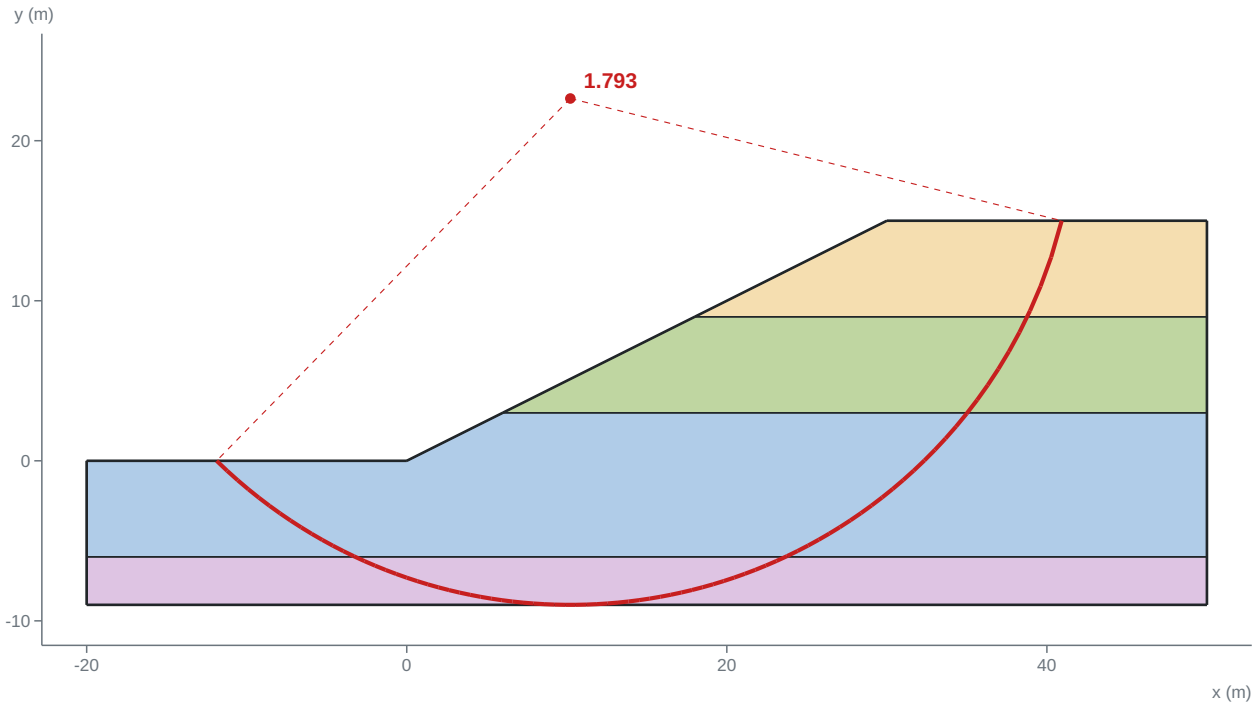


Figure 159. V096a: critical circle (Bishop simplified, FOS = 1.793).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Layer 1 (top)	Mohr-Coulomb	18.80	c' = 20.00	18.00°
Layer 2	Mohr-Coulomb	18.50	c' = 40.00	22.00°
Layer 3	Mohr-Coulomb	18.40	c' = 25.00	26.00°
Layer 4 (bottom)	Mohr-Coulomb	18.00	c' = 10.00	12.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.804	1.793
Spencer (M-P, constant f(x))	Commercial program A	1.804	1.794

Notes

- Entry on the upper bench (x = 33.1–50) and exit on the lower bench (x = −20 to −4.3), as in the source.
- Zhu & Lee (surface 3): Spencer 1.836, M-P 1.823. program D 1.796 (Bishop) without the crack. V096c checks the reference circle itself.

3.160 V096b — Zhu & Lee heterogeneous slope with a water table, deep circle

Zhu & Lee (2002).

Critical circle search · Units: SI (m, kPa, kN/m³)

As V096a (dry 1.015 m crack) with the water table of Table 52.2.

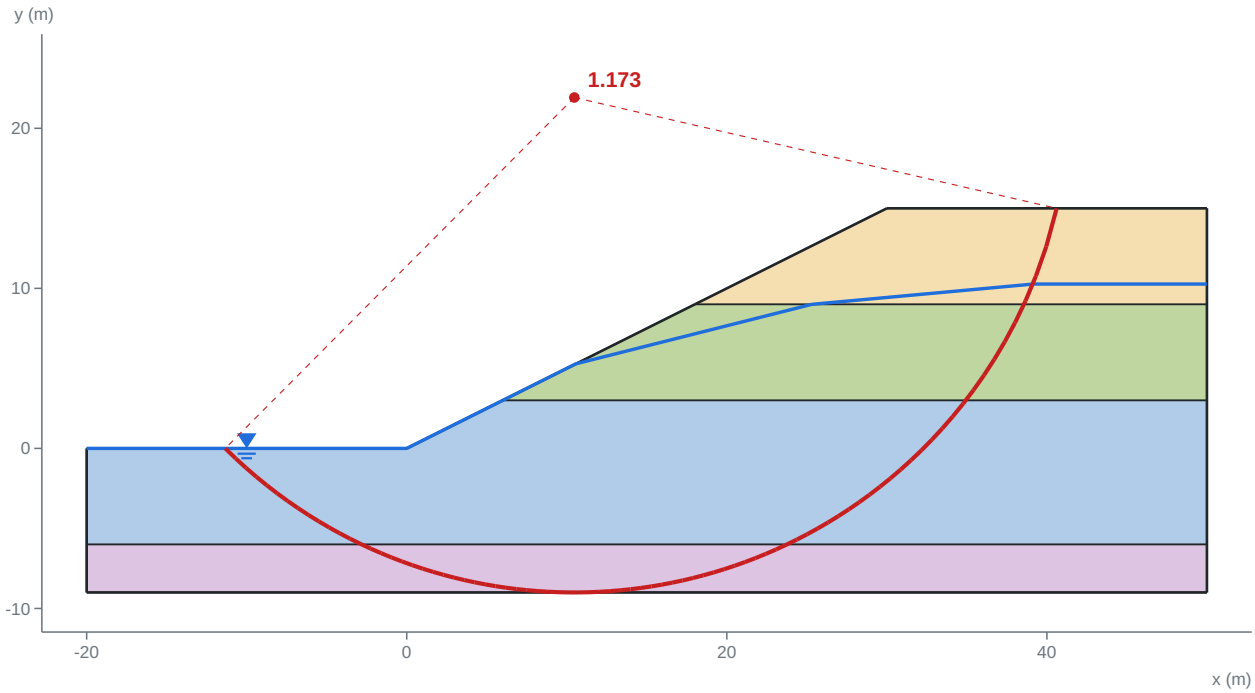


Figure 160. V096b: critical circle (Bishop simplified, FOS = 1.173).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Layer 1 (top)	Mohr-Coulomb	18.80	c' = 20.00	18.00°
Layer 2	Mohr-Coulomb	18.50	c' = 40.00	22.00°
Layer 3	Mohr-Coulomb	18.40	c' = 25.00	26.00°
Layer 4 (bottom)	Mohr-Coulomb	18.00	c' = 10.00	12.00°

Method	Published by	Reference	cSlope
Bishop simplified	Commercial program A	1.176	1.173
Spencer (M-P, constant f(x))	Commercial program A	1.189	1.187

Notes

- Entry on the upper bench (x = 33.1–50) and exit on the lower bench (x = −20 to −4.3), as in the source.
- Zhu & Lee (surface 3): Spencer 1.211, M-P 1.197. program D 1.189 (Spencer) without the crack.

3.161 V096c — Zhu & Lee slope, surface 3 (the reference circle), dry, 1 m crack

Zhu & Lee (2002). Verification manual of a commercial slope stability program.

Fixed circle · Units: SI (m, kPa, kN/m³)

The V096a slope with the dry 1.015 m crack on the reference surface 3 circle (10.911, 23.208, R = 32.106; Fig. 52.4).

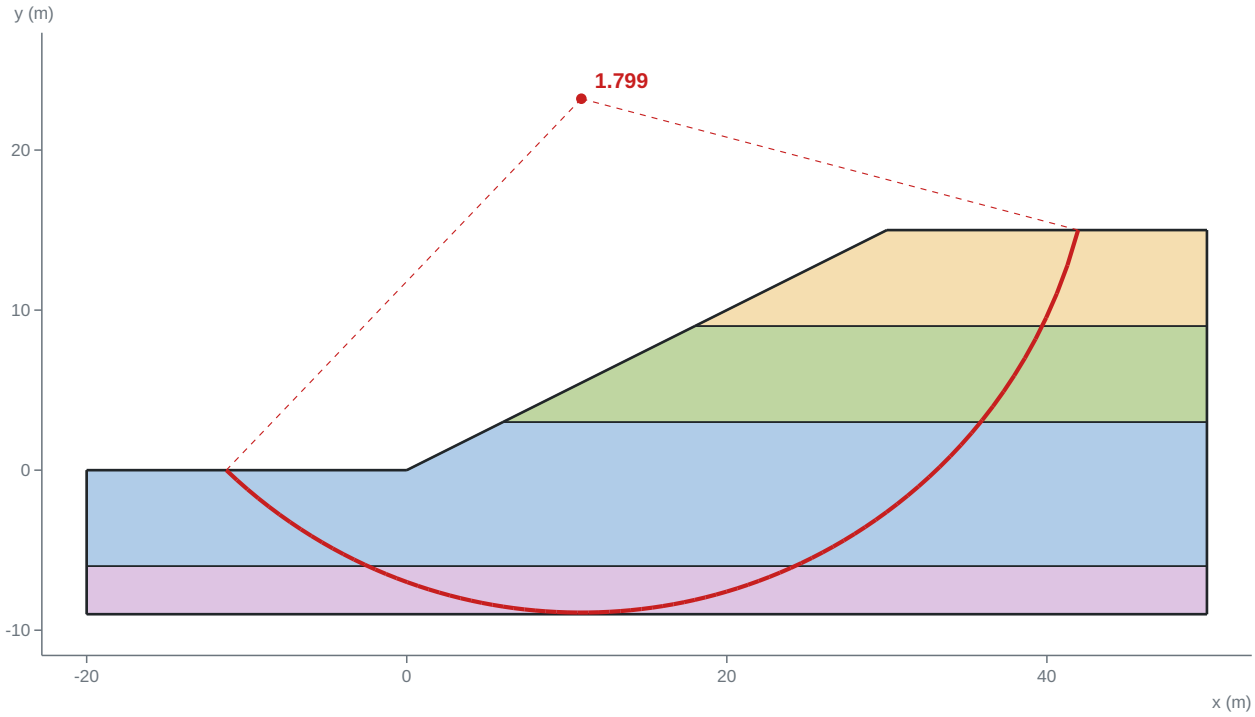


Figure 161. V096c: analysed circle (Spencer (M-P, constant f(x)), FOS = 1.799).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Layer 1	Mohr-Coulomb	18.80	c' = 20.00	18.00°
Layer 2	Mohr-Coulomb	18.50	c' = 40.00	22.00°
Layer 3	Mohr-Coulomb	18.40	c' = 25.00	26.00°
Layer 4	Mohr-Coulomb	18.00	c' = 10.00	12.00°

Method	Published by	Reference	cSlope
Spencer (M-P, constant f(x))	Commercial program A	1.804	1.799
Bishop simplified	Commercial program A	1.804	1.799

Notes

- Crack depth from commercial program A's right endpoint (41.663, 13.985) under the 15 m crest.
- Zhu & Lee: Spencer 1.836, M-P 1.823.

3.162 V097a — Eurocode 7 cutting with surcharge

Verification manual of a commercial slope stability program.

Fixed circle · Units: SI (m, kPa, kN/m³)

Cutting in clay with an inclined piezometric line and a 35 kPa surcharge on the crest, analysed on the published circle.

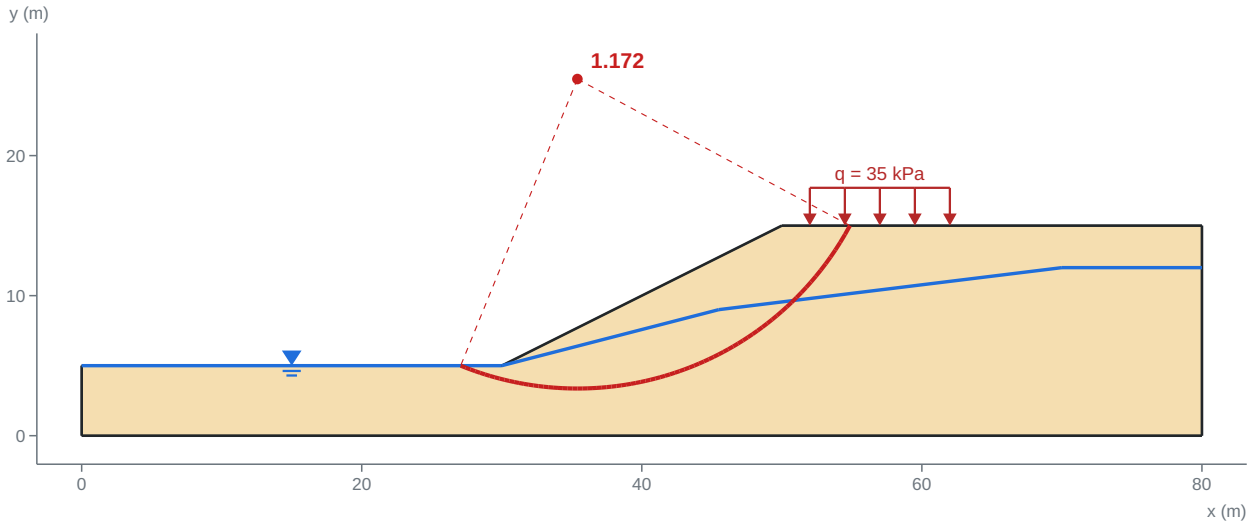


Figure 162. V097a: analysed circle (Bishop simplified (vertical head), FOS = 1.172).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Clay	Mohr-Coulomb	20.00	c' = 8.00	23.04°

Method	Published by	Reference	cSlope
Bishop simplified (vertical head)	Commercial program B	1.173	1.172

3.163 V097b — Eurocode 7 cutting, partial factors applied by cSlope

Verification manual of a commercial slope stability program.

Fixed circle · Units: SI (m, kPa, kN/m³)

As V097a, but with the characteristic strength c' = 10 kPa, φ' = 28° and Design Approach 3 material factors (set M2: γc' = γφ' = 1.25) applied by the analysis; the surcharge is permanent (factor 1.0). The FOS is the over-design factor on design values.

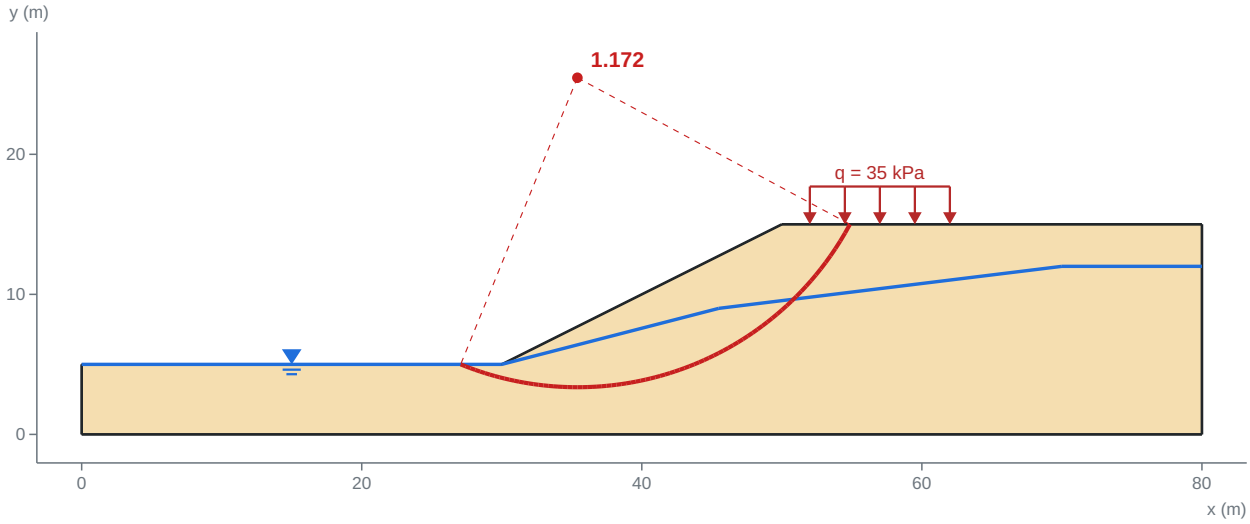


Figure 163. V097b: analysed circle (Bishop simplified, design values, FOS = 1.172).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Clay	Mohr-Coulomb	20.00	c' = 10.00	28.00°

Method	Published by	Reference	cSlope
Bishop simplified, design values	Commercial program B	1.173	1.172
Spencer, design values	Commercial program B	1.174	1.174

3.164 V098 — Example slope without seismic load

Verification manual of a commercial slope stability program.

Critical circle search · Units: SI (m, kPa, kN/m³)

10 m, 2:1 slope in three layers over a horizontal base, dry, no seismic load; Spencer search.

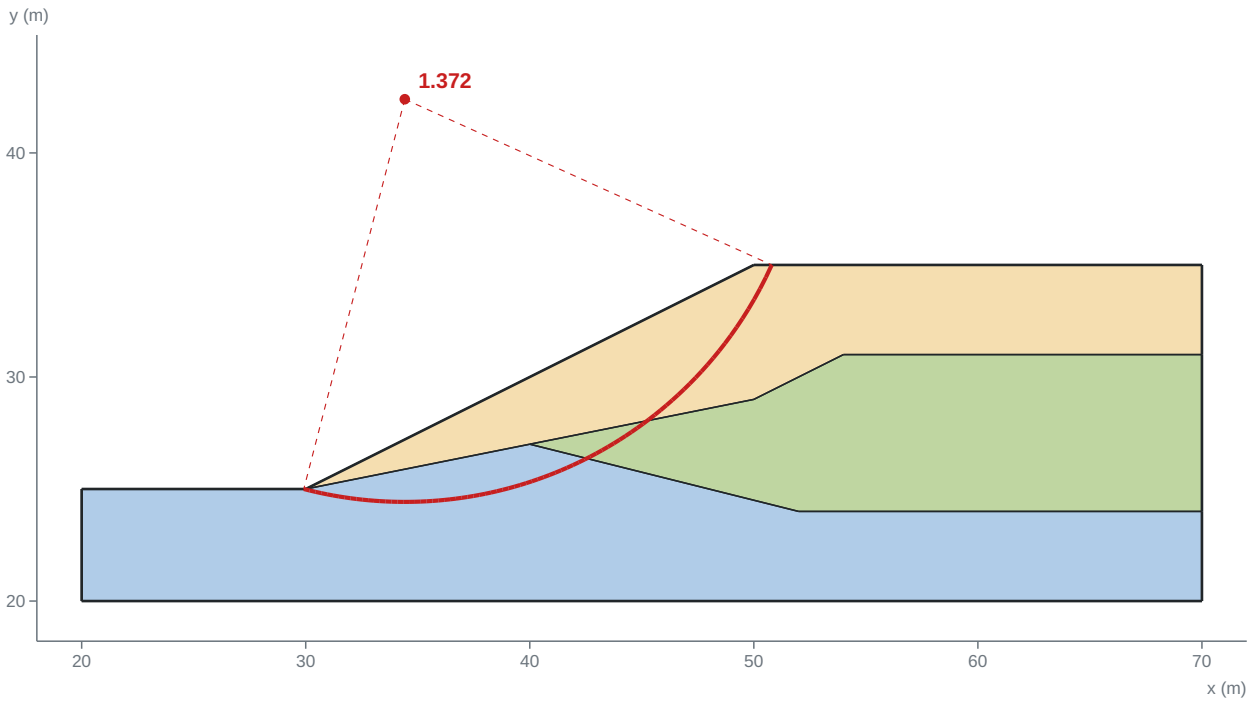


Figure 164. V098: critical circle (Spencer (M-P, constant f(x)), FOS = 1.372).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
soil1	Mohr-Coulomb	19.50	c' = 0.00	38.00°
soil2	Mohr-Coulomb	19.50	c' = 5.30	23.00°
soil3	Mohr-Coulomb	19.50	c' = 7.20	20.00°

Method	Published by	Reference	cSlope
Spencer (M-P, constant f(x))	Commercial program A	1.360	1.372
Spencer (M-P, constant f(x))	Program D	1.372	1.372

Notes

- commercial program A multi-modal optimisation 1.359. The k = 0.15 and yield-acceleration cases need a seismic coefficient.

4 Benchmark models bundled with cSlope

These models are shipped with the program (Help → Benchmarks). They are run through the same request builder as the user interface, with the analysis options saved in each model.

4.1 T-ACADS Simple

Bundled model "T-ACADS Simple" · Units: SI (m, kPa, kN/m³) · Critical circle search · 30 slices, 1000 + 500 search iterations

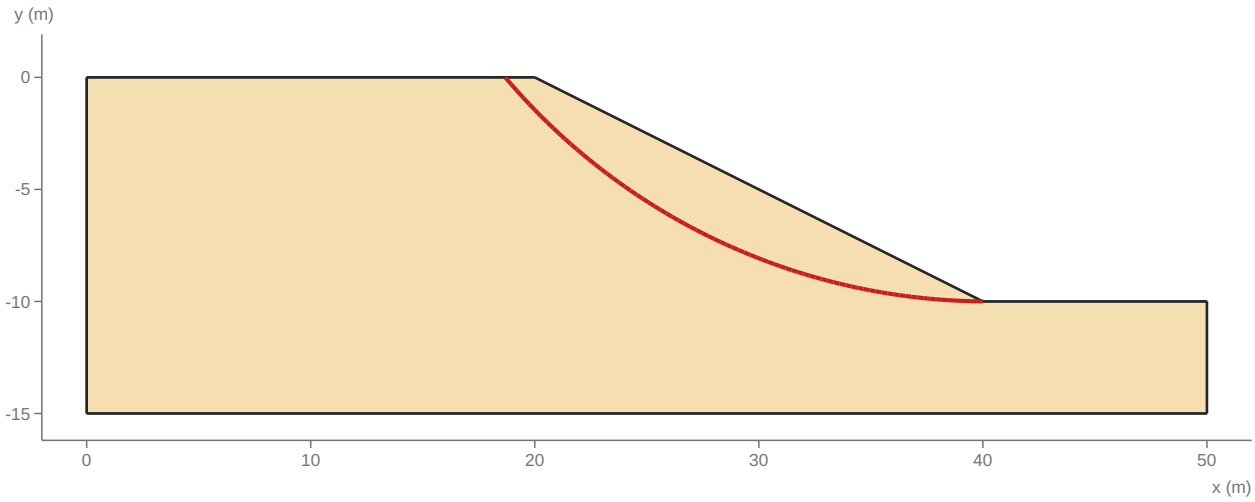


Figure 165. T-ACADS Simple: critical circle (Morgenstern-Price, FOS = 0.984).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
M1	Mohr-Coulomb	20.00	c' = 3.00	19.60°

Method	Reference	cSlope
Morgenstern-Price	0.984	0.984
Bishop simplified	0.985	0.984
Janbu simplified	0.938	0.935

4.2 T-ACADS Non-Homogeneous

Bundled model "T-ACADS Non-Homogeneous" · Units: SI (m, kPa, kN/m³) · Critical circle search · 30 slices, 1000 + 500 search iterations

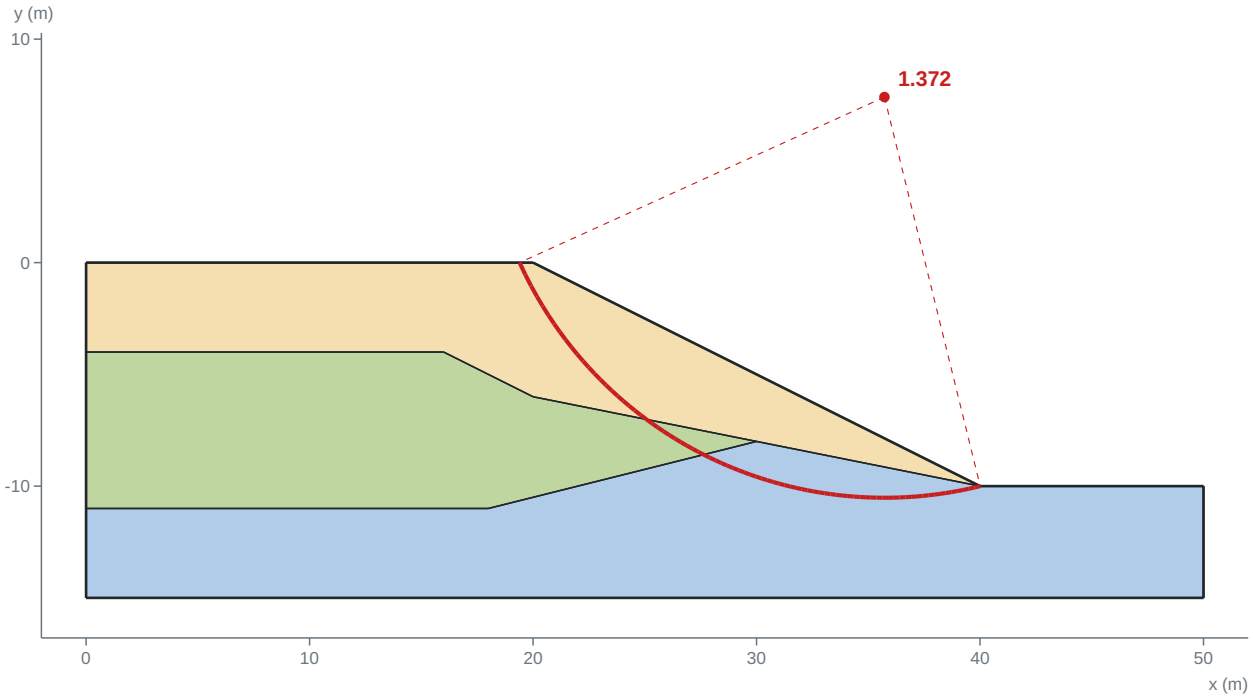


Figure 166. T-ACADS Non-Homogeneous: critical circle (Morgenstern-Price, FOS = 1.372).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil #1	Mohr-Coulomb	19.50	c' = 0.00	38.00°
Soil #2	Mohr-Coulomb	19.50	c' = 5.30	23.00°
Soil #3	Mohr-Coulomb	19.50	c' = 7.20	20.00°

Method	Reference	cSlope
Morgenstern-Price	1.373	1.372

4.3 Arai & Tagyo (1985)

Bundled model "Arai & Tagyo (1985)" · Units: SI (m, kPa, kN/m³) · Critical circle search · 30 slices, 1000 + 500 search iterations

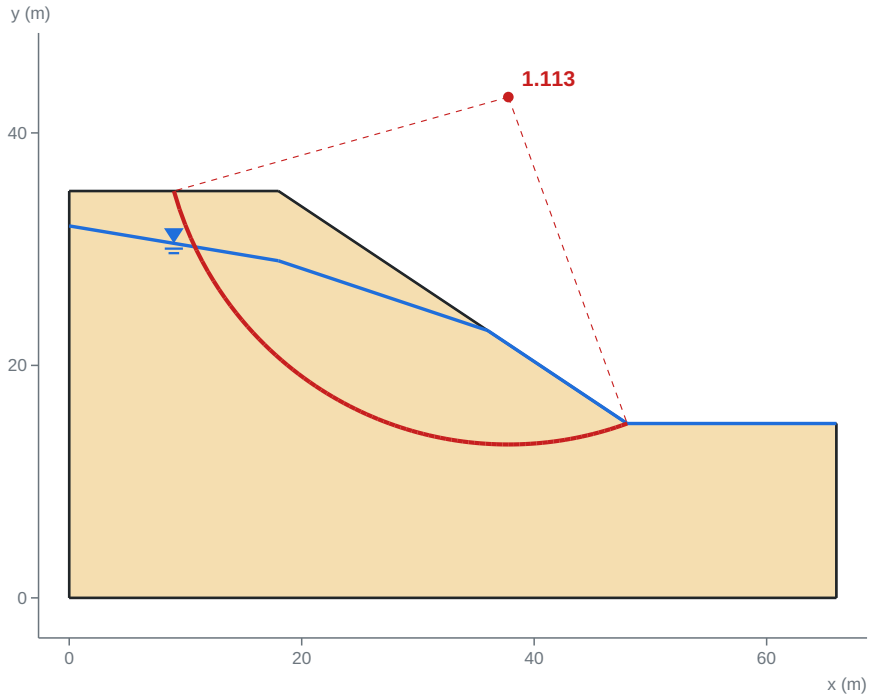


Figure 167. Arai & Tagyo (1985): critical circle (Bishop simplified, FOS = 1.113).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	18.82	c' = 41.65	15.00°

Method	Reference	cSlope
Bishop simplified	1.118	1.113

4.4 Arai & Tagyo (1985), weak layer

Bundled model "Arai & Tagyo (1985), weak layer" · Units: SI (m, kPa, kN/m³) · Critical circle search · 30 slices, 1000 + 500 search iterations

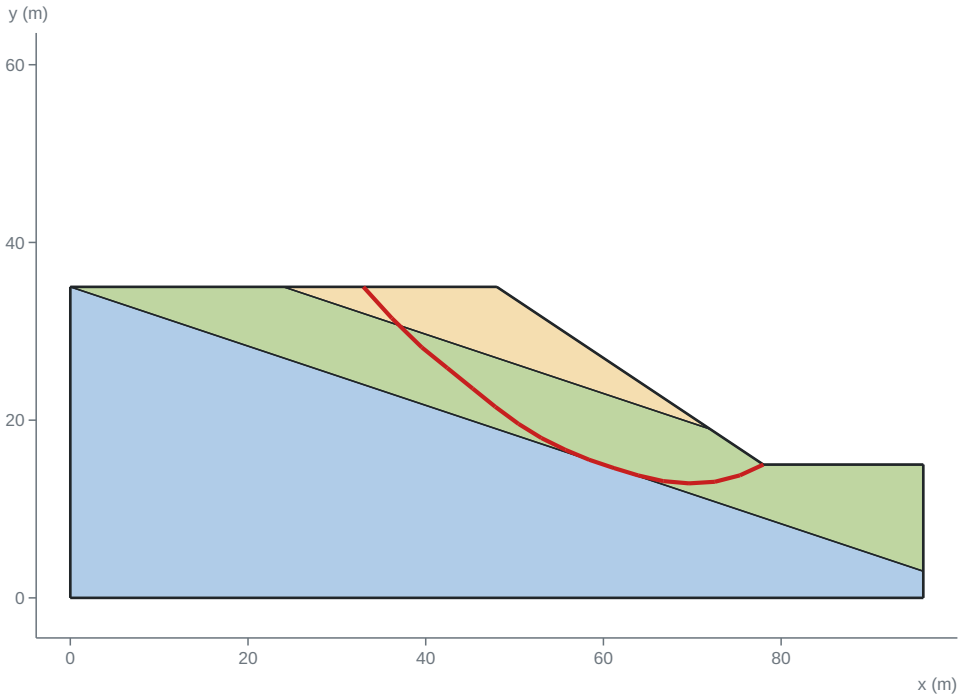


Figure 168. Arai & Tagyo (1985), weak layer: critical circle (Morgenstern-Price, FOS = 0.412).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Upper	Mohr-Coulomb	18.82	c' = 29.40	12.00°
Weak layer	Mohr-Coulomb	18.82	c' = 9.80	5.00°
Lower	Mohr-Coulomb	18.82	c' = 294.0	40.00°

Method	Reference	cSlope
Morgenstern-Price	0.414	0.412

4.5 Talbingo Dam

Bundled model "Talbingo Dam" · Units: SI (m, kPa, kN/m³) · Critical circle search · 30 slices, 1000 + 500 search iterations

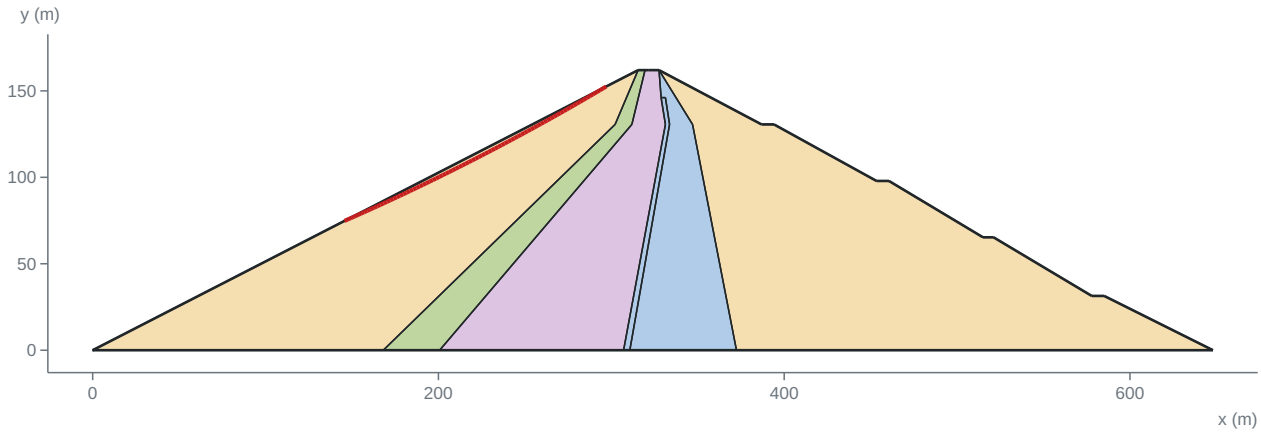


Figure 169. Talbingo Dam: critical circle (Morgenstern-Price, FOS = 1.951).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Rockfill	Mohr-Coulomb	20.40	c' = 0.00	45.00°
Transition	Mohr-Coulomb	20.40	c' = 0.00	45.00°
Filter	Mohr-Coulomb	20.40	c' = 0.00	45.00°
Core	Mohr-Coulomb	18.10	c' = 85.00	23.00°

Method	Reference	cSlope
Morgenstern-Price	1.950	1.951

4.6 Talbingo Dam (Single Circle)

Bundled model "Talbingo Dam (Single Circle)" · Units: SI (m, kPa, kN/m³) · Fixed circle · 30 slices, 1000 + 500 search iterations

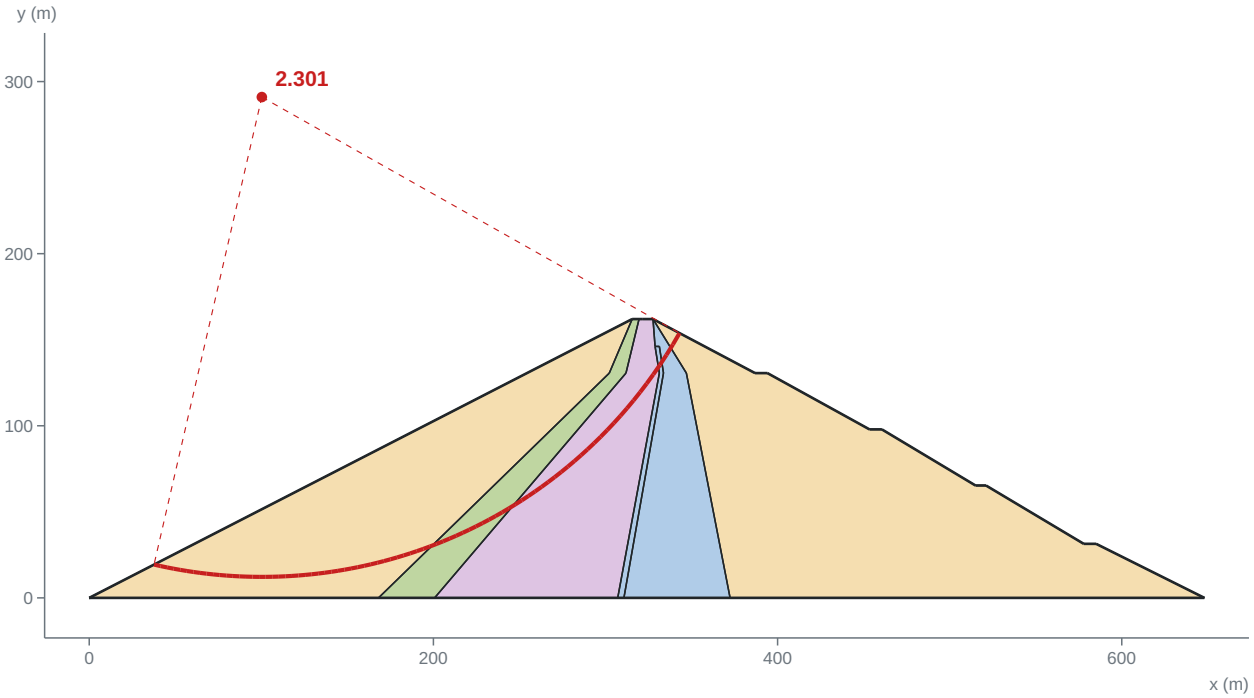


Figure 170. Talbingo Dam (Single Circle): critical circle (Morgenstern-Price, FOS = 2.301).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Rockfill	Mohr-Coulomb	20.40	c' = 0.00	45.00°
Transition	Mohr-Coulomb	20.40	c' = 0.00	45.00°
Filter	Mohr-Coulomb	20.40	c' = 0.00	45.00°
Core	Mohr-Coulomb	18.10	c' = 85.00	23.00°

Method	Reference	cSlope
Morgenstern-Price	2.290	2.301